
بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ
IN PC
PRESS CRTL+F
SO U CAN SEARCH!!
MTH
501
"Always Push Yourself
a little and watch your
Potential rise

1. **If A is an invertible square matrix then**

None of the given option….confirm

If A is invertible, then $\det(A)\det(A^{-1})=1$.

► True

2. **If A=transpose of A(where A is a square matrix), then which of the following is the most appropriate option for A?**

A is symmetric matrix…..confirm

If A is equal to the transpose of A (A = A^T), then A is a symmetric matrix.

3. **IF AB = I = BA for matrices A, B and I where I is an identity matrix, then**

All of the given options….confirm

If AB = I = BA for matrices A, B and I, where I is an identity matrix, then

1. B is inverse of A
2. A is inverse of B
3. $A^{(-1)} = B$, $B^{(-1)} = A$,
4. All of the given

4. **IF A =** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$**, then which of the following is |A|.**

ad - bc….confirm

To find the determinant (|A|) of a 2×2 matrix A, you can use the following formula:

|A| = ad - bc

Given the matrix A:

5. **A partitioned square matrix "a" is said to be block diagonal matrix if the matrices on the main diagonal are square and all matrices above the main diagonal are _____.**

Zero…..confirm

Block Upper Triangular Matrices

A partitioned square matrix A is said to be block upper triangular if the matrices on the main diagonal are square and all matrices below the main diagonal are zero; that is, the matrix is partitioned as

6. **A system of linear equations is said to be homogenous if the constant terms are all.**

Zero….confirm

A system of linear equations is said to be homogeneous if the constant terms are all

1. One
2. Zero
3. Both (a) and (b)
4. None of the above

7. **Which of the following is true for the partitioned matrices Y = | Q R| and Z = |U V| where sub-matrices Q and R have the same sizes as U and V.**

$$Y+Z=\begin{vmatrix} Q+U & R+U \end{vmatrix}$$ **....confirm**

The true statement among the options provided is:

Y+Z=|Q+U R+V|

When adding two partitioned matrices Y and Z, the addition is performed element-wise between corresponding sub-matrices. The resulting matrix will have sub-matrices that are sums of the corresponding sub-matrices from Y and Z.

Therefore, the correct expression for Y+Z is:
Y+Z=|Q+U R+V|

8. **Multiplication of a partitioned matrix by a sector is also computed _____.**

Block by block……confirm

Multiplication of a partitioned matrix by a scalar is also computed block by block.

9. **If A is invertible and b in R^n be any vector. Then we must have a matrix $A^{-1}b$. which is a solution of _____.**

$$\text{i.e. } Ax = A(A^{-1}b) = Ib = b.$$

….confirm

Proof

Since A is invertible and $b \in R^n$ be any vector. Then, we must have a matrix $A^{-1}b$ which is a solution of eq. (1) .i.e. $Ax = A(A^{-1}b) = Ib = b.$

10. A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a ____ matrix.

Blocked Toeplitz matrix or blocked diagonal constant……confirm

Block Toeplitz matrix

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a blocked Toeplitz matrix.

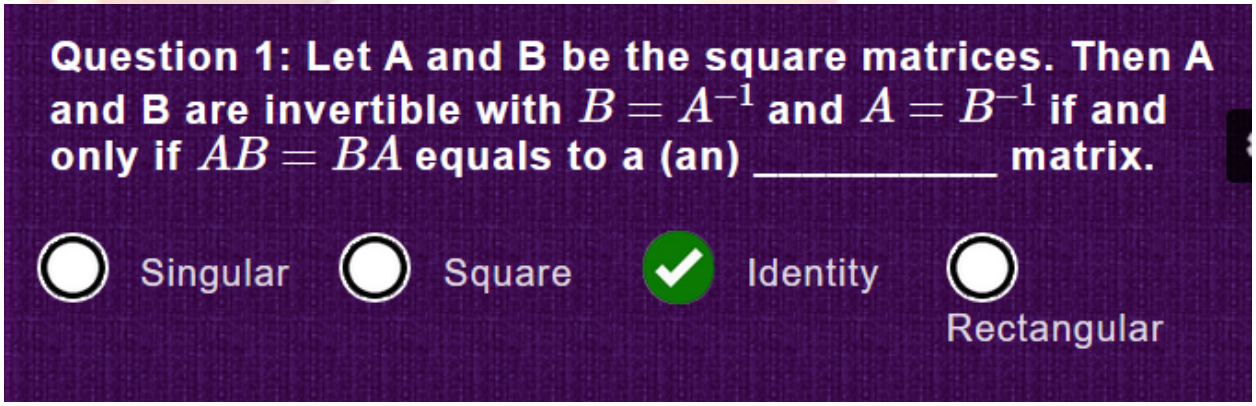
11. Which one of the following is a diagonal matrix?

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$ **……confirm**

For example $\begin{bmatrix} 5 & 0 \\ 0 & 7 \end{bmatrix}$ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 11 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ are all diagonal matrices.

12. Let A and B be the square matrices. The A and B are invertible with B = A^{-1} and A = B^{-1} if and only if AB=BA equals to a(An) _____matix.

Identiy….confirm



13. If the inverse of a linear transformation exists then it is _____.

Unique……confirm

Important Note

If the inverse of a linear transformation exits then it is unique.

14. The matrix of coefficients either have no solution or have infinite solution is system of equation are

Linearly dependent or inconsistent…..confirm

Now, if the matrix of coefficients either has no solution or has infinite solutions, it means that the system of equations is either linearly dependent or inconsistent. Linearly Dependent: A set of equations is linearly dependent if one or more of the equations can be expressed as a linear combination of the others.

15. Let T be a linear Transformation and A be a standard matrix for 1, then T is invertible if and only if A is a (an) _____ matrix.

Invertible…..confirm

17. Let $T : R^n \rightarrow R^n$ be a linear transformation and let A be the standard matrix for T. Then T is invertible if and only if A is an invertible matrix.

16. A matrix can be subdivided into sub matrices in various ways by inserting lines between selected _____ and columns.

Rows…..confirm

General Partitioning of a Matrix -

A matrix can be **partitioned** (subdivided) into **sub matrices** (also called **blocks**) in various ways by inserting lines between selected rows and columns.

17. If A is an invertible square matrix then

None of options……confirm

Mathematically, if A is an invertible square matrix, then:

A * A^(-1) = A^(-1) * A = I

18. A partitioned matrix is a partition of a matrix into rectangular smaller matrices called _____.

Blocks…..confirm

A **block matrix** or a **partitioned matrix** is a partition of a matrix into rectangular smaller matrices called **blocks**.Partitioned matrices appear often in modern applications

19. A matrix in which each descending diagonal from left to right is constant is called a _____ matrix.

Toeplitz matrix or diagonal-constant matrix……confirm

Toeplitz matrix

A matrix in which each descending diagonal from left to right is constant is called a **Toeplitz matrix** or **diagonal-constant matrix**

20. A homogeneous linear system always has the trivial solution; there are only two possibilities for its solutions:

The system has for infinitely many solutions in addition to the trivial solutiom……confirm

Question 1: A homogeneous linear system always has the trivial solution: there are only two possibilities for its solutions:

- The system has only the trivial solution
- ✓ The system has infinitely many solutionsnin addition to rivial solution

21. A partitioned square matrix "a" is said to be _____ matrix if the matrices on the main diagonal are square and all matrices above the main diagonal are zero.

Block diagonal or Block upper triangle…….confirm

Block Upper Triangular Matrices

A partitioned square matrix A is said to be block upper triangular if the matrices on the main diagonal are square and all matrices below the main diagonal are zero; that is, the matrix is partitioned as

22. Let T: R^2 ->R^2 be transformation T(r_1, r_2)=(x,0). The null space (or kernel), N9(T) of T is

(0, x_2)……confirm

Let T:R^2 -> R^2 be the transformation T(x_1,x_2) = (x_1,0). The null space (or kernel) N(T) of T is

a) (x_1,0) : x_1 is real
b) 1
c) (0,1)
d) (0,x_2) : x_2 is real

Correct answer is option 'D'. Can you explain this answer?

Most Upvoted Answer

Let T:R2-> R2be the transformation T(x1,x2) = (x1,0). The null spac...

B/z for every x belong kernel T iff T (x) =0 now putting (x1,0) we ge x1=0while x2 can not be defined so d is correct

23. A null space vector space.

True……confirm

A null space is a vector space.

- ▶ True
- ▶ False

24. In matrics, rows are denoted by _____.

R

25. $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

26. The elementary row operation $R_2^1 -> R_2 + 4R_1 \; and R_3^1 -> R_3 - 6R_1$ **are performed on to get** $\begin{pmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{pmatrix}$ **_____?**

$\begin{pmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{pmatrix}$

27. We can add the matrices of _____.

Same order……confirm

No We can not. For matrix arithmetic(addition, subtraction) both the matrices need to be of same order.

28. The determinant of a triangular matrix is the sum of the entries of the main diagonal.

False….confirm

False - The determinant of a triangular matrix is the PRODUCT of the entries on the main diagonal.

29.Non square matrices do not have inverse.

True…..confirm

Non-square matrices (m-by-n matrices for which m ≠ n) do not have an inverse. However, in some cases such a matrix may have left inverse or right inverse. If A is m-by-n and the rank of A is

30.If determinant of a matrix is not zero then which of the following is true for that system?

Inversion method applies…..confirm

. Inverse of a matrix exists if its determinant is not equal to 0.

31.If determinant of a matrix is zero then which of the following is true for that system?

The inversion method falls…..confirm

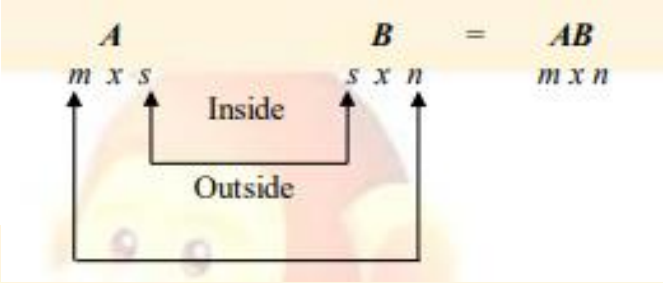
32.If A = transpose of A (where A is a square matrix) then which of the following is the most appropriate option for A?

A is invertible matrix…..confirm

c. If A is an invertible matrix, then so is A^T; and the inverse of A^T is the transpose of A^{-1}. That is $(A^T)^{-1} = (A^{-1})^T$

33.Let A be the matrix of order 3x2, B be the matrix of order 2x4 and C be the matrix of order 4x5 then which of the following is the order of the matrix ABC?

3x5….confirm



34. Which one of the following is a null matrix?

Null Matrix or Zero Matrix

An $m \times n$ matrix whose entries are all zero is a Null or **zero matrix** and is always written as O. A null matrix may be of any order.

For example $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (3 x 3) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ (3 x 2) $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ (4 x 5)

are all Zero Matrices

….confirm

35. Let A be a square n x n matrix, then which of the following statement(s) is (are) equivalent?

(d) option (A to c)..All options are confirm

Theorem (Invertible Matrix Theorem) Let A be a square $n \times n$ matrix. Then the following statements are equivalent. (Means if any one holds then all are true).

(a) A is an invertible matrix.

(b) A is row equivalent to the $n \times n$ identity matrix.

(c) A has **n** pivot positions.

(d) The equation $Ax = 0$ has only the trivial solution.

(e) The columns of A form a linearly independent set.

(f) The linear transformation $x \rightarrow Ax$ is one-to-one.

(g) The equation $Ax = b$ has at least one solution for each b in R^n.

(h) The columns of A span R^n.

(i) The linear transformation $x \rightarrow Ax$ maps R^n onto R^n.

(j) There is a $n \times n$ matrix C such that $CA = I$.

(k) There is a $n \times n$ matrix D such that $AD = I$.

(l) A^T is an invertible matrix.

36. What is the maximum possible number of pivots a 3 by 3 matrix?

3 pivots…..confirm

What is the maximum possible number of pivots in a 3 by 3 matrix?

3 pivots

37. What is the maximum possible number of pivots a 6 by 6 matrix?

6 pivots…..confitm

What is the maximum possiblle number of pivots in a 6 × 6 matrix?

1. 0
2. 2
3. 4
4. 6

38. What is the maximum possible number of pivots in a 4x6 matrix?

4…..confirm

The largest possible number of pivots a 4 x 6 matrix can have is 4. This is because a pivot can only occur in each row once, and there are 4 rows in the matrix.

39.A _____ in a matrix A is a location in A that corresponds to be leading entry in an echelon form of A.

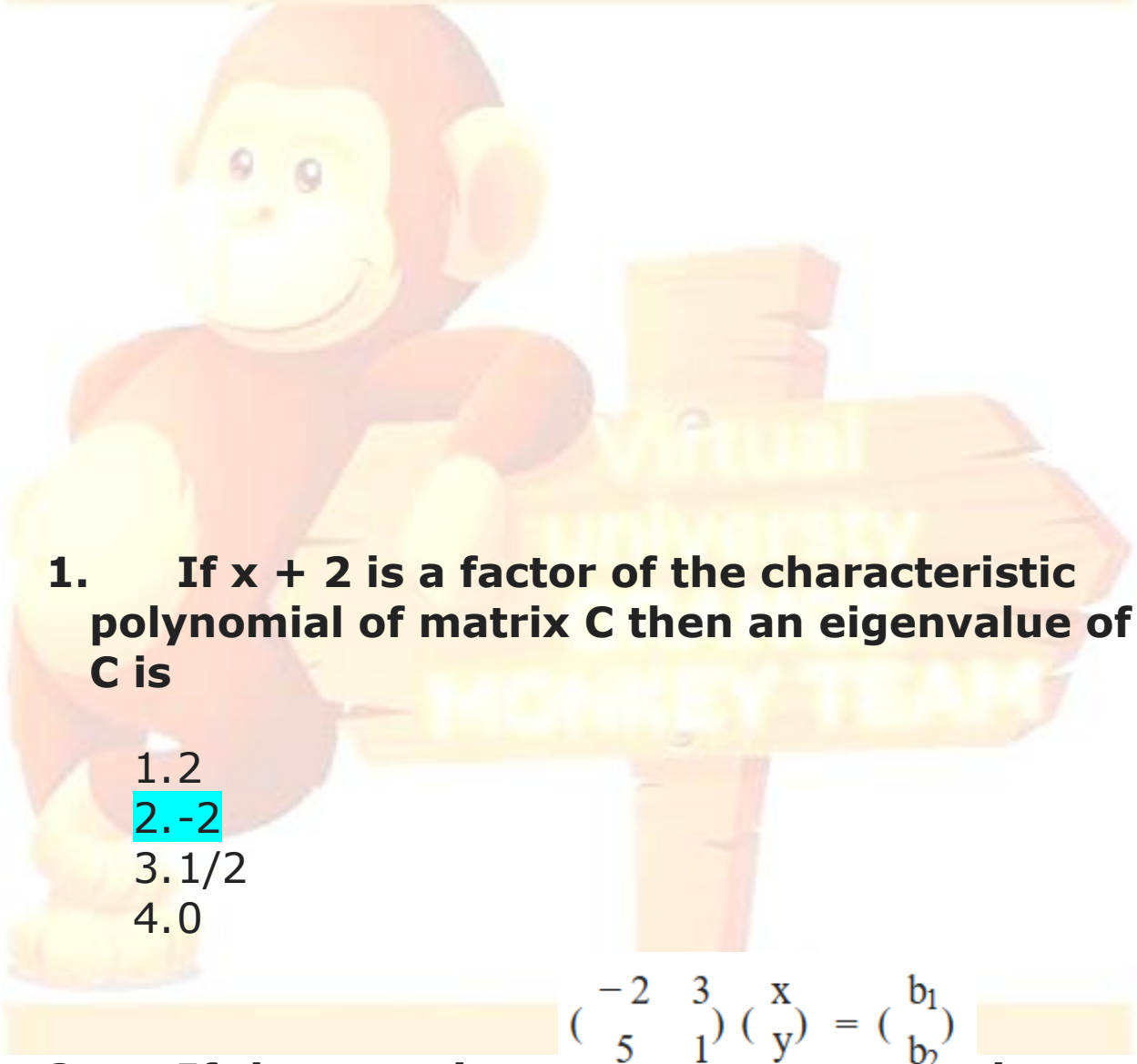
Pivot position…..confirm

Pivot Positions
A pivot position in a matrix A is a location in A that corresponds to a leading entry in an echelon form of A.

40.A _____ is a column of A that contains a pivot position.

Pivot column…..confirm

A **pivot column** is a column of A that contains a pivot position.



1. If x + 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

1. 2
2. -2
3. 1/2
4. 0

2. If the equation: $\begin{pmatrix} -2 & 3 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ **has the solution for all ,** $b_1, b_2 \in R$**, then** $\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \in$ _____.

$$\text{Span}\left\{\begin{pmatrix}-2\\5\end{pmatrix},\begin{pmatrix}3\\1\end{pmatrix}\right\}$$

3. In A is a square matrix, then the minor of entry ith row and jth column is to be the determinant of the sub matrix that remains when the ith row and jth column of A are

1. added
2. deleted
3. multiplied
4. divided

4. A system of linear equations is said to be homogeneous if the constant terms are all

1. One
2. Zero
3. Both (a) and (b)
4. None of the above

5. Let A be n × n matrix, then A is invertible if and only if

1. det A is not zero
2. det A is zero

6. Let a matrix A has both negative and positive eigen values, so in this case origin behaves as a _____ point.

1. Saddle
2. Critical

7. Let A be the matrix of order 2x3 and B be the matrix of order 3x5, then which of the following is the order of the matrix AB?

1. 2x3
2. 3x5
3. 3x3
4. 2x5

8. Let A and B be the square matrices. Then A and B are invertible with $B=A^{-1}$ and $A=B^{-1}$ if and only if equals to a (an) _____ matrix.

1. Singular
2. Square
3. Identity
4. Rectangular

9. If $\vec{v_1}, \vec{v_2}$ and $\vec{v_3}$ are in R^m then which of the following is equivalent to $[v_1 \quad v_2 \quad v_3]\begin{bmatrix} 2 \\ -7 \\ 5 \end{bmatrix}$

$2\vec{v_1} - 7\vec{v_2} + 5\vec{v_3}$

10. If AB = I = BA for matrices A, B and I, where I is an identity matrix, then

1. B is inverse of A
2. A is inverse of B

3. $A^{(-1)} = B$, $B^{(-1)} = A$,
4. All of the given

11. If $v_1^r = (2, 1), v_2^r = (3, 4)$ and $v_3^r = (7, 8)$ then which of the following is true?

$\{v_1^r, v_2^r, v_3^r\}$ is linearly dependent.

12. Since every linear transformation $T : R^n \rightarrow R^m$ is actually matrix transformation, then which of the following is the alternate notation for the transformation?

$T(x^r)$ a Ax^r

13. A sufficient condition for the jacobi's method to converge for the linear system Ax=b

1. A is diagonally dominant
2. A-I is diagonally dominant
3. A is non-singular
4. None of the given

14. If u + v = u + w, then:

1. v + w
2. v ≠ w
3. v = w

4. None of the given

15. If $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1+1 & 2-1 \\ 2+2 & 4-1 \end{bmatrix}$, then which of the following is true for A and B?

1. A and B are equal matrices.
2. A is the transpose of B.
3. B is the transpose of A.
4. B is the multiplicative inverse of A.

16. A square matrix A is said to be diagonal if A is similar to a matrix

1. Column matrix
2. Zero matrix
3. Diagonal matrix
4. None of the given

17. Let 'Ax = 0' be a homogeneous linear system of 'n' equations and 'n' unknowns. Then, the coefficient matrix 'A' is invertible if and only if this system has _____ solution.

1. No
2. trivial
3. non-trivial
4. infinite many

18. If one of the eigenvalues of $[A]_{n\times n}$ is zero, it implies _____

The determinant of [A] is zero

19. If $Ax^r = b^r$ and factorization of A is LU, then which of the following pair of equations can be used to solve for value of ''?

$Ux^r = y^r$ and $Ly^r = b^r$

20. What is Eigen value?

A scalar associated with a given linear transformation

21. A 3 × 3 identity matrix have three and _____ eigen values.

1. same
2. distinct

22. Which of the following is the most appropriate operation(s) for the linear transformation

1. Scalar multiplication
2. Vector addition and scalar multiplication
3. Vector addition
4. Vector and scalar multiplications

23. Which of the following is true about the existence of free variables (parameter) in a system of linear equations?

1. They guarantee the Consistency.
2. They guarantee the Inconsistency.
3. They do not guarantee the Consistency.
4. None of the given.

24. If a homogeneous system Ax=0 has a trivial solution, then which of the following is(are) the value(s) of the vector x?

1. -1
2. 0
3. 1
4. 2

25. Why inverse of the matrix A= [1 2] is NOT possible?

1. Because it is a saquare matrix.
2. Because it is a zero matrix.
3. Because it is an identity matrix,
4. Because it is rectahular matrix.

26. What is the maximum possiblle number of pivots in a 6 × 6 matrix?

1. 0
2. 2
3. 4
4. 6

27. If the determinant of the

$$: A = \begin{bmatrix} 4 & 3 & 5 \\ 3 & 1 & 1 \\ 5 & 7 & 7 \end{bmatrix}$$

matrix is 32 and the matrix B is obtained by multiplying any row of A with an integer value 4, then which of the following is true for the matrix B?

1. Its determinant is 18.
2. Its determinant is -32.
3. Its determinant is 128.
4. The information is not sufficient to calculate the determinant.

28. Which of the following Elementary Row operations would perform in order to get

$$\begin{pmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{pmatrix} ?$$

?

$$R_2' \to R_2 + 4R_1, R_3' \to R_3 - 6R_1$$

29. $9x^2+3x+4$ is _____.

1. an equation

2. a term
3. an algebraic expression
4. quadratic equation

30. The Invertible Matrix Theorem applies only to _____ matrices.

1. Rectangular
2. Square
3. Identity
4. Scalar

31. If A be the standard matrix of linear transformation , then which of the following is true for the mapping from R^n onto R^m ?

The columns of A span R^n .

32. The Elementary Row operations: and are performed on to get _____?

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & 26 \\ 0 & -9 & -26 \end{bmatrix}$$

33. Which one of the following is a null matrix?

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

34. If a system of equations is solved using the Gauss-Seidel method, then which of the following is NOT true about the matrix M that is derived from the coefficient matrix?

1. All of its entries below the diagonal must be zero.
2. All of its entries above the diagonal must be zero.
3. Its determinant is non-zero.
4. It is an invertible matrix.

35. Which of the following is the coefficient matrix for the system

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 7x_3 &= 0 \\ -4x_1 + 3x_2 + 9x_3 &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -7 \\ -4 & 3 & 9 \end{bmatrix}$$

36. A homogeneous linear system always has the trivial solution: there are only two possibilities for its solutions:

1. The system has only the trivial solution
2. The system has infinitely many solutionsnin addition to trivial solution
3. Both (a) and (b)
4. None of the above

37. If T be a transformation, then which of the following is true for its linearity?

$T(cu^r + dv^r) = cT(u^r)$; whre 'c' and 'd' are scalars

38. Gauss-Seidel method is also termed as a method of

1. Elimination Method
2. False Position Method
3. Successive Displacement
4. Iteration Method

39. If , then which of the following is true for A?

1. det(A) = 1
2. det(A) = - 1
3. det(A) = 0
4. det(A) = ±1

40. The equation: will have the solution only if _____.

$$\begin{pmatrix} -7 \\ 4 \end{pmatrix} = x \begin{pmatrix} -2 \\ 5 \end{pmatrix} + y \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

41. Let V be a five-dimensional vector space, and let S be a subset of V which spans V. Then S

1. Must be linearly dependent

2. Must be a basis for V
3. Must have infinitely many elements
4. Must have at most five elements

42. If the determinant of the matrix A = $\begin{bmatrix} 2 & 3 & 4 \\ 1 & 1 & 2 \\ 3 & 4 & 5 \end{bmatrix}$ is -1 and the matrix B is obtained by adding 2 times of the second row in the first row of the matrix A, then which of the following is true for the matrix B?

1. Its determinant is -1.
2. Its determinant is 1.
3. Its determinant can not be evaluated.
4. The information is not sufficient to calculate the determinant.

43. If X=$\begin{bmatrix} M \\ N \end{bmatrix}$ and Y = [Q p] (Where M, N, Q and P are saqure sub-matrices of same size), then Which of the following is possible?

1. The product and both are not defined
2. The product and both are defined
3. The product XY is defied but YX is not defined
4. None of the given

44. 7x is an algebraic term in which 7 is a _____ and x is a _____.

1. term, expression
2. coefficient, variable
3. variable, coefficient
4. numerical, alphabet

45. If x - 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

1. 2
2. -2
3. 1/2
4. 0

46. A system of linear equations is said to be homogeneous if it can be written in the form _____.

1. AX=B
2. AX=0

47. If λ is an eigenvector of A, then every nonzero vector x such that Ax = λx is called an _____ of A corresponding to _____.

1. Eigenvalue, λ
2. Eigenvector, λ
3. Eigenvalue, A
4. Eigenvector, A

48. Which of the following is the simplified form of

[3 1]

49. If and span , then which of the following is true for any arbitrary ?

$(b_1,b_2,b_3)= k_1(2,2,2) + k_2 (2,2,2) + k_3(0,1,1)$

50. An n × n matrix A is said to be diagonalizable if and only if A has n _____ eigenvectors.

1. Linearly dependent
2. Linearly Independent

51. Two simultaneous linear equations in two variables have no solution if their corresponding lines are _____.

1. parallel and distinct
2. intersecting
3. coincident
4. perpendicular

52. How many Pivot partitions the matrix: will have?

1. 1
2. 2
3. 3
4. 4

53. If the equation: has the solution for all , then and will span _____.

1. R^2 space
2. space
3. R space
4. Nothing

54. If λ + 2 is a factor of the characteristic polynomial of matrix C, then which of the following is the Eigenvalue of C?

1. 2
2. -2
3. 3
4. 0

55. If then Which of the following is the determinant of the matrix M?

1. 1
2. [1]
3. 3
4. [3]

56. Which of the following will be the Matrix Product corresponding to Linear Combination: ?

$$\begin{pmatrix} -2 & 3 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

57. Which of the following will be the Linear Combination corresponding to ?

$$\begin{pmatrix} -2 \\ 5 \end{pmatrix} x + \begin{pmatrix} 3 \\ 1 \end{pmatrix} y$$

58. Which of the following is true for the matrix ?

1. It is a null matrix.
2. It is a scalar matrix.
3. It is a diagonal matrix.
4. It is an identity matrix.

59. If , then which of the following is true for the matrix?

1. det(A) = 1
2. det(A) = k - 1
3. det(A) = k
4. det(A) = k + 1

60. If , then which of the following is the value of det(A)?

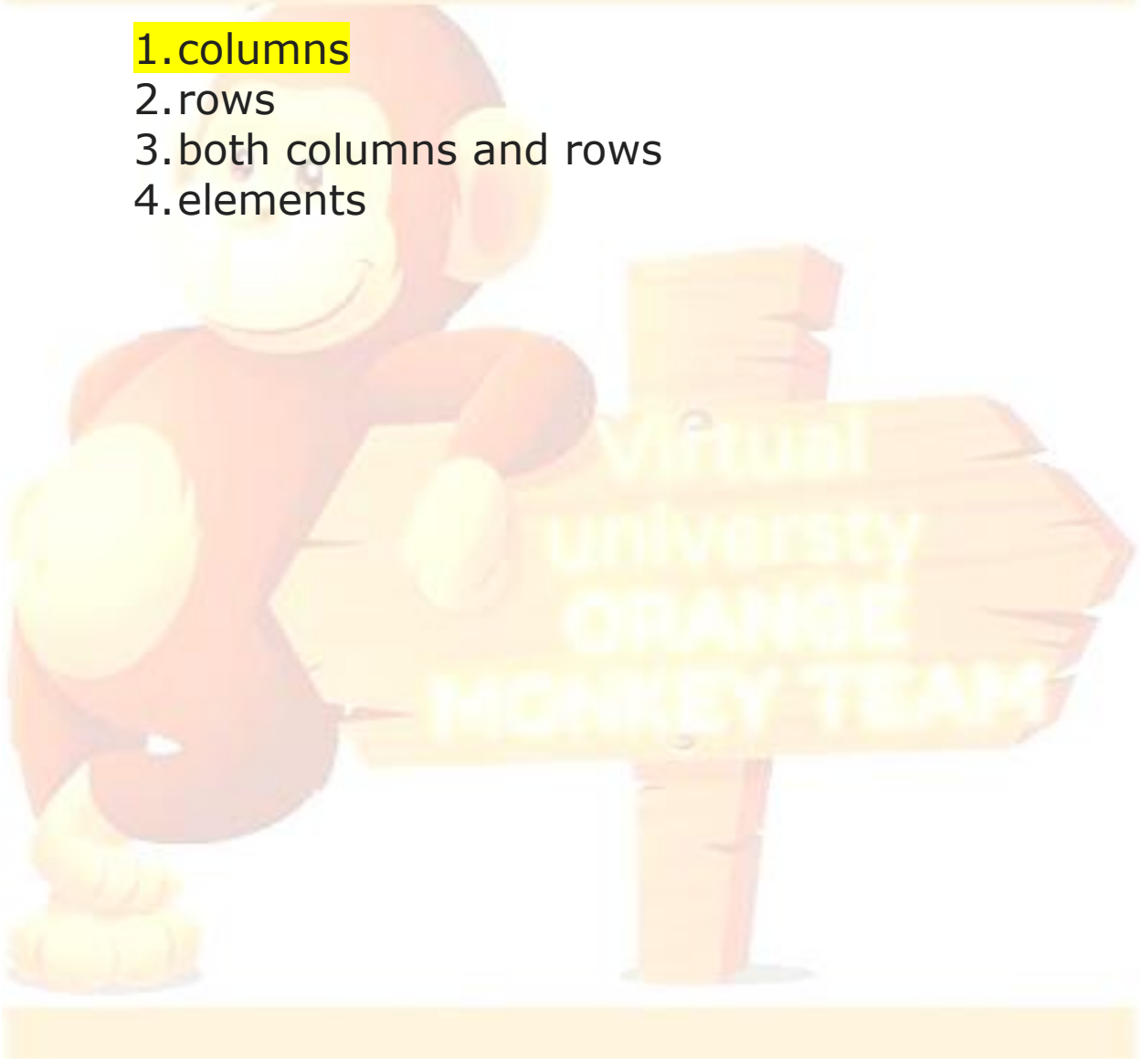
1. 6
2. 18
3. 24
4. 36

61. An real matrix is invertible if and only if the span of the rows of A is

1. True
2. False

62. The solution of Ax = b exists if and only if b can be written as a linear combination of _____ of A.

1. columns
2. rows
3. both columns and rows
4. elements



MTH-501 Quiz-1 BY Attiq Kundi

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MCSQ

AND Many MoRe

1. If the determinant of the matrix A = $\begin{pmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 3 & 4 & 5 \end{pmatrix}$ adding 2 times of the second row in the first row of the matrix A, then which of the following is true for the matrix B?

Its determinant is -1.

2. The solution of Ax = b exists if and only if b can be written as a linear combination of of A.

Columns

3. A sufficient condition for the jacobi's method to converge for the linear system Ax=b

A is diagonally dominant

4. Two simultaneous linear equations in two variables have no solution if their corresponding lines are .

parallel and distinct

5. Which of the following will be the Matrix Product corresponding to Linear Combination: $\begin{pmatrix} -2 \\ 5 \end{pmatrix} x + \begin{pmatrix} 3 \\ 1 \end{pmatrix} y$?

$\begin{pmatrix} -2 & 3 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

6. Which of the following will be the Linear Combination corresponding to $\begin{pmatrix} -2 & 3 \\ 5 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} -2 \\ 5 \end{pmatrix}x + \begin{pmatrix} 3 \\ 1 \end{pmatrix}y$?

7. If a homogeneous system Ax = 0 has atrivial solution, then which of the following is(are) the value (s)of the vector x?

0

8. A square matrix A is said to be diagonal if A is similar to a matrix

Diagonal matrix

9. Why inverse of the matrix A= [1 2] is NOT possible?

Because it is a saquare matrix

10. Which of the following Elementary Row operations would perform in order to $\begin{pmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -5 \\ 0 & 9 & -36 \\ 0 & -9 & 26 \end{pmatrix}$?

$[R_2 \to R_2 + 4R_1, R_3 \to R_3 \to 6R_1]$

11. What is the maximum possiblle number of pivots in a 6 × 6 matrix?

6

12. A homogeneous linear system always has the trivial solution: there are only two possibilities for its solutions:

The system has infinitely much solutionsnin addition to trivial solution

13. A system of linear equations is said to be homogeneous if the constant terms are all

Zero

14. In A is a square matrix, then the minor of entry ith row and jth column is to be the determinant of the sub matrix that remains when the ith row and jth column of A are

Added

15. 7x is an algebraic term in which 7 is a and x is a .

coefficient, variable

16. Which of the following is the coefficient matrix for the system $\begin{matrix} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 7x = 8 \\ -4x_1 + 3x_2 + 9x_3 = \end{matrix}$

$$\begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -7 \\ -4 & 3 & 9 \end{pmatrix}$$

17. Let 'Ax = 0' be a homogeneous linear system of 'n' equations and 'n' unknowns. Then, the coefficient matrix 'A' is invertible if and only if this system has solution.

Trivial

18. Two simultaneous linear equations in two variables have no solution if their corresponding lines are .

Parallel and distinct

19. The solution of Ax = b exists if and only if b can be written as a linear combination of of A.

Columns

20. Let V be a five-dimensional vector space, and let S be a subset of V which spans

V. Then S

Must have at most five elements

21. Gauss-Seidel method is also termed as a method of

Successive Displacement

22. If A be the standard matrix of linear transformation T : Rn → Rm , then which of the following is true for the mapping from R^n onto R^m ?

The columns of A R^m span .

23. If T be a transformation, then which of the following is true for its linearity?

$T(cu^r + dv^r) : cT^r(u^r) + dT(v^r)$

where 'c' and 'd' are scalars

24. Which of the following is true for the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

It is a diagonal matrix.

25. $9x^2 + 3x + 4$ is .

Quadratic equation

26. An n × n matrix A is said to be diagonalizable if and only if A has n _ eigenvectors.

Linearly Independent

27. 7x is an algebraic term in which 7 is a and x is a .

Coefficient, variable

28. Let A be n × n matrix, then A is invertible if and only if

det A is not zero

29. What is Eigen value?

A scalar associated with a given linear transformation

30. If one of the eigenvalues of [A] n x n is zero, it implies

TheDeterminant [A] of is zero

31. Which of the following is true about the existence of free variables (parameter) in a system of linear equations?

They guarantee the Consistency.

32. If a matrix equation is solved using the Jacobi's method. Then which of the following is true about its main diagonal?

It has two zeros along its main diagonal.

It has at least one zero along its main diagonal

It has more than one zero along its main diagonal

It has no zero along its diagonal.

33. As soon as an approximate solution of a linear system is sufficiently accurate for practical work than which of the following is true for an iterative process?

The process may be stopped.

The reduced row-echelon method may be used.

The row-echelon method may be used.

The process may be stopped.

34. If determinant of a matrix is not zero then which of the following is true for that system?

The inversion method is applies.

he inversion method is fails.

35. A partitioned matrix "A" is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are

Non zero skew symmetric

Nonzero symmetric

Unit

Zero

36. If A is strictly diagonal dominant matrix. Then which of the following is true for A.

It is a singular matrix.

It is a rectangular matrix.

It is an invertible matrix.

It is a null matrix.

37. A matrix can be subdivided into sub matrices in various ways by inserting lines between selected _ and columns.

Elements

Rows

Blocks

Diagonals.

38. The determinant of a triangular matrix is the sum of the entries of the main diagonal

True

False

39. What is the maximum possible number of pivots in a '3by 3' matrix?

0

3

1

5

40. Let A be a square n×n matrix. Then which of the following statement(s) is(are) equivalent?

A row is equivalent to the n×n identity matrix.

A has n pivot positions.

Options (b to d).

A is an invertible matrix.

41. What is the maximum possible number of pivots in a 4×6 matrix?

4

8

6

10

42. In LU-decomposition, if A is the square matrix of order m, then the order of both lower and upper triangular matrices will be _.

m^2

$m + 1$

2

m

43. A sufficient condition for the Jacobi's method to converge for the linear system Ax=b

A is diagonally dominant

A is non singular

None of the above

A-I is diagonally dominant

44. An m×m matrix is only square matrix that is both unit lower triangular and upper triangular.

Scalar

Null

Identity

Diagonal

45. homogeneous linear system always has the trivial solution; there are only two possibilities for its solutions:

None of the given.

Both c and d.

The system has only the trivial solution.

The system has infinitely many solutions in addition to the trivial solution.

46. 15.A null space is the vector space.

False

True

47. LU-factorization is used to solve system of linear equations are calculate the .

Row

Column

Diagonal

Determinant

48. Let A be the matrix of order 2×3 and B be the matrix of order 3×5, then which of the following is the order of the matrix AB?

2×3

3×5

3×3

2×5

49. LU-decomposition converts the problem of solving the single system A x=b into the problem of solving the two systems. Ly=b and U x=y. these systems are easy to solve because their co-efficient matrices are_ .

Symmetric

Identity

Diagonal

Triangular

50. The computational efficiency of the LU factorization depends on knowing matrix (matrices).

Upper triangular

Lower triangular

Diagonal

Diagonal both a and b

51. In matrix factorization A=LU , where L is a lower triangular matrix with any on the diagonal and U is an upper triangular matrix.

Fixed constant

Constant

Variable

Fixed variable

52. Which one of the following is null matrix?

1 0 0

0 0 0

53. If A is invertible and b in □□ be any vector. Then we must have a matrix □–□ b,which is solution of .

$Ax = b$

$A^{-1}x = b$

$A^{t}x = b$

$A^{2}x = b$

54. If A is an invertible square matrix then

$(A^{T})^{T} = (A^{-1})^{T}$

$(AT)^{-1}$ = $(A^{-1})^{T}$

$(A^{T})^{-1} = (A^{-1})^{-1}$

None of the above

55. For which of the matrix; the Gauss-seidel method is applicable?

Rectangular matrix

Null matrix

Diagonally dominant matrix

Strictly diagonally dominant matrix

56. Multiplication of a partitioned matrix by a scalar is also computed .

Block by block

Column by column

Diagonal by diagonal

Row by row

57. Gauss-seidel method is similar to the

Newton Raphson Method

Regula-Falsi Method

Jacobi Method

Iteration Method

58. Let T: R2→ R2be transformation T (x1,x2) =(x,0). The null space (or ker nel)N(T) of T is

(0,1)

(1,0)

(0,x2)

(x_1,0)

59. Which of the following system will have a trivial solution?

a. x = y = 1

b. x = 1, y = 0

c. x = 0, y = 1

d. x = y = 0

60. The system of equations:

x1=0=x2

61. can be expressed in the form

Ax=0

Ax=b

Ax=0, By=1

Ax=1, By=0

62. Which of the following is Row - Equivalent of $\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$?

$\begin{matrix} 2 & -4 \\ -1 & 3 \end{matrix}$

$\begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix}$

$\begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

63. The system of equations : y=0 and x=5 has solution(s).

No

Unique

Infinite many

distinct multiple

64. The equation : 0x-0y=-5 has solution(s)

No

Unique

Infinite many

distinct finite

65. Which of the following is an example of Matrix in reduced Echelon form?

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

66. Under which of the following condition, a system of Linear Equations whose Row-Reduce

form is $\left(\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 0 & h-3k \end{array}\right)$ has Infinite many solutions?

h=3k

h ≠ 3k

(h,k) = (0,0)

(h,k) ≠(0,0)

67. For the following system of Equation: x+2y = 0, 2x-y=0; the solution set is

Real line $\mathbb{R}$

Line through origin in $\mathbb{R}^2$

Line through origin in $\mathbb{R}^3$

Any arbitrary line in $\mathbb{R}2$

68. Which of the following corresponding Matrix form of the Linear equation x+y=3?

x |1| + y |1| = 3|1|

69. .Cant be expressed in Matrix Form

$\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -66 \\ 99 \end{pmatrix}$ are linearly?

Dependent

Independent

70. Which of the following is true about the existence of free variables (parameter) in a system of Linear Equations?

They guarantee the Consistency

They guarantee the inconsistency

They do not guarantee the consistency

None

71. Slope and y-intercept of the line: 2x+3y=-1 respectively

a. -2/3 and -1/3

2/3 and -1/3

-2/3 and 1/3

2/3 and 1/3

72. The equation 0x-0y=0 has solution(s).

Infinite many

73. If the determinant of the matrix A = $\begin{pmatrix} 1 & 3 & 5 \\ 3 & 1 & 1 \\ 5 & 7 & 7 \end{pmatrix}$

is 32 and the matrix B is obtained by multiplying any row of A with an integer value 4, then which of the following istrue for the matrix B?

Its determinant is 128.

74. The elementary Row operations: $[R_2 \to R_2 \to 4R_1]$ and $[R_3 \to R_3 \to 6R_1]$ are performed on to get $\begin{pmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 & -5 \\ 0 & 1 & 26 \\ 0 & -9 & -26 \end{pmatrix}$

75. Let A and B be the square matrices. Then A and B are invertible

With $B = A^{-1}$ and $A = B^{-1}$ if and only if $AB = AB$ equals to a (an)___matrix.

Identity

76. If λ is an eigenvector of A, then every nonzero vector x such that Ax = λx is calledan _of A corresponding to_.

Eigenvector, λ

77. If x + 2 is a factor of the characteristic polynomial of matrix C then an eigenvalueof C is

-2

78. Let A be n × n matrix, then A is invertible if and only if

det A is not zero

79. Gauss-Seidel method is also termed as a method of

Successive Displacement

80. If one of the eigenvalues of is zero, $[A]_{n \times n}$ it implies __.

The determinant of | A | is Zero

81. If u + v = u + w, then:

v = w

Linear Algebra
MTH 501



Virtual University of Pakistan

Knowledge beyond the boundaries

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Lecture 1

Introduction and Overview

What is Algebra?

History

Algebra is named in honor of Mohammed Ibn-e- Musa al-Khowârizmî. Around 825, he wrote a book entitled Hisb al-jabr u'l muqubalah, ("the science of reduction and cancellation"). His book, Al-jabr, presented rules for solving equations.

Algebra is a branch of Mathematics that uses mathematical statements to describe relationships between things that vary over time. These variables include things like the relationship between supply of an object and its price. When we use a mathematical statement to describe a relationship, we often use letters to represent the quantity that varies, since it is not a fixed amount. These letters and symbols are referred to as variables.

Algebra is a part of mathematics in which unknown quantities are found with the help of relations between the unknown and known.

In algebra, letters are sometimes used in place of numbers.

The mathematical statements that describe relationships are expressed using algebraic terms, expressions, or equations (mathematical statements containing letters or symbols to represent numbers). Before we use algebra to find information about these kinds of relationships, it is important to first introduce some basic terminology.

Algebraic Term

The basic unit of an algebraic expression is a term. In general, a term is either a product of a number and with one or more variables.

For example 4x is an algebraic term in which 4 is coefficient and x is said to be variable.

Study of Algebra

Today, algebra is the study of the properties of operations on numbers. Algebra generalizes arithmetic by using symbols, usually letters, to represent numbers or unknown quantities. Algebra is a problem-solving tool. It is like a tractor, which is a

farmer's tool. Algebra is a mathematician's tool for solving problems. Algebra has applications to every human endeavor. From art to medicine to zoology, algebra can be a tool. People who say that they will never use algebra are people who do not know about algebra. Learning algebra is a bit like learning to read and write. If you truly learn algebra, you will use it. Knowledge of algebra can give you more power to solve problems and accomplish what you want in life. Algebra is a mathematicians' shorthand!

Algebraic Expressions

An expression is a collection of numbers, variables, and +ve sign or –ve sign, of operations that must make mathematical and logical behaviour.

For example $8x^2 + 9x - 1$ is an algebraic expression.

What is Linear Algebra?

One of the most important problems in mathematics is that of solving systems of linear equations. It turns out that such problems arise frequently in applications of mathematics in the physical sciences, social sciences, and engineering. Stated in its simplest terms, the world is not linear, but the only problems that we know how to solve are the linear ones. What this often means is that only recasting them as linear systems can solve non-linear problems. A comprehensive study of linear systems leads to a rich, formal structure to analytic geometry and solutions to 2x2 and 3x3 systems of linear equations learned in previous classes.

It is exactly what the name suggests. Simply put, it is the algebra of systems of linear equations. While you could solve a system of, say, five linear equations involving five unknowns, it might not take a finite amount of time. With linear algebra we develop techniques to solve m linear equations and n unknowns, or show when no solution exists. We can even describe situations where an infinite number of solutions exist, and describe them geometrically.

Linear algebra is the study of linear sets of equations and their transformation properties.

Linear algebra, sometimes disguised as matrix theory, considers sets and functions, which preserve linear structure. In practice this includes a very wide portion of mathematics!

Thus linear algebra includes axiomatic treatments, computational matters, algebraic structures, and even parts of geometry; moreover, it provides tools used for analyzing differential equations, statistical processes, and even physical phenomena.

Linear Algebra consists of studying matrix calculus. It formalizes and gives geometrical interpretation of the resolution of equation systems. It creates a formal link between matrix calculus and the use of linear and quadratic transformations. It develops the idea of trying to solve and analyze systems of linear equations.

Applications of Linear algebra

Linear algebra makes it possible to work with large arrays of data. It has many applications in many diverse fields, such as

- Computer Graphics,
- Electronics,
- Chemistry,
- Biology,
- Differential Equations,
- Economics,
- Business,
- Psychology,
- Engineering,
- Analytic Geometry,
- Chaos Theory,
- Cryptography,
- Fractal Geometry,
- Game Theory,
- Graph Theory,
- Linear Programming,
- Operations Research

It is very important that the theory of linear algebra is first understood, the concepts are cleared and then computation work is started. Some of you might want to just use the

computer, and skip the theory and proofs, but if you don't understand the theory, then it can be very hard to appreciate and interpret computer results.

Why using Linear Algebra?

Linear Algebra allows for formalizing and solving many typical problems in different engineering topics. It is generally the case that (input or output) data from an experiment is given in a discrete form (discrete measurements). Linear Algebra is then useful for solving problems in such applications in topics such as Physics, Fluid Dynamics, Signal Processing and, more generally Numerical Analysis.

Linear algebra is not like algebra. It is mathematics of linear spaces and linear functions. So we have to know the term "linear" a lot. Since the concept of linearity is fundamental to any type of mathematical analysis, this subject lays the foundation for many branches of mathematics.

Objects of study in linear algebra

Linear algebra merits study at least because of its ubiquity in mathematics and its applications. The broadest range of applications is through the concept of vector spaces and their transformations. These are the central objects of study in linear algebra

1. The solutions of homogeneous systems of linear equations form paradigm examples of vector spaces. Of course they do not provide the only examples.
2. The vectors of physics, such as force, as the language suggests, also provide paradigmatic examples.
3. Binary code is another example of a vector space, a point of view that finds application in computer sciences.
4. Solutions to specific systems of differential equations also form vector spaces.
5. Statistics makes extensive use of linear algebra.
6. Signal processing makes use of linear algebra.
7. Vector spaces also appear in number theory in several places, including the study of field extensions.
8. Linear algebra is part of and motivates much abstract algebra. Vector spaces form the basis from which the important algebraic notion of module has been abstracted.

9. Vector spaces appear in the study of differential geometry through the tangent bundle of a manifold.
10. Many mathematical models, especially discrete ones, use matrices to represent critical relationships and processes. This is especially true in engineering as well as in economics and other social sciences.

There are two principal aspects of linear algebra: theoretical and computational. A major part of mastering the subject consists in learning how these two aspects are related and how to move from one to the other.

Many computations are similar to each other and therefore can be confusing without reasonable level of grasp of their theoretical context and significance. It will be very tempting to draw false conclusions.

On the other hand, while many statements are easier to express elegantly and to understand from a purely theoretical point of view, to apply them to concrete problems you will need to "get your hands dirty". Once you have understood the theory sufficiently and appreciate the methods of computation, you will be well placed to use software effectively, where possible, to handle large or complex calculations.

Course Segments

The course is covered in 45 Lectures spanning over six major segments, which are given below;

1. Linear Equations
2. Matrix Algebra
3. Determinants
4. Vector spaces
5. Eigen values and Eigenvectors, and
6. Orthogonal sets

Course Objectives

The main purpose of the course is to introduce the concept of linear algebra, to explain the underline theory, the computational techniques and then try to apply them on real life problems. Mayor course objectives are as under;

- To master techniques for solving systems of linear equations
- To introduce matrix algebra as a generalization of the single-variable algebra of high school.
- To build on the background in Euclidean space and formalize it with vector space theory.
- To develop an appreciation for how linear methods are used in a variety of applications.
- To relate linear methods to other areas of mathematics such as calculus and, differential equations.

Recommended Books and Supported Material

I am indebted to several authors whose books I have freely used to prepare the lectures that follow. The lectures are based on the material taken from the books mentioned below.

1. **Linear Algebra and its Applications** (3rd Edition) by David C. Lay.
2. **Contemporary Linear Algebra** by Howard Anton and Robert C. Busby.
3. **Introductory Linear Algebra** (8th Edition) by Howard Anton and Chris Rorres.
4. **Introduction to Linear Algebra** (3rd Edition) by L. W. Johnson, R.D. Riess and J.T. Arnold.
5. **Linear Algebra** (3rd Edition) by S. H. Friedberg, A.J. Insel and L.E. Spence.
6. **Introductory Linear Algebra with Applications** (6th Edition) by B. Kolman.

I have taken the structure of the course as proposed in the book of David C. Lay. I would be following this book. I suggest that the students should purchase this book, which is easily available in the market and also does not cost much. For further study and supplement, students can consult any of the above mentioned books.

I strongly suggest that the students should also browse on the Internet; there is plenty of supporting material available. In particular, I would suggest the website of David C. Lay; *www.laylinalgebra.com*, where the entire material, study guide, transparencies are readily available. Another very useful website is *www.wiley.com/college/anton*, which contains a variety of useful material including the data sets. A number of other books are also available in the market and on the internet with free access.

I will try to keep the treatment simple and straight. The lectures will be presented in simple Urdu and easy English. These lectures are supported by the handouts in the form of lecture notes. The theory will be explained with the help of examples. There will be enough exercises to practice with. Students are advised to go through the course on daily basis and do the exercises regularly.

Schedule and Assessment

The course will be spread over 45 lectures. Lectures one and two will be introductory and the Lecture 45 will be the summary. The first two lectures will lay the foundations and would provide the overview of the course. These are important from the conceptual point of view. I suggest that these two lectures should be viewed again and again.

The course will be interesting and enjoyable, if the student will follow it regularly and completes the exercises as they come along. To follow the tradition of a semester system or of a term system, there will be a series of assignments (Max eight assignments) and a mid term exam. Finally there will be terminal examination.

The assignments have weights and therefore they have to be taken seriously.

Lecture 2
Background

Introduction to Matrices

Matrix A matrix is a collection of numbers or functions arranged into rows and columns.

Matrices are denoted by capital letters $A, B, \ldots, Y, Z$. The numbers or functions are called elements of the matrix. The elements of a matrix are denoted by small letters $a, b, \ldots, y, z$.

Rows and Columns The horizontal and vertical lines in a matrix are, respectively, called the rows and columns of the matrix.

Order of a Matrix The size (or dimension) of matrix is called as order of matrix. Order of matrix is based on the number of rows and number of columns. It can be written as $r \times c$; r means no. of row and c means no. of columns.

If a matrix has m rows and n columns then we say that the size or order of the matrix is $m \times n$. If A is a matrix having m rows and n columns then the matrix can be written as

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The element, or entry, in the ith row and jth column of a $m \times n$ matrix A is written as a_{ij}

For example: The matrix $A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \end{pmatrix}$ has two rows and three columns. So order of A will be 2×3

Square Matrix A matrix with equal number of rows and columns is called square matrix.

For Example The matrix $A = \begin{pmatrix} 4 & 7 & -8 \\ 9 & 3 & 5 \\ 1 & -1 & 2 \end{pmatrix}$ has three rows and three columns. So it is a square matrix of order 3.

Equality of matrices

The two matrices will be equal if they must have

a) The same dimensions (i.e. same number of rows and columns)
b) Corresponding elements must be equal.

Example The matrices $A=\begin{pmatrix} 4 & 7 & -8 \\ 9 & 3 & 5 \\ 1 & -1 & 2 \end{pmatrix}$ and $B=\begin{pmatrix} 4 & 7 & -8 \\ 9 & 3 & 5 \\ 1 & -1 & 2 \end{pmatrix}$ equal matrices

(i.e A = B) because they both have same orders and same corresponding elements.

Column Matrix A column matrix X is any matrix having n rows and only one column. Thus the column matrix X can be written as

$$X=\begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \\ \vdots \\ b_{n1} \end{pmatrix}=[b_{i1}]_{n\times 1}$$

A column matrix is also called a column vector or simply a vector.

Multiple of matrix A multiple of a matrix A by a nonzero constant k is defined to be

$$kA=\begin{bmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ ka_{21} & ka_{22} & \cdots & ka_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ ka_{m1} & ka_{m2} & \cdots & ka_{mn} \end{bmatrix}=[ka_{ij}]_{m\times n}$$

Notice that the product kA is same as the product Ak. Therefore, we can write $kA = Ak$.

It implies that if we multiply a matrix by a constant *k*, then each element of the matrix is to be multiplied by *k*.

Example 1

(a)

$$5\cdot\begin{bmatrix} 2 & -3 \\ 4 & -1 \\ 1/5 & 6 \end{bmatrix}=\begin{bmatrix} 10 & -15 \\ 20 & -5 \\ 1 & 30 \end{bmatrix}$$

(b) $$e^t \cdot \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} e^t \\ -2e^t \\ 4e^t \end{bmatrix}$$

Since we know that $kA = Ak$. Therefore, we can write

$$e^{-3t} \cdot \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} e^{-3t}$$

Addition of Matrices Only matrices of the same order may be added by adding corresponding elements.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are two $m \times n$ matrices then $A + B = [a_{ij} + b_{ij}]$

Obviously order of the matrix A + B is $m \times n$

Example 2 Consider the following two matrices of order 3×3

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 6 \\ -6 & 10 & -5 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 7 & -8 \\ 9 & 3 & 5 \\ 1 & -1 & 2 \end{pmatrix}$$

Since the given matrices have same orders, therefore, these matrices can be added and their sum is given by

$$A + B = \begin{pmatrix} 2+4 & -1+7 & 3+(-8) \\ 0+9 & 4+3 & 6+5 \\ -6+1 & 10+(-1) & -5+2 \end{pmatrix} = \begin{pmatrix} 6 & 6 & -5 \\ 9 & 7 & 11 \\ -5 & 9 & -3 \end{pmatrix}$$

Example 3 Write the following single column matrix as the sum of three column vectors

$$\begin{pmatrix} 3t^2 - 2e^t \\ t^2 + 7t \\ 5t \end{pmatrix}$$

Solution

$$\begin{pmatrix} 3t^2-2e^t \\ t^2+7t \\ 5t \end{pmatrix} = \begin{pmatrix} 3t^2 \\ t^2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 7t \\ 5t \end{pmatrix} + \begin{pmatrix} -2e^t \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} t^2 + \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix} t + \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} e^t$$

Difference of Matrices The difference of two matrices A and B of same order $m \times n$ is defined to be the matrix $A - B = A + (-B)$

The matrix $-B$ is obtained by multiplying the matrix B with -1. So that $-B = (-1)B$

Multiplication of Matrices We can multiply two matrices if and only if, the number of columns in the first matrix equals the number of rows in the second matrix.
Otherwise, the product of two matrices is not possible.
OR
If the order of the matrix A is $m \times n$ then to make the product AB possible order of the matrix B must be $n \times p$. Then the order of the product matrix AB is $m \times p$. Thus

$$A_{m\times n} \cdot B_{n\times p} = C_{m\times p}$$

If the matrices A and B are given by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}b_{11}+a_{12}b_{21}+\cdots+a_{1n}b_{n1} & \cdots & a_{11}b_{1p}+a_{12}b_{2p}+\cdots+a_{1n}b_{np} \\ a_{21}b_{11}+a_{22}b_{21}+\cdots+a_{2n}b_{n1} & \cdots & a_{21}b_{1p}+a_{22}b_{2p}+\cdots+a_{2n}b_{np} \\ \vdots & \cdots & \vdots \\ a_{m1}b_{11}+a_{m2}b_{21}+\cdots+a_{mn}b_{n1} & \cdots & a_{m1}b_{1p}+a_{m2}b_{2p}+\cdots+a_{mn}b_{np} \end{bmatrix}$$

$$= \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right)_{n \times p}$$

Example 4 If possible, find the products AB and BA, when

(a) $$A = \begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix}$$

(b) $$A = \begin{pmatrix} 5 & 8 \\ 1 & 0 \\ 2 & 7 \end{pmatrix}, \quad B = \begin{pmatrix} -4 & -3 \\ 2 & 0 \end{pmatrix}$$

Solution (a) The matrices A and B are square matrices of order 2. Therefore, both of the products AB and BA are possible.

$$AB = \begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix}\begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix} = \begin{pmatrix} 4\cdot 9 + 7\cdot 6 & 4\cdot(-2) + 7\cdot 8 \\ 3\cdot 9 + 5\cdot 6 & 3\cdot(-2) + 5\cdot 8 \end{pmatrix} = \begin{pmatrix} 78 & 48 \\ 57 & 34 \end{pmatrix}$$

Similarly $$BA = \begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix}\begin{pmatrix} 4 & 7 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 9\cdot 4 + (-2)\cdot 3 & 9\cdot 7 + (-2)\cdot 5 \\ 6\cdot 4 + 8\cdot 3 & 6\cdot 7 + 8\cdot 5 \end{pmatrix} = \begin{pmatrix} 30 & 53 \\ 48 & 82 \end{pmatrix}$$

Note From above example it is clear that generally a matrix multiplication is not commutative i.e. $AB \neq BA$.

(b) The product AB is possible as the number of columns in the matrix A and the number of rows in B is 2. However, the product BA is not possible because the number of column in the matrix B and the number of rows in A is not same.

$$AB = \begin{pmatrix} 5 & 8 \\ 1 & 0 \\ 2 & 7 \end{pmatrix}\begin{pmatrix} -4 & -3 \\ 2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 5\cdot(-4) + 8\cdot 2 & 5\cdot(-3) + 8\cdot 0 \\ 1\cdot(-4) + 0\cdot 2 & 1\cdot(-3) + 0\cdot 0 \\ 2\cdot(-4) + 7\cdot 2 & 2\cdot(-3) + 7\cdot 0 \end{pmatrix} = \begin{pmatrix} -4 & -15 \\ -4 & -3 \\ 6 & -6 \end{pmatrix}$$

$$AB = \begin{pmatrix} 78 & 48 \\ 57 & 34 \end{pmatrix}, \quad BA = \begin{pmatrix} 30 & 53 \\ 48 & 82 \end{pmatrix}$$

Clearly $AB \neq BA$.

$$AB = \begin{pmatrix} -4 & -15 \\ -4 & -3 \\ 6 & -6 \end{pmatrix}$$

However, the product BA is not possible.

Example 5

(a) $$\begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ 1 & -7 & 9 \end{pmatrix}\begin{pmatrix} -3 \\ 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 2\cdot(-3)+(-1)\cdot 6+3\cdot 4 \\ 0\cdot(-3)+4\cdot 6+5\cdot 6 \\ 1\cdot(-3)+(-7)\cdot 6+9\cdot 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 44 \\ -9 \end{pmatrix}$$

(b) $$\begin{pmatrix} -4 & 2 \\ 3 & 8 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4x+2y \\ 3x+8y \end{pmatrix}$$

Multiplicative Identity For a given any integer n, the $n \times n$ matrix

$$I = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \square & \square & \square & \cdots & \square \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

is called the multiplicative identity matrix. If A is a matrix of order $n \times n$, then it can be verified that $I \cdot A = A \cdot I = A$

Example $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are identity matrices of orders 2 x 2 and 3 x 3 respectively and If $B = \begin{pmatrix} 9 & -2 \\ 6 & 8 \end{pmatrix}$ then we can easily prove that BI = IB = B

Zero Matrix or Null matrix A matrix whose all entries are zero is called zero matrix or null matrix and it is denoted by O.

For example $O = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$; $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$; $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

and so on. If A and O are the matrices of same orders, then $A + O = O + A = A$

Associative Law The matrix multiplication is associative. This means that if A, B and C are $m \times p$, $p \times r$ and $r \times n$ matrices, then $A(BC) = (AB)C$

The result is a $m \times n$ matrix. This result can be verified by taking any three matrices which are confirmable for multiplication.

Distributive Law If B and C are matrices of order $r \times n$ and A is a matrix of order $m \times r$, then the distributive law states that

$$A(B + C) = AB + AC$$

Furthermore, if the product $(A + B)C$ is defined, then

$$(A + B)C = AC + BC$$

Remarks

It is important to note that some rules arithmetic for real numbers do not carry over the matrix arithmetic.

For example, $\forall a, b, c$ and $d \in$

i) if $ab = cd$ and $a \neq 0$, then $b = c$ (Law of Cancellation)

ii) if $ab = 0$, then least one of the factors a or b (or both) are zero.

However the following examples shows that the corresponding results are not true in case of matrices.

Example

Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 7 \\ 0 & 0 \end{bmatrix}$, then one can easily check that $AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$. But $B \neq C$.

Similarly neither A nor B are zero matrices but $AD = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

But if D is diagonal say $D = \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix}$, then $AD \neq DA$.

Determinant of a Matrix Associated with every square matrix A of constants, there is a number called the determinant of the matrix, which is denoted by $\det(A)$ or $|A|$. There is a special way to find the determinant of a given matrix.

Example 6 Find the determinant of the following matrix $A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$

Solution The determinant of the matrix A is given by

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

We expand the $\det(A)$ by first row, we obtain

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix} = 3\begin{vmatrix} 5 & 1 \\ 2 & 4 \end{vmatrix} - 6\begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} + 2\begin{vmatrix} 2 & 5 \\ -1 & 2 \end{vmatrix}$$

or $$\det(A) = 3(20 - 2) - 6(8 + 1) + 2(4 + 5) = 18$$

Transpose of a Matrix The transpose of $m \times n$ matrix A is denoted by A^{tr} and it is obtained by interchanging rows of A into its columns. In other words, rows of A become the columns of A^{tr}. Clearly A^{tr} is $n \times m$ matrix.

If $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$, then $A^{tr} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \cdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$

Since order of the matrix A is $m \times n$, the order of the transpose matrix A^{tr} is $n \times m$.

Properties of the Transpose

The following properties are valid for the transpose;

- The transpose of the transpose of a matrix is the matrix itself:`
- The transpose of a matrix times a scalar (k) is equal to the constant times the transpose of the matrix: $(\underline{ABC})^T = \underline{C}^T \underline{B}^T \underline{A}^T$ $(k\underline{A})^T = k\underline{A}^T$
- The transpose of the sum of two matrices is equivalent to the sum of their transposes: $(\underline{A} + \underline{B})^T = \underline{A}^T + \underline{B}^T$
- The transpose of the product of two matrices is equivalent to the product of their transposes in reversed order: $(\underline{AB})^T = \underline{B}^T \underline{A}^T$
- The same is true for the product of multiple matrices: $(\underline{ABC})^T = \underline{C}^T \underline{B}^T \underline{A}^T$

Example 7 (a) The transpose of matrix $A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$ is $A^T = \begin{pmatrix} 3 & 2 & -1 \\ 6 & 5 & 2 \\ 2 & 1 & 4 \end{pmatrix}$

(b) If $X = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$, then $X^T = \begin{bmatrix} 5 & 0 & 3 \end{bmatrix}$

Multiplicative Inverse Suppose that A is a square matrix of order $n \times n$. If there exists an $n \times n$ matrix B such that $AB = BA = I$, then B is said to be the multiplicative inverse of the matrix A and is denoted by $B = A^{-1}$.

For example: If $A = \begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}$ then the matrix $B = \begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}$ is multiplicative inverse of A because $AB = \begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}\begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

Similarly we can check that BA = I

Singular and Non-Singular Matrices A square matrix A is said to be a **non-singular** matrix if $\det(A) \neq 0$, otherwise the square matrix A is said to be **singular**. Thus for a singular matrix A we must have $\det(A) = 0$

Example: $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$

$$|A| = 2(5-0) - 3(5-0) - 1(-3-2)$$
$$= 10 - 15 + 5 = 0$$

which means that A is singular.

Minor of an element of a matrix

Let A be a square matrix of order n x n. Then minor M_{ij} of the element $a_{ij} \in A$ is the determinant of $(n-1) \times (n-1)$ matrix obtained by deleting the *ith* row and *jth* column from A.

Example If $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$ is a square matrix. The Minor of $3 \in A$ is denoted by M_{12} and is defined to be $M_{12} = \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} = 5\text{-}0 = 5$

Cofactor of an element of a matrix

Let A be a non singular matrix of order $n \times n$ and let C_{ij} denote the cofactor (signed minor) of the corresponding entry $a_{ij} \in A$, then it is defined to be $C_{ij} = (-1)^{i+j} M_{ij}$

Example If $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$ is a square matrix. The cofactor of $3 \in A$ is denoted by C_{12} and is defined to be $C_{12} = = (-1)^{1+2} \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} = -(5 - 0) = -5$

Theorem If A is a square matrix of order $n \times n$ then the matrix has a multiplicative inverse A^{-1} if and only if the matrix A is non-singular.

Theorem Then inverse of the matrix A is given by $A^{-1} = \frac{1}{\det(A)} (C_{ij})^{tr}$

1. For further reference we take $n = 2$ so that A is a 2×2 non-singular matrix given by

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Therefore $C_{11} = a_{22}$, $C_{12} = -a_{21}$, $C_{21} = -a_{12}$ and $C_{22} = a_{11}$. So that

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{21} \\ -a_{12} & a_{11} \end{pmatrix}^{tr} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

2. For a 3×3 non-singular matrix $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

$$C_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, C_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \quad C_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \text{ and so on.}$$

Therefore, inverse of the matrix A is given by $A^{-1} = \dfrac{1}{\det A}\begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$.

Example 8 Find, if possible, the multiplicative inverse for the matrix $A = \begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}$.

Solution The matrix A is non-singular because $\det(A) = \begin{vmatrix} 1 & 4 \\ 2 & 10 \end{vmatrix} = 10 - 8 = 2$

Therefore, A^{-1} exists and is given by $A^{-1} = \dfrac{1}{2}\begin{pmatrix} 10 & -4 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}$

Check
$$AA^{-1} = \begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}\begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix} = \begin{pmatrix} 5-4 & -2+2 \\ 10-10 & -4+5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$AA^{-1} = \begin{pmatrix} 5 & -2 \\ -1 & 1/2 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix} = \begin{pmatrix} 5-4 & 20-20 \\ -1+1 & -4+5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Example 9 Find, if possible, the multiplicative inverse of the following matrix

$$A = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}$$

Solution The matrix is singular because

$$\det(A) = \begin{vmatrix} 2 & 2 \\ 3 & 3 \end{vmatrix} = 2 \cdot 3 - 2 \cdot 3 = 0$$

Therefore, the multiplicative inverse A^{-1} of the matrix does not exist.

Example 10 Find the multiplicative inverse for the following matrix

$$A = \begin{pmatrix} 2 & 2 & 0 \\ -2 & 1 & 1 \\ 3 & 0 & 1 \end{pmatrix}.$$

Solution Since $\det(A) = \begin{vmatrix} 2 & 2 & 0 \\ -2 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix} = 2(1-0) - 2(-2-3) + 0(0-3) = 12 \neq 0$

Therefore, the given matrix is non singular. So, the multiplicative inverse A^{-1} of the matrix A exists. The cofactors corresponding to the entries in each row are

$$C_{11} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1, \quad C_{12} = -\begin{vmatrix} -2 & 1 \\ 3 & 1 \end{vmatrix} = 5, \quad C_{13} = \begin{vmatrix} -2 & 1 \\ 3 & 0 \end{vmatrix} = -3$$

$$C_{21} = -\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = -2, \quad C_{22} = \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} = 2, \quad C_{23} = -\begin{vmatrix} 2 & 2 \\ 3 & 0 \end{vmatrix} = 6$$

$$C_{31} = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2, \quad C_{32} = -\begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix} = -2, \quad C_{33} = \begin{vmatrix} 2 & 2 \\ -2 & 1 \end{vmatrix} = 6$$

Hence $A^{-1} = \frac{1}{12}\begin{pmatrix} 1 & -2 & 2 \\ 5 & 2 & -2 \\ -3 & 6 & 6 \end{pmatrix} = \begin{pmatrix} 1/12 & -1/6 & 1/6 \\ 5/12 & 1/6 & -1/6 \\ -1/4 & 1/2 & 1/2 \end{pmatrix}$

We can also verify that $A \cdot A^{-1} = A^{-1} \cdot A = I$

Derivative of a Matrix of functions

Suppose that

$$A(t) = \lceil a_{ij}(t) \rceil_{m \times n}$$

is a matrix whose entries are functions those are differentiable in a common interval, then derivative of the matrix $A(t)$ is a matrix whose entries are derivatives of the corresponding entries of the matrix $A(t)$. Thus

$$\frac{dA}{dt} = \left\lceil \frac{da_{ij}}{dt} \right\rceil_{m \times n}$$

The derivative of a matrix is also denoted by $A'(t)$.

Integral of a Matrix of Functions

Suppose that $A(t) = \left(a_{ij}(t)\right)_{m \times n}$ is a matrix whose entries are functions those are continuous on a common interval containing t, then integral of the matrix $A(t)$ is a matrix whose entries are integrals of the corresponding entries of the matrix $A(t)$. Thus

$$\int_{t_0}^{t} A(s)ds = \left(\int_{t_0}^{t} a_{ij}(s)ds \right)_{m \times n}$$

Example 11 Find the derivative and the integral of the following matrix $X(t) = \begin{pmatrix} \sin 2t \\ e^{3t} \\ 8t-1 \end{pmatrix}$

Solution The derivative and integral of the given matrix are, respectively, given by

$$X'(t) = \begin{pmatrix} \frac{d}{dt}(\sin 2t) \\ \frac{d}{dt}(e^{3t}) \\ \frac{d}{dt}(8t-1) \end{pmatrix} = \begin{pmatrix} 2\cos 2t \\ 3e^{3t} \\ 8 \end{pmatrix} \quad \text{and} \quad \int_0^t X(s)ds = \begin{pmatrix} \int_0^t \sin 2s ds \\ \int_0^t e^{3s} ds \\ \int_0^t 8s-1 ds \end{pmatrix} = \begin{pmatrix} -1/2\cos 2t + 1/2 \\ 1/3e^{3t} - 1/3 \\ 4t^2 - t \end{pmatrix}$$

Exercise

Write the given sum as a single column matrix

1. $3t\begin{pmatrix} 2 \\ t \\ -1 \end{pmatrix} + (t-1)\begin{pmatrix} -1 \\ -t \\ 3 \end{pmatrix} - 2\begin{pmatrix} 3t \\ 4 \\ -5t \end{pmatrix}$

2. $\begin{pmatrix} 1 & -3 & 4 \\ 2 & 5 & -1 \\ 0 & -4 & -2 \end{pmatrix}\begin{pmatrix} t \\ 2t-1 \\ -t \end{pmatrix} + \begin{pmatrix} -t \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 8 \\ -6 \end{pmatrix}$

Determine whether the given matrix is singular or non-singular. If singular, find A^{-1}.

3. $A = \begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 0 \\ -2 & 5 & -1 \end{pmatrix}$

4. $A = \begin{pmatrix} 4 & 1 & -1 \\ 6 & 2 & -3 \\ -2 & -1 & 2 \end{pmatrix}$

Find $\frac{dX}{dt}$

5. $X = \begin{pmatrix} 1 \\ \sin 2t - 4\cos 2t \\ 2 \\ -3\sin 2t + 5\cos 2t \end{pmatrix}$

6. If $A(t) = \begin{pmatrix} e^{4t} & \cos \pi t \\ 2t & 3t^2 - 1 \end{pmatrix}$ then find (a) $\int_0^2 A(t)dt$, (b) $\int_0^t A(s)ds$.

7. Find the integral $\int_1^2 B(t)dt$ if $B(t) = \begin{pmatrix} 6t & 2 \\ 1/t & 4t \end{pmatrix}$

Lecture 3

Systems of Linear Equations

In this lecture we will discuss some ways in which systems of linear equations arise, how to solve them, and how their solutions can be interpreted geometrically.

Linear Equations

We know that the equation of a straight line is written as $y = mx + c$, where m is the slope of line(Tan of the angle of line with x-axis) and c is the y-intercept(the distance at which the straight line meets y-axis from origin).
Thus a line in $\boldsymbol{R^2}$ (2-dimensions) can be represented by an equation of the form $a_1x + a_2y = b$ (where a_1, a_2 not both zero). Similarly a plane in $\boldsymbol{R^3}$ (3-dimensional space) can be represented by an equation of the form $a_1x + a_2y + a_3z = b$ (where a_1, a_2, a_3 not all zero).

A linear equation in n variables $x_1, x_2, \cdots, x_n$ can be expressed in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \text{ (hyper plane in } \mathbb{R}^n \text{)} \text{ ------- (1)}$$

where $a_1, a_2, \cdots, a_n$ and b are constants and the *"a's"* are not all zero.

Homogeneous Linear equation

In the special case if $b = 0$, Equation (1) has the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$ (2)
This equation is called homogeneous linear equation.

Note A linear equation does not involve any products or square roots of variables. All variables occur only to the first power and do not appear, as arguments of trigonometric, logarithmic, or exponential functions.

Examples of Linear Equations

(1) The equations

$$2x_1 + 3x_2 + 2 = x_3 \quad \textit{and} \quad x_2 = 2\left(\sqrt{5} + x_1\right) + 2x_3 \text{ are both linear}$$

(2) The following equations are also linear

$$x + 3y = 7 \qquad x_1 - 2x_2 - 3x_3 + x_4 = 0$$

$$\tfrac{1}{2}x - y + 3z = -1 \qquad x_1 + x_2 + \cdots + x_n = 1$$

(3) The equations $3x_1 - 2x_2 = x_1x_2$ *and* $x_2 = 4\sqrt{x_1} - 6$

are ***not linear*** because of the presence of x_1x_2 in the first equation and $\sqrt{x_1}$ in the second.

System of Linear Equations

A finite set of linear equations is called a system of linear equations or ***linear system***. The variables in a linear system are called the ***unknowns***.

For example,

$$4x_1 - x_2 + 3x_3 = -1$$
$$3x_1 + x_2 + 9x_3 = -4$$

is a linear system of two equations in three unknowns x_1, x_2, and x_3.

General System of Linear Equations

A general linear system of *m* equations in n-unknowns $x_1, x_2, \cdots, x_n$ can be written as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \quad & \quad \vdots \quad \quad \vdots \quad \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{3}$$

Solution of a System of Linear Equations

A solution of a linear system in the unknowns $x_1, x_2, \cdots, x_n$ is a sequence of *n* numbers $s_1, s_2, \cdots, s_n$ such that when substituted for $x_1, x_2, \cdots, x_n$ respectively, makes every equation in the system a true statement. The set of all such solutions $\{s_1, s_2, \cdots, s_n\}$ of a linear system is called its ***solution set***.

Linear System with Two Unknowns

When two lines intersect in $\boldsymbol{R^2}$, we get system of linear equations with two unknowns

For example, consider the linear system

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

The graphs of these equations are straight lines in the *xy-plane*, so a solution (*x*, *y*) of this system is infact a point of intersection of these lines.

Note that there are three possibilities for a pair of straight lines in xy-plane:

1. The lines may be parallel and distinct, in which case there is no intersection and consequently ***no solution***.
2. The lines may intersect at only one point, in which case the system has exactly ***one solution***.
3. The lines may coincide, in which case there are infinitely many points of intersection (the points on the common line) and consequently ***infinitely many solutions.***

Consistent and inconsistent system

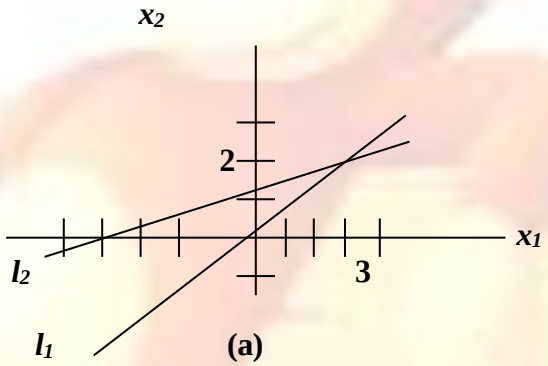
A linear system is said to be **consistent** if it has at least one solution and it is called **inconsistent** if it has no solutions.

Thus, a consistent linear system of two equations in two unknowns has either one solution or infinitely many solutions – there is no other possibility.

Example consider the system of linear equations in two variables $x_1 - 2x_2 = -1,\ -x_1 + 3x_2 = 3$

Solve the equation simultaneously:

Adding both equations we get $x_2 = 2$, Put $x_2 = 2$ in any one of the above equation we get $x_1 = 3$. So the solution is the single point **(3, 2)**. See the graph of this linear system

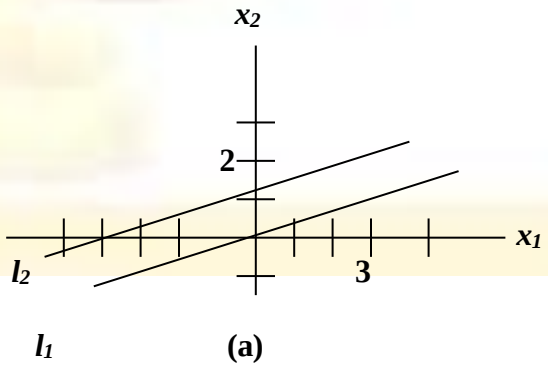


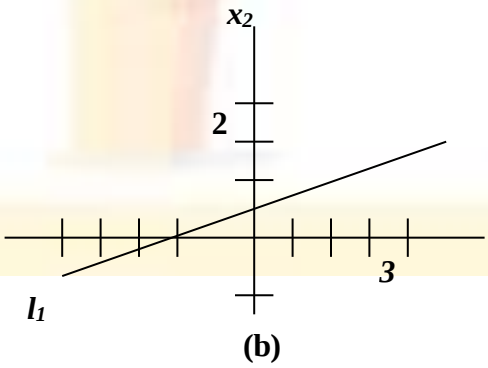
This system has exactly one solution

See the graphs to the following linear systems:

$(a)\quad x_1 - 2x_2 = -1$
$\quad\quad -x_1 + 2x_2 = 3$

$(b)\quad x_1 - 2x_2 = -1$
$\quad\quad -x_1 + 2x_2 = 1$





(a) No solution.

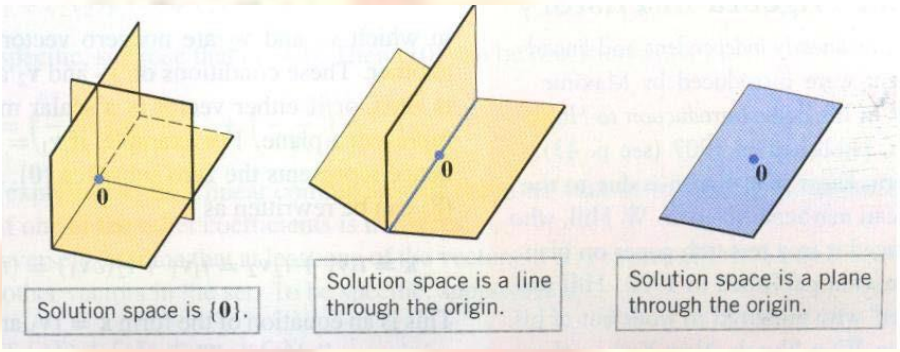
(b) Infinitely many solutions.

Linear System with Three Unknowns

Consider r a linear system of three equations in three unknowns:

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_1z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

In this case, the graph of each equation is a plane, so the solutions of the system, If any correspond to points where all three planes intersect; and again we see that there are only three possibilities – no solutions, one solution, or infinitely many solutions as shown in figure.



Theorem 1 Every system of linear equations has zero, one or infinitely many solutions; there are no other possibilities.

Example 1 Solve the linear system

$$x - y = 1$$
$$2x + y = 6$$

Solution

Adding both equations, we get $x = \frac{7}{3}$. Putting this value of x in 1st equation, we get $y = \frac{4}{3}$. Thus, the system has the ***unique solution*** $x = \frac{7}{3}, y = \frac{4}{3}$.

Geometrically, this means that the lines represented by the equations in the system intersect at a single point $\left(\frac{7}{3}, \frac{4}{3}\right)$ and thus has a unique solution.

Example 2 Solve the linear system

$$x + y = 4$$
$$3x + 3y = 6$$

Solution

Multiply first equation by 3 and then subtract the second equation from this. We obtain

$$0 = 6$$

This equation is contradictory.

Geometrically, this means that the lines corresponding to the equations in the original system are parallel and distinct. So the given system has ***no solution.***

Example 3 Solve the linear system

$$4x - 2y = 1$$
$$16x - 8y = 4$$

Solution

Multiply the first equation by -4 and then add in second equation.

$$\begin{array}{r} -16x + 8y = -4 \\ 16x - 8y = 4 \\ \hline 0 \quad = 0 \end{array}$$

Thus, the solutions of the system are those values of x and y that satisfy the single equation $4x - 2y = 1$

Geometrically, this means the lines corresponding to the two equations in the original system coincide and thus the system has infinitely many solutions.

Parametric Representation

It is very convenient to describe the solution set in this case is to ***express it parametrically.*** We can do this by letting $y = t$ and solving for x in terms of t, or by letting $x = t$ and solving for y in terms of t.

The first approach yields the following parametric equations (by taking *y=t in the equation* $4x - 2y = 1$)

$$4x - 2t = 1, \quad y = t$$
$$x = \frac{1}{4} + \frac{1}{2}t, \quad y = t$$

We can now obtain some solutions of the above system by substituting some numerical values for the parameter.

Example For $t = 0$ the solution is $(\frac{1}{4}, 0)$. For $t = 1$, the solution is $(\frac{3}{4}, 1)$ and for $t = -1$ the solution is $(-\frac{1}{4}, -1)$ *etc.*

Example 4 Solve the linear system

$$x - y + 2z = 5$$
$$2x - 2y + 4z = 10$$
$$3x - 3y + 6z = 15$$

Solution

Since the second and third equations are multiples of the first.

Geometrically, this means that the three planes coincide and those values of *x*, *y* and *z* that satisfy the equation $x - y + 2z = 5$ automatically satisfy all three equations.

We can express the solution set parametrically as

$$x = 5 + t_1 - 2t_2, y = t_1, \quad z = t_2$$

Some solutions can be obtained by choosing some numerical values for the parameters.

For example if we take $y = t_1 = 2$ and $z = t_2 = 3$ then

$$x = 5 + t_1 - 2t_2$$
$$= 5 + 2 - 2(3)$$
$$= 1$$

Put these values of x, y, and z in any equation of linear system to verify

$$x - y + 2z = 5$$
$$1 - 2 + 2(3) = 5$$
$$1 - 2 + 6 = 5$$
$$5 = 5$$

Hence x = 1, y = 2, z = 3 is the solution of the system. Verified.

Matrix Notation

The essential information of a linear system can be recorded compactly in a rectangular array called a **matrix**.

Given the system

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned}$$

With the coefficients of each variable aligned in columns, the matrix $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ -4 & 5 & 9 \end{bmatrix}$ is called the coefficient matrix (or matrix of coefficients) of the system.

An augmented matrix of a system consists of the coefficient matrix with an added column containing the constants from the right sides of the equations. It is always denoted by A_b

$$A_b = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix}$$

Solving a Linear System

In order to solve a linear system, we use a number of methods. 1st of them is given below.

Successive elimination method In this method the x_1 term in the first equation of a system is used to eliminate the x_1 terms in the other equations. Then we use the x_2 term in the second equation to eliminate the x_2 terms in the other equations, and so on, until we finally obtain a very simple equivalent system of equations.

Example 5 Solve

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned}$$

Solution We perform the elimination procedure with and without matrix notation, and place the results side by side for comparison:

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned} \qquad \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix}$$

To eliminate the x_1 term from third equation add 4 times equation 1 to equation 3,

$$\begin{aligned} 4x_1 - 8x_2 + 4x_3 &= 0 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \\ \hline -3x_2 + 13x_3 &= -9 \end{aligned}$$

The result of the calculation is written in place of the original third equation:

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -3x_2 + 13x_3 &= -9 \end{aligned} \qquad \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{bmatrix}$$

Next, multiply equation 2 by ½ in order to obtain 1 as the coefficient for x_2

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 4x_3 &= 4 \\ -3x_2 + 13x_3 &= -9 \end{aligned} \qquad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right]$$

To eliminate the x_2 term from third equation add 3 times equation 2 to equation 3,

The new system has a triangular form

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 4x_3 &= 4 \\ x_3 &= 3 \end{aligned} \qquad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Now using 3rd equation eliminate the x_3 term from first and second equation i.e. multiply 3rd equation with 4 and add in second equation. Then subtract the third equation from first equation we get

$$\begin{aligned} x_1 - 2x_2 &= -3 \\ x_2 &= 16 \\ x_3 &= 3 \end{aligned} \qquad \left[\begin{array}{cccc} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Adding 2 times equation 2 to equation 1, we obtain the result

$$\begin{cases} x_1 = 29 \\ x_2 = 16 \\ x_3 = 3 \end{cases} \qquad \left[\begin{array}{cccc} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

This completes the solution.
Our work indicates that the only solution of the original system is (29, 16, 3).

To verify that (29, 16, 3) is a solution, substitute these values into the left side of the original system for x_1, x_2 and x_3 and after computing, we get

$$(29) - 2(16) + (3) = 29 - 32 + 3 = 0$$
$$2(16) - 8(3) = 32 - 24 = 8$$
$$-4(29) + 5(16) + 9(3) = -116 + 80 + 27 = -9$$

The results agree with the right side of the original system, so (29, 16, 3) is a solution of the system.

This example illustrates how operations on equations in a linear system correspond to operations on the appropriate rows of the augmented matrix. The three basic operations listed earlier correspond to the following operations on the augmented matrix.

Elementary Row Operations

1. (Replacement) Replace one row by the sum of itself and a nonzero multiple of another row.
2. (Interchange) Interchange two rows.
3. (Scaling) Multiply all entries in a row by a nonzero constant.

Row equivalent matrices

A matrix B is said to be row equivalent to a matrix A of the same order if B can be obtained from A by performing a finite sequence of elementary row operations of A.
If A and B are row equivalent matrices, then we write this expression mathematically as A B.

For example $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{bmatrix}$ are row equivalent matrices because we add 4 times of 1st row in 3rd row in 1st matrix.

Note If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Row operations are extremely easy to perform, but they have to be learnt and practice.

Two Fundamental Questions

1. Is the system consistent; that is, does at least one solution exist?
2. If a solution exists is it the only one; that is, is the solution unique?

We try to answer these questions via row operations on the augmented matrix.

Example 6 Determine if the following system of linear equations is consistent

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned}$$

Solution

First obtain the triangular matrix by removing x_1 and x_2 term from third equation and removing x_2 from second equation.

First divide the second equation by 2 we get

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 4x_3 &= 4 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned} \qquad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ -4 & 5 & 9 & -9 \end{array}\right]$$

Now multiply equation 1 with 4 and add in equation 3 to eliminate x_1 from third equation.

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 4x_3 &= 4 \\ -3x_2 + 13x_3 &= -9 \end{aligned} \qquad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right]$$

Now multiply equation 2 with 3 and add in equation 3 to eliminate x_2 from third equation.

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 4x_3 &= 4 \\ x_3 &= 3 \end{aligned} \qquad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Put value of x_3 in second equation we get

$$x_2 - 4(3) = 4$$

$$x_2 = 16$$

Now put these values of x_2 and x_3 in first equation we get

$$x_1 - 2(16) + 3 = 0$$

$$x_1 = 29$$

So a solution exists and the system is consistent and has a unique solution.

Example 7 Solve if the following system of linear equations is consistent.

$$\begin{aligned} x_2 - 4x_3 &= 8 \\ 2x_1 - 3x_2 + 2x_3 &= 1 \\ 5x_1 - 8x_2 + 7x_3 &= 1 \end{aligned}$$

Solution The augmented matrix is

$$\left[\begin{array}{cccc} 0 & 1 & -4 & 8 \\ 2 & -3 & 2 & 1 \\ 5 & -8 & 7 & 1 \end{array}\right]$$

To obtain x_1 in the first equation, interchange rows 1 and 2:

$$\left[\begin{array}{cccc} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 5 & -8 & 7 & 1 \end{array}\right]$$

To eliminate the $5x_1$ term in the third equation, add –5/2 times row 1 to row 3:

$$\left[\begin{array}{cccc} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & -1/2 & 2 & -3/2 \end{array}\right]$$

Next, use the x_2 term in the second equation to eliminate the $-(1/2)\,x_2$ term from the third equation. Add ½ times row 2 to row 3:

$$\left[\begin{array}{cccc} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{array}\right]$$

The augmented matrix is in triangular form.
To interpret it correctly, go back to equation notation:

$$\begin{aligned} 2x_1 - 3x_2 + 2x_3 &= 1 \\ x_2 - 4x_3 &= 8 \\ 0 &= 2.5 \end{aligned}$$

There are no values of x_1, x_2, x_3 that will satisfy because the equation 0 = 2.5 is never true.
Hence original ***system is inconsistent (i.e., has no solution)***.

Exercises

1. State in words the next elementary "row" operation that should be performed on the system in order to solve it. (More than one answer is possible in (a).)

a.
$$\begin{aligned} x_1 + 4x_2 - 2x_3 + 8x_4 &= 12 \\ x_2 - 7x_3 + 2x_4 &= -4 \\ 5x_3 - x_4 &= 7 \\ x_3 + 3x_4 &= -5 \end{aligned}$$

b.
$$\begin{aligned} x_1 - 3x_2 + 5x_3 - 2x_4 &= 0 \\ x_2 + 8x_3 &= -4 \\ 2x_3 &= 7 \\ x_4 &= 1 \end{aligned}$$

2. The augmented matrix of a linear system has been transformed by row operations into the form below. Determine if the system is consistent.

$$\left[\begin{array}{cccc} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ & & & \\ 0 & 0 & 5 & 0 \end{array}\right]$$

3. Is (3, 4, –2) a solution of the following system?

$$\begin{aligned} 5x_1 - x_2 + 2x_3 &= 7 \\ -2x_1 + 6x_2 + 9x_3 &= 0 \\ -7x_1 + 5x_2 - 3x_3 &= -7 \end{aligned}$$

4. For what values of *h* and *k* is the following system consistent?

$$\begin{aligned} 2x_1 - x_2 &= h \\ -6x_1 + 3x_2 &= k \end{aligned}$$

Solve the systems in the exercises given below;

5.
$$\begin{aligned} x_2 + 5x_3 &= -4 \\ x_1 + 4x_2 + 3x_3 &= -2 \\ 2x_1 + 7x_2 + x_3 &= -1 \end{aligned}$$

6.
$$\begin{aligned} x_1 - 5x_2 + 4x_3 &= -3 \\ 2x_1 - 7x_2 + 3x_3 &= -2 \\ 2x_1 - x_2 - 7x_3 &= 1 \end{aligned}$$

7.
$$\begin{aligned} x_1 + 2x_2 &= 4 \\ x_1 - 3x_2 - 3x_3 &= 2 \\ x_2 + x_3 &= 0 \end{aligned}$$

8.
$$\begin{aligned} 2x_1 - 4x_3 &= -10 \\ x_2 + 3x_3 &= 2 \\ 3x_1 + 5x_2 + 8x_3 &= -6 \end{aligned}$$

Determine the value(s) of h such that the matrix is augmented matrix of a consistent linear system.

9. $\begin{bmatrix} 1 & & \\ -2 & -3 & h \\ & 6 & -5 \end{bmatrix}$

10. $\begin{bmatrix} 1 & & \\ -4 & h & -2 \\ & 2 & 10 \end{bmatrix}$

Find an equation involving g, h, and that makes the augmented matrix correspond to a consistent system.

11. $\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & k \end{bmatrix}$

12. $\begin{bmatrix} 2 & 5 & -3 & g \\ 4 & 7 & -4 & h \\ -6 & -3 & 1 & k \end{bmatrix}$

Find the elementary row operations that transform the first matrix into the second, and then find the reverse row operation that transforms the second matrix into first.

13. $\begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & -4 \\ 0 & -3 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -3 & 4 \end{bmatrix}$

14. $\begin{bmatrix} 0 & 5 & -3 \\ 1 & 5 & -2 \\ 2 & 1 & 8 \end{bmatrix}, \begin{bmatrix} 1 & 5 & -2 \\ 0 & 5 & -3 \\ 2 & 1 & 8 \end{bmatrix}$

15. $\begin{bmatrix} 1 & 3 & -1 & 5 \\ 0 & 1 & -4 & 2 \\ 0 & 2 & -5 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 3 & -1 & 5 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 3 & -5 \end{bmatrix}$

Lecture 4

Row Reduction and Echelon Forms

To analyze system of linear equations, we shall discuss how to refine the row reduction algorithm. While applying the algorithm to any matrix, we begin by introducing a non zero row or column (i.e. contains at least one nonzero entry) in a matrix,

Echelon form of a matrix

A rectangular matrix is in *echelon form* (or row echelon form) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zero.

Reduced Echelon Form of a matrix

If a matrix in echelon form satisfies the following additional conditions, then it is in *reduced echelon form* (or reduced row echelon form):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

Examples of Echelon Matrix form

The following matrices are in echelon form. The leading entries (□) may have any nonzero value; the started entries (*) may have any values (including zero).

1. $\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5/2 \end{bmatrix}$

2. $\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 0 & \square & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \square & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \square & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \square & * \end{bmatrix}$

4. $\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6. $\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Examples of Reduced Echelon Form

The following matrices are in reduced echelon form because the leading entries are 1's, and there are 0's below and above each leading 1.

1. $\begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6. $\begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Note A matrix may be row reduced into more than one matrix in echelon form, using different sequences of row operations. However, the reduced echelon form obtained from a matrix, is unique.

Theorem 1 (Uniqueness of the Reduced Echelon Form) Each matrix is row equivalent to one and only one reduced echelon matrix.

Pivot Positions

A pivot position in a matrix ***A*** is a location in ***A*** that corresponds to a leading entry in an echelon form of ***A***.

Note When row operations on a matrix produce an echelon form, further row operations to obtain the reduced echelon form do not change the positions of the leading entries.

Pivot column

A **pivot column** is a column of ***A*** that contains a pivot position.

Example 2 Reduce the matrix ***A*** below to echelon form, and locate the pivot columns

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Solution Leading entry in first column of above matrix is zero which is the pivot position. A nonzero entry, or pivot, must be placed in this position. So interchange first and last row.

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

Pivot

Pivot Column

Since all entries in a column below a leading entry should be zero. For this add row 1 in row 2, and multiply row 1 by 2 and add in row 3.

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 5 & 10 & -15 & -15 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

Pivot

Next pivot column

$$R_1 + R_2$$
$$2R_1 + R_3$$

Add –5/2 times row 2 to row 3, and add 3/2 times row 2 to row 4.

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 \end{bmatrix} \quad \begin{matrix} -\frac{5}{2}R_2 + R_3 \\ 3R_2 + R_4 \end{matrix}$$

Interchange rows 3 and 4, we can produce a leading entry in column 4.

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot

Pivot column

General form

$$\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This is in echelon form and thus columns 1, 2, and 4 of ***A*** are pivot columns.

$$\begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Pivot positions

Pivot columns

Pivot element

A pivot is a nonzero number in a pivot position that is used as needed to create zeros via row operations

The Row Reduction Algorithm consists of four steps, and it produces a matrix in echelon form. A fifth step produces a matrix in reduced echelon form.

The algorithm is explained by an example.

Example 3 Apply elementary row operations to transform the following matrix first into echelon form and then into reduced echelon form.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Solution

STEP 1 Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Pivot column

STEP 2 Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position

Interchange rows 1 and 3. (We could have interchanged rows 1 and 2 instead.)

$$\begin{bmatrix} \boxed{3} & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Pivot

STEP 3 Use row replacement operations to create zeros in all positions below the pivot

Subtract Row 1 from Row 2. i.e. $R_2 - R_1$

$$\begin{bmatrix} \boxed{3} & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Pivot

STEP 4 Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1 –3 to the sub-matrix, which remains. Repeat the process until there are no more nonzero rows to modify.

With row 1 covered, step 1 shows that column 2 is the next pivot column; for step 2, we'll select as a pivot the "top" entry in that column.

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Pivot

Next pivot column

According to step 3 "*All entries in a column below a leading entry are zero*". For this subtract 3/2 time R_2 from R_3

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad R_3 - \frac{3}{2} R_2$$

When we cover the row containing the second pivot position for step 4, we are left with a new sub matrix having only one row:

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Pivot

This is the Echelon form of the matrix.
To change it in reduced echelon form we need to do one more step:

STEP 5 Make the leading entry in each nonzero row 1. Make all other entries of that column to 0.

Divide first Row by 3 and 2nd Row by 2

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \frac{1}{2} R_2 \;,\; \frac{1}{3} R_1$$

Multiply second row by 3 and then add in first row.

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 5 & -4 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad 3R_2 + R_1$$

Subtract row 3 from row 2, and multiply row 3 by 5 and then subtract it from first row

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \begin{matrix} R_2 - R_3 \\ R_1 - 5R_3 \end{matrix}$$

This is the matrix is in reduced echelon form.

Solutions of Linear Systems

When this algorithm is applied to the augmented matrix of the system it gives solution set of linear system.

Suppose, for example, that the augmented matrix of a linear system has been changed into the equivalent reduced echelon form

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are three variables because the augmented matrix has four columns. The associated system of equations is

$$\begin{aligned} x_1 \quad & -5x_3 = 1 \\ x_2 + x_3 &= 4 \qquad (1) \\ 0 = 0 \quad & \text{which means } x_3 \text{ is free} \end{aligned}$$

The variables x_1 and x_2 corresponding to pivot columns in the above matrix are called basic variables. The other variable, x_3 is called a free variable.

Whenever a system is consistent, the solution set can be described explicitly by solving the reduced system of equations for the basic variables in terms of the free variables. This operation is possible because the reduced echelon form places each basic variable in one and only one equation.

In (4), we can solve the first equation for x_1 and the second for x_2. (The third equation is ignored; it offers no restriction on the variables.)

$$\begin{aligned} x_1 &= 1 + 5x_3 \\ x_2 &= 4 - x_3 \qquad (2) \\ x_3 \quad & \textit{is} \quad \textit{free} \end{aligned}$$

By saying that x_3 is "free", we mean that we are free to choose any value for x_3. When $x_3 = 0$, the solution is (1, 4, 0); when $x_3 = 1$, the solution is (6, 3, 1 etc).

Note The solution in (2) is called a **general solution** of the system because it gives an explicit description of all solutions.

Example 4 Find the general solution of the linear system whose augmented matrix has been reduced to

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

Solution The matrix is in echelon form, but we want the reduced echelon form before solving for the basic variables. The symbol "~" before a matrix indicates that the matrix is row equivalent to the preceding matrix.

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

By $R_1 + 2R_3$ and $R_2 + R_3$ Weget

$$\sim \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 2 & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

By $\frac{1}{2}R_2$ we get

$$\sim \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

By $R_1 - 2R_2$ we get

$$\sim \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

The matrix is now in reduced echelon form.
The associated system of linear equations now is

$$\begin{aligned} x_1 + 6x_2 \quad + 3x_4 &= 0 \\ x_3 - 4x_4 &= 5 \\ x_5 &= 7 \end{aligned} \qquad (6)$$

The pivot columns of the matrix are 1, 3 and 5, so the basic variables are x_1, x_3, and x_5. The remaining variables, x_2 and x_4, must be free.

Solving for the basic variables, we obtain the general solution:

$$\begin{cases} x_1 = -6x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 4x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases} \qquad (7)$$

Note that the value of x_5 is already fixed by the third equation in system (6).

Exercise

1. Find the general solution of the linear system whose augmented matrix is

$$\begin{bmatrix} 1 & -3 & -5 & 0 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

2. Find the general solution of the system

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= 0 \\ -2x_1 + 4x_2 + 5x_3 - 5x_4 &= 3 \\ 3x_1 - 6x_2 - 6x_3 + 8x_4 &= 2 \end{aligned}$$

Find the general solutions of the systems whose augmented matrices are given in Exercises 3-12

3. $\begin{bmatrix} 1 & 0 & 2 & 5 \\ 2 & 0 & 3 & 6 \end{bmatrix}$

4. $\begin{bmatrix} 1 & -3 & 0 & -5 \\ -3 & 7 & 0 & 9 \end{bmatrix}$

5. $\begin{bmatrix} 0 & 3 & 6 & 9 \\ -1 & 1 & -2 & -1 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 3 & -3 & 7 \\ 3 & 9 & -4 & 1 \end{bmatrix}$

7. $\begin{pmatrix} 1 & 2 & -7 \\ -1 & -1 & 1 \\ 2 & 1 & 5 \end{pmatrix}$

8. $\begin{pmatrix} 1 & 2 & 4 \\ -2 & -3 & -5 \\ 2 & 1 & -1 \end{pmatrix}$

9. $\begin{pmatrix} 2 & -4 & 3 \\ -6 & 12 & -9 \\ 4 & -8 & 6 \end{pmatrix}$

10. $\begin{pmatrix} 1 & 0 & -9 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$

11. $\begin{pmatrix} 1 & -2 & 0 & 0 & 7 & -3 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

12. $\begin{pmatrix} 1 & 0 & -5 & 0 & -8 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

Determine the value(s) of h such that the matrix is the augmented matrix of a consistent linear system.

13. $\begin{bmatrix} 1 & 4 & 2 \\ -3 & h & -1 \end{bmatrix}$

14. $\begin{bmatrix} 1 & h & 3 \\ 2 & 8 & 1 \end{bmatrix}$

Choose h and k such that the system has (a) no solution, (b) a unique solution, and (c) many solutions. Give separate answer for each part.

15. $x_1 + hx_2 = 1$
$2x_1 + 3x_2 = k$

16. $x_1 - 3x_2 = 1$
$2x_1 + hx_2 = k$

Lecture 5

Vector Equations

This lecture is devoted to connect equations involving vectors to ordinary systems of equations. The term vector appears in a variety of mathematical and physical contexts, which we will study later, while studying "Vector Spaces". Until then, we will use vector to mean a list of numbers. This simple idea enables us to get interesting and important applications as quickly as possible.

Column Vector

"*A matrix with only one column is called column vector or simply a vector".*

e.g. $u = \begin{bmatrix} 3 & -1 \end{bmatrix}^T = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, $v = \begin{bmatrix} 2 & 3 & 5 \end{bmatrix}^T = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$, $w = \begin{bmatrix} w_1 & w_2 & w_3 & w_4 \end{bmatrix}^T$ are all column vectors or simply vectors.

Vectors in R^2

If $\Re$ is the set of all real numbers then the set of all vectors with two entries is denoted by $\Re^2 = \Re \times \Re$.

For example: the vector $u = \begin{bmatrix} 3 & -1 \end{bmatrix}^T = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \in \Re^2$

Here real numbers are appeared as entries in the vectors, and the exponent **2** indicates that the vectors contain only two entries.

Similarly ***R³*** and ***R⁴*** contain all vectors with three and four entries respectively. The entries of the vectors are always taken from the set of real numbers ***R.*** The entries in vectors are assumed to be the elements of a set, called a ***Field***. It is denoted by F.

Algebra of Vectors

Equality of vectors in $\Re^2$

*Two vectors in **R²** are equal if and only if their corresponding entries are equal.*

If $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \Re^2$ *then* $u = v$ *iff* $\boxed{u_1 = v_1} \wedge \boxed{u_2 = v_2}$

So $\begin{bmatrix} 4 \\ 6 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ as $4 = 4$ but $6 \neq 3$

Note In fact, vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ in $\boldsymbol{R^2}$ are nothing but ordered pairs (x, y) of real numbers both representing the position of a point with respect to origin.

Addition of Vectors

Given two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{R^2}$, their sum is the *vector* $\boldsymbol{u} + \boldsymbol{v}$ *obtained by adding corresponding entries of the vectors* $\boldsymbol{u}$ *and* $\boldsymbol{v}$, which is again a vector in $\mathbb{R}^2$

For $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2$ Then $u + v = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} \in \mathbb{R}^2$

For example, $\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1+2 \\ -2+5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$

Scalar Multiplication of a vector

Given a vector $\boldsymbol{u}$ and a real number c, the scalar multiple of $\boldsymbol{u}$ by c is the *vector* $c\boldsymbol{u}$ *obtained by multiplying each entry in* $\boldsymbol{u}$ *by* c.

For example, if $u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ *and* $c = 5$, *then* $cu = 5\begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$

Notations The number c in $c\boldsymbol{u}$ is a **scalar**; it is written in lightface type to distinguish it from the boldface vector $\boldsymbol{u}$.

Example 1 Given $u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ *and* $v = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$, find $4\boldsymbol{u}$, $(-3)\,\boldsymbol{v}$, and $4u + (-3)\,\boldsymbol{v}$

Solution $4u = 4\begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \times 1 \\ 4 \times (-2) \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \end{bmatrix}$, $(-3)v = (-3)\begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} -6 \\ 15 \end{bmatrix}$

And $4u + (-3)v = \begin{bmatrix} 4 \\ -8 \end{bmatrix} + \begin{bmatrix} -6 \\ 15 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$

Note: Sometimes for our convenience, we write a column vector $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$ in the form $(3, -1)$. In this case, we use *parentheses and a comma to distinguish the vector* $(3, -1)$ *from the* 1×2 *row matrix* $[3 \quad -1]$, written with brackets and no comma.

Thus $\begin{bmatrix} 3 \\ -1 \end{bmatrix} \neq [3 \quad -1]$ but $\begin{bmatrix} 3 \\ -1 \end{bmatrix} = (3, -1)$

Geometric Descriptions of R^2

Consider a rectangular coordinate system in the plane. Because each point in the plane is determined by an ordered pair of numbers, *we can identify a geometric point (a, b) with the column vector* $\begin{bmatrix} a \\ b \end{bmatrix}$. So we may regard $\boldsymbol{R^2}$ as the set of all points in the plane.

See Figure 1.

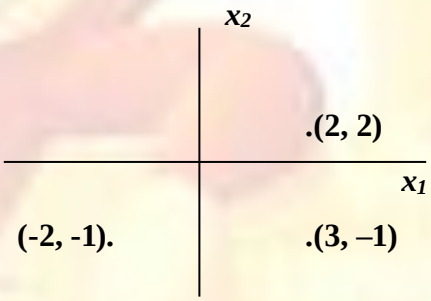


Figure 1 Vectors as points.

Vectors in R^3

Vectors in $\boldsymbol{R^3}$ are 3×1 column matrices with three entries. They are represented geometrically by points in a three-dimensional coordinate space, with arrows from the origin sometimes included for visual clarity.

Vectors in R^n

If **n** is a positive integer, $\boldsymbol{R^n}$ (read "r-n") denotes the collection of all lists (or ordered n- tuples) of **n** real numbers, usually written as $n\times 1$ column matrices, such as

$$u = \begin{bmatrix} u_1 & u_2 & \cdots u_n \end{bmatrix}^T$$

The vector whose all entries are zero is called the **zero vector** and is denoted by **O.** (The number of entries in **O** will be clear from the context.)

Algebraic Properties of R^n

For all ***u, v, w*** in R^n and all scalars *c* and *d*:

(i) $u + v = v + u$ (Commutative)

(ii) $(u + v) + w = u + (v + w)$ (Associative)

(iii) $u + 0 = 0 + u = u$ (Additive Identity)

(iv) $u + (-u) = (-u) + u = 0$ (Additive Inverse)

where $-u$ denotes $(-1)u$

(v) $c(u + v) = cu + cv$ (Scalar Distribution over Vector Addition)

(vi) $(c + d)u = cu + du$ (Vector Distribution over Scalar Addition)

(vii) $c(du) = (cd)u$

(viii) $1u=u$

Linear Combinations Given vectors $v_1, v_2, \ldots, v_p$ in R^n and given scalars $c_1, c_2, \ldots, c_p$ the vector defined by

$$y = c_1v_1 + c_2v_2 + \cdots + c_pv_p$$

is called a **linear combination** of $v_1, \ldots, v_p$ using weights $c_1, \ldots, c_p$.

Property (ii) above permits us to omit parenthesis when forming such a linear combination. The weights in a linear combination can be any real numbers, including zero.

Example

For $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, if $w = \frac{5}{2}v_1 - \frac{1}{2}v_2$ then we say that w is a linear combination of v_1 and v_2.

Example As (3, 5 , 2) = 3(1, 0 , 0) + 5(0, 1 , 0) + 2(0, 0 , 1)

$(3, 5 , 2) = 3v_1 + 5v_2 + 2v_3$ where $v_1 = (1, 0 , 0)$, $v_2 = (0, 1 , 0)$ $v_3 = (0, 0 , 1)$

So (3, 5 , 2) is a vector which is linear combination of v_1, v_2, v_3

Example 5 Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, a_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}, and \quad b = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}.$

Determine whether $\boldsymbol{b}$ can be generated (or written) as a linear combination of $\boldsymbol{a_1}$ and $\boldsymbol{a_2}$. That is, determine whether weights $\boldsymbol{x_1}$ and $\boldsymbol{x_2}$ exist such that

$$x_1\,\boldsymbol{a_1} + x_2\,\boldsymbol{a_2} = \boldsymbol{b} \qquad (1)$$

If the vector equation (1) has a solution, find it.

Solution Use the definitions of scalar multiplication and vector addition to rewrite the vector equation

$$x_1\begin{bmatrix}1\\-2\\-5\end{bmatrix} + x_2\begin{bmatrix}2\\5\\6\end{bmatrix} = \begin{bmatrix}7\\4\\-3\end{bmatrix}$$

$\boldsymbol{a_1}$ $\boldsymbol{a_2}$ $\boldsymbol{b}$

$$\Rightarrow \begin{bmatrix}x_1\\-2x_1\\-5x_1\end{bmatrix} + \begin{bmatrix}2x_2\\5x_2\\6x_2\end{bmatrix} = \begin{bmatrix}7\\4\\-3\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}x_1+2x_2\\-2x_1+5x_2\\-5x_1+6x_2\end{bmatrix} = \begin{bmatrix}7\\4\\-3\end{bmatrix} \qquad (2)$$

$$\Rightarrow \begin{aligned} x_1+2x_2 &= 7\\ -2x_1+5x_2 &= 4\\ -5x_1+6x_2 &= -3 \end{aligned} \qquad (3)$$

We solve this system by row reducing the augmented matrix of the system as follows:

$$\begin{bmatrix}1 & 2 & 7\\ -2 & 5 & 4\\ -5 & 6 & -3\end{bmatrix}$$

$By\ R_2+2R_1\ ;R_3+5R_1$

$$\sim \begin{bmatrix}1 & 2 & 7\\ 0 & 9 & 18\\ 0 & 16 & 32\end{bmatrix}$$

$By\ \left(\frac{1}{9}\right)R_2; \left(\frac{1}{16}\right)R_3$

$$\sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

$$By\ R_3 - R_2; R_1 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

The solution of (3) is $x_1 = 3$ and $x_2 = 2$. Hence $\boldsymbol{b}$ is a linear combination of $\boldsymbol{a_1}$ and $\boldsymbol{a_2}$, with weights $x_1 = 3$ and $x_2 = 2$.

Spanning Set

If $\boldsymbol{v_1, \ldots, v_p}$ are in $\mathbf{R^n}$, then the set of all linear combinations of $\boldsymbol{v_1, \ldots, v_p}$ is denoted by Span $\boldsymbol{\{ v_1, \ldots, v_p \}}$ and is called the **subset of $\mathbf{R^n}$ spanned** (or **generated**) by $\boldsymbol{v_1, \ldots, v_p}$. That is, Span $\boldsymbol{\{ v_1, \ldots, v_p\}}$ is the collection of all vectors that can be written in the form of $c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + \ldots + c_p\boldsymbol{v_p}$, with $c_1, \ldots, c_p$ scalars.

If we want to check whether a vector **b** is in Span $\boldsymbol{\{v_1, \ldots, v_p\}}$ then we will see whether the vector equation

$x_1\boldsymbol{v_1} + x_2\boldsymbol{v_2} + \ldots + x_p\boldsymbol{v_p} = \boldsymbol{b}$ has a solution, or

Equivalently, whether the linear system with augmented matrix $\boldsymbol{[v_1, \ldots, v_p \quad b]}$ has a solution.

Note

(1) The set Span $\boldsymbol{\{ v_1, \ldots, v_p\}}$ contains every scalar multiple of $\boldsymbol{v_1}$

because $c\boldsymbol{v_1} = c\boldsymbol{v_1} + 0\boldsymbol{v_2} + \ldots + 0\boldsymbol{v_p}$ i.e every $c\boldsymbol{v_i}$ can be written as a linear combination of $\boldsymbol{v_1, \ldots, v_p}$

(2) Zero vector $= 0 \in Span\{v_1, v_2, \cdots v_n\}$ as 0 can be written as the linear combination of $v_1, v_2, \cdots v_n$ that is $0_v = 0_F v_1 + 0_F v_2 + \cdots + 0_F v_n$ here for the convenience it is mentioned that 0_v is the vector(zero vector) while 0_F is zero scalar (weight of all $v_1, v_2, \cdots v_n$) and in particular not to make confusion that 0_v and 0_F are same!

A Geometric Description of Span {*v*} and Span {*u*, *v*}

Let $\boldsymbol{v}$ be a nonzero vector in $\boldsymbol{R^3}$. Then Span **{*v*}** is the set of all linear combinations of $\boldsymbol{v}$ or in particular set of scalar multiples of $\boldsymbol{v}$, and we visualize it as the set of points on the line in $\boldsymbol{R^3}$ through $\boldsymbol{v}$ and $\boldsymbol{0}$.

If $\boldsymbol{u}$ and $\boldsymbol{v}$ are nonzero vectors in $\boldsymbol{R^3}$, *with* $\boldsymbol{v}$ *not a multiple of* $\boldsymbol{u}$, then Span **{*u*, *v*}** is the plane in $\boldsymbol{R^3}$ that contains $\boldsymbol{u}$, $\boldsymbol{v}$ and **0**. In particular, Span **{*u*, *v*}** contains the line in $\boldsymbol{R^3}$ through $\boldsymbol{u}$ and **0** and the line through $\boldsymbol{v}$ and **0**.

Example 6 Let $a_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, a_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}, and\ b = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}.$

Then Span $\boldsymbol{\{a_1, a_2\}}$ is a plane through the origin in $\boldsymbol{R^3}$. Does $\boldsymbol{b}$ lie in that plane?

Solution First we see the equation $x_1\boldsymbol{a_1} + x_2\boldsymbol{a_2} = \boldsymbol{b}$ has a solution?

To answer this, row-reduce the augmented matrix $[\boldsymbol{a_1}\ \ \boldsymbol{a_2}\ \ \boldsymbol{b}]$:

$$\begin{bmatrix} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{bmatrix}$$

$By\ R_2 + 2R_1$

$$\sim \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 18 & 10 \end{bmatrix}$$

$By\ R_3 + 6R_2$

$$\sim \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

Last row $\Rightarrow 0x_2 = -2$ which can not be true for any value of $x_2 \in$ ✾

$\Rightarrow$ Given system has no solution

$\therefore b \notin Span\{a_1, a_2\}$ and

in geometrical meaning, vector b does not lie in the plane spanned by vectors $a_1\ and\ a_2$

Linear Combinations in Applications

The final example shows how scalar multiples and linear combinations can arise when a quantity such as "cost" is broken down into several categories. The basic principle for the example concerns the cost of producing several units of an item when the cost per unit is known:

$$\begin{Bmatrix}\text{number}\\ \text{of units}\end{Bmatrix} \cdot \begin{Bmatrix}\text{cost}\\ \text{per unit}\end{Bmatrix} = \begin{Bmatrix}\text{total}\\ \text{cost}\end{Bmatrix}$$

Example 7 A Company manufactures two products. For one dollar's worth of product B, the company spends \$0.45 on materials, \$0.25 on labor, and \$0.15 on overhead. For one dollar's worth of product C, the company spends \$0.40 on materials, \$0.30 on labor and \$0.15 on overhead.

Let $b = \begin{bmatrix} .45 \\ .25 \\ .15 \end{bmatrix}$ *and* $c = \begin{bmatrix} .40 \\ .30 \\ .15 \end{bmatrix}$, then ***b*** and ***c*** represent the "costs per dollar of income" for the two products.

a) What economic interpretation can be given to the vector $100\boldsymbol{b}$?

b) Suppose the company wishes to manufacture x_1 dollars worth of product B and x_2 dollars worth of product C. Give a vector that describes the various costs the company will have (for materials, labor and overhead).

Solution

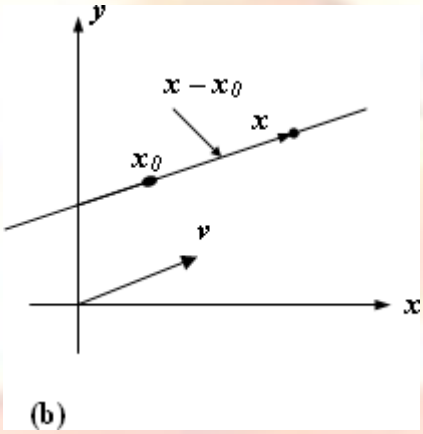
(a) We have $100b = 100\begin{bmatrix} .45 \\ .25 \\ .15 \end{bmatrix} = \begin{bmatrix} 45 \\ 25 \\ 15 \end{bmatrix}$

The vector $100\boldsymbol{b}$ represents a list of the various costs for producing \$100 worth of product B, namely, \$45 for materials, \$25 for labor, and \$15 for overhead.

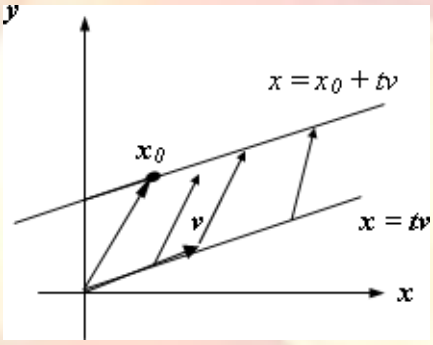
(b) The costs of manufacturing x_1 dollars worth of B are given by the vector $x_1\boldsymbol{b}$ and the costs of manufacturing x_2 dollars worth of C are given by $x_2\boldsymbol{c}$. Hence the total costs for both products are given by the vector $x_1\boldsymbol{b} + x_2\boldsymbol{c}$.

Vector Equation of a Line

Let $\mathbf{x_0}$ be a fixed point on the line and $\mathbf{v}$ be a nonzero vector that is parallel to the required line. Thus, if $\mathbf{x}$ is a variable point on the line through $\mathbf{x_0}$ that is parallel to $\mathbf{v}$, then the vector $\mathbf{x} - \mathbf{x_0}$ is a vector parallel to $\mathbf{v}$ as shown in fig below,



(b)



So by definition of parallel vectors $\boldsymbol{x - x_0 = tv}$ *for some scalar t.*

it is also called a ***parameter*** which varies from $-\infty$ to $+\infty$. The variable point x traces out the line, so the line can be represented by the equation

$$\boldsymbol{x - x_0 = tv} \text{ ----------(1)} \qquad (-\infty < t < +\infty)$$

This is a ***vector equation of the line*** through $\boldsymbol{x_0}$ and parallel to $\boldsymbol{v}$.

In the special case, where $\boldsymbol{x_0 = 0}$, the line passes through the origin, it simplifies to

$$\boldsymbol{x = tv} \qquad (-\infty < t < +\infty)$$

Parametric Equations of a Line in R^2

Let $\boldsymbol{x} = (x, y) \in R^2$ be a general point of the line through $\boldsymbol{x_0} = (x_0, y_0) \in R^2$ which is parallel to

$\boldsymbol{v} = (a, b) \in R^2$, then eq. 1 takes the form

$$(x, y) - (x_0, y_0) = t(a, b) \qquad (-\infty < t < +\infty)$$

$$\Rightarrow (x - x_0 , y - y_0) = (ta , tb) \qquad (-\infty < t < +\infty)$$

$$\Rightarrow x = x_0 + at, \quad y = y_0 + bt \quad (-\infty < t < +\infty)$$

These are called ***parametric equations*** of the line *in* R^2 .

Parametric Equations of a Line in R^3

Similarly, if we let $\boldsymbol{x} = (x, y, z) \in R^3$ be a general point on the line through $\boldsymbol{x_0} = (x_0, y_0 , z_0) \in R^3$ that is parallel to $\boldsymbol{v} = (a, b, c) \in R^3$, then again eq. 1 takes the form

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c) \ (-\infty < t < +\infty)$$

$$\Rightarrow \quad x= x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct \quad (-\infty < t < +\infty)$$

These are the ***parametric equations*** of the line *in* R^3

Example 8

(a) Find a vector equation and parametric equations of the line in $\boldsymbol{R^2}$ that passes through the origin and is parallel to the vector $\boldsymbol{v} = (–2, 3)$.

(b) Find a vector equation and parametric equations of the line in $\boldsymbol{R^3}$ that passes through the point $P_0(1, 2, –3)$ and is parallel to the vector $\boldsymbol{v} = (4, –5, 1)$.

(c) Use the vector equation obtained in part (b) to find two points on the line that are different from P_0.

Solution

(a) We know that a vector equation of the line passing through origin is $\boldsymbol{x} = t\boldsymbol{v}$.

Let $\boldsymbol{x} = (x, y)$. Then this equation can be expressed in component form as

$$(x, y) = t\,(–2, 3)$$

This is the vector equation of the line.

Equating corresponding components on the two sides of this equation yields the parametric equations $x = – 2t, \quad y = 3t$

(b) The vector equation of the line is $\boldsymbol{x} = \boldsymbol{x_0} + t\boldsymbol{v}$.

Let $\boldsymbol{x} = (x, y, z)$, Here $\boldsymbol{x_0} = (1, 2, -3)$ and $\boldsymbol{v} = (4, -5, 1)$, then above equation can be expressed in component form as

$$(x, y, z) = (1, 2, -3) + t\,(4, -5, 1)$$

Equating corresponding components on the two sides of this equation yields the parametric equations

$$x = 1 + 4t, \quad y = 2 - 5t, \quad z = -3 + t$$

(c) Specific points on a line can be found by substituting numerical values for the parameter t.

For example, if we take $t = 0$ in part (b), we obtain the point $(x, y, z) = (1, 2, -3)$, which is the given point P_0.

$t = 1$ yields the point $(5, -3, -2)$ and

$t = -1$ yields the point $(-3, 7, -4)$.

Vector Equation of a Plane

Let x_0 be a fixed point on the required plane W and $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ be two nonzero vectors that are parallel to W and are not scalar multiples of one another. If x is any variable point in the plane W. Suppose $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ have their initial points at x_0, we can create a parallelogram with adjacent side's $t_1\boldsymbol{v_1}$ and $t_2\boldsymbol{v_2}$ in which $\boldsymbol{x} - \boldsymbol{x_0}$ is the diagonal given by the sum

$$\boldsymbol{x} - \boldsymbol{x_0} = t_1\boldsymbol{v_1} + t_2\boldsymbol{v_2}$$

or, equivalently, $\boldsymbol{x} = \boldsymbol{x_0} + t_1\boldsymbol{v_1} + t_2\boldsymbol{v_2}$ ----------**(1)**

where t_1 and t_2 *are parameters* vary independently from $-\infty$ to $+\infty$,

This is a ***vector equation of the plane*** through $\boldsymbol{x_0}$ and parallel to the vectors $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$. In the special case where $x_0 = 0$, then vector equation of the plane passes through the origin takes the form

$$\boldsymbol{x} = t_1\boldsymbol{v_1} + t_2\boldsymbol{v_2} \qquad (-\infty < t_1 < +\infty, -\infty < t_2 < +\infty)$$

Parametric Equations of a Plane

Let $\boldsymbol{x} = (x, y, z)$ be a general or variable point in the plane passes through a fixed point $\boldsymbol{x_0} = (x_0, y_0, z_0)$ and parallel to the vectors $\boldsymbol{v_1} = (a_1, b_1, c_1)$ and $\boldsymbol{v_2} = (a_2, b_2, c_2)$, then the component form of eq. 1 will be

$$(x, y, z) = (x_0, y_0, z_0) + t_1(a_1, b_1, c_1) + t_2(a_2, b_2, c_2)$$

Equating corresponding components, we get

$$x = x_0 + a_1t_1 + a_2t_2$$

$$y = y_0 + b_1t_1 + b_2t_2 \qquad (-\infty < t_1 < +\infty, -\infty < t_2 < +\infty)$$

$$z = z_0 + c_1t_1 + c_2t_2$$

These are called the parametric equations for this plane.

Example 9 (Vector and Parametric Equations of Planes)

(a) Find vector and parametric equations of the plane that passes through the origin of $\boldsymbol{R^3}$ and is parallel to the vectors $\boldsymbol{v_1} = (1, -2, 3)$ and $\boldsymbol{v_2} = (4, 0, 5)$.

(b) Find three points in the plane obtained in part (a).

Solution

(a) As vector equation of the plane passing through origin is $x = t_1v_1 + t_2v_2$.

Let $\boldsymbol{x} = (x, y, z)$ then this equation can be expressed in component form as

$$(x, y, z) = t_1(1, -2, 3) + t_2(4, 0, 5)$$

This is the vector equation of the plane.

Equating corresponding components, we get

$$x = t_1 + 4t_2, \quad y = -2t_1, \quad z = 3t_1 + 5t_2$$

These are the parametric equations of the plane.

(b) Points in the plane can be obtained by assigning some real values to the parameters t_1 and t_2:

$t_1 = 0$ and $t_2 = 0$ produces the point (0, 0, 0)

$t_1 = -2$ and $t_2 = 1$ produces the point (2, 4, –1)

$t_1 = ½$ and $t_2 = ½$ produces the point (5/2, –1, 4)

Vector equation of Plane through Three Points

If x_0, x_1 and x_2 are three non collinear points in the required plane, then, obviously, the vectors $\boldsymbol{v_1} = \boldsymbol{x_1} - \boldsymbol{x_0}$ and $\boldsymbol{v_2} = \boldsymbol{x_2} - \boldsymbol{x_0}$ are parallel to the plane. So, a vector equation of the plane is

$$\boldsymbol{x} = \boldsymbol{x_0} + t_1(\boldsymbol{x_1} - \boldsymbol{x_0}) + t_2(\boldsymbol{x_2} - \boldsymbol{x_0})$$

Example Find vector and parametric equations of the plane that passes through the points. $P(2, -4, 5)$, $Q(-1, 4, -3)$ and $R(1, 10, -7)$.

Solution

Let $\boldsymbol{x} = (x, y, z)$, and if we take $\boldsymbol{x_0}$, $\boldsymbol{x_1}$ and $\boldsymbol{x_2}$ to be the points P, Q and R respectively, then $x_1 - x_0 = PQ = (-3,8,-8)$ and $x_2 - x_0 = PR = (-1,14,-12)$

So the component form will be

$$(x, y, z) = (2,-4,5) + t_1(-3,8,-8) + t_2(-1,14,-12)$$

This is the required vector equation of the plane.

By equating corresponding components, we get

$$x = 2 - 3t_1 - t_2, \quad y = -4 + 8t_1 + 14t_2, \quad z = 5 - 8t_1 - 12t_2$$

These are the parametric equations of the required plane.

Question: How can you tell that the points P, Q and R are not collinear?

Finding a Vector Equation from Parametric Equations

Example 11 Find a vector equation of the plane whose parametric equations are

$$x = 4 + 5t_1 - t_2, \quad y = 2 - t_1 + 8t_2, \quad z = t_1 + t_2$$

Solution First we rewrite the three equations as the single vector equation

$$(x, y, z) = (4 + 5t_1 - t_2, 2 - t_1 + 8t_2, t_1 + t_2)$$

$$\Rightarrow (x, y, z) = (4, 2, 0) + (5t_1, -t_1, t_1) + (-t_2, 8t_2, t_2)$$

$$\Rightarrow (x, y, z) = (4, 2, 0) + t_1(5, -1, 1) + t_2(-1, 8, 1)$$

This is a vector equation of the plane that passes through the point (4, 2, 0) and is parallel to the vectors $\boldsymbol{v_1} = (5, -1, 1)$ and $\boldsymbol{v_2} = (-1, 8, 1)$.

Finding Parametric Equations from a General Equation

Example 12 Find parametric equations of the plane $x - y + 2z = 5$.

Solution First we solve the given equation for x in terms of y and z

$$x = 5 + y - 2z$$

Now make y and z into parameters, and then express x in terms of these parameters.

Let $y = t_1$ and $z = t_2$

Then the parametric equations of the given plane are

$$x = 5 + t_1 - 2t_2, \quad y = t_1, \quad z = t_2$$

Exercises

1. Prove that $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ for any $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{R^n}$.

2. For what value(s) of $\boldsymbol{h}$, $\boldsymbol{y}$ belongs to Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$? Where

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix} \quad v_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, and \quad y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$

3. Determine whether $\boldsymbol{b}$ is a linear combination of $\boldsymbol{a_1}$, $\boldsymbol{a_2}$, and $\boldsymbol{a_3}$.

i). $a_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, a_2 = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}, a_3 = \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix}, b = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$

ii). $a_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, a_2 = \begin{bmatrix} -4 \\ 3 \\ 8 \end{bmatrix}, a_3 = \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix}, b = \begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix}$

4. Determine if ***b*** is a linear combination of the vectors formed from the columns of the matrix ***A***.

i). $A = \begin{bmatrix} 1 & 0 & 2 \\ -2 & 5 & 0 \\ 2 & 5 & 8 \end{bmatrix}, b = \begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix}$

ii). $A = \begin{bmatrix} 1 & 0 & 5 \\ -2 & 1 & -6 \\ 0 & 2 & 8 \end{bmatrix}, b = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$

In exercises 7-10, list seven vectors in Span **{v_1, v_2}**. For each vector, show that the weights on v_1 and v_2 used to generate the vector and list the three entries of the vector. Give also geometric description of the Span **{v_1, v_2}**.

7. $v_1 = \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix}$

8. $v_1 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$

9. $v_1 = \begin{pmatrix} 2 \\ 6 \\ -4 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -3 \\ -9 \\ 6 \end{pmatrix}$

10. $v_1 = \begin{pmatrix} -3.7 \\ -0.4 \\ 11.2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 5.8 \\ 2.1 \\ 5.3 \end{pmatrix}$

11. Let $a_1 = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}, a_2 = \begin{bmatrix} -5 \\ -8 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 3 \\ -5 \\ h \end{bmatrix}$. For what value(s) of h is ***b*** in the plane spanned by $\boldsymbol{a_1}$ and $\boldsymbol{a_2}$?

12. Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}, and\ y = \begin{bmatrix} h \\ -3 \\ -5 \end{bmatrix}$. For what value(s) of h is ***y*** in the plane generated by $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$?

13. Let $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix} and \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Show that $\begin{bmatrix} h \\ k \end{bmatrix}$ is in Span{u, v} for all h and k.

Lecture 6

Matrix Equations

A fundamental idea in linear algebra is to view a linear combination of vectors as the product of a matrix and a vector. The following definition will permit us to rephrase some of the earlier concepts in new ways.

Definition If A is an $m \times n$ matrix, with columns $a_1, a_2, \ldots, a_n$ and if x is in R^n, then the product of A and x denoted by Ax, is the linear combination of the columns of A using the corresponding entries in x as weights, that is,

$$Ax = \begin{bmatrix} a_1 & a_2 & \ldots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + x_2 a_2 + \ldots + x_n a_n$$

Note that Ax is defined only if the number of columns of A equals the number of entries in x.

Example 1

a) $$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = 4\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 2 \\ -5 \end{bmatrix} + 7\begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 6 \\ -15 \end{bmatrix} + \begin{bmatrix} -7 \\ 21 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

b) $$\begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = 4\begin{bmatrix} 2 \\ 8 \\ -5 \end{bmatrix} + 7\begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 32 \\ -20 \end{bmatrix} + \begin{bmatrix} -21 \\ 0 \\ 14 \end{bmatrix} = \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix}$$

Example 2 For v_1, v_2, v_3 in R^m, write the linear combination $3v_1 - 5v_2 + 7v_3$ as a matrix times a vector.

Solution Place v_1, v_2, v_3 into the columns of a matrix A and place the weights 3, -5, and 7 into a vector x.

That is, $$3v_1 - 5v_2 + 7v_3 = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} = Ax$$

We know how to write a system of linear equations as a vector equation involving a linear combination of vectors. For example, we know that the system

$$\begin{matrix} x_1 + 2x_2 - x_3 = 4 \\ -5x_2 + 3x_3 = 1 \end{matrix}$$ is equivalent to $$x_1\begin{bmatrix}1\\0\end{bmatrix} + x_2\begin{bmatrix}2\\-5\end{bmatrix} + x_3\begin{bmatrix}-1\\3\end{bmatrix} = \begin{bmatrix}4\\1\end{bmatrix}$$

Writing the linear combination on the left side as a matrix times a vector, we get

$$\begin{bmatrix}1 & 2 & -1\\ 0 & -5 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}4\\1\end{bmatrix}$$

Which has the form $\boldsymbol{Ax = b}$, and we shall call such an equation a **matrix equation**, to distinguish it from a vector equation.

Theorem 1 If $\boldsymbol{A}$ is an $m \times n$ matrix, with columns $\boldsymbol{a_1, a_2, \ldots, a_n}$ and if $\boldsymbol{b}$ is in $\boldsymbol{R^m}$, the matrix equation $\boldsymbol{Ax = b}$ has the same solution set as the vector equation

$$x_1\boldsymbol{a_1} + x_2\boldsymbol{a_2} + \ldots + x_n\boldsymbol{a_n} = \boldsymbol{b}$$

which, in turn, has the same solution set as the system of linear equations whose augmented matrix is $\begin{bmatrix}a_1 & a_2 & \ldots & a_n & b\end{bmatrix}$

Existence of Solutions The equation $\boldsymbol{Ax = b}$ has a solution if and only if $\boldsymbol{b}$ is a linear combination of the columns of $\boldsymbol{A}$.

Example 3 Let $A = \begin{bmatrix}1 & 3 & 4\\ -4 & 2 & -6\\ -3 & -2 & -7\end{bmatrix}$ and $b = \begin{bmatrix}b_1\\ b_2\\ b_3\end{bmatrix}$.

Is the equation $\boldsymbol{Ax = b}$ consistent for all possible b_1, b_2, b_3?

Solution Row reduce the augmented matrix for $\boldsymbol{Ax = b}$:

$$\begin{bmatrix}1 & 3 & 4 & b_1\\ -4 & 2 & -6 & b_2\\ -3 & -2 & -7 & b_3\end{bmatrix} \xrightarrow{4R_1 + R_2,\, 3R_1 + R_3} \begin{bmatrix}1 & 3 & 4 & b_1\\ 0 & 14 & 10 & b_2 + 4b_1\\ 0 & 7 & 5 & b_3 + 3b_1\end{bmatrix}$$

$$R_3 - \frac{1}{2}R_2$$

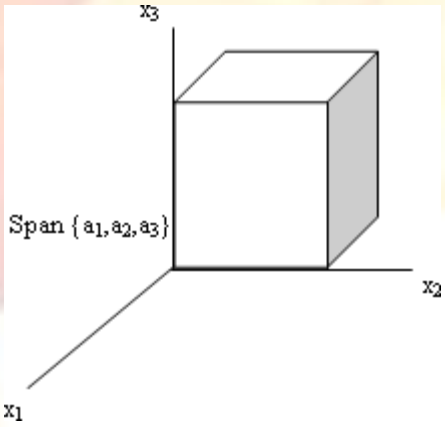
$$\begin{bmatrix} 1 & -2 & -1 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{bmatrix}$$

The third entry in the augmented column is $b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1)$

The equation $\boldsymbol{Ax} = \boldsymbol{b}$ is not consistent for every $\boldsymbol{b}$ because some choices of $\boldsymbol{b}$ can make $b_1 - \frac{1}{2}b_2 + b_3$ nonzero.

The entries in $\boldsymbol{b}$ must satisfy $b_1 - \frac{1}{2}b_2 + b_3 = 0$

This is the equation of a plane through the origin in $\boldsymbol{R^3}$. The plane is the set of all linear combinations of the three columns of $\boldsymbol{A}$. See figure below.



The equation $\boldsymbol{Ax} = \boldsymbol{b}$ fails to be consistent for all $\boldsymbol{b}$ because the echelon form of $\boldsymbol{A}$ has a row of zeros. If $\boldsymbol{A}$ had a pivot in all three rows, we would not care about the calculations in the augmented column because in this case an echelon form of the augmented matrix could not have a row such as [0 0 0 1].

Example 4 Which of the following are linear combinations of

$$A = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

(a) $\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ **(c)** $\begin{bmatrix} 6 & 0 \\ 3 & 8 \end{bmatrix}$

Solution

(a) $\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} = a\,A + b\,B + c\,C$

$$= a\begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} + b\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} + c\begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4a+b & -b+2c \\ -2a+2b+c & -2a+3b+4c \end{bmatrix}$$

$\Rightarrow$ $4a + b = 6$ (1)

$-b + 2c = -8$ (2)

$-2a + 2b + c = -1$ (3)

$-2a + 3b + 4c = -8$ (4)

Subtracting equation (4) from equation (3), we obtain

$-b - 3c = 7$ (5)

Subtracting equation (5) from equation (2):

$5c = -15 \Rightarrow c = -3$

From (2), $-b + 2(-3) = -8 \Rightarrow b = 2$

From (3), $-2a + 2(2) - 3 = -1 \Rightarrow a = 1$

Now we check whether these values satisfy equation (1).

$4(1) + 2 = 6$

It means that $\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix}$ is the linear combination of **A**, **B** and **C**.

Thus

$$\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} = 1A + 2B - 3C$$

(b)

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = a\,A + b\,B + c\,C$$

$$= a\begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} + b\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} + c\begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4a+b & -b+2c \\ -2a+2b+c & -2a+3b+4c \end{bmatrix}$$

$\Rightarrow$

$$4a + b = 0 \qquad (1)$$

$$-b + 2c = 0 \qquad (2)$$

$$-2a + 2b + c = 0 \qquad (3)$$

$$-2a + 3b + 4c = 0 \qquad (4)$$

Subtracting equation (3) from equation (4) we get

$$b + 3c = 0 \qquad (5)$$

Adding equation (2) and equation (5), we get

$$5c = 0 \Rightarrow c = 0$$

Put c = 0 in equation (5), we get b = 0

Put b = c = 0 in equation (3), we get a = 0

$\Rightarrow$ a = b = c =0

It means that $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is the linear combination of **A**, **B** and **C**.

Thus $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0A + 0B + 0C$

(c)

$$\begin{bmatrix} 6 & 0 \\ 3 & 8 \end{bmatrix} = a\,A + b\,B + c\,C$$

$$= a\begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} + b\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} + c\begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4a+b & -b+2c \\ -2a+2b+c & -2a+3b+4c \end{bmatrix}$$

$\Rightarrow$ $4a + b = 6$ (1)

$-b + 2c = 0$ (2)

$-2a + 2b + c = 3$ (3)

$-2a + 3b + 4c = 8$ (4)

Subtracting (4) from (3), we obtain

$-b - 3c = -5$ (5)

Subtracting (5) from (2):

$5c = 5 \Rightarrow c = 1$

From (2), $-b + 2(1) = 0 \Rightarrow b = 2$

From (3), $-2a + 2(2) + 1 = 3 \Rightarrow a = 1$

Now we check whether these values satisfy (1).

$4(1) + 2 = 6$

It means that $\begin{bmatrix} 6 & 0 \\ 3 & 8 \end{bmatrix}$ is the linear combination of **A**, **B** and **C**.

Thus $\begin{bmatrix} 6 & 0 \\ 3 & 8 \end{bmatrix} = 1\mathbf{A} + 2\mathbf{B} + 1\mathbf{C}$

Theorem 2 Let $\boldsymbol{A}$ be an $m \times n$ matrix. Then the following statements are logically equivalent.

(a) For each $\boldsymbol{b}$ in $\boldsymbol{R^m}$, the equation $\boldsymbol{Ax = b}$ has a solution.

(b) The columns of $\boldsymbol{A}$ Span $\boldsymbol{R^m}$.

(c) A has a pivot position in every row.

This theorem is one of the most useful theorems. It is about a coefficient matrix, not an augmented matrix. If an augmented matrix [***A*** ***b***] *has a pivot position in every row, then the equation* ***Ax*** **=** ***b*** *may or may not be consistent.*

Example 4 Compute ***Ax***, where $A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$ *and* $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Solution From the definition,

$$\begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1\begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} + x_2\begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} + x_3\begin{bmatrix} 4 \\ -3 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1 \\ -x_1 \\ 6x_1 \end{bmatrix} + \begin{bmatrix} 3x_2 \\ 5x_2 \\ -2x_2 \end{bmatrix} + \begin{bmatrix} 4x_3 \\ -3x_3 \\ 8x_3 \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{bmatrix}$$

Note

In above example the first entry in ***Ax*** is a sum of products (sometimes called a **dot product**), using the first row of ***A*** and the entries in ***x***.

That is $\begin{bmatrix} 2 & 3 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \end{bmatrix}$

Examples

In each part determine whether the given vector span R^3

(*a*) $v_1 = (2, 2, 2)$, $v_2 = (0, 0, 3)$, $v_3 = (0, 1, 1)$

(*b*) $v_1 = (3, 1, 4)$, $v_2 = (2, -3, 5)$, $v_3 = (5, -2, 9)$, $v_4 = (1, 4, -1)$

(*c*) $v_1 = (1, 2, 6)$, $v_2 = (3, 4, 1)$, $v_3 = (4, 3, 1)$, $v_4 = (3, 3, 1)$

Solutions

(a) We have to determine whether arbitrary vectors $b=(b_1, b_2, b_3)$ in R^3 can be expressed as a linear combination $b=k_1v_1+k_2v_2+k_3v_3$ of the vectors v_1, v_2, v_3

Expressing this in terms of components given by

$$(b_1, b_2, b_3)=k_1(2,2,2)+k_2(0,0,3)+k_3(0,1,1)$$

$$(b_1, b_2, b_3)=(2k_1+0k_2+0k_3, 2k_1+0k_2+k_3, 2k_1+3k_2+k_3)$$

$$2k_1+0k_2+0k_3=b_1$$

$$2k_1+0k_2+k_3=b_2$$

$$2k_1+3k_2+k_3=b_3$$

$$A=\begin{bmatrix}2 & 0 & 0\\ 2 & 0 & 1\\ 2 & 3 & 1\end{bmatrix}$$ has a non zero determinant

Now

$$\det(A)=-6\neq 0$$

Therefore v_1, v_2, v_3 span R^3

(b) The set $S\{v_1, v_2, v_3, v_4\}$ of vectors in R^3 spans $V=R^3$ if

$$c_1v_1+c_2v_2+c_3v_3+c_4v_4=d_1w_1+d_2w_2+d_3w_3 \quad \text{..... (1)}$$

with

$$w_1=(1,0,0)$$

$$w_2=(0,1,0)$$

$$w_3=(0,0,1)$$

With our vectors v_1, v_2, v_3, v_4 equation (1) becomes

$$c_1(3,1,4)+c_2(2,-3,5)+c_3(5,-2,9)+c_4(1,4,-1)=d_1(1,,0,0)+d_2(0,1,0)+d_3(0,0,1)$$

Rearranging the left hand side yields

$$3c_1+2c_2+5c_3+1c_4=1d_1+0d_2+0d_3$$

$$1c_1-3c_2-2c_3+4c_4=0d_1+1d_2+0d_3$$

$$4c_1+5c_2+9c_3-1c_4=0d_1+0d_2+1d_3$$

$$\begin{bmatrix} 3 & 2 & 5 & 1 & 1 & 0 & 0 \\ 1 & -3 & -2 & 4 & 0 & 1 & 0 \\ 4 & 5 & 9 & -1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 1 & -1 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 & -3 & -2 \end{bmatrix}$$

The reduce row echelon form

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & \frac{5}{17} & \frac{3}{17} \\ 0 & 1 & 1 & -1 & 0 & \frac{-4}{17} & \frac{1}{17} \\ 0 & 0 & 0 & 0 & 1 & \frac{-7}{17} & -\frac{11}{17} \end{bmatrix}$$

Corresponds to the system of equations

$$1c_1 + 1c_3 + 1c_4 = \left(\frac{5}{17}\right)d_2 + \left(\frac{3}{17}\right)d_3$$

$$1c_2 + 1c_3 + -1c_4 = \left(\frac{-4}{17}\right)d_2 + \left(\frac{1}{17}\right)d_3 \quad \ldots\ldots\ldots\ldots\ldots(2)$$

$$0 = 1d_1 + \left(\frac{-7}{17}\right)d_2 + \left(-\frac{11}{17}\right)d_3$$

So this system is inconsistent. The set S does not span the space V.

Similarly Part C can be solved by the same way.

Exercise

1. Let $A = \begin{bmatrix} 1 & 5 & -2 & 0 \\ -3 & 1 & 9 & -5 \\ 4 & -8 & -1 & 7 \end{bmatrix}$, $x = \begin{bmatrix} 3 \\ -2 \\ 0 \\ -4 \end{bmatrix}$, *and* $b = \begin{bmatrix} -7 \\ 9 \\ 0 \end{bmatrix}$.

It can be shown that $\boldsymbol{Ax = b}$. Use this fact to exhibit $\boldsymbol{b}$ as a specific linear combination of the columns of $\boldsymbol{A}$.

2. Let $A=\begin{bmatrix}2 & 5\\ 3 & 1\end{bmatrix}, u=\begin{bmatrix}4\\ -1\end{bmatrix}, and \quad v=\begin{bmatrix}-3\\ 5\end{bmatrix}$. Verify $\boldsymbol{A(u+v) = Au + Av}$.

3. Solve the equation $\boldsymbol{Ax = b}$, with $A=\begin{bmatrix}2 & 4 & -6\\ 0 & 1 & 3\\ -3 & -5 & 7\end{bmatrix}, b=\begin{bmatrix}2\\ 5\\ -3\end{bmatrix}$.

4. Let $u=\begin{bmatrix}-5\\ -3\\ -6\end{bmatrix} and\ A=\begin{bmatrix}3 & 5\\ 1 & 1\\ -2 & -8\end{bmatrix}$. Is $\boldsymbol{u}$ belongs to the plane in R^3 spanned by the columns of A? Why or why not?

5. Let $u=\begin{bmatrix}8\\ 2\\ 3\end{bmatrix} and\ A=\begin{bmatrix}4 & 3 & 5\\ 0 & 1 & -1\\ 1 & 2 & 0\end{bmatrix}$. Is $\boldsymbol{u}$ in the subset of R^3 spanned by the columns of A? Why or why not?

6. Let $A=\begin{bmatrix}-3 & 1\\ 6 & -2\end{bmatrix} and\ b=\begin{bmatrix}b_1\\ b_2\end{bmatrix}$. Show that the equation $\boldsymbol{Ax = b}$ is not consistent for all possible $\boldsymbol{b}$, and describe the set of all $\boldsymbol{b}$ for which $\boldsymbol{Ax = b}$ is consistent.

7. How many rows of $A=\begin{bmatrix}1 & 3 & -2 & -2\\ 0 & 1 & -1 & 5\\ -1 & -2 & 1 & 7\\ 1 & 1 & 0 & -6\end{bmatrix}$ contain pivot positions?

In exercises 8 to 13 , explain how your calculations justify your answer, and mention an appropriate theorem.

8. Do the columns of the matrix $A=\begin{bmatrix}1 & 3 & -4\\ 3 & 2 & -6\\ -5 & -1 & 8\end{bmatrix}$ span R^3?

9. Do the columns of the matrix $A=\begin{bmatrix} 1 & 3 & -2 & -2 \\ 0 & 1 & -1 & 5 \\ -1 & -2 & 1 & 7 \\ 1 & 1 & 0 & -6 \end{bmatrix}$ span R^4?

10. Do the columns of the matrix $A=\begin{bmatrix} 0 & 0 & 2 \\ 0 & -5 & 1 \\ 4 & 6 & -3 \end{bmatrix}$ span R^3?

11. Do the columns of the matrix $A=\begin{bmatrix} 3 & 5 \\ 1 & 1 \\ -2 & -8 \end{bmatrix}$ span R^3?

12. Let $v_1=\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, v_2=\begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, v_3=\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$. Does $\{v_1, v_2, v_3\}$span R^4?

13. Let $v_1=\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, v_2=\begin{bmatrix} -1 \\ 3 \\ 7 \end{bmatrix}, v_3=\begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix}$. Does $\{ v_1, v_2, v_3\}$ span R^3?

14. It can be shown that $\begin{bmatrix} 4 & 1 & 2 \\ -2 & 0 & 8 \\ 3 & 5 & -6 \end{bmatrix}\begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}=\begin{bmatrix} 4 \\ 18 \\ 5 \end{bmatrix}$. Use this fact(and no row operations)

to find scalars c_1, c_2, c_3 such that $\begin{bmatrix} 4 \\ 18 \\ 5 \end{bmatrix}=c_1\begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}+c_2\begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}+c_3\begin{bmatrix} 2 \\ 8 \\ -6 \end{bmatrix}$.

15. Let $u = \begin{bmatrix} 3 \\ 8 \\ -4 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, and\ w = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$. It can be shown that 2u – 5v – w = 0. Use this fact(and no row operations) to solve the equation $\begin{bmatrix} 3 & 1 \\ 8 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$.

Determine if the columns of the matrix span R^4.

16. $\begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix}$

17. $\begin{bmatrix} 12 & -7 & 11 & -9 & 5 \\ -9 & 4 & -8 & 7 & -3 \\ -6 & 11 & -7 & 3 & -9 \\ 4 & -6 & 10 & -5 & 12 \end{bmatrix}$

Lecture 7

Solution Sets of Linear Systems

Solution Set

A solution of a linear system is an assignment of values to the variables $x_1, x_2, \ldots, x_n$ such that each of the equations in the linear system is satisfied. The set of all possible solutions is called the Solution Set

Homogeneous Linear System

A system of linear equations is said to be **homogeneous** if it can be written in the form $\boldsymbol{Ax = 0}$, where $\boldsymbol{A}$ is an $m \times n$ matrix and $\boldsymbol{0}$ is the zero vector in $\boldsymbol{R^m}$.

Trivial Solution

A homogeneous system $\boldsymbol{Ax = 0}$ always has at least one solution, namely, $\boldsymbol{x = 0}$ (the zero vector in $\boldsymbol{R^n}$). This zero solution is usually called the trivial solution of the homogeneous system.

Nontrivial solution

A solution of a linear system other than trivial is called its nontrivial solution.
i.e the solution of a homogenous equation $\boldsymbol{Ax = 0}$ such that $\boldsymbol{x \neq 0}$ is called **nontrivial solution**, that is, a nonzero vector $\boldsymbol{x}$ that satisfies $\boldsymbol{Ax = 0}$.

Existence and Uniqueness Theorem

The homogeneous equation $\boldsymbol{Ax = 0}$ has a nontrivial solution if and only if the equation has at least one free variable.

Example 1 Find the solution set of the following system

$$3x_1 + 5x_2 - 4x_3 = 0$$
$$3x_1 + 2x_2 - 4x_3 = 0$$
$$6x_1 + x_2 - 8x_3 = 0$$

Solution

Let $A = \begin{bmatrix} 3 & 5 & -4 \\ 3 & 2 & -4 \\ 6 & 1 & -8 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

The augmented matrix is

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ 3 & 2 & -4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix}$$

For solution set, row reduce to reduced echelon form

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & -9 & 0 & 0 \end{bmatrix} \qquad -1R_1 + R_2, -2R_1 + R_3$$

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad -3R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & -\frac{4}{3} & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad 1/3R_1, 1/3R_2, 5/3R_2 + R_1$$

$$\begin{bmatrix} 1 & 0 & -\frac{4}{3} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad (-1)R_2$$

$$x_1 - \frac{4}{3}x_3 = 0$$

$$x_2 = 0$$

$$0 = 0$$

It is clear that x_3 is a free variable, so $Ax = 0$ has nontrivial solutions (one for each choice of x_3). From above equations we have,

$$x_1 = \frac{4}{3}x_3, \quad x_2 = 0, \text{ with } x_3 \text{ free.}$$

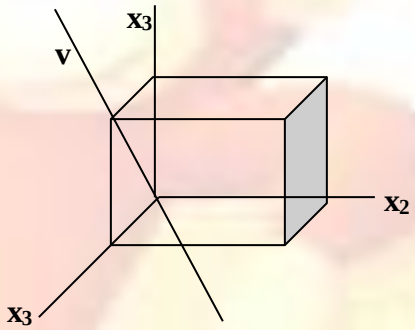
As a vector, the general solution of $Ax = 0$ is given by:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix} = x_3 v, \qquad \text{where} \quad v = \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$

This shows that every solution of $\boldsymbol{Ax = 0}$ in this case is a scalar multiple of $\boldsymbol{v}$ (it means that v generate or spans the whole general solution).The trivial solution is obtained by choosing $\boldsymbol{x_3 = 0}$.

Geometric Interpretation

Geometrically, the solution set is a line through $\boldsymbol{0}$ in $\boldsymbol{R^3}$, as given in the **Figure below:**



Note: A nontrivial solution $\boldsymbol{x}$ can have some zero entries so long as not all of its entries are zero.

Example 2

Solve the following system

$$10x_1 - 3x_2 - 2x_3 = 0 \qquad (1)$$

Solution

Solving for the basic variable $\boldsymbol{x_1}$ in terms of the free variables, dividing eq. 1 by 10 and solve for x

$\boldsymbol{x_1 = 0.3x_2 + 0.2x_3}$ *where* $\boldsymbol{x_2}$ and $\boldsymbol{x_3}$ free variables.

As a vector, the general solution is:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.3x_2 + 0.2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.3x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.2x_3 \\ 0 \\ x_3 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} 0.3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0.2 \\ 0 \\ 1 \end{bmatrix} \qquad (2)$$

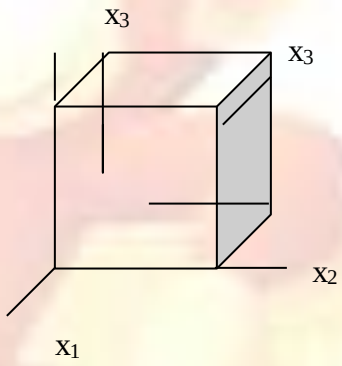
$\boldsymbol{u}$ $\boldsymbol{v}$

This calculation shows that every solution of (1) is a linear combination of the vector ***u, v*** shown in (2). That is, the solution set is Span {***u, v***}

.

Geometric Interpretation

Since neither ***u*** nor ***v*** is a scalar multiple of the other, so these are not parallel, the solution set is a plane through the origin, see the Figure below:



Note:

Above examples illustrate the fact that the solution set of a homogeneous equation $Ax = 0$ can be expressed explicitly as Span **{v_1, v_2, … , v_p}** for suitable vectors $v_1, v_2, \ldots, v_p$ (because solution sets can be written in the form of linear combination of these vectors). If the only solution is the zero-vector then the solution set is **Span {0}**.

Example 3 (For Practice) Find the solution set of the following homogenous system:

$$x_1 + 3x_2 + x_3 = 0$$
$$-4x_1 - 9x_2 + 2x_3 = 0$$
$$-3x_2 - 6x_3 = 0$$

Solution:

Let $A = \begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

The augmented matrix is:

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ -4 & -9 & 2 & 0 \\ 0 & -3 & -6 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & -3 & -6 & 0 \end{bmatrix} \qquad 4R_1 + R_2,$$

$$\sim \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad R_2 + R_3$$

$$\sim \begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \frac{1}{2}R_2,\ (-3)R_2 + R_1$$

SO

$$\begin{aligned} x_1 - 5x_3 &= 0 \\ x_2 + 2x_3 &= 0 \\ 0 &= 0 \end{aligned}$$

From above results, it is clear that x_3 is a free variable, so $\boldsymbol{Ax = 0}$ has nontrivial solutions (one for each choice of x_3).

From above equations we have,

$$x_1 = 5x_3, \quad x_2 = -2x_3, \text{ with } x_3 \text{ a free variable.}$$

As a vector, the general solution of $\boldsymbol{Ax = 0}$ is given by

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} = x_3 v, \qquad \textit{where} \quad v = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

Parametric Vector Form of the solution

Whenever a solution set is described explicitly with vectors, we say that the solution is in **parametric vector form**

The equation

$$\boldsymbol{x} = s\boldsymbol{u} + t\boldsymbol{v} \qquad (s, t \text{ in } \boldsymbol{R})$$

is called a **parametric vector equation** of the plane. It is written in this form to emphasize that the parameters vary over all real numbers.

Similarly, the equation $\boldsymbol{x} = x_3\boldsymbol{v}$ (with x_3 free), or $\boldsymbol{x} = t\boldsymbol{v}$ (with t in $\boldsymbol{R}$), is a parametric vector equation of a line.

Solutions of Non-homogeneous Systems

When a non-homogeneous linear system has many solutions, the general solution can be written in parametric vector form as one vector plus an arbitrary linear combination of vectors that satisfy the corresponding homogeneous system.
To clear this concept consider the following examples,

Example: 5 Describe all solutions of $\boldsymbol{Ax = b}$, where

$$A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$$

Solution
Row operations on **[A b]** produce

$$\begin{bmatrix} 3 & 5 & -4 & 7 \\ -3 & -2 & 4 & -1 \\ 6 & 1 & -8 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & 5 & -4 & 7 \\ 0 & 3 & 0 & 6 \\ 0 & -9 & 0 & -18 \end{bmatrix} \qquad R_1 + R_2, -2R_1 + R_3$$

$$\sim \begin{bmatrix} 3 & 5 & -4 & 7 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad 3R_2 + R_3, \frac{1}{3}R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -\frac{4}{3} & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad -5R_2 + R_1, \frac{1}{3}R_1$$

$$x_1 - \frac{4}{3}x_3 = -1$$

$$OR \quad x_2 = 2$$

$$0 = 0$$

Thus $x_1 = -1 + \frac{4}{3}x_3$, $x_2 = 2$, and $\boldsymbol{x_3}$ is free.

As a vector, the general solution of $\boldsymbol{Ax = b}$ has the form

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 + \frac{4}{3}x_3 \\ 2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$

$\boldsymbol{p}$ $\boldsymbol{v}$

The equation $\boldsymbol{x = p + x_3 v}$, or, writing $\boldsymbol{t}$ as a general parameter,

$$\boldsymbol{x = p + tv} \quad (\boldsymbol{t} \text{ in } \boldsymbol{R}) \qquad (3)$$

Note

We know that the solution set of this question when $\boldsymbol{Ax = 0}$ (example 1) has the parametric vector equation

$$\boldsymbol{x = tv} \quad (\boldsymbol{t} \text{ in } \boldsymbol{R}) \qquad (4)$$

With the same $\boldsymbol{v}$ that appears in equation (3) in above example.
Thus the solutions of $\boldsymbol{Ax = b}$ are obtained by adding the vector $\boldsymbol{p}$ to the solutions of $\boldsymbol{Ax = 0}$. The vector $\boldsymbol{p}$ itself is just one particular solution of $\boldsymbol{Ax = b}$ (corresponding to $\boldsymbol{t = 0}$ in (3)).

The following theorem gives the precise statement.

Theorem

Suppose the equation $Ax = b$ is consistent for some given b, and let p be a solution. Then the solution set of $Ax = b$ is the set of all vectors of the form $w = p + v_h$, where v_h is any solution of the homogeneous equation Ax = 0.

Example 6: (For practice)

$$\begin{aligned} x_1 + 3x_2 + x_3 &= 1 \\ -4x_1 - 9x_2 + 2x_3 &= -1 \\ -3x_2 - 6x_3 &= -3 \end{aligned}$$

Solution

Let $A = \begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix}$

The augmented matrix is

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ -4 & -9 & 2 & -1 \\ 0 & -3 & -6 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 3 & 6 & 3 \\ 0 & -3 & -6 & -3 \end{bmatrix} \qquad 4R_1 + R_2,$$

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 3 & 6 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad R_2 + R_3$$

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \frac{1}{3}R_2$$

$$\begin{bmatrix} 1 & 0 & -5 & -2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad (-3)R_2 + R_1$$

SO

$$x_1 - 5x_3 = -2$$
$$x_2 + 2x_3 = 1$$
$$0 = 0$$

Thus $x_1 = -2 + 5x_3$, $x_2 = 1 - 2x_3$, and x_3 is free.

As a vector, the general solution of $\boldsymbol{Ax = b}$ has the form

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 + 5x_3 \\ 1 - 2x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

$\boldsymbol{p}$ $\qquad$ $\boldsymbol{v}$

So we can write solution set in parametric vector form as

$$x = p + x_3 v$$

Steps of Writing a Solution Set (of a Consistent System) in a Parametric Vector Form

Step 1:
Row reduces the augmented matrix to reduced echelon form.

Step 2:
Express each basic variable in terms of any free variables appearing in an equation.

Step 3:
Write a typical solution ***x*** as a vector whose entries depend on the free variables if any.

Step 4:
Decompose ***x*** into a linear combination of vectors (with numeric entries) using the free variables as parameters.

Exercise

Determine if the system has a nontrivial solution. Try to use as few row operations as possible.

1. $x_1 - 5x_2 + 9x_3 = 0$
$-x_1 + 4x_2 - 3x_3 = 0$
$2x_1 - 8x_2 + 9x_3 = 0$

2. $3x_1 + 6x_2 - 4x_3 - x_4 = 0$
$-5x_1 + 8x_3 + 3x_4 = 0$
$8x_1 - x_2 + 7x_4 = 0$

3. $5x_1 - x_2 + 3x_3 = 0$
$4x_1 - 3x_2 + 7x_3 = 0$

Write the solution set of the given homogeneous system in parametric vector form.

4. $x_1 - 3x_2 - 2x_3 = 0$
$x_2 - x_3 = 0$
$-2x_1 + 3x_2 + 7x_3 = 0$

5. $x_1 + 2x_2 - 7x_3 = 0$
$-2x_1 - 3x_2 + 9x_3 = 0$
$-2x_2 + 10x_3 = 0$

In exercises 6-8, describe all solutions of $\boldsymbol{Ax = 0}$ in parametric vector form where A is row equivalent to the matrix shown.

6. $\begin{bmatrix} 1 & -5 & 0 & 2 & 0 & -4 \\ 0 & 0 & 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 6 & 0 & 8 & -1 & -2 \\ 0 & 0 & 1 & -3 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

8. $\begin{bmatrix} 1 & -5 & 0 & 4 \end{bmatrix}$

9. Describe the solution set in R^3 of $x_1 - 4x_2 + 3x_3 = 0$, compare it with the solution set of $x_1 - 4x_2 + 3x_3 = 7$.

10. Find the parametric equation of the line through ***a*** parallel to ***b***.

$$a = \begin{bmatrix} 3 \\ -8 \end{bmatrix}, b = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$$

11. Find a parametric equation of the line *M* through ***p*** and ***q***.

$$p = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, q = \begin{bmatrix} 0 \\ 7 \end{bmatrix}$$

12. Given $A = \begin{bmatrix} 5 & 10 \\ -8 & -16 \\ 7 & 14 \end{bmatrix}$, find one nontrivial solution of $\boldsymbol{Ax = 0}$ by inspection.

13. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 3 & 9 \end{bmatrix}$, find one nontrivial solution of $\boldsymbol{Ax = 0}$ by inspection.

Lecture 8

Linear Independence

Definition

An indexed set of vectors $\{v_1, v_2, \ldots, v_p\}$ in R^n is said to be **linearly independent** if the vector equation $x_1v_1 + x_2v_2 + \cdots + x_pv_p = 0$ has only the trivial solution. The set $\{v_1, v_2, \ldots, v_p\}$ is said to be **linearly dependent** if there exist weights $c_1, \ldots, c_p$, not all zero, such that $c_1v_1 + c_2v_2 + \cdots + c_pv_p = 0$ (1)

Equation (1) is called a **linear dependence relation** among $v_1, \ldots, v_p$, when the weights are not all zero.

Example 1

Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, $v_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

(a) Determine whether the set of vectors $\{v_1, v_2, v_3\}$ is linearly independent or not.
(b) If possible, find a linear dependence relation among v_1, v_2, v_3.

Solution

(a) Row operations on the associated augmented matrix show that

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -6 & -6 & 0 \end{bmatrix} \qquad (-2)R_1 + R_2,\ (-3)R_1 + R_3$$

$$\sim \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad R_2 + R_3 \qquad (2)$$

Clearly, x_1 and x_2 are basic variables and x_3 is free. Each nonzero value of x_3 determines a nontrivial solution.
Hence v_1, v_2, v_3 are linearly dependent (and not linearly independent).

(b) To find a linear dependence relation among v_1, v_2, v_3, completely row reduce the augmented matrix and write the new system:

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad -1R_2 + 3$$

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_1 - 4R_2$$

$$\Rightarrow \begin{aligned} x_1 \quad & -2x_3 = 0 \\ x_2 \quad & +x_3 = 0 \\ 0 \quad & = 0 \end{aligned}$$

Thus $x_1 = 2x_3$, $x_2 = -x_3$, and x_3 is free.

Choose any nonzero value for x_3, say, $x_3 = 5$, then $x_1 = 10$, and $x_2 = -5$.
Substitute these values into $x_1v_1 + x_2v_2 + x_3v_3 = 0$

$$\Rightarrow \quad 10v_1 - 5v_2 + 5v_3 = 0$$

This is one (out of infinitely many) possible linear dependence relation among v_1, v_2, v_3.

Example (for practice)

Check whether the vectors are linearly dependent or linearly independent.

$v_1 = (3, -1)$ $v_2 = (-2, 2)$

Solution

Consider two constants C_1 and C_2. Suppose:

$$c_1(3, -1) + c_2(-2, 2) = 0$$

$$(3c_1 - 2c_2, -c_1 + 2c_2) = (0, 0)$$

Now, set each of the components equal to zero to arrive at the following system of equations:

$$3c_1 - 2c_2 = 0$$
$$-c_1 + 2c_2 = 0$$

Solving this system gives the following solution,

$$c_1 = 0 \qquad c_2 = 0$$

The trivial solution is the only solution, so these two vectors are linearly independent.

Linear Independence of Matrix Columns

Suppose that we begin with a matrix $A = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix}$ instead of a set of vectors. The matrix equation $\boldsymbol{Ax = 0}$ can be written as $x_1a_1 + x_2a_2 + \cdots + x_na_n = 0$

Each linear dependence relation among the columns of ***A*** corresponds to a nontrivial solution of $\boldsymbol{Ax = 0}$.

Thus we have the following important fact.

The columns of a matrix A are linearly independent if and only if the equation $Ax = 0$ has only the trivial solution.

Example 2 Determine whether the columns of $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

Solution To study $\boldsymbol{Ax = 0}$, row reduce the augmented matrix:

$$\begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \qquad R_{12}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{bmatrix} \qquad (-5)R_1 + R_3$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{bmatrix} \qquad (2)R_2 + R_3$$

At this point, it is clear that there are three basic variables and no free variables. So the equation $\boldsymbol{Ax = 0}$ has only the trivial solution, and the columns of ***A*** are linearly independent.

Sets of One or Two Vectors

A set containing only one vector (say, ***v***) is linearly independent if and only if ***v*** is not the zero vector. This is because the vector equation $\boldsymbol{x_1v = 0}$ has only the trivial solution when $v \neq 0$. The zero vector is linearly dependent because $\boldsymbol{x_10 = 0}$ has many nontrivial solutions.

Example 3

Check the following sets for linearly independence and dependence.

a. $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

b. $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

Solution

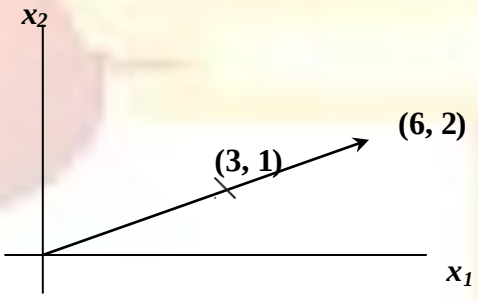
a) Notice that $\boldsymbol{v_2}$ is a multiple of $\boldsymbol{v_1}$, namely, $\boldsymbol{v_2 = 2v_1}$.
Hence $\boldsymbol{-2v_1 + v_2 = 0}$, which shows that $\boldsymbol{\{v_1, v_2\}}$ is linearly dependent.

b) $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are certainly not multiples of one another. Could they be linearly dependent?

Suppose $\boldsymbol{c}$ and $\boldsymbol{d}$ satisfy $\boldsymbol{cv_1 + dv_2 = 0}$

If $c \neq 0$, then we can solve for $\boldsymbol{v_1}$ in terms of $\boldsymbol{v_2}$, namely, $\boldsymbol{v_1 = (-d/c)\, v_2}$. This result is impossible because $\boldsymbol{v_1}$ is not a multiple of $\boldsymbol{v_2}$. So, $\boldsymbol{c}$ must be zero. Similarly, $\boldsymbol{d}$ must also be zero.
Thus $\boldsymbol{\{v_1, v_2\}}$ is a linearly independent set.

Note A set of two vectors $\{v_1, v_2\}$ is linearly dependent if and only if one of the vectors is a multiple of the other.
In geometric terms, two vectors are linearly dependent if and only if they lie on the same line through the origin. Figure 1, shows the vectors from Example 3.



Linearly dependent

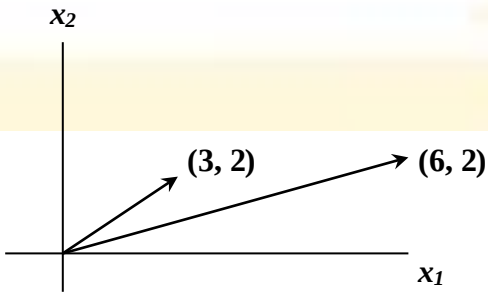


Figure 1 Linearly independent

Sets of Two or More Vectors

Theorem (Characterization of Linearly dependent Sets)

An indexed set $s = \{v_1, v_2, \cdots, v_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in ***S*** is a linear combination of the others. In fact, if **S** is linearly dependent, and $v \neq 0$, then some v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \cdots, v_{j-1}$.

Proof

If some v_j in ***S*** equals a linear combination of the other vectors, then v_j can be subtracted from both sides of the equation, producing a linear dependence relation with a nonzero weight (–1) on v_j.

For instance, if $\boldsymbol{v_1 = c_2v_2 + c_3v_3}$, then $\boldsymbol{0 = (-1)v_1 + c_2v_2 + c_3v_3 + 0v_4 + \ldots + 0v_p}$.
Thus ***S*** is linearly dependent.

Conversely, suppose ***S*** is linearly dependent. If $\boldsymbol{v_1}$ is zero, then it is a (trivial) linear combination of the other vectors in ***S***.

If $v \neq 0$ and there exist weights $\boldsymbol{c_1, \ldots, c_p}$, not all zero(because vectors are linearly dependent), such that

$$\boldsymbol{c_1v_1 + c_2v_2 + \ldots + c_pv_p = 0}$$

Let j be the largest subscript for which $c_j \neq 0$. If $j = 1$, then $\boldsymbol{c_1v_1 = 0}$, which is impossible because $v_j \neq 0$.

So $j > 1$, and $c_1v_1 + \cdots + c_jv_j + 0v_{j+1} + \cdots + 0v_p = 0$

$$c_jv_j = -c_1v_1 - c_2v_2 - \cdots - c_{j-1}v_{j-1}$$

$$v_j = \left(-\frac{c_1}{c_j}\right)v_1 + \left(-\frac{c_2}{c_j}\right)v_2 + \cdots + \left(\frac{c_{j-1}}{c_j}\right)v_{j-1}$$

Note: This theorem does not say that every vector in a linearly dependent set is a linear combination of the preceding vectors. A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

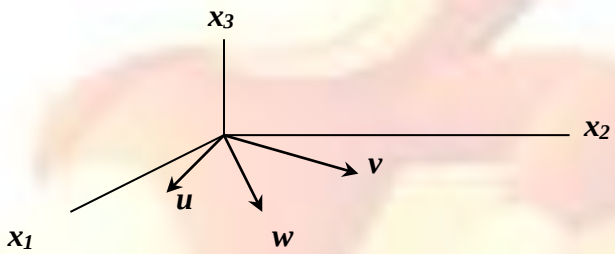
Example 4 Let $u = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}$. Describe the set spanned by ***u*** and ***v***, and prove that a vector ***w*** is in Span **{*u, v*}** if and only if **{*u, v, w*}** is linearly dependent.

Solution

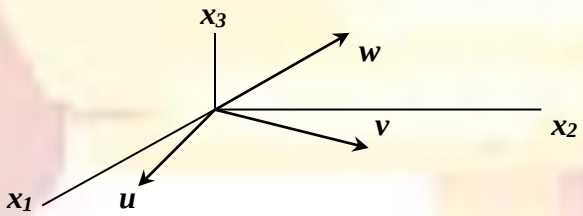
The vectors ***u*** and ***v*** are linearly independent because neither vector is a multiple of the other, nor so they span a plane in $\boldsymbol{R^3}$. In fact, Span **{*u, v*}** is the x_1x_2-**plane** (with $x_3 = 0$). If ***w*** is a linear combination of ***u*** and ***v***, then **{*u, v, w*}** is linearly dependent.

Conversely, suppose that **{*u, v, w*}** is linearly dependent.

Some vector in **{*u, v, w*}** is a linear combination of the preceding vectors (since $u \neq 0$). That vector must be ***w***, since ***v*** is not a multiple of ***u***. So ***w*** is in Span **{*u, v*}**



Linearly dependent ***w*** in Span **{*u, v*}**.



Linearly independent ***w*** not in **Span {*u, v*}**

Figure 2: Linear dependence in $\boldsymbol{R^3}$.

This example generalizes to any set **{*u, v, w*}** in $\boldsymbol{R^3}$ with ***u*** and ***v*** linearly independent. The set **{*u, v, w*}** will be linearly dependent if and only if ***w*** is in the plane spanned by ***u*** and ***v***.

Theorem

If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1, v_2, \ldots, v_p\}$ in R^n is linearly dependent if $p > n$.

Example 5 The vectors $\begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}4\\-1\end{bmatrix}, \begin{bmatrix}-2\\2\end{bmatrix}$ are linearly dependent, because there are three vectors in the set and there are only two entries in each vector.

Notice, however, that none of the vectors is a multiple of one of the other vectors. See Figure 4.

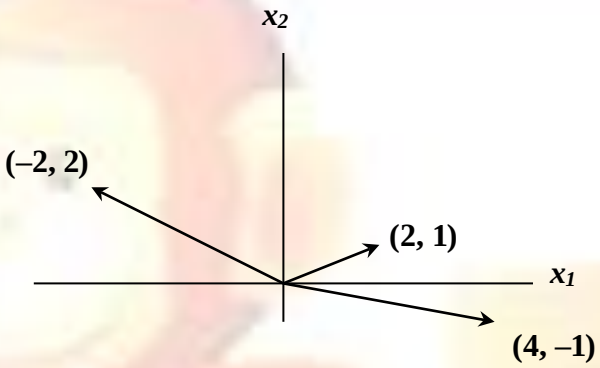


Figure 4 A linearly dependent set in $\boldsymbol{R^2}$.

Theorem

If a set $S = \{v_1, v_2, \ldots, v_p\}$ in R^n contains the zero vector, then the set is linearly dependent**.**

Proof

By renumbering the vectors, we may suppose that $\boldsymbol{v_1 = 0}$.
Then (1)$\boldsymbol{v_1 + 0v_2 + \ldots + 0v_p = 0}$ shows that $\boldsymbol{S}$ is linearly dependent(because in this relation coefficient of v_1 is non zero).

Example 6 Determine by inspection if the given set is linearly dependent.

a. $\begin{bmatrix}1\\7\\6\end{bmatrix}, \begin{bmatrix}2\\0\\9\end{bmatrix}, \begin{bmatrix}3\\1\\5\end{bmatrix}, \begin{bmatrix}4\\1\\8\end{bmatrix}$ **b.** $\begin{bmatrix}2\\3\\5\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\8\end{bmatrix}$ **c.** $\begin{bmatrix}-2\\4\\6\\10\end{bmatrix}, \begin{bmatrix}3\\-6\\-9\\15\end{bmatrix}$

Solution

a) The set contains four vectors that each has only three entries. So,the set is linearly dependent by the Theorem above.
b) The same theorem does not apply here because the number of vectors does not exceed the number of entries in each vector. Since the zero vector is in the set, the set is linearly dependent by the next theorem.

c) As we compare corresponding entries of the two vectors, the second vector seems to be –3/2 times the first vector. This relation holds for the first three pairs of entries, but fails for the fourth pair. Thus neither of the vectors is a multiple of the other, and hence they are linearly independent.

Exercise

1. Let $u = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}, v = \begin{bmatrix} -6 \\ 1 \\ 7 \end{bmatrix}, w = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}, and \quad z = \begin{bmatrix} 3 \\ 7 \\ -5 \end{bmatrix}.$

(i) Are the sets **{*u, v*}, {*u, w*}, {*u, z*}, {*v, w*}, {*v, z*},** and **{*w, z*}** each linearly independent? Why or why not?
(ii) Does the answer to Problem (i) imply that **{*u, v, w, z*}** is linearly independent?
(iii) To determine if **{*u, v, w, z*}** is linearly dependent, is it wise to check if, say ***w*** is a linear combination of ***u, v*** and ***z***?
(iv) Is **{*u, v, w, z*}** linear dependent?

Decide if the vectors are linearly independent. Give a reason for each answer.

2. $\begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 0 \end{bmatrix}$

3. $\begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ -6 \end{bmatrix}$

Determine if the columns of the given matrix form a linearly dependent set.

4. $\begin{bmatrix} 1 & 3 & -2 & 0 \\ 3 & 10 & -7 & 1 \\ -5 & -5 & 3 & 7 \end{bmatrix}$

5. $\begin{bmatrix} 3 & 4 & 3 \\ -1 & -7 & 7 \\ 1 & 3 & -2 \\ 0 & 2 & -6 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 1 & 0 & 4 \\ -1 & 0 & 3 & -1 \\ 0 & -2 & 1 & 1 \\ 1 & 0 & -1 & 3 \end{bmatrix}$

7. $\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & 1 & 3 \end{bmatrix}$

For what values of h is v_3 in span $\{v_1, v_2\}$ and for what values of h is $\{v_1, v_2, v_3\}$ linearly dependent?

8. $v_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} -2 \\ -6 \\ 4 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 2 \\ h \end{bmatrix}$

9. $v_1 = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 9 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} -2 \\ -6 \\ h \end{bmatrix}$

Find the value(s) of h for which the vectors are linearly dependent.

10. $\begin{bmatrix}1\\3\\-3\end{bmatrix}, \begin{bmatrix}-2\\-4\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\h\end{bmatrix}$

11. $\begin{bmatrix}1\\-5\\-2\end{bmatrix}, \begin{bmatrix}-3\\8\\6\end{bmatrix}, \begin{bmatrix}4\\h\\-8\end{bmatrix}$

Determine by inspection whether the vectors are linearly independent. Give reasons for your answers.

12. $\begin{bmatrix}5\\5\end{bmatrix}, \begin{bmatrix}6\\1\end{bmatrix}, \begin{bmatrix}2\\4\end{bmatrix}, \begin{bmatrix}3\\-6\end{bmatrix}$

13. $\begin{bmatrix}2\\-5\\1\end{bmatrix}, \begin{bmatrix}-6\\5\\3\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}$

14. $\begin{bmatrix}6\\2\\-8\end{bmatrix}, \begin{bmatrix}3\\1\\-2\end{bmatrix}$

15. Given $A = \begin{bmatrix}2 & 3 & 5\\-5 & 1 & -4\\-3 & -1 & -4\\1 & 0 & 1\end{bmatrix}$, observe that the third column is the sum of the first two columns. Find a nontrivial solution of $\boldsymbol{Ax = 0}$ without performing row operations.

Each statement in exercises 16-18 is either true(in all cases) or false(for at least one example). If false, construct a specific example to show that the statement is not always true. If true, give a justification.

16. If $v_1, \ldots, v_4$ are in R^4, and $v_3 = 2v_1 + v_2$, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.

17. If v_1 and v_2 are in R^4, and v_1 is not a scalar multiple of v_2, then $\{v_1, v_2\}$ is linearly independent.

18. . If $v_1, \ldots, v_4$ are in R^4, and$\{v_1, v_2, v_3\}$ is linearly dependent, then $\{v_1, v_2, v_3, v_4\}$ is also linearly dependent.

19. Use as many columns of $A = \begin{bmatrix}8 & -3 & 0 & -7 & 2\\-9 & 4 & 5 & 11 & -7\\6 & -2 & 2 & -4 & 4\\5 & -1 & 7 & 0 & 10\end{bmatrix}$ as possible to construct a matrix B with the property that equation $\boldsymbol{Bx = 0}$ has only the trivial solution. Solve

Bx = 0 to verify your work.



Lecture 9

Linear Transformations

Outlines

- Matrix Equation
- Transformation, Examples, Matrix as Transformations
- Linear Transformation, Examples, Some Properties

Matrix Equation

An equation ***Ax = b*** is called a matrix equation in which a matrix ***A*** acts on a vector ***x*** by multiplication to produce a new vector called ***b.***

For instance, the equations

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$A \qquad x \qquad b$

and

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$A \qquad u \qquad \underline{o}$

Solution of Matrix Equation

Solution of the ***Ax = b*** consists of those vectors ***x*** in ***the domain*** that are transformed into the vector ***b*** in ***range***.

Matrix equation ***Ax = b*** is an important example of transformation we would see later in the lecture.

Transformation or Function or Mapping

A transformation (or **function** or **mapping**) ***T*** from R^n to R^m is a rule that assigns to each vector ***x*** in R^n, an image vector ***T(x)*** in R^m.

$$T : R^n \rightarrow R^m$$

The set R^n is called the **domain** of ***T***, and R^m is called the **co-domain** of ***T***. For ***x*** in R^n the set of all images ***T(x)*** is called the **range** of ***T***.

Note

To define a mapping or function, domain and co-domain are the ordinary sets. However to define a linear transformation, the domain and co-domain has to be $\mathbb{R}^m (or\ \mathbb{R}^n)$. Moreover a map $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a linear transformation if for any two vectors say $u, v \in \mathbb{R}^m$ and the scalars c_1, c_2 the following equation is satisfied

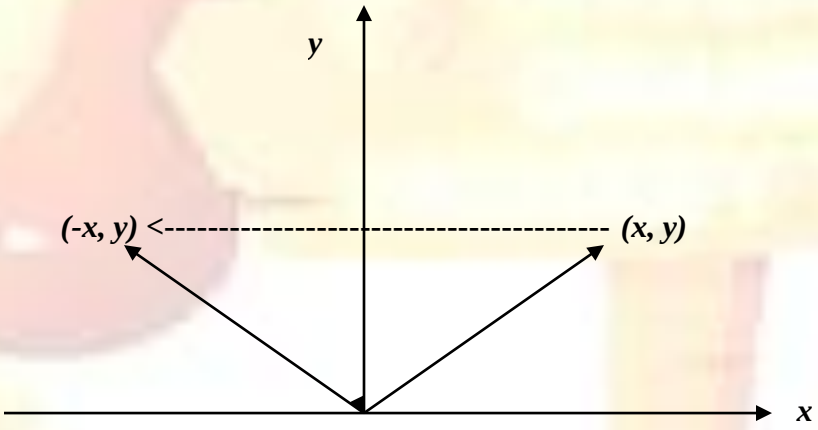
$$T(c_1u + c_2v) = c_1T(u) + c_2T(v)$$

Example1 Consider a mapping $T : R^2 \rightarrow R^2$ defined by $T(x, y) = (-x, y)$. This transformation is a reflection about y-axis in xy plane.

Here $T(1, 2) = (-1, 2)$. T has transformed vector (1,2) into another vector (-1,2).

In matrix form:

$$Tv = Av = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$



Further the projection transformation $T : R^2 \rightarrow R^2$ defined by $T(x, y) = (x, 0)$ is given as:

$$Tv = Av = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

Example 2 Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$,

and define a transformation $T : R^2 \to R^3$ by $\boldsymbol{T(x) = Ax}$, so that

$$T(x) = Ax = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

a) Find ***T (u)***, the image of ***u*** under the transformation ***T***.
b) Find an ***x*** in $\boldsymbol{R^2}$, whose image under ***T*** is ***b***.
c) Is there more than one ***x*** whose image under ***T*** is ***b***?
d) Determine if ***c*** is in the range of the transformation ***T***.

Solution (a)

$$T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

Here $T(u) = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$

Here the matrix transformation has transformed $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ into another vector $\begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$

(b) We have to find an x such that **T (x) = b or Ax = b**

i. e $$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \qquad (1)$$

Now row reduced augmented matrix will be:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \xrightarrow{-3R_1 + R_2,\ R_1 + R_3} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \xrightarrow{\frac{1}{14}R_2,\ -4R_2 + R_3} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{3R_2 + R_1} \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence $x_1 = 1.5$, $x_2 = -0.5$, and $x = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}$.

The image of this x under T is the given vector b.

(c) From (2) it is clear that equation (1) has a unique solution. So, there is exactly one x whose image is b.

(d) The vector c is in the range of T if c is the image of some x in R^2, that is, if $c = T(x)$ for some x. This is just another way of asking if the system $Ax = c$ is consistent. To find the answer, we will row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \xrightarrow{-3R_1 + R_2,\ R_1 + R_3} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \xrightarrow{\frac{1}{14}R_2,\ R_3 - 4R_2} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 10 \end{bmatrix} \xrightarrow{14R_2 + R_3} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 14 & -7 \end{bmatrix} \xrightarrow{\frac{1}{10}R_3} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 1 \end{bmatrix}$$

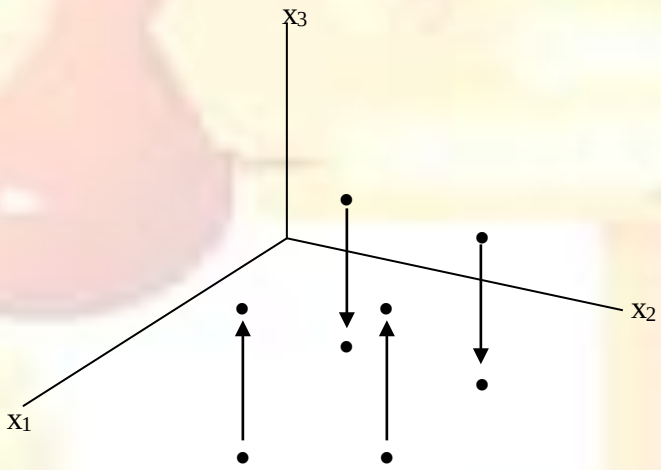
$$x_1 - 3x_2 = 3$$
$$0x_1 + x_2 = 2$$
$$0x_1 + 0x_2 = -35 \Rightarrow 0 = 35 \; but \; 0 \neq 35$$

Hence the system is inconsistent. So **c** is not in the range of **T**.

So from the above example we can view a transformation in the form of a matrix. We'll see that a transformation $T : R^n \to R^m$ can be transformed into a matrix of order $m \times n$ and every matrix of order $m \times n$ can be viewed as a linear transformation.

The next two matrix transformations can be viewed geometrically. They reinforce the dynamic view of a matrix as something that transforms vectors into other vectors.

Example 3 If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then the transformation $x \to Ax$ projects points in $\boldsymbol{R^3}$ onto the $x_1x_2 - coordinate \; plane$ because $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$



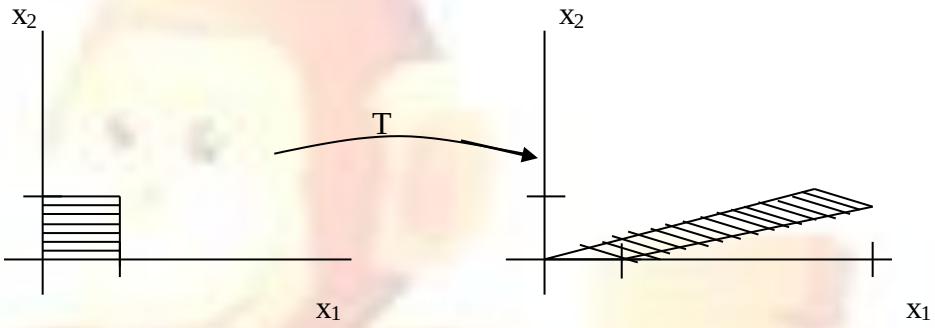
A projection transformation

Example 4 Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$., the transformation $T : R^2 \to R^2$ defined by $\boldsymbol{T(x) = Ax}$ is called a **shear** transformation.

The image of the point $u = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ *is* $T(u) = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$,

and the image of $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ *is* $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$.

Here, ***T*** deforms the square as if the top of the square was pushed to the right while the base is held fixed. Shear transformations appear in physics, geology and crystallography.



A shear transformation

Linear Transformations

We know that if ***A*** is $m \times n$ matrix, then the transformation $x \to Ax$ has the properties $A(u+v) = Au + Av$ *and* $A(cu) = cAu$ for all ***u, v*** in $\boldsymbol{R^n}$ and all scalars ***c***.
These properties for a transformation identify the most important class of transformations in linear algebra.

Definition A transformation (or mapping) ***T*** is linear if:

1. ***T(u + v) = T(u) + T(v)*** for all ***u, v*** in the domain of ***T***;
2. ***T(cu) = cT(u)*** for all ***u*** and all scalars ***c***.

Example 5 Every matrix transformation is a linear transformation.

Example 6 Let $L : R^3 \to R^2$ be defined by $L(x, y, z) = (x, y)$.

we let $\boldsymbol{u} = (x_1, y_1, z_1)$ and $\boldsymbol{v} = (x_2, y_2, z_2)$.

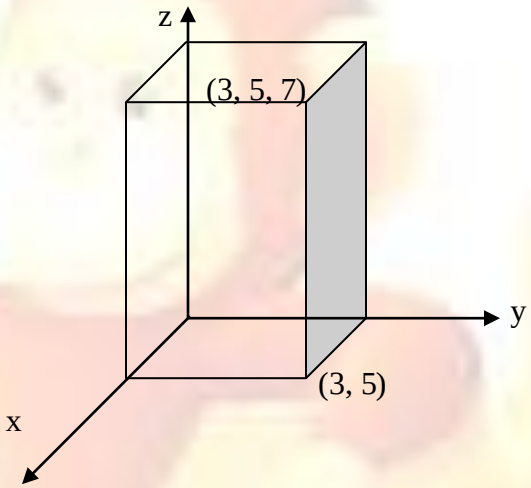
$$L(\boldsymbol{u}+\boldsymbol{v}) = L((x_1, y_1, z_1) + (x_2, y_2, z_2))$$
$$= L(x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$= (x_1 + x_2, y_1 + y_2)$$
$$= (x_1, y_1) + (x_2, y_2) = L(\boldsymbol{u}) + L(\boldsymbol{v})$$

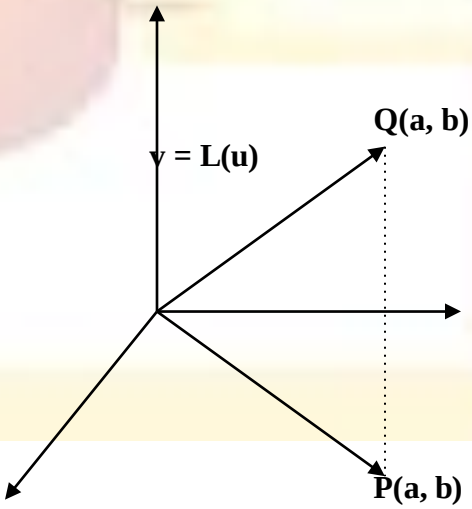
Also, if k is a real number, then

$$L(k\boldsymbol{u}) = L(kx_1, ky_1, kz_1) = (kx_1, ky_1) = kL(\boldsymbol{u})$$

Hence, ***L*** is a linear transformation, which is called a projection. The image of the vector (or point) (3, 5, 7) is the vector (or point) (3, 5) in xy-plane. See figure below:



Geometrically the image under ***L*** of a vector (a, b, c) in $\boldsymbol{R^3}$ is (a, b) in $\boldsymbol{R^2}$ can be found by drawing a line through the end point P(a, b, c) of ***u*** and perpendicular to $\boldsymbol{R^2}$, the *xy*-plane. The intersection Q(a, b)of this line with the *xy*-plane will give the image under ***L***. See the figure below:



Example 7 Let $L : R \to R$ be defined by $L(x) = x^2$

Let x and y in R and

$$L(x+y) = (x+y)^2 = x^2 + y^2 + 2xy \neq x^2 + y^2 = L(x) + L(y)$$
$$\Rightarrow L(x+y) \neq L(x) + L(y)$$

So we conclude that the function ***L*** is not a linear transformation.

Linear transformations preserve the operations of vector addition and scalar multiplication.

Properties

If ***T*** is a linear transformation, then

1. ***T(0) = 0***
2. ***T(cu +dv) = cT(u) + dT(v)***
3. ***T(c_1v_1 + …+ c_pv_p) = c_1T(v_1) + … + c_pT(v_p)***

for all vectors ***u, v*** in the domain of ***T*** and all scalars *c*, *d*.

Proof

1. By the definition of Linear Transformation we have ***T(cu) = cT(u)*** for all ***u*** and all scalars ***c***. Put $c = 0$ we'll get ***T(0u)= 0T(u)*** This implies ***T(0) = 0***

2. Just apply the definition of linear transformation. i. e

T(cu + dv) = T(cu) + T(dv) = cT(u) + dT(v)

Property (3) follows from (ii), because ***T(0) = T (0u) =0T(u) = 0***.
Property (4) requires both (i) and (ii):

OBSERVATION Observe that if a transformation satisfies property 2 for all ***u, v*** and *c*, *d*, it must be linear (Take $c = d = 1$ for preservation of addition, and take $d = 0$)

3. Generalizing Property 2 we'll get 3

$$T(c_1v_1 + \ldots + c_pv_p) = c_1T(v_1) + \ldots + c_pT(v_p)$$

Applications in Engineering

In engineering and physics, property 3 is referred to as a superposition principle. Think of $v_1, \ldots, v_p$ as signals that go into a system or process and $T(v_1), \ldots, T(v_p)$ as the responses of that system to the signals. The system satisfies the superposition principle if an input is expressed as a linear combination of such signals, the system's response is the same linear combination of the responses to the individual signals.

Example 8 Given a scalar r, define $T : R \to R$ by

$$T(x) = x+1.$$

T is not a linear transformation (why!) because $T(0) \neq 0$ (by property 3)

Example 9 Given a scalar r, define $T : R^2 \to R^2$ by $T(x) = rx$.
T is called a **contraction** when $0 \leq r < 1$
and a **dilation** when $r \geq 1$.

Let $r = 3$ and show that T is a linear transformation.

Solution Let u, v be in R^2 and let c, d be scalars, then

$$\begin{aligned} T(cu + dv) &= 3(cu + dv) && \text{Definition of } T \\ &= 3cu + 3dv \\ &= c(3u) + d(3v) && \text{Vector arithmetic} \\ &= cT(u) + dT(v) \end{aligned}$$

Thus T is a linear transformation because it satisfies (4).

Example 10 Define a linear transformation $T : R^2 \to R^2$ by

$$T(x) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

Find the images under T of $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, *and* $u + v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.

Solution
$$T(u)=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}4\\ 1\end{bmatrix}=\begin{bmatrix}-1\\ 4\end{bmatrix},\qquad T(v)=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}2\\ 3\end{bmatrix}=\begin{bmatrix}-3\\ 2\end{bmatrix}$$
$$T(u+v)=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}6\\ 4\end{bmatrix}=\begin{bmatrix}-4\\ 6\end{bmatrix}$$

In the above example, ***T*** rotates ***u, v*** and ***u + v*** counterclockwise through 90^0.
In fact, ***T*** transforms the entire parallelogram determined by ***u*** and ***v*** into the one determined by ***T (u)*** and ***T (v)***

Example 11 Let L: $R^3 \to R^2$ be a linear transformation for which:
L (1, 0, 0) = (2, -1),
L (0, 1, 0) = (3, 1), and
L (0, 0, 1) = (-1, 2).
Then find L (-3, 4, 2).

Solution Since (-3, 4, 2) = -3**i** + 4**j** + 2**k**,

$$L(-3,4,2)=L(-3\boldsymbol{i}+4\boldsymbol{j}+2\boldsymbol{k})=-3L(\boldsymbol{i})+4L(\boldsymbol{j})+2L(\boldsymbol{k})$$
$$=-3(2,-1)+4(3,1)+2(-1,2)=(4,11)$$

Exercise

1. Suppose that $T: R^5 \to R^2$ and ***T(x) = Ax*** for some matrix ***A*** and each ***x*** in $\boldsymbol{R^5}$. How many rows and columns do ***A*** have?
2. Let $A=\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$. Give a geometric description of the transformation $x \to Ax$.
3. The line segment from **0** to a vector ***u*** is the set of points of the form ***tu***, where $0 \le t \le 1$. Show that a linear transformation ***T*** maps this segment into the segment between **0** and ***T(u)***.
4. Let $A=\begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 2\end{bmatrix}, u=\begin{bmatrix}1\\ 0\\ -3\end{bmatrix}, and \begin{bmatrix}5\\ -1\\ 4\end{bmatrix}$. Define T:$R^3 \to R^3$ by T (x) = Ax. Find T (u) and T (v).

In exercises 5 and 6, with T defined by T (x) = Ax, find an x whose image under T is b, and determine if x is unique.

5. $A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 1 & -5 \\ -4 & 2 & 1 \end{bmatrix}, b = \begin{bmatrix} 0 \\ -5 \\ -6 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -4 \\ 3 & 2 & 1 \\ -2 & -1 & -2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ -5 \\ -7 \\ 3 \end{bmatrix}$

Find all x in R^4 that are mapped into the zero vector by the transformation $x \to Ax$.

7. $A = \begin{bmatrix} 1 & 2 & -7 & 5 \\ 0 & 1 & -4 & 0 \\ 1 & 0 & 1 & 6 \\ 2 & -1 & 6 & 8 \end{bmatrix}$

8. $\begin{bmatrix} 1 & 3 & 4 & -3 \\ 0 & 1 & 3 & -2 \\ 3 & 7 & 6 & -5 \end{bmatrix}$

9. Let $b = \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix}$ and let A be the matrix in exercise 8. Is b in the range of the linear transformation $x \to Ax$?

10. Let $b = \begin{bmatrix} 9 \\ 5 \\ 0 \\ -9 \end{bmatrix}$ and let A be the matrix in exercise 7. Is b in the range of the linear transformation $x \to Ax$?

Let T (x) = Ax for x in R^2.

(a) On a rectangular coordinate system, plot the vectors u, v, T (u) and T (v).

(b) Give a geometric description of what T does to a vector x in R^2.

11. $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, u = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, and\ v = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

12. $A = \begin{bmatrix} .5 & 0 \\ 0 & .5 \end{bmatrix}, u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, and\ v = \begin{bmatrix} -5 \\ -2 \end{bmatrix}$

13. Let T: $R^2 \to R^2$ be a linear transformation that maps $u = \begin{bmatrix} 1 \\ 5 \end{bmatrix} into \begin{bmatrix} 2 \\ 0 \end{bmatrix} and\ maps\ v = \begin{bmatrix} 3 \\ 1 \end{bmatrix} into \begin{bmatrix} 1 \\ -4 \end{bmatrix}$. Use the fact that T is linear find the images under T of 2u, 3v, and 2u + 3v.

14. Let $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, y_1 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}, and\ y_2 = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$. Let T: $R^2 \to R^2$ be a linear transformation that maps e_1 into y_1 and maps e_2 into y_2. Find the images of $\begin{bmatrix} 7 \\ 6 \end{bmatrix} and \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

15. Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, v_1 = \begin{bmatrix} -7 \\ 4 \end{bmatrix} and\ v_2 = \begin{bmatrix} 3 \\ -8 \end{bmatrix}$. Let T: $R^2 \to R^2$ be a linear transformation that maps x into $x_1v_1 + x_2v_2$. Find a matrix A such that T (x) is Ax for each x.

Lecture 10

The Matrix of a Linear Transformation

Outline

- Matrix of a Linear Transformation.
- Examples, Geometry of Transformation, Reflection and Rotation
- Existence and Uniqueness of solution of ***T(x)=0***

In the last lecture we discussed that every linear transformation from $\boldsymbol{R^n}$ to $\boldsymbol{R^m}$ is actually a matrix transformation $x \to Ax$, where A is a matrix of order $m \times n$. First see an example

Example 1 The columns of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Suppose ***T*** is a linear transformation from $\boldsymbol{R^2}$ into $\boldsymbol{R^3}$ such that

$$T(e_1) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} \quad and \quad T(e_2) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}$$

with no additional information, find a formula for the image of an arbitrary $\boldsymbol{x}$ in $\boldsymbol{R^2}$.

Solution Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x_1 e_1 + x_2 e_2$

Since ***T*** is a linear transformation, $T(x) = x_1 T(e_1) + x_2 T(e_2)$

$$T(x) = x_1 \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix} = \begin{bmatrix} 5x_1 - 3x_2 \\ -7x_1 + 8x_2 \\ 2x_1 + 0 \end{bmatrix}$$

$$Hence \quad T(x) = \begin{bmatrix} 5x_1 - 3x_2 \\ -7x_1 + 8x_2 \\ 2x_1 + 0 \end{bmatrix}$$

Theorem Let $T : R^n \to R^m$ be a linear transformation. Then there exists a unique matrix ***A*** such that $T(x) = Ax$ for all $\boldsymbol{x}$ in $\boldsymbol{R^n}$

In fact, ***A*** is the $m \times n$ matrix whose *jth* column is the vector ***T (e$_j$)***, where e_j is the *jth* column of the identity matrix in $\boldsymbol{R^n}$.

$$A = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}$$

Proof Write

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \\ \vdots \\ 0 \end{bmatrix} + \dots + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$= x_1 e_1 + \dots + x_n e_n = \begin{bmatrix} e_1 & \dots & e_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} e_1 & \dots & e_n \end{bmatrix} x$$

Since T is Linear, So

$$T(x) = T(x_1 e_1 + \dots + x_n e_n) = x_1 T(e_1) + \dots + x_n T(e_n)$$

$$= \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = Ax \qquad (1)$$

The matrix ***A*** in (1) is called the **standard matrix for the linear transformation *T***. We know that every linear transformation from $\boldsymbol{R^n}$ to $\boldsymbol{R^m}$ is a matrix transformation and vice versa.

The term linear transformation focuses on a property of a mapping, while matrix transformation describes how such a mapping is implemented, as the next three examples illustrate.

Example 2 Find the standard matrix ***A*** for the dilation transformation ***T (x) = 3x***, $x \in R^2$.

Solution Write

$$T(e_1) = 3e_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad and \quad T(e_2) = 3e_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

Example 3 Let $L: R^3 \to R^3$ is the linear operator defined by $L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y-z \\ x+z \end{bmatrix}$.

Find the standard matrix representing ***L*** and verify ***L (x) = Ax***.

Solution

The standard matrix ***A*** representing ***L*** is the 3 x 3 matrix whose columns are ***L*** (e_1), ***L*** (e_2), and ***L*** (e_3) respectively. Thus

$$L(e_1) = L\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1+0 \\ 0-0 \\ 1+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = col_1(A)$$

$$L(e_2) = L\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0+1 \\ 1-0 \\ 0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = col_2(A)$$

$$L(e_3) = L\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0+0 \\ 0-1 \\ 0+1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = col_3(A)$$

Hence $$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$$

Now $$\boldsymbol{Ax} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ y-z \\ x+z \end{bmatrix} = L(x)$$

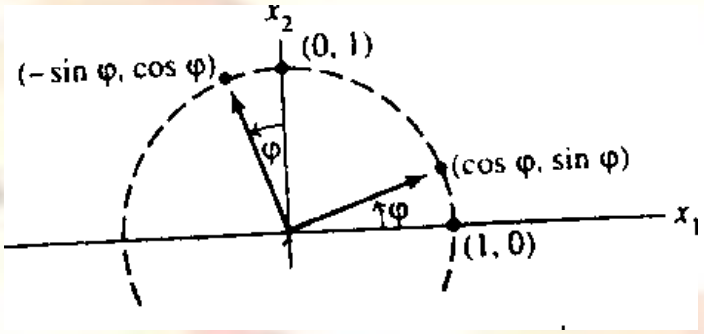
Hence verified.

Example 4 Let $T: R^2 \to R^2$ be the transformation that rotates each point in $\boldsymbol{R^2}$ through an angle ϕ, with counterclockwise rotation for a positive angle. We could show geometrically that such a transformation is linear. Find the standard matrix ***A*** of this transformation.

Solution $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ rotates into $\begin{bmatrix} \cos\phi \\ \sin\phi \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ rotates into $\begin{bmatrix} -\sin\phi \\ \cos\phi \end{bmatrix}$.

See figure below.

By above theorem $A = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$



A rotation transformation

Example 5 A reflection with respect to the ***x***-axis of a vector ***u*** in $\boldsymbol{R^2}$ is defined by the linear operator $L(u) = L\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ -a_2 \end{bmatrix}$.

Then $L(e_1) = L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $L(e_2) = L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

Hence the standard matrix representing ***L*** is $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Thus we have $L(u) = Au = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ -a_2 \end{bmatrix}$

To illustrate a reflection with respect to the x-axis in computer graphics, let the triangle ***T*** have vertices (-1, 4), (3, 1), and (2, 6).
To reflect ***T*** with respect to ***x***-axis, we let $u_1 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, u_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and compute the images ***L*** $\boldsymbol{(u_1)}$, ***L*** $\boldsymbol{(u_2)}$, and ***L*** $\boldsymbol{(u_3)}$ by forming the products

$$Au_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ -4 \end{bmatrix},$$

$$Au_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix},$$

$$Au_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \end{bmatrix}.$$

Thus the image of ***T*** has vertices (-1, -4), (3, -1), and (2, -6).

Geometric Linear Transformations of R^2

Examples 3-5 illustrate linear transformations that are described geometrically. In example 4 transformations is a rotation in the plane. It rotates each point in the plane through an angle ϕ. Example 5 is reflection in the plane.

Existence and Uniqueness of the solution of *T(x)=b*

The concept of a linear transformation provides a new way to understand existence and uniqueness questions asked earlier. The following two definitions give the appropriate terminology for transformations.

Definition A mapping $T : R^n \to R^m$ is said to be onto $\boldsymbol{R^m}$ if each ***b*** in $\boldsymbol{R^m}$ is the image of at least one ***x*** in $\boldsymbol{R^n}$.

OR

Equivalently, ***T*** is onto $\boldsymbol{R^m}$ if for each ***b*** in $\boldsymbol{R^m}$ there exists at least one solution of ***T (x) = b***. "Does ***T*** map $\boldsymbol{R^n}$ onto $\boldsymbol{R^m}$?" is an existence question.

The mapping ***T*** is not onto when there is some ***b*** in $\boldsymbol{R^m}$ such that the equation ***T (x) = b*** has no solution.

Definition A mapping $T : R^n \to R^m$ is said to be one-to-one (or 1:1) if each ***b*** in $\boldsymbol{R^m}$ is the image of at most one ***x*** in $\boldsymbol{R^n}$.

OR

Equivalently, **T** is one-to-one if for each $\boldsymbol{b}$ in $\boldsymbol{R^m}$ the equation $\boldsymbol{T(x) = b}$ has either a unique solution or none at all, "Is $\boldsymbol{T}$ one-to-one?" is a uniqueness question.

The mapping $\boldsymbol{T}$ is not one-to-one when some $\boldsymbol{b}$ in $\boldsymbol{R^m}$ is the image of more than one vector in $\boldsymbol{R^n}$. If there is no such $\boldsymbol{b}$, then $\boldsymbol{T}$ is one-to-one.

Example 6 Let $\boldsymbol{T}$ be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does $\boldsymbol{T}$ map $\boldsymbol{R^4}$ onto $\boldsymbol{R^3}$? Is $\boldsymbol{T}$ a one-to-one mapping?

Solution Since $\boldsymbol{A}$ happens to be in echelon form, we can see at once that $\boldsymbol{A}$ has a pivot position in each row.

We know that for each $\boldsymbol{b}$ in $\boldsymbol{R^3}$, the equation $\boldsymbol{Ax = b}$ is consistent. In other words, the linear transformation $\boldsymbol{T}$ maps $\boldsymbol{R^4}$ (its domain) onto $\boldsymbol{R^3}$.

However, since the equation $\boldsymbol{Ax = b}$ has a free variable (because there are four variables and only three basic variables), each $\boldsymbol{b}$ is the image of more than one $\boldsymbol{x}$. That is, $\boldsymbol{T}$ is not one-to-one.

Theorem Let $T : R^n \to R^m$ be a linear transformation. Then $\boldsymbol{T}$ is one-to-one if and only if the equation $\boldsymbol{T(x) = 0}$ has only the trivial solution.

Proof: Since $\boldsymbol{T}$ is linear, $\boldsymbol{T(0) = 0}$ if $\boldsymbol{T}$ is one-to-one, then the equation $\boldsymbol{T(x) = 0}$ has at most one solution and hence only the trivial solution. If $\boldsymbol{T}$ is not one-to-one, then there is a $\boldsymbol{b}$ that is the image of at least two different vectors in $\boldsymbol{R^n}$ (say, $\boldsymbol{u}$ and $\boldsymbol{v}$).
That is, $\boldsymbol{T(u) = b}$ and $\boldsymbol{T(v) = b}$.

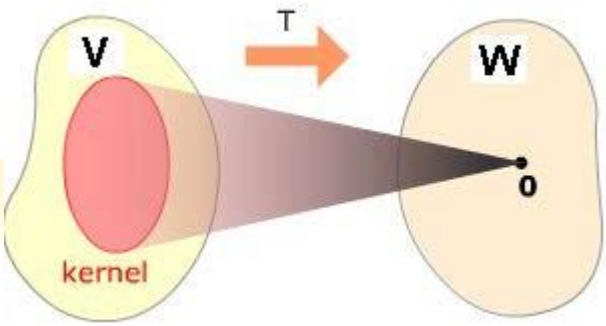
But then, since $\boldsymbol{T}$ is linear $T(u - v) = T(u) = b - b = 0$

The vector $\boldsymbol{u - v}$ is not zero, since $u \neq v$. Hence the equation $\boldsymbol{T(x) = 0}$ has more than one solution. So either the two conditions in the theorem are both true or they are both false.

Kernel of a Linear Transformation

Let $T : V \to W$ be a linear transformation. Then kernel of T (usually written as Ker$T$), is the set of those elements in V which maps onto the zero vector in W .Mathematically:

$$Ker T = \{ v \in V \mid T(v) = 0 \text{ in } W \}$$



Remarks

i) Ker$T$ is subspace of V

ii) T is one-one iff $KerT = 0$ in V

One-One Linear Transformation

Let $T : V \to W$ be a linear transformation. Then T is said to be one-one if for any $u, v \in V$ with $u \neq v$ implies $Tu \neq Tv$. Equivalently if $Tu = Tv$ then $u = v$.

T is said to be one-to-one or bijective if

i) T is one-one

ii)T is onto

Theorem Let $T : R^n \to R^m$ be a linear transformation and let ***A*** be the standard matrix for ***T***. Then

(a) ***T*** maps $\boldsymbol{R^n}$ onto $\boldsymbol{R^m}$ if and only if the columns of ***A*** span $\boldsymbol{R^m}$;

(b) ***T*** is one-to-one if and only if the columns of **A** are linearly independent.

Proof:

(a) The columns of ***A*** span $\boldsymbol{R^m}$ if and only if for each ***b*** the equation ***Ax = b*** is consistent – in other words, if and only if for every ***b***, the equation ***T(x) = b*** has at least one solution. This is true if and only if ***T*** maps $\boldsymbol{R^n}$ onto $\boldsymbol{R^m}$.

(b) The equations ***T (x) = 0*** and ***Ax = 0*** are the same except for notation. So ***T*** is one-to-one if and only if ***Ax = 0*** has only the trivial solution. This happens if and only if the columns of ***A*** are linearly independent.

We can also write column vectors in rows, using parentheses and commas. Also, when we apply a linear transformation ***T*** to a vector – say, $x = \begin{bmatrix} x_1 \\ \\ x_2 \end{bmatrix} = (x_1, x_2)$ we write

$T(x_1, x_2)$ instead of the more formal $T\big((x_1, x_2)\big)$.

Example 7 Let $\boldsymbol{T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)}$.

Show that ***T*** is a one-to-one linear transformation.

Does ***T*** map $\boldsymbol{R^2}$ onto $\boldsymbol{R^3}$?

Solution When x and $T(x)$ are written as column vectors, it is easy to see that T is described by the equation

$$\begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x^1 \\ x^2 \end{bmatrix} = \begin{bmatrix} 3x_1 + x_2 \\ 5x^1 + 7x^2 \\ x^1 + 3x^2 \end{bmatrix} \qquad (4)$$

so T is indeed a linear transformation, with its standard matrix A shown in (4). The columns of A are linearly independent because they are not multiples. Hence T is one-to-one. To decide if T is onto R^3, we examine the span of the columns of A. Since A is 3×2, the columns of A span R^3 if and only if A has 3 pivot positions. This is impossible, since A has only 2 columns. So the columns of A do not span R^3 and the associated linear transformation is not onto R^3.

Exercises

1. Let $T: R^2 \to R^2$ be transformation that first performs a horizontal shear that maps e_2 into $e_2 - .5e_1$ (but leaves e_1 unchanged) and then reflects the result in the x_2 – axis. Assuming that ***T*** is linear, find its standard matrix.

Assume that T is a linear transformation. Find the standard matrix of T.

2. $T: R^2 \to R^3, T(1,0) = (4,-1,2)\ and\ T(0,1) = (-5,3,-6)$

3. $T: R^3 \to R^2, T(e_1) = (1,4), T(e_2) = (-2,9), and\ T(e_3) = (3,-8)$, where e_1, e_2, and e_3 are the columns of the identity matrix.

4. $T: R^2 \to R^2$ rotates points clockwise through π radians.

5. $T: R^2 \to R^2$ is a "vertical shear" transformation that maps e_1 into $e_1 + 2e_2$ but leaves the vector e_2 unchanged.

6. $T: R^2 \to R^2$ is a "horizontal shear" transformation that maps e_2 into $e_2 - 3e_1$ but leaves the vector e_1 unchanged.

7. $T: R^3 \to R^3$ projects each point (x_1, x_2, x_3) vertically onto the x_1x_2-plane (where x_3=0).

8. $T: R^2 \to R^2$ first performs a vertical shear mapping e_1 into $e_1 - 3e_2$(leaving e_2 unchanged) and then reflects the result in the x_2-axis.

9. $T: R^2 \to R^2$ first rotates points counterclockwise through $\pi/4$ radians and then reflects the result in the x_2-axis.

Show that T is a linear transformation by finding a matrix that implements the mapping. Note that x_1, x_2, … are not vectors but are entries in vectors.

10. $T(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2 + x_3, x_3 + x_4, 0)$

11. $T(x_1, x_2, x_3) = (3x_2 - x_3, x_1 + 4x_2 + x_3)$

12. $T(x_1, x_2, x_3, x_4) = 3x_1 - 4x_2 + 8x_4$

13. Let $T: R^2 \to R^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, 4x_1 + 7x_2)$. Find ***x*** such that T (x) =(-2, -5).

13. Let $T : R^2 \to R^3$ be a linear transformation such that
$T(x_1, x_2) = (x_1 + 2x_2, -x_1 - 3x_2, -3x_1 - 2x_2)$. Find ***x*** such that T (x) =(-4, 7, 0).

In exercises 14 and 15, let T be the linear transformation whose standard matrix is given.

14. Decide if T is one-to-one mapping. Justify your answer.

$$\begin{bmatrix} -5 & 10 & -5 & 4 \\ 8 & 3 & -4 & 7 \\ 4 & -9 & 5 & -3 \\ -3 & -2 & 5 & 4 \end{bmatrix}$$

15. Decide if T maps R^5 onto R^5. Justify your answer.

$$\begin{bmatrix} 4 & -7 & 3 & 7 & 5 \\ 6 & -8 & 5 & 12 & -8 \\ -7 & 10 & -8 & -9 & 14 \\ 3 & -5 & 4 & 2 & -6 \\ -5 & 6 & -6 & -7 & 3 \end{bmatrix}$$

Lecture 11

Matrix Operations

(i-j)th Element of a matrix

Let A be an $m \times n$ matrix, where m and n are number of rows and number of columns respectively, then a_{ij} represents the ***i***-th row and ***j***-th column entry of the matrix. For example a_{12} represents 1st row and 2nd column entry.

Similarly a_{32} represents 3rd row and 2nd column entry. The columns of ***A*** are vectors in $\boldsymbol{R^m}$ and are denoted by (boldface) $\boldsymbol{a_1, a_2, \cdots, a_n}$.

These columns are $A = \begin{bmatrix} \boldsymbol{a_1} & \boldsymbol{a_2} & \ldots & \boldsymbol{a_n} \end{bmatrix}$

The number a_{ij} is the ***i-th*** entry (from the top) of ***j-th*** column vector a_j.

$$\begin{array}{c} \\ \\ Row\ i \end{array} \begin{array}{c} \begin{array}{cccccc} & & & Column & & \\ & & & j & & \end{array} \\ \begin{bmatrix} a_{11} & \ldots & a_{1j} & \ldots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \ldots & a_{ij} & \ldots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \ldots & a_{mj} & \ldots & a_{mn} \end{bmatrix} = A \\ \begin{array}{ccccc} \uparrow & & \uparrow & & \uparrow \\ \boldsymbol{a_1} & & \boldsymbol{a_j} & & \boldsymbol{a_n} \end{array} \end{array}$$

Figure 1 Matrix notation.

Definitions

A **diagonal matrix** is a square matrix whose non-diagonal entries are zero.

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{bmatrix}$$

The diagonal entries in $A = \begin{bmatrix} a_{ij} \end{bmatrix}$ are $a_{11}, a_{22}, a_{33}, \cdots$ and they form the **main diagonal** of ***A.***

For example $\begin{bmatrix} 5 & 0 \\ 0 & 7 \end{bmatrix}$ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 11 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ are all diagonal matrices.

Null Matrix or Zero Matrix

An $m \times n$ matrix whose entries are all zero is a Null or **zero matrix** and is always written as ***O***. A null matrix may be of any order.

For example $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

3 x 3 3 x 2 4 x 5

are all Zero Matrices

Equal Matrices

Two matrices are said to be **equal** if they have the same size (i.e., the same number of rows and columns) and same corresponding entries.

Example 1 Consider the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 3 & x+1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \end{bmatrix}$$

The matrices ***A*** and ***B*** are equal if and only if $x+1 = 5$ or $x = 4$. There is no value of x for which ***A*** = ***C***, since ***A*** and ***C*** have different sizes.

If ***A*** and B are $m \times n$ matrices, then the **sum, *A* + *B***, is the $m \times n$ matrix whose columns are the sums of the corresponding columns in ***A*** and ***B***. Each entry in ***A*** + ***B*** is the sum of the corresponding entries in ***A*** and ***B***. The sum ***A*** + ***B*** is defined only when ***A*** and ***B*** are of the same size.

If r is a scalar and ***A*** is a matrix, then the **scalar multiple** r***A*** is the matrix whose columns are r times the corresponding columns in ***A***.

Note: Negative of a matrix A is defined as $-A$ to mean $(-1)A$ and the difference of A and B is written as $A-B$, which means $A + (-1)B$.

Example 2 Let $A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix}$, $C = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$

Then $A + B = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$

But ***A + C*** is not defined because ***A*** and ***C*** have different sizes.

$$2B = 2\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix}$$

$$A - 2B = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 3 \\ -7 & -7 & -12 \end{bmatrix}$$

Theorem 1 Let ***A, B,*** and ***C*** are matrices of the same size, and let *r* and *s* are scalars.

a.	***A + B = B + A***	d.	***r (A + B)= r A + r B***
b.	***(A + B) + C = A + (B + C)***	e.	***(r + s) A = r A + s A***
c.	***A + 0 = A***	f.	***r (sA) = (rs) A***

Each equality in Theorem 1 can be verified by showing that the matrix on the left side has the same size as the matrix on the right and that corresponding columns are equal. Size is no problem because ***A, B,*** and ***C*** are equal in size. The equality of columns follows immediately from analogous properties of vectors.

For instance, if the *jth* columns of ***A, B,*** and ***C*** are a_j, b_j and c_j, respectively, then the *jth* columns of ***(A + B) + C*** and ***A + (B + C)*** are

$$(a_j + b_j) + c_j \qquad and \qquad a_j + (b_j + c_j)$$

respectively. Since these two vector sums are equal for each *j*, property (b) is verified.

Because of the associative property of addition, we can simply write ***A + B + C*** for the sum, which can be computed either as ***(A + B) + C*** or ***A + (B + C)***. The same applies to sums of four or more matrices.

Matrix Multiplication

Multiplying an m×n matrix with an n×p matrix results in an m×p matrix. If many matrices are multiplied together, and their dimensions are written in a list in order, e.g. m×n, n×p, p×q, q×r, the size of the result is given by the first and the last numbers (m×r).

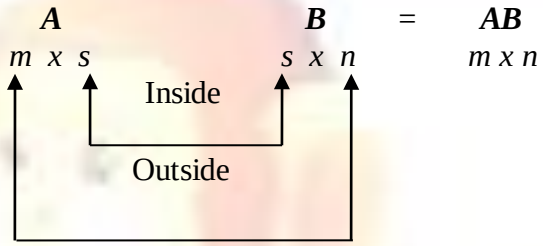
It is important to keep in mind that this definition requires the number of columns of the first factor A to be the same as the number of rows of the second factor B. When this condition is satisfied, the sizes of A and B are said to conform for the product AB. If the sizes of A and B do not conform for the product AB, then this product is undefined.

Definition If ***A*** is an $m \times n$ matrix, and if ***B*** is an $n \times p$ matrix with columns $b_1, \cdots, b_p$, then the product ***AB*** is the $m \times p$ matrix whose columns are $Ab_1, \cdots, Ab_p$.

That is $$AB = A\begin{bmatrix} b_1 & b_2 & \dots & b_p \end{bmatrix} = \begin{bmatrix} Ab_1 & Ab_2 & \dots & Ab_p \end{bmatrix}$$

This definition makes equation (1) true for all $\boldsymbol{x}$ in $\boldsymbol{R^p}$. Equation (1) proves that the composite mapping ***(AB)*** is a linear transformation and that its standard matrix is ***AB***. Multiplication of matrices corresponds to composition of linear transformations.

A convenient way to determine whether ***A*** and ***B*** conform for the product ***AB*** and, if so, to find the size of the product is to write the sizes of the factors side by side as in Figure below (the size of the first factor on the left and the size of the second factor on the right).



If the inside numbers are the same, then the product ***AB*** is defined and the outside numbers then give the size of the product.

Example 3 Compute ***AB***, where $A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$ *and* $B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}$

Solution: Here $\boldsymbol{B = [b_1 \quad b_2 \quad b_3]}$, therefore

$$Ab_1 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad Ab^2 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}\begin{bmatrix} 3 \\ -2 \end{bmatrix}, \quad Ab_3 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}\begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 11 \\ -1 \end{bmatrix} \qquad = \begin{bmatrix} 0 \\ 13 \end{bmatrix} \qquad = \begin{bmatrix} 21 \\ -9 \end{bmatrix}$$

Then

$$AB = A[b_1 \quad b_2 \quad b_3] = \begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{bmatrix}$$

$\boldsymbol{Ab_1} \quad \boldsymbol{Ab_2} \quad \boldsymbol{Ab_3}$

Note from the definition of ***AB*** that its first column, $\boldsymbol{Ab_1}$, is a linear combination of the columns of ***A***, using the entries in $\boldsymbol{b_1}$ as weights. The same holds true for each column of ***AB***. Each column of ***AB*** is a linear combination of the columns of ***A*** using weights from the corresponding column of ***B***.

Example 4 Find the product ***AB*** for

$$A=\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix} \quad \text{and} \quad B=\begin{bmatrix}4 & 1 & 4 & 3\\ 0 & -1 & 3 & 1\\ 2 & 7 & 5 & 2\end{bmatrix}$$

Solution It follows from definition that the product ***AB*** is formed in a column-by-column manner by multiplying the successive columns of ***B*** by ***A***. The computations are

$$\begin{matrix} & c_1 & c_2 & c_3\end{matrix}$$

$$\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix}\begin{bmatrix}4\\ 0\\ 2\end{bmatrix} = 4c_1 + 0c_2 + 2c_3 = (4)\begin{bmatrix}1\\ 2\end{bmatrix} + (0)\begin{bmatrix}2\\ 6\end{bmatrix} + (2)\begin{bmatrix}4\\ 0\end{bmatrix} = \begin{bmatrix}12\\ 8\end{bmatrix}$$

Similarly,

$$\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix}\begin{bmatrix}1\\ -1\\ 7\end{bmatrix} = (1)\begin{bmatrix}1\\ 2\end{bmatrix} + (-1)\begin{bmatrix}2\\ 6\end{bmatrix} + (7)\begin{bmatrix}4\\ 0\end{bmatrix} = \begin{bmatrix}27\\ -4\end{bmatrix}$$

$$\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix}\begin{bmatrix}4\\ 3\\ 5\end{bmatrix} = (4)\begin{bmatrix}1\\ 2\end{bmatrix} + (3)\begin{bmatrix}2\\ 6\end{bmatrix} + (5)\begin{bmatrix}4\\ 0\end{bmatrix} = \begin{bmatrix}30\\ 26\end{bmatrix}$$

$$\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix}\begin{bmatrix}3\\ 1\\ 2\end{bmatrix} = (3)\begin{bmatrix}1\\ 2\end{bmatrix} + (1)\begin{bmatrix}2\\ 6\end{bmatrix} + (2)\begin{bmatrix}4\\ 0\end{bmatrix} = \begin{bmatrix}13\\ 12\end{bmatrix}$$

Thus,

$$AB=\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix}\begin{bmatrix}4 & 1 & 4 & 3\\ 0 & -1 & 3 & 1\\ 2 & 7 & 5 & 2\end{bmatrix} = \begin{bmatrix}12 & 27 & 30 & 13\\ 8 & -4 & 26 & 12\end{bmatrix}$$

Example 5 (An Undefined Product) Find the product ***BA*** for the matrices

$$A=\begin{bmatrix}1 & 2 & 4\\ 2 & 6 & 0\end{bmatrix} \quad \textit{and} \quad B=\begin{bmatrix}4 & 1 & 4 & 3\\ 0 & -1 & 3 & 1\\ 2 & 7 & 5 & 2\end{bmatrix}$$

Solution The number of columns of B is not equal to number of rows of A so BA multiplication is not possible.

The matrix ***B*** has size 3×4 and the matrix ***A*** has size 2×3. The "inside" numbers are not the same, so the product ***BA*** is undefined.

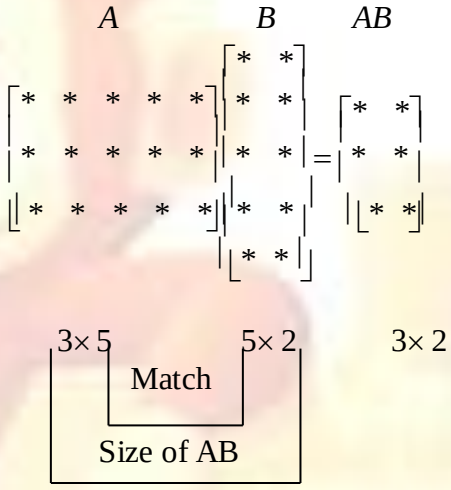
Obviously, the number of columns of ***A*** must match the number of rows in ***B*** in order for a linear combination such as $\boldsymbol{Ab_1}$ to be defined. Also, the definition of ***AB*** shows that ***AB*** has the same number of rows as ***A*** and the same number of columns as ***B***.

Example 6 If ***A*** is a 3×5 matrix and ***B*** is a 5×2 matrix, what are the sizes of ***AB*** and ***BA***, if they are defined?

Solution The product of matrices A and B of orders 3×5 and 5×2 will result in 3×2 matrix AB.

But for BA we have 5×2 and 3×5, here number of columns in1st matrix are 2 which is not equal to number of rows in 2nd matrix. So BA is not possible.

Since ***A*** has 5 columns and ***B*** has 5 rows, the product ***AB*** is defined and is a 3×2 matrix:



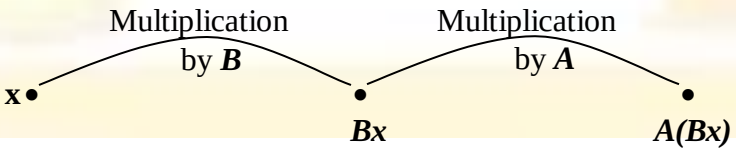
The product ***BA*** is not defined because the 2 columns of ***B*** do not match the 3 rows of ***A***.

The definition of ***AB*** is important for theoretical work and applications, but the following rule provides a more efficient method for calculating the individual entries in ***AB*** when working small problems by hand.

Row-Column Rule for Computing *AB*

Explanation

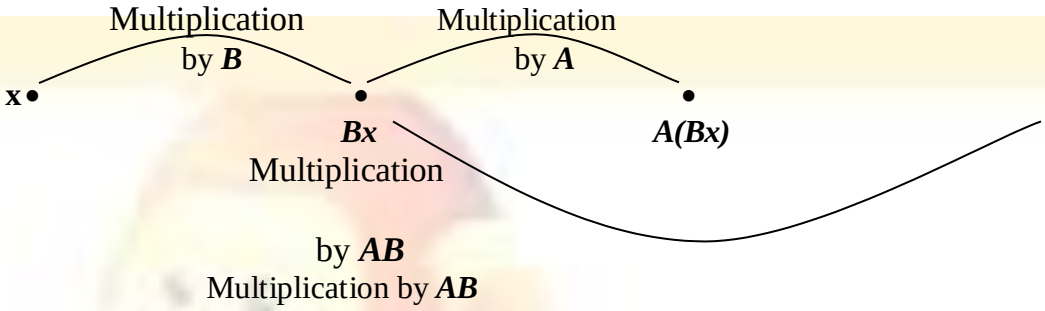
If a matrix ***B*** is multiplied with a vector ***x***, it transforms ***x*** into a vector ***Bx***. If this vector is then multiplied in turn by a matrix ***A***, the resulting vector is ***A (Bx)***.



Multiplication by ***B*** and then ***A***

Thus ***A(Bx)*** is produced from ***x*** by a composition of mappings. Our goal is to represent this composite mapping as multiplication by a single matrix, denoted by ***AB***, so that

$$A\,(Bx) = (AB)\,x \text{----------(1)}$$



If A is $m \times n$, B is $n \times p$, and x is in R^p, denote the columns of B by $b_1, \cdots, b_p$ and the entries in x by $x_1, \cdots, x_p$, then $Bx = x_1 b_1 + x_2 b_2 + \cdots + x_p b_p$

By the linearity of multiplication by A,

$$\begin{aligned} A(Bx) &= A(x_1 b_1) + A(x_2 b_2) + \cdots + A(x_p b_p) \\ &= x_1 Ab_1 + x_2 Ab_2 + \cdots + x_p Ab_p \end{aligned}$$

The vector $A\,(Bx)$ is a linear combination of the vectors $Ab_1, \cdots, Ab_p$, using the entries in x as weights. If we rewrite these vectors as the columns of a matrix, we have

$$A(Bx) = \begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_p \end{bmatrix} x$$

Thus multiplication by $\begin{bmatrix} Ab_1 & Ab_2 & \cdots & Ab_p \end{bmatrix}$ transforms x into $A(Bx)$.
We have found the matrix we sought!

Row-Column Rule for Computing *AB*

If the product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B. If $(AB)_{ij}$ denotes the (i, j) – entry in AB, and if A is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1} b_{1j} + a_{i2} b_{2j} + \ldots + a_{in} b_{nj}$$

To verify this rule, let $B = \begin{bmatrix} b_1 & \cdots & b_p \end{bmatrix}$. Column j of AB is Ab_j, and we can compute Ab_j. The ith entry in Ab_j is the sum of the products of corresponding entries from row i of A and the vector b_j, which is precisely the computation described in the rule for computing the (i, j) – entry of AB.

Finding Specific Entries in a Matrix Product Sometimes we will be interested in finding a specific entry in a matrix product without going through the work of computing the entire column that contains the entry.

Example 7 Use the row-column rule to compute two of the entries in ***AB*** for the matrices in Example 3.

Solution: To find the entry in row 1 and column 3 of ***AB***, consider row 1 of ***A*** and column 3 of ***B***. Multiply corresponding entries and add the results, as shown below:

$$AB = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} \square & \square & 2(6)+3(3) \\ \square & \square & \square \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & \square & \square \end{bmatrix}$$

For the entry in row 2 and column 2 of ***AB***, use row 2 of ***A*** and column 2 of ***B***:

$$\begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 1(3)+-5(-2) & \square \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 13 & \square \end{bmatrix}$$

Example 8 Use the dot product rule to compute the individual entries in the product of ***AB*** where $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}$ *and* $B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$.

Solution Since ***A*** has size 2×3 and ***B*** has size 3×4, the product ***AB*** is a 2×4 matrix of the form

$$AB = \begin{bmatrix} r_1(A)\times c_1(B) & r_1(A)\times c_2(B) & r_1(A)\times c_3(B) & r_1(A)\times c_4(B) \\ r_2(A)\times c_1(B) & r_2(A)\times c_2(B) & r_2(A)\times c_3(B) & r_2(A)\times c_4(B) \end{bmatrix}$$

where $r_1(A)$ and $r_2(A)$ are the row vectors of ***A*** and $c_1(B), c_2(B), c_3(B)$ *and* $c_4(B)$ are the column vectors of ***B***. For example, the entry in row 2 and column 3 of ***AB*** can be computed as

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}\begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} \square & \square & \square & \square \\ \square & \square & \boxed{26} & \square \end{bmatrix}$$

$$(2\times 4)+(6\times 3)+(0\times 5) = 26$$

and the entry in row 1 and column 4 of ***AB*** can be computed as

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}\begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} \square & \square & \square & \boxed{13} \\ \square & \square & \square & \square \end{bmatrix}$$

$$(1\times 3)+(2\times 1)+(4\times 2)=13$$

Here is the complete set of computations:

$$(AB)_{11} = (1\times 4)+(2\times 0)+(4\times 2)=12$$
$$(AB)_{12} = (1\times 1)+(2\times \text{-}1)+(4\times 7)=27$$
$$(AB)_{13} = (1\times 4)+(2\times 3)+(4\times 5)=30$$
$$(AB)_{14} = (1\times 3)+(2\times 1)+(4\times 2)=13$$
$$(AB)_{21} = (2\times 4)+(6\times 0)+(0\times 2)=8$$
$$(AB)_{22} = (2\times 1)+(6\times \text{-}1)+(0\times 7)=\text{-}4$$
$$(AB)_{23} = (2\times 4)+(6\times 3)+(0\times 5)=26$$
$$(AB)_{24} = (2\times 3)+(6\times 1)+(0\times 2)=12$$

Finding Specific Rows and Columns of a Matrix Product

The specific column of ***AB*** is given by the formula

$$AB = A\begin{bmatrix} \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_n \end{bmatrix} = \begin{bmatrix} A\boldsymbol{b}_1 & A\boldsymbol{b}_2 & \cdots & A\boldsymbol{b}_n \end{bmatrix}$$

Similarly, the specific row of ***AB*** is given by the formula $AB = \begin{bmatrix} a_1 \\ a_2 \\ \square \\ a_m \end{bmatrix} B = \begin{bmatrix} a_1B \\ a_2B \\ \square \\ a_mB \end{bmatrix}$

Example 9 Find the entries in the second row of ***AB***, where

$$A = \begin{bmatrix} 2 & -5 & 0 \\ -1 & 3 & -4 \\ 6 & -8 & -7 \\ -3 & 0 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -6 \\ 7 & 1 \\ 3 & 2 \end{bmatrix}$$

Solution: By the row-column rule, the entries of the second row of ***AB*** come from row 2 of ***A*** (and the columns of ***B***):

$$\begin{matrix} & & \downarrow & \downarrow \\ \rightarrow & \begin{bmatrix} 2 & -5 & 0 \\ -1 & 3 & -4 \\ 6 & -8 & -7 \\ -3 & 0 & 9 \end{bmatrix} & \begin{bmatrix} 4 & -6 \\ 7 & 1 \\ 3 & 2 \end{bmatrix} & \end{matrix} = \begin{bmatrix} \square & \square \\ -4+21-12 & 6+3-8 \\ \square & \square \\ \square & \square \end{bmatrix} = \begin{bmatrix} \square & \square \\ 5 & 1 \\ \square & \square \\ \square & \square \end{bmatrix}$$

Example 10 (Finding a Specific Row and Column of AB)

Let $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$

Find the second column and the first row of ***AB***.

Solution

$$c_2(AB) = Ac_2(B) = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

$$r_1(AB) = r_1(A)B = \begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 27 & 30 & 13 \end{bmatrix}$$

Properties of Matrix Multiplication

These are standard properties of matrix multiplication. Remember that I_m represents the $m \times m$ identity matrix and $I_m x = x$ for all ***x belong to*** $\boldsymbol{R^m}$.

Theorem 2 Let ***A*** be $m \times n$, and let ***B*** and ***C*** have sizes for which the indicated sums and products are defined.

a.	***A (BC) = (AB) C***	(associative law of multiplication)
b.	***A (B + C) = AB + AC***	(left distributive law)
c.	***(B + C)A = BA + CA***	(right distributive law)
d.	***r (AB) = (r A)B = A(r B)***	(for any scalar *r*)
e.	$I_m A = A = A I_m$	(identity for matrix multiplication)

Proof. Properties (b) to (e) are considered exercises for you. We start property (a) follows from the fact that matrix multiplication corresponds to composition of linear transformations (which are functions), and it is known (or easy to check) that the composition of functions is associative.

Here is another proof of (a) that rests on the "column definition" of the product of two matrices. Let $C = \begin{bmatrix} c_1 & \cdots & c_p \end{bmatrix}$

By definition of matrix multiplication $BC = \begin{bmatrix} Bc_1 & \cdots & Bc_p \end{bmatrix}$

$$A(BC) = \begin{bmatrix} A(Bc_1) & \cdots & A(Bc_p) \end{bmatrix}$$

From above, we know that ***A(Bx) = (AB)x*** for all ***x***, so

$$A(BC) = \begin{bmatrix} (AB)c_1 & \cdots & (AB)c_p \end{bmatrix} = (AB)C$$

The associative and distributive laws say essentially that pairs of parentheses in matrix expressions can be inserted and deleted in the same way as in the algebra of real numbers. In particular, we can write ***ABC*** for the product, which can be computed as ***A(BC)*** or as ***(AB)C***. Similarly, a product ***ABCD*** of four matrices can be computed as

A(BCD) or ***(ABC)D*** or ***A(BC)D***, and so on. It does not matter how we group the matrices when computing the product, so long as the left-to-right order of the matrices is preserved.

The left-to-right order in products is critical because, in general, ***AB*** and ***BA*** are not the same. This is not surprising, because the columns of ***AB*** are linear combinations of the columns of ***A***, whereas the columns of ***BA*** are constructed from the columns of ***B***.

If ***AB = BA***, we say that ***A*** and ***B* commute** with one another.

Example 11 Let $A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}$ *and* $B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$

Show that these matrices don not commute, i.e. $AB \neq BA$.

Solution:

$$AB = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 10+4 & 0+3 \\ 6-8 & 0-6 \end{bmatrix} = \begin{bmatrix} 14 & 3 \\ -2 & -6 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 10+0 & 2-0 \\ 20+9 & 4-6 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ 29 & -2 \end{bmatrix}$$

For emphasis, we include the remark about commutativity with the following list of important differences between matrix algebra and ordinary algebra of real numbers.

WARNINGS

1. In general, $AB \neq BA$. Clear from the Example # 11.

2. The cancellation laws do not hold for matrix multiplication. That is, if $AB = AC$, then it is not true in general that $B = C$.

For example: Consider the following three matrices

$$A = \begin{bmatrix} -3 & 2 \\ -6 & 4 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 2 \\ 3 & -2 \end{bmatrix} \quad C = \begin{bmatrix} -1 & 2 \\ 3 & -2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 9 & -10 \\ 18 & -20 \end{bmatrix} = AC \quad \text{But } B \neq C$$

3. If a product ***AB*** is the zero matrix, you cannot conclude in general that either ***A = 0*** *or* ***B = 0***.

For example:

$$If\ AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\ then\ it\ can\ be\ either$$

$$A=\begin{bmatrix}1 & 4\\ 6 & 2\end{bmatrix} \text{ and } B=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix} \text{ or}$$

$$A=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix} \text{ and } B=\begin{bmatrix}1 & 4\\ 6 & 2\end{bmatrix}$$

Example

$$AB=\begin{bmatrix}5 & 1\\ 3 & -2\end{bmatrix}\begin{bmatrix}2 & 0\\ 4 & 3\end{bmatrix}=\begin{bmatrix}14 & 3\\ -2 & -6\end{bmatrix}$$

$$AB=\begin{bmatrix}5 & 1\\ 3 & -2\end{bmatrix}\begin{bmatrix}3 & 0\\ -1 & 3\end{bmatrix}=\begin{bmatrix}15-1 & 0+3\\ 6-8 & 0-6\end{bmatrix}=\begin{bmatrix}14 & 3\\ -2 & -6\end{bmatrix}$$

Powers of a Matrix If $\boldsymbol{A}$ is an $n\times n$ matrix and if k is a positive integer, $\boldsymbol{A^k}$ denotes the product of k copies of $\boldsymbol{A}$, $A^k = A.....A$ Also, we interpret $\boldsymbol{A^0}$ as $\boldsymbol{I}$.

Transpose of a Matrix Given an $m\times n$ matrix $\boldsymbol{A}$, the **transpose** of $\boldsymbol{A}$ is the $n\times m$ matrix, denoted by $\boldsymbol{A^t}$, whose columns are formed from the corresponding rows of $\boldsymbol{A}$.
OR, if $\boldsymbol{A}$ is an *m x n* matrix, then transpose of $\boldsymbol{A}$ is denoted by $\boldsymbol{A^t}$, is defined to be the *nxm* matrix that is obtained by making the rows of $\boldsymbol{A}$ into columns; that is, the first column of $\boldsymbol{A^t}$ is the first row of $\boldsymbol{A}$, the second column of $\boldsymbol{A^t}$ is the second row of $\boldsymbol{A}$, and so forth.

Example 12 (Transpose of a Matrix)
The following is an example of a matrix and its transpose.

$$A=\begin{bmatrix}2 & 3\\ 1 & 4\\ 5 & 6\end{bmatrix}$$

$$A^t=\begin{bmatrix}2 & 1 & 5\\ 3 & 4 & 6\end{bmatrix}$$

Example 13 Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$, $B=\begin{bmatrix}-5 & 2\\ 1 & -3\\ 0 & 4\end{bmatrix}$, $C=\begin{bmatrix}1 & \frac{1}{5} & 1 & \frac{1}{7}\\ -3 & 5 & -2 & 7\end{bmatrix}$

Then $A^t=\begin{bmatrix}a & c\\ b & d\end{bmatrix}$, $B^t=\begin{bmatrix}-5 & 1 & 0\\ 2 & -3 & 4\end{bmatrix}$, $C^t=\begin{bmatrix}1 & -3\\ 1 & 5\\ 1 & -2\\ 1 & 7\end{bmatrix}$

Theorem 3 Let $\boldsymbol{A}$ and $\boldsymbol{B}$ denote matrices whose sizes are appropriate for the following sums and products.

a. $(A^t)^t = A$

b. $(A+B)^t = A^t + B^t$

c. *For any scalar* $r, (rA)^t = rA^t$

d. $(AB)^t = B^t A^t$

The generalization of (d) to products of more than two factors can be stated in words as follows.

"The transpose of a product of matrices equals the product of their transposes in the reverse order."



Lecture 12

The Inverse of a Matrix

In this lecture and the next, we will consider only square matrices and we will investigate the matrix analogue of the reciprocal or multiplicative inverse of a nonzero real number.

Inverse of a square Matrix

If ***A*** is an $n \times n$ matrix, A matrix C of order $n \times n$ is called multiplicative inverse of A if

$AC = CA = I$ where I is the $n \times n$ identity matrix.

Invertible Matrix

If the inverse of a square matrix exists, it is called an invertible matrix.

In this case, we say that ***A*** is **invertible** and we call ***C*** an **inverse** of ***A***.

Note: If ***B*** is another inverse of ***A***, then we would have

$$B = BI = B(AC) = (BA)C = IC = C.$$

Thus when ***A*** is **invertible**, its inverse is unique.

The inverse of ***A*** is denoted by $\boldsymbol{A^{-1}}$, so that

$$AA^{-1} = I \qquad and \qquad A^{-1}A = I$$

Note: A matrix that is not invertible is sometimes called a **singular** matrix, and an invertible matrix is called a **non-singular** matrix.

Warning

By No means $A^{-1}.A = A.A^{-1} = I$ *The identity matrix* $A^{-1} = \frac{1}{A}$, as in case of real number, we have $3^{-1} = \frac{1}{3}$.

A^{-1} is, in fact the $n \times n$ matrix corresponding to the $n \times n$ matrix A, which satisfies the property

$$A^{-1}.A = A.A^{-1} = I \quad The\ identity\ matrix$$

Example 1 If $A=\begin{bmatrix}2 & 5\\ -3 & -7\end{bmatrix}$ *and* $C=\begin{bmatrix}-7 & -5\\ 3 & 2\end{bmatrix}$, then

$$AC=\begin{bmatrix}2 & 5\\ -3 & -7\end{bmatrix}\begin{bmatrix}-7 & -5\\ 3 & 2\end{bmatrix}=\begin{bmatrix}-14+15 & -10+10\\ 21-21 & 15-14\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} \quad \textit{and}$$

$$CA=\begin{bmatrix}-7 & -5\\ 3 & 2\end{bmatrix}\begin{bmatrix}2 & 5\\ -3 & -7\end{bmatrix}=\begin{bmatrix}-14+15 & -35+35\\ 6-6 & 15-14\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Thus $\mathbf{C} = \boldsymbol{A}^{-1}$.

Theorem Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$.

If $ad-bc\neq 0$, then $\boldsymbol{A}$ is invertible or non singular and $A^{-1}=\dfrac{1}{ad-bc}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$

If $ad-bc=0$, then $\boldsymbol{A}$ is not invertible or singular.

The quantity $ad-bc$ is called the **determinant of *A***, and we write

$$\det A = ad-bc$$

This implies that a 2×2 matrix $\boldsymbol{A}$ is invertible if and only if $\det A\neq 0$.

Example 2 Find the inverse of $A=\begin{bmatrix}3 & 4\\ 5 & 6\end{bmatrix}$.

Solution We have det $\boldsymbol{A}$ = 3(6) – 4(5) = –2 ≠ 0.

Hence $\boldsymbol{A}$ is invertible $A^{-1}=\dfrac{1}{-2}\begin{bmatrix}6 & -4\\ -5 & 3\end{bmatrix}=\begin{bmatrix}6/(-2) & -4/(-2)\\ -5/(-2) & 3/(-2)\end{bmatrix}=\begin{bmatrix}-3 & 2\\ 5/2 & -3/2\end{bmatrix}$

The next theorem provides three useful facts about invertible matrices.

Theorem

a. If $\boldsymbol{A}$ is an invertible matrix, then $\boldsymbol{A^{-1}}$ is invertible and $(A^{-1})^{-1}=A$

b. If $\boldsymbol{A}$ and $\boldsymbol{B}$ are $n\times n$ invertible matrices, then so is $\boldsymbol{AB}$, and the inverse of $\boldsymbol{AB}$ is the product of the inverses of $\boldsymbol{A}$ and $\boldsymbol{B}$ in the reverse order. That is $(AB)^{-1}=B^{-1}A^{-1}$

c. If $\boldsymbol{A}$ is an invertible matrix, then so is $\boldsymbol{A^T}$, and the inverse of $\boldsymbol{A^T}$ is the transpose of $\boldsymbol{A^{-1}}$. That is $(A^T)^{-1}=(A^{-1})^T$

Proof

(a) We must find a matrix ***C*** such that $A^{-1}C = I$ *and* $CA^{-1} = I$

However, we already know that these equations are satisfied with ***A*** in place of ***C***. Hence is invertible and ***A*** is its inverse.

(b) We use the associative law for multiplication:

$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\ &= AIA^{-1} \\ &= AA^{-1} \\ &= I\end{aligned}$$

A similar calculation shows that $(B^{-1}A^{-1})(AB) = I$.

Hence ***AB*** is invertible, and its inverse is $B^{-1}A^{-1}$ i.e $(AB)^{-1} = B^{-1}A^{-1}$

Generalization

Similarly we can prove the same results for more than two matrices i.e

$$\left((A_1)(A_2)(A_3)\ldots(A_n)\right)^{-1} = A_n^{-1}A_{n-1}^{-1}\ldots A_3^{-1}A_2^{-1}A_1^{-1}$$

The product of $n \times n$ invertible matrices is invertible, and the inverse is the product of their inverses in the reverse order.

Example 3 (Inverse of a Transpose). Consider a general 2×2 invertible matrix and its transpose:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \textit{and} \quad A^t = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Since A is invertible, its determinant *(ad – bc)* is nonzero. But the determinant of $\boldsymbol{A^t}$ is also *(ad – bc)*, so $\boldsymbol{A^t}$ is also invertible. It follows that

$$(A^t)^{-1} = \begin{bmatrix} \dfrac{d}{ad-bc} & -\dfrac{c}{ad-bc} \\ -\dfrac{b}{ad-bc} & \dfrac{a}{ad-bc} \end{bmatrix} \text{---------- (1)}$$

Now $$A^{-1} = \begin{bmatrix} \dfrac{d}{ad-bc} & -\dfrac{b}{ad-bc} \\ \dfrac{c}{ad-bc} & \dfrac{a}{ad-bc} \end{bmatrix}$$

Therefore, $$(A^{-1})^t = \frac{\begin{bmatrix} \frac{d}{ad-bc} & -\frac{c}{ad-bc} \end{bmatrix}}{\begin{bmatrix} -\frac{b}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}} \quad \text{---------- (2)}$$

From (1) and (2), we have

$$(A^t)^{-1} = (A^{-1})^t .$$

Example 4 (The Inverse of a Product). Consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

Here

$$AB = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix},$$

$$(AB)^{-1} = \frac{1}{|AB|} Adj(AB) = \frac{1}{-2}\begin{bmatrix} 8 & -6 \\ -9 & 7 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} Adj(A) = \frac{1}{1}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix},$$

$$B^{-1} = \frac{1}{|B|} Adj(B) = \frac{1}{2}\begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix},$$

$$B^{-1}A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

Thus, $(AB)^{-1} = B^{-1}A^{-1}$

Theorem: If A is invertible and n is a non-negative integer, then:
(a) $\boldsymbol{A^n}$ is invertible and $\boldsymbol{(A^n)^{-1} = A^{-n} = (A^{-1})^n}$
(b) $k\boldsymbol{A}$ is invertible for any nonzero scalar k, and $(k\boldsymbol{A})^{-1} = k^{-1}\boldsymbol{A}^{-1}$.

Example 5 (Related to the above theorem)

(a) Let

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \qquad then\ A^{-1} = \frac{1}{|A|} Adj(A) = \frac{1}{1}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

Now

$$A^{-3} = \left(A^{-1}\right)^3 = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix}$$

Also,

$$A^3 = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 30 \\ 15 & 41 \end{bmatrix}$$

$$(A^3)^{-1} = \frac{1}{(11)(41)-(30)(15)}\begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix} = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix} = (A^{-1})^3$$

(b)

Take $A=\begin{bmatrix}1 & 2\\ 3 & 1\end{bmatrix}$ and k=3

$kA=3A=\begin{bmatrix}3 & 6\\ 9 & 3\end{bmatrix}$, $(kA)^{-1}=(3A)^{-1}=\frac{1}{9-54}\begin{bmatrix}3 & -6\\ -9 & 3\end{bmatrix}=\begin{bmatrix}-1/15 & 2/15\\ 1/5 & -1/15\end{bmatrix}$ -----(1)

$A^{-1}=-\frac{1}{5}\begin{bmatrix}1 & -2\\ -3 & 1\end{bmatrix}$

$A=$ So $k^{-1}A^{-1}=3^{-1}A^{-1}=\frac{1}{3}.\frac{-1}{5}\begin{bmatrix}1 & -2\\ -3 & 1\end{bmatrix}=\begin{bmatrix}-1/15 & 2/15\\ 1/5 & -1/15\end{bmatrix}$ ---------- (2)

From (1) and (2), we have

$(3A)^{-1}=3^{-1}A^{-1}$

There is an important connection between invertible matrices and row operations that leads to a method for computing inverses. As we shall see, an invertible matrix ***A*** is row equivalent to an identity matrix, and we can find A^{-1} by watching the row reduction of ***A*** to ***I***.

Elementary Matrices

As we have studied that there are three types of elementary row operations that can be performed on a matrix:

There are three types of elementary operations

- Interchanging of any two rows
- Multiplication to a row by a nonzero constant
- Adding a multiple of one row to another

Elementary matrix

An elementary matrix is a matrix that results from applying a single elementary row operation to an identity matrix.

Some examples are given below:

$\begin{bmatrix}1 & 0\\ 0 & -3\end{bmatrix}, \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\end{bmatrix}, \begin{bmatrix}1 & 0 & 3\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}, \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

Multiply the second row of I_2 by -3.

Interchange the second and fourth rows of I_4.

Add 3 times the third row of I_3 to the first row.

Multiply the first row of I_3 by 1.

From Def it is clear that elementary matrices are always square.

Elementary matrices are important because they can be used to execute elementary row operations by matrix multiplication.

Theorem If $\boldsymbol{A}$ is an $n \times n$ identity matrix, and if the elementary matrix $\boldsymbol{E}$ results by performing a certain row operation on the identity matrix, then the product EA is the matrix that results when the same row operation is performed on $\boldsymbol{A}$.

In short, this theorem states that an elementary row operation can be performed on a matrix $\boldsymbol{A}$ using a left multiplication by and appropriate elementary matrix.

Example 6 (Performing Row Operations by Matrix Multiplication). Consider the matrix $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$

Find an elementary matrix $\boldsymbol{E}$ such that $\boldsymbol{EA}$ is the matrix that results by adding 4 times the first row of $\boldsymbol{A}$ to the third row.

Solution: The matrix $\boldsymbol{E}$ must be 3×3 to conform to the product $\boldsymbol{EA}$. Thus, we obtain $\boldsymbol{E}$ by adding 4 times the first row of I_3 to the third row. This gives us $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$

As a check, the product $\boldsymbol{EA}$ is $EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 5 & 4 & 12 & 12 \end{bmatrix}$

So left multiplication by $\boldsymbol{E}$ does, in fact, add 4 times the first row of $\boldsymbol{A}$ to the third row.

If an elementary row operation is applied to an identity matrix $\boldsymbol{I}$ to produce an elementary matrix $\boldsymbol{E}$, then there is a second row operation that, when applied to $\boldsymbol{E}$, produces $\boldsymbol{I}$ back again.

For example, if $\boldsymbol{E}$ is obtained by multiplying the *i-th* row of $\boldsymbol{I}$ by a nonzero scalar c, then $\boldsymbol{I}$ can be recovered by multiplying the *i-th* row of $\boldsymbol{E}$ by 1/c. The following table explains how to recover the identity matrix from an elementary matrix for each of the three elementary row operations. The operations on the right side of this table are called the **inverse operations** of the corresponding operations on the left side.

Row operation on I that produces E	**Row operation on E that reproduces I**
Multiply row *i* by $c \neq 0$	Multiply row *i* by 1/c
Interchange rows *i* and *j*	Interchange rows *i* and *j*
Add c times row *i* to row *j*	Add –c times row *i* to row *j*

Example 7 (Recovering Identity Matrices from Elementary Matrices). Here are three examples that use inverses of row operations to recover the identity matrix from

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiply the second row by 7.

Multiply the second row by 1/7.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Interchange the first and second rows.

Interchange the first and second rows.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Add 5 times the second row to the first.

Add -5 times the second row to the first.

The next theorem is the basic result on the invertibility of elementary matrices.

Theorem An elementary matrix is invertible and the inverse is also an elementary matrix.

Example 8 Let $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Compute E_1A, E_2A, E_3A and describe how these products can be obtained by elementary row operations on A.

Solution We have

$$E_1A = \begin{bmatrix} a & b & c \\ d & e & f \\ g-4a & h-4b & i-4c \end{bmatrix}, \quad E_2A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix}, \quad E_3A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}$$

Addition of (–4) times row 1 of $\boldsymbol{A}$ to row 3 produces $\boldsymbol{E_1A}$. (This is a row replacement operation.) An interchange of rows 1 and 2 of $\boldsymbol{A}$ produces $\boldsymbol{E_2A}$ and multiplication of row 3 of $\boldsymbol{A}$ by 5 produces $\boldsymbol{E_3A}$.

Left-multiplication (that is, multiplication on the left) by $\boldsymbol{E_1}$ in Example 8 has the same effect on any $3\times n$ matrix. It adds – 4 times row 1 to row 3. In particular, since $\boldsymbol{E_1\ I = E_1}$, we see that $\boldsymbol{E_1}$ itself is produced by the same row operation on the identity. Thus Example 8 illustrates the following general fact about elementary matrices.

Note: Since row operations are reversible, elementary matrices are invertible, for if $\boldsymbol{E}$ is produced by a row operation on $\boldsymbol{I}$, then there is another row operation of the same type that changes $\boldsymbol{E}$ back into $\boldsymbol{I}$. Hence there is an elementary matrix $\boldsymbol{F}$ such that $\boldsymbol{FE = I}$. Since $\boldsymbol{E}$ and $\boldsymbol{F}$ correspond to reverse operations, $\boldsymbol{EF = I}$.

Each elementary matrix $\boldsymbol{E}$ is invertible. The inverse of $\boldsymbol{E}$ is the elementary matrix of the same type that transforms $\boldsymbol{E}$ back into $\boldsymbol{I}$.

Example Find the inverse of $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$.

Solution: To transform $\boldsymbol{E_1}$ into $\boldsymbol{I}$, add + 4 times row 1 to row 3.

The elementary matrix which does that is $E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ +4 & 0 & 1 \end{bmatrix}$

Theorem An $n\times n$ matrix $\boldsymbol{A}$ is invertible if and only if $\boldsymbol{A}$ is row equivalent to $\boldsymbol{I_n}$, and in this case, any sequence of elementary row operations that reduces $\boldsymbol{A}$ to $\boldsymbol{I_n}$ also transforms $\boldsymbol{I_n}$ into $\boldsymbol{A^{-1}}$.

An Algorithm for Finding $\boldsymbol{A^{-1}}$ If we place $\boldsymbol{A}$ and $\boldsymbol{I}$ side-by-side to form an augmented matrix $\boldsymbol{[A\ \ I]}$, then row operations on this matrix produce identical operations on $\boldsymbol{A}$ and $\boldsymbol{I}$. Then either there are row operations that transform $\boldsymbol{A}$ to $\boldsymbol{I_n}$, and $\boldsymbol{I_n}$ to $\boldsymbol{A^{-1}}$, or else $\boldsymbol{A}$ is not invertible.

Algorithm for Finding $\boldsymbol{A^{-1}}$

Row reduce the augmented matrix $\boldsymbol{[A\ \ I]}$. If $\boldsymbol{A}$ is row equivalent to $\boldsymbol{I}$, then $\boldsymbol{[A\ \ I]}$ is row equivalent to $\boldsymbol{[I\ \ A^{-1}]}$. Otherwise, $\boldsymbol{A}$ does not have an inverse.

Example 9 Find the inverse of the matrix $A=\begin{bmatrix}0 & 1 & 2\\ 1 & 0 & 3\\ 4 & -3 & 8\end{bmatrix}$, if it exists.

Solution

$$[A \;\; I]=\begin{bmatrix}0 & 1 & 2 & 1 & 0 & 0\\ 1 & 0 & 3 & 0 & 1 & 0\\ 4 & -3 & 8 & 0 & 0 & 1\end{bmatrix} \to \begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 4 & -3 & 8 & 0 & 0 & 1\end{bmatrix} R_{12}$$

$$\xrightarrow{-4R_1+R_3} \begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & -3 & -4 & 0 & -4 & 1\end{bmatrix} \xrightarrow{3R_2+R_3} \begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & 0 & 2 & 3 & -4 & 1\end{bmatrix}$$

$$\xrightarrow{-2R_3+R_2} \begin{bmatrix}1 & 0 & 3 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 0\\ 0 & 0 & 1 & 3/2 & -2 & 1/2\end{bmatrix} \xrightarrow{-3R_3+R_1} \begin{bmatrix}1 & 0 & 0 & -9/2 & 7 & -3/2\\ 0 & 1 & 0 & -2 & 4 & -1\\ 0 & 0 & 1 & 3/2 & -2 & 1/2\end{bmatrix}$$

Since $\boldsymbol{A} \to \mathbf{I}$, we conclude that $\boldsymbol{A}$ is invertible, and $A^{-1}=\begin{bmatrix}-9/2 & 7 & -3/2\\ -2 & 4 & -1\\ 3/2 & -2 & 1/2\end{bmatrix}$

It is a good idea to check the final answer:

$$AA^{-1}=\begin{bmatrix}0 & 1 & 2\\ 1 & 0 & 3\\ 4 & -3 & 8\end{bmatrix}\begin{bmatrix}-9/2 & 7 & -3/2\\ -2 & 4 & -1\\ 3/2 & -2 & 1/2\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

It is not necessary to check that $\boldsymbol{A^{-1}A = I}$ since $\boldsymbol{A}$ is invertible.

Example 10 Find the inverse of the matrix $A=\begin{bmatrix}1 & 2 & -3 & 1\\ -1 & 3 & -3 & -2\\ 2 & 0 & 1 & 5\\ 3 & 1 & -2 & 5\end{bmatrix}$, if it exists.

Consider $detA = \begin{vmatrix} 1 & 2 & -3 & 1 \\ -1 & 3 & -3 & -2 \\ 2 & 0 & 1 & 5 \\ 3 & 1 & -2 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 2 & -3 & 1 \\ 0 & 5 & -6 & -1 \\ 0 & -4 & 7 & 3 \\ 0 & -5 & 7 & 2 \end{vmatrix}$

operating $R_2 + R_1, R_3 - 2R_1, R_4 - 3R_1$

Expand from first column $= \begin{vmatrix} 5 & -6 & -1 \\ -4 & 7 & 3 \\ -5 & 7 & 2 \end{vmatrix} = \begin{vmatrix} 5 & -6 & -1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = 5(1-2) + 6(1-0) - 1(1-0) = 0$

As the given matrix is singular, so it is not invertible.

Example 11 Find the inverse of the given matrix if possible $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

Solution $\det A = \begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{vmatrix} = -1$

As the given matrix is non-singular therefore, inverse is possible.

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$R_2 - 2R_1, R_3 - 3R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 & 1 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 \end{array}\right]$$

Multiply R_3 *by* -1

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 1 & 1 & -1 \end{array}\right]$$

$R_1 - R_3, R_2 + R_3$

Hence the inverse of matrix A is $A^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$

Example 12 Find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 5 & 2 & 3 \end{bmatrix}$

Solution $\det A = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 5 & 2 & 3 \end{vmatrix} = 6$

As the given matrix is non-singular, therefore, inverse of the matrix is possible.
We reduce it to reduce echelon form.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 0 \\ 5 & 2 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -2 & -1 \\ 0 & -8 & -7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$$

$R_2 - 2R_1, R_3 - 5R_1$

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1/2 \\ 0 & -8 & -7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1/2 & 0 \\ -5 & 0 & 1 \end{bmatrix}$$

multiply 2nd row by -1/2

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1/2 \\ 0 & 0 & -3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1/2 & 0 \\ 3 & -4 & 1 \end{bmatrix}$$

$R_3 + 8R_2$

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1/2 & 0 \\ -1 & 4/3 & -1/3 \end{bmatrix}$$

Multiply 3rd row by -1/3

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 3 & -\frac{8}{3} & \frac{2}{3} \\ 3/2 & -7/6 & 1/6 \\ -1 & 4/3 & -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1/3 & 1/3 \\ 3/2 & -7/6 & 1/6 \\ -1 & 4/3 & -1/3 \end{bmatrix}$$

$R_2 - (1/2)R_3, R_1 - 2R_3$

Hence the inverse of the original matrix $A^{-1} = \begin{bmatrix} 0 & -1/3 & 1/3 \\ 3/2 & -7/6 & 1/6 \\ -1 & 4/3 & -1/3 \end{bmatrix}$

Exercises

In exercises 1 to 4, find the inverses of the matrices, if they exist. Use elementary row operations.

1. $\begin{bmatrix} 1 & 2 \\ 5 & 9 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 0 & 5 \\ 1 & 1 & 0 \\ 3 & 2 & 6 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 4 & -3 \\ -2 & -7 & 6 \\ 1 & 7 & -2 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 2 \\ -1 & 0 & 8 \end{bmatrix}$

5. $\begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$

6. $\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$

7. Let $A = \begin{bmatrix} -1 & -5 & -7 \\ 2 & 5 & 6 \\ 1 & 3 & 4 \end{bmatrix}$. Find the third column of A^{-1} without computing the other columns.

8. Let $A = \begin{bmatrix} -25 & -9 & -27 \\ 546 & 180 & 537 \\ 154 & 50 & 149 \end{bmatrix}$. Find the second and third columns of A^{-1} without computing the first column.

9. Find an elementary matrix E that satisfies the equation.
(a) EA = B (b) EB = A (c) EA = C (d) EC = A
where $A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$.

10. Consider the matrix $A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

(a) Find elementary matrices E_1 and E_2 such that $E_2E_1A=I$.
(b) Write A^{-1} as a product of two elementary matrices.

(c) Write A as a product of two elementary matrices.

In exercises 11 and 12, express A and A^{-1} as products of elementary matrices.

11. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

12. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

13. Factor the matrix $A = \begin{bmatrix} 0 & 1 & 7 & 8 \\ 1 & 3 & 3 & 8 \\ -2 & -5 & 1 & -8 \end{bmatrix}$ as A = EFGR, where E, F, and G are elementary matrices and R is in row echelon form.

Lecture 13

Characterizations of Invertible Matrices

This chapter involves a few techniques of solving the system of n linear equations in **n** unknowns and transformation associated with a matrix.

Solving Linear Systems by Matrix Inversion

Theorem

Let $\boldsymbol{A}$ be an $n \times n$ invertible matrix. For any $\boldsymbol{b} \in \boldsymbol{R^n}$, the equation $\boldsymbol{Ax = b}$ …..(1) has the unique solution i.e. $\boldsymbol{x = A^{-1}b}$.

Proof

Since A is invertible and $\boldsymbol{b} \in \boldsymbol{R^n}$ be any vector. Then, we must have a matrix $A^{-1}b$ which is a solution of eq. (1) .i.e. $\boldsymbol{Ax = A\,(A^{-1}b) = Ib = b.}$

Uniqueness

For uniqueness, we assume that there is another solution $\boldsymbol{u}$. Indeed, it is a solution of eq.(1) so it must be $\boldsymbol{u = A^{-1}b,}$ it means x = $\boldsymbol{A^{-1}b = u.}$ This shows that $\boldsymbol{u = x.}$

Theorem

Let A and B be the square matrices such that AB = I. Then, A and B are invertible with $B = A^{-1}$ and $A = B^{-1}$

Example 1

Solve the system of linear equations

$$2x_1 + x_2 + x_3 = 1$$
$$5x_1 + x_2 + 3x_3 = 3$$
$$x_1 + \quad 4x_3 = 6$$

Solution

Consider the linear system

$$2x_1 + x_2 + x_3 = 1$$
$$5x_1 + x_2 + 3x_3 = 3$$
$$x_1 + \quad 4x_3 = 6$$

The Matrix form of system is $\boldsymbol{Ax = b}$, where

$$A=\begin{bmatrix}2 & 1 & 1\\ 5 & 1 & 3\\ 1 & 0 & 4\end{bmatrix},\quad x=\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix},\quad b=\begin{bmatrix}1\\ 3\\ 6\end{bmatrix}$$

Here, det ($\boldsymbol{A}$) = 2(4) – 1 (20 – 3) + 1(0 – 1) = 8 – 17 – 1 = -10 ≠ 0

So, $\boldsymbol{A}$ is invertible. Now, we apply the inversion algorithm:

$$\left[\begin{array}{ccc:ccc}2 & 1 & 1 & 1 & 0 & 0\\ 5 & 1 & 3 & 0 & 1 & 0\\ 1 & 0 & 4 & 0 & 0 & 1\end{array}\right]$$

$$\left[\begin{array}{ccc:ccc}1 & 0 & 4 & 0 & 0 & 1\\ 5 & 1 & 3 & 0 & 1 & 0\\ 2 & 1 & 1 & 1 & 0 & 0\end{array}\right]$$

$$\left[\begin{array}{ccc:ccc}1 & 0 & 4 & 0 & 0 & 1\\ 0 & 1 & -17 & 0 & 1 & -5\\ 0 & 1 & -7 & 1 & 0 & -2\end{array}\right],\quad -5R_1+R_2,\quad -2R_1+R_3$$

$$\left[\begin{array}{ccc:ccc}1 & 0 & 4 & 0 & 0 & 1\\ 0 & 1 & -17 & 0 & 1 & -5\\ 0 & 0 & 10 & 1 & -1 & 3\end{array}\right],\quad -R_2+R_3,$$

$$\left[\begin{array}{ccc:ccc}1 & 0 & 4 & 0 & 0 & 1\\ 0 & 1 & -17 & 0 & 1 & -5\\ 0 & 0 & 1 & \frac{1}{10} & \frac{-1}{10} & \frac{3}{10}\end{array}\right],\quad \frac{R_3}{10}$$

$$\left[\begin{array}{ccc:ccc}1 & 0 & 0 & \frac{-2}{5} & \frac{2}{5} & \frac{-1}{5}\\ 0 & 1 & 0 & \frac{17}{10} & -7 & 1\\ 0 & 0 & 1 & \frac{1}{10} & -1 & 3\end{array}\right],\quad -4R_3+R_1,\quad -17R_3+R_2$$

Hence, $$A^{-1} = \begin{bmatrix} \frac{-2}{5} & \frac{2}{5} & \frac{-1}{5} \\ \frac{17}{10} & \frac{-7}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{-1}{10} & \frac{3}{10} \end{bmatrix}$$

Thus, the solution of the linear system is $x = A^{-1}b = A^{-1} = \begin{bmatrix} \frac{-2}{5} & \frac{2}{5} & \frac{-1}{5} \\ \frac{17}{10} & \frac{-7}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{-1}{10} & \frac{3}{10} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{-2}{5} \\ \frac{1}{1} \\ 5 \\ 8 \\ \frac{-}{5} \end{bmatrix}$

Example # 2 Solve the system $\begin{cases} x_1 + 2x_2 + 3x_3 = 5 \\ 2x_1 + 5x_2 + 3x_3 = 3 \\ x_1 + 8x_3 = 17 \end{cases}$ by inversion method.

Solution Consider the linear system $\begin{cases} x_1 + 2x_2 + 3x_3 = 5 \\ 2x_1 + 5x_2 + 3x_3 = 3 \\ x_1 + 8x_3 = 17 \end{cases}$

This system can be written in matrix form as $\boldsymbol{Ax = b}$, where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix}$$

Here, $[\det(\boldsymbol{A})] = 40 - 2(16 - 3) + 3(0 - 5) = 40 - 26 - 15 = -1 \neq 0$
Therefore, $\boldsymbol{A}$ is invertible.

Now, we apply the inversion algorithm:

$$\left[\begin{array}{ccc:ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc:ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right] \begin{matrix} \\ -2R_1 + R_2, \\ -1R_1 + R_3 \end{matrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right] \quad 2R_2 + R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \quad -1R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \quad 3R_3 + R_2,\ -3R_3 + R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \quad -2R_2 + R_1$$

Hence, $$A^{-1} = \left[\begin{array}{ccc} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{array}\right]$$

Thus, the solution of the linear system is $x = A^{-1}b = \left[\begin{array}{ccc} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{array}\right]\left[\begin{array}{c} 5 \\ 3 \\ 17 \end{array}\right] = \left[\begin{array}{c} 1 \\ -1 \\ 2 \end{array}\right]$

Thus $x_1 = 1, x_2 = -1, x_3 = 2$.

Note: It is only applicable when the number of equations = number of unknown and fails if given matrix is not invertible.

Example 3

Solve the system of linear equation

$$x_1 + 6x_2 + 4x_3 = 2$$
$$2x_1 + 4x_2 - x_3 = 3$$
$$-x_1 + 2x_2 + 5x_3 = 3$$

Solution

The matrix coefficient

$$A = \left[\begin{array}{ccc} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{array}\right]$$

det (***A***) = 1(20 + 2) – 6(10 – 1) + 4 (4+4)

$= 22 - 54 + 32$

$= 0$

Thus, A is not invertible. Hence, the inversion method fails.

Solution for the system of a $n \times n$ Homogeneous linear equations with an Invertible Coefficient Matrix:-

Let see if the system is considered as homogeneous then what does the above theorem say?

Theorem:-

Let ***Ax = 0*** be a homogeneous linear system of **n** equations and **n** unknowns. Then, the coefficient matrix ***A*** is invertible iff this system has only a trivial solution.

Example 4

State whether the following system of linear equation has a solution or not?

$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0 \\ x_1 - 3x_2 + x_3 &= 0 \\ x_1 - 4x_3 &= 0 \end{aligned}$$

Solution

We see

$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -3 & 1 \\ 1 & 0 & -4 \end{bmatrix}$ is an invertible matrix (det (A) $\neq 0$)

Thus, this homogeneous linear system has only the trivial solution.

Example 5

Solve $\begin{cases} x_1 + x_2 + x_3 = 8 \\ 2x_2 + 3x_3 = 24 \\ 5x_1 + 5x_2^2 + x_3^3 = 8 \end{cases}$

Solution This system can be written in matrix form as ***Ax = b***, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 5 & 5 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 8 \\ 24 \\ 8 \end{bmatrix}$$

Here [det (***A***)] = 1(2 – 15) – 1(0 – 15) + 1(0 – 10) = -13 + 15 – 10 = -8 ≠ 0
Therefore, ***A*** is invertible.

Now, we apply the inversion algorithm:

$$\begin{array}{c|c} A & I_3 \\ \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & 0 & 1 & 0 \\ 5 & 5 & 1 & 0 & 0 & 1 \end{array}\right] & \\ \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & 0 & 1 & 0 \\ 0 & 0 & -4 & -5 & 0 & 1 \end{array}\right] & -5R_1 + R_3 \\ \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & \frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -4 & -5 & 0 & 1 \end{array}\right] & \frac{1}{2}R_2 \\ \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -4 & -5 & 0 & 1 \end{array}\right] & -1R_2 + R_1 \\ \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{3}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{5}{4} & 0 & -\frac{1}{4} \end{array}\right] & -\frac{1}{4}R_3 \\ \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & 0 & 1 & \frac{5}{4} & 0 & -\frac{1}{4} \end{array}\right] & -\frac{3}{2}R_3 + R_2 \\ \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{13}{8} & -\frac{1}{2} & -\frac{1}{8} \\ 0 & 1 & 0 & -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & 0 & 1 & \frac{5}{4} & 0 & -\frac{1}{4} \end{array}\right] & \frac{1}{2}R_3 + R_1 \end{array}$$

Hence, $$A^{-1} = \begin{bmatrix} \frac{13}{8} & -\frac{1}{2} & -\frac{1}{8} \\ -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\ \frac{5}{4} & 0 & -\frac{1}{4} \end{bmatrix}$$

Thus, the solution of the linear system is

$$x = A^{-1}b = \begin{bmatrix} \frac{13}{8} & -\frac{1}{2} & -\frac{1}{8} \\ -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\ \frac{5}{4} & 0 & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 8 \\ 24 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix}$$

Thus $x_1 = 0, x_2 = 0, x_3 = 8$.

Theorem (Invertible Matrix Theorem) Let ***A*** be a square $n \times n$ matrix. Then the following statements are equivalent. (Means if any one holds then all are true).

(a) ***A*** is an invertible matrix.

(b) ***A*** is row equivalent to the $n \times n$ identity matrix.

(c) ***A*** has **n** pivot positions.

(d) The equation ***Ax = 0*** has only the trivial solution.

(e) The columns of ***A*** form a linearly independent set.

(f) The linear transformation $x \to Ax$ is one-to-one.

(g) The equation ***Ax = b*** has at least one solution for each ***b*** in $\boldsymbol{R^n}$.

(h) The columns of ***A*** span $\boldsymbol{R^n}$.

(i) The linear transformation $x \to Ax$ maps $\boldsymbol{R^n}$ onto $\boldsymbol{R^n}$.

(j) There is a $n \times n$ matrix ***C*** such that ***CA = I***.

(k) There is a $n \times n$ matrix ***D*** such that ***AD = I.***

(l) $\boldsymbol{A^T}$ is an invertible matrix.

Example 6

Show that the matrix $A = \begin{bmatrix} 1 & 0 & -4 \\ 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}$ is invertible by using Invertible Matrix Theorem

Solution

By row equivalent,

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}, -R_1 + R_2$$

$$\begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 9 \\ 0 & 1 & 2 \end{bmatrix}, -R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 9 \\ 0 & 0 & -7 \end{bmatrix}$$

It shows that A has three pivot positions and hence is invertible, by the Invertible Matrix Theorem (c).

Example 7

Use the Invertible Matrix Theorem to decide if ***A*** is invertible, where

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \square \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$ Here, ***A*** has three pivot positions and hence is invertible by the Invertible Matrix Theorem (c).

Example 7 Find A^t and show that A^t is an invertible matrix.

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Solution

$$A^t = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Now, by row equivalent of A,

$$A=\begin{bmatrix}1&2&1&0\\2&4&1&1\\1&0&1&1\\0&1&1&1\end{bmatrix}\xrightarrow{R_2-2R_1,\,R_3-R_1}\begin{bmatrix}1&2&1&0\\0&0&-1&1\\0&-2&0&1\\0&1&1&1\end{bmatrix}\xrightarrow{R_{23}}\begin{bmatrix}1&2&1&0\\0&-2&0&1\\0&0&-1&1\\0&1&1&1\end{bmatrix}$$

$$\xrightarrow{-\frac{1}{2}R_2}\begin{bmatrix}1&2&1&0\\0&1&0&-\frac{1}{2}\\0&0&-1&1\\0&1&1&1\end{bmatrix}\xrightarrow{R_4-R_2}\begin{bmatrix}1&2&1&0\\0&1&0&-\frac{1}{2}\\0&0&-1&1\\0&0&1&\frac{3}{2}\end{bmatrix}\xrightarrow{(-1)R_3}\begin{bmatrix}1&2&1&0\\0&1&0&-\frac{1}{2}\\0&0&1&-1\\0&0&1&\frac{3}{2}\end{bmatrix}\xrightarrow{-R_3+R_4}\begin{bmatrix}1&2&1&0\\0&1&0&-\frac{1}{2}\\0&0&1&-1\\0&0&0&\frac{5}{2}\end{bmatrix}$$

Here ***A*** has 4 pivot positions so by Invertible Matrix Theorem (c) A is invertible. Thus, by (l) A^t is invertible.

Example 8

Use the Invertible Matrix Theorem to decide if ***A*** is invertible, where

$$A=\begin{bmatrix}1&2&1&0\\2&4&1&2\\1&0&1&2\\0&1&1&2\end{bmatrix}$$

Solution

$$A=\begin{bmatrix}1&2&1&0\\2&4&1&2\\1&0&1&2\\0&1&1&2\end{bmatrix}\xrightarrow{R_2-2R_1,\,R_3-R_1}\begin{bmatrix}1&2&1&0\\0&0&-1&2\\0&-2&0&2\\0&1&1&2\end{bmatrix}\xrightarrow{R_{23}}\begin{bmatrix}1&2&1&0\\0&-2&0&2\\0&0&-1&2\\0&1&1&2\end{bmatrix}$$

$$\xrightarrow{-\frac{1}{2}R_2}\begin{bmatrix}1&2&1&0\\0&1&0&-1\\0&0&-1&2\\0&1&1&2\end{bmatrix}\xrightarrow{R_4-R_2}\begin{bmatrix}1&2&1&0\\0&1&0&-1\\0&0&-1&2\\0&0&1&3\end{bmatrix}\xrightarrow{(-1)R_3}\begin{bmatrix}1&2&1&0\\0&1&0&-1\\0&0&1&-2\\0&0&1&3\end{bmatrix}\xrightarrow{-R_3+R_4}\begin{bmatrix}1&2&1&0\\0&1&0&-1\\0&0&1&-2\\0&0&0&5\end{bmatrix}$$

Here, ***A*** has 4 pivot positions and hence is invertible by Invertible Matrix Theorem (c).

Solving Multiple Linear Systems with a Common Coefficient Matrix

This technique is used in solving a sequence of linear systems

$$Ax_1 = b_1,\ Ax_2 = b_2,\ \cdots,\ Ax_k = b_k \qquad (1)$$

where coefficient matrix ***A*** *remains* same and off course if it is invertible, then we have a sequence of solutions. i.e.

$x_1 = A^{-1}b_1,\ x_2 = A^{-1}b_2,\ \cdots,\ x_k = A^{-1}b_k$.

Find a Matrix from Linear Transformation

We can find a matrix corresponding to every transformation. In this section we will learn how to find a matrix attached with a linear transformation.

Example 9

Let L be the linear transformation from R^2 to P_2 (Polynomials of order 2) defined by

$T(x, y) = x\ y\ t + (x + y)t^2$

Find the matrix representing T with respect to the standard bases.

Solution

Let A = {(1,0),(0,1)} be the basis of R^2, then

$T(1,0) = t^2 = (0,0,1)$ (This triple represents the coefficients of polynomial t^2)

i.e. $t^2 = 0.1 + 0.t + 1.t^2$

Similarly, $T(0, 1) = t^2 = (0,0,1)$. Hence, the matrix is given by

$$A = \begin{pmatrix}0&0\\0&0\\1&1\end{pmatrix}$$

Now, we will proceed with a more complicated example.

Example 10

Let T be the linear transformation from R^2 to R^2 such that T(x, y) = (x , y + 2x). Find a matrix A for T.

Solution

This matrix is found by finding T (1, 0) = (1, 2) and T (0, 1) = (0, 1)The matrix is $A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$.

Important Note

It should be clear that the Invertible Matrix Theorem applies only to square matrices. For example, if the columns of a 4×3 matrix are linearly independent, we cannot use the Invertible Matrix Theorem to conclude any thing about the existence or nonexistence of solutions to equations of the form $\boldsymbol{Ax = b}$.

Invertible Linear Transformations

Recall that matrix multiplication corresponds to composition of linear transformations. When a matrix $\boldsymbol{A}$ is invertible, the equation $\boldsymbol{A^{-1}Ax = x}$ can be viewed as a statement about linear transformations. See Figure 2.

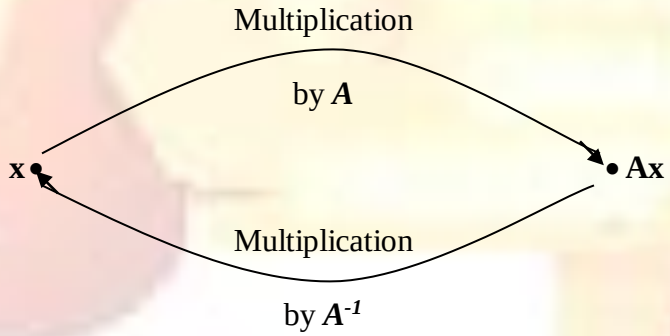


Figure 2 $\boldsymbol{A^{-1}}$ transforms $\boldsymbol{Ax}$ back to $\boldsymbol{x}$

Definition

A linear transformation $T : R^n \rightarrow R^n$ be a linear Transformation and A be a standard matrix for T. Then, T is invertible if and only if A is an invertible matrix in that case linear transformation S given by S(x) = A^{-1}x is a unique function satisfying (1) and (2)

$S(T(x)) = x \ \forall \quad x \in R^n$ (1)

$T(S(x)) = x \ \forall \quad x \in R^n$ (2)

Important Note

If the inverse of a linear transformation exits then it is unique.

Proposition

Let $T : R^n \to R^m$ be linear transformation, given as $T(x) = Ax \quad , \forall\ x \in R^n$, where A is a $m \times n$ matrix. The mapping T is invertible if the system $y = Ax$ has a unique solution.

Case 1:

If $m < n$, then the system $Ax = y$ has either no solution or infinitely many solutions, for any y in $R^{m.}$ Therefore, $y = Ax$ is non-invertible.

Case 2:

If $m = n$, then the system $Ax = y$ has a unique solution if and only if Rank (A) = n.

Case 3:

If $m > n$, then the transformation $y = Ax$ is non-invertible because we can find a vector y in R^m such that $Ax = y$ is inconsistent.

Exercises

1. Solve the system of linear equations by inverse matrix method.

$$x_1 + x_2 + 4x_3 = 2$$
$$2x_2 + 3x_3 = 4$$
$$5x_1 + x_2 - x_3 = 3$$

2. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}, b_1 = \begin{bmatrix} 5 \\ 7 \end{bmatrix}, b_2 = \begin{bmatrix} -2 \\ -2 \end{bmatrix}, b_3 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \text{and } b_4 = \begin{bmatrix} 1 \\ 7 \end{bmatrix}.$

(a) Find A^{-1} and use it to solve the equations $Ax = b_1$, $Ax = b_2$, $Ax = b_3$, $Ax = b_4$.

(b) Solve the four equations in part (a) by row reducing the augmented matrix

$[A \quad b_1 \quad b_2 \quad b_3 \quad b_4]$.

3. (a) Solve the two systems of linear equations

$$x_1 + 2x_2 + x_3 = -1$$
$$x_1 + 3x_2 + 2x_3 = 3$$
$$x_2 + 2x_3 = 4$$

&

$$x_1 + 2x_2 + x_3 = 0$$
$$x_1 + 3x_2 + 2x_3 = 0$$
$$x_2 + 2x_3 = 4$$

by row reduction.

(b) Write the systems in (a) as $Ax = b_1$ and $Ax = b_2$, and then solve each of them by the method of inversion.

Determine which of the matrices in exercises 4 to 10 are invertible?

4. $\begin{bmatrix} -4 & 16 \\ 3 & -9 \end{bmatrix}$

5. $\begin{bmatrix} 5 & 0 & 3 \\ 7 & 0 & 2 \\ 9 & 0 & 1 \end{bmatrix}$

6. $\begin{bmatrix} 2 & 3 & 4 \\ 2 & 3 & 4 \\ 2 & 3 & 4 \end{bmatrix}$

7. $\begin{bmatrix} 5 & -9 & 3 \\ 0 & 3 & 4 \\ 1 & 0 & 3 \end{bmatrix}$

8. $\begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & -2 & -1 \\ -2 & -6 & 3 & 2 \\ 3 & 5 & 8 & -3 \end{bmatrix}$ 9. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 5 & 0 & 0 \\ 3 & 6 & 8 & 0 \\ 4 & 7 & 9 & 10 \end{bmatrix}$ 10. $\begin{bmatrix} 7 & -6 & -4 & 1 \\ -5 & 1 & 0 & -2 \\ 10 & 11 & 7 & -3 \\ 19 & 9 & 7 & 1 \end{bmatrix}$

11. $\begin{bmatrix} 5 & 4 & 3 & 6 & 3 \\ 7 & 6 & 5 & 9 & 5 \\ 8 & 6 & 4 & 10 & 4 \\ 9 & -8 & 9 & -5 & 8 \\ 10 & 8 & 7 & -9 & 7 \end{bmatrix}$

12. Suppose that ***A*** and ***B*** are $n \times n$ matrices and the equation ***ABx = 0*** has a nontrivial solution. What can you say about the matrix ***AB***?

13. What can we say about a one-to-one linear transformation ***T*** from $\boldsymbol{R^n}$ into $\boldsymbol{R^n}$?

14. Let $T : R^2 \to R^2$ be a linear transformation given as $T(x) = 5x$, then find a matrix A of linear transformation T.

In exercises 15 and 16, T is a linear transformation from R^2 into R^2. Show that T is invertible.

15. $T(x_1, x_2) = (-5x_1 + 9x_2, 4x_1 - 7x_2)$

16. $T(x_1, x_2) = (6x_1 - 8x_2, -5x_1 + 7x_2)$

17. Let $T : R^n \to R^n$ be a linear transformation and let ***A*** be the standard matrix for ***T***. Then ***T*** is invertible if and only if ***A*** is an invertible matrix.

Lecture 14

Partitioned Matrices

A **block matrix** or a **partitioned matrix** is a partition of a matrix into rectangular smaller matrices called **blocks**.Partitioned matrices appear often in modern applications of linear algebra because the notation simplifies many discussions and highlights essential structure in matrix calculations. This section provides an opportunity to review matrix algebra and use of the Invertible Matrix Theorem.

General Partitioning of a Matrix -

A matrix can be **partitioned** (subdivided) into **sub matrices** (also called **blocks**) in various ways by inserting lines between selected rows and columns.

Example 1

The matrix

$$P = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 \\ 3 & 3 & 4 & 4 \\ 3 & 3 & 4 & 4 \end{bmatrix}$$

can be partitioned into four 2×2 blocks

$$P_{11} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, P_{12} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, P_{21} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}, P_{22} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix},$$

The partitioned matrix can then be written as

$$P = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix}$$

Example 2

The matrix $A = \left[\begin{array}{ccc:cc:c} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ \hline -8 & -6 & 3 & 1 & 7 & -4 \end{array}\right]$

can also be written as the 2×3 partitioned (or block) matrix

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

whose entries are the blocks or sub matrices.

Note:-

It is important to know that in how many ways to block up an ordinary matrix A. See the following example in which a matrix A is block up into three different ways.

Example 3

Let A be a general matrix of 5×3 order, we have

Partition (a)

$$A = \left[\begin{array}{c|cc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \hline a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

In this case we partitioned the matrix into four sub matrices. Also notice that we simplified the matrix into a more compact form and in this compact form we've mixed and matched some of our notation. The partitioned matrix can be thought of as a smaller matrix with four entries, except this time each of the entries are matrices instead of numbers and so we used capital letters to represent the entries and subscripted each one with the location in the partitioned matrix.

Be careful not to confuse the location subscripts on each of the sub matrices with the size of each sub matrix. In this case A_{11} is a 2×1 sub matrix of A, A_{12} is a 2×2 sub matrix of A, A_{21} is a 3×1 sub matrix of A and A_{22} is a 3×3 sub matrix of A.

Partition (b)

$$A=\left[\begin{array}{c|c|c} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \end{array}\right]=\left[\mathbf{c}_1 \mid \mathbf{c}_2 \mid \mathbf{c}_3\right]$$

In this case, we partitioned *A* into three column matrices each representing one column in the original matrix. Again, note that we used the standard column matrix notation (the bold face letters) and subscripted each one with the location in the partitioned matrix. The c_i in the partitioned matrix are sometimes called the **column matrices of *A***.

Partition (c)

$$A=\left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ \hline a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ \hline a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \end{array}\right]=\left[\begin{array}{c} \mathbf{r}_1 \\ \hline \mathbf{r}_2 \\ \mathbf{r}_3 \\ \hline \mathbf{r}_4 \\ \mathbf{r}_5 \end{array}\right]$$

Just as we can partition a matrix into each of its columns as we did in the previous part we can also partition a matrix into each of its rows. The r_i in the partitioned matrix are sometimes called the **row matrices** of ***A*.**

Addition of Blocked Matrices

If matrices ***A*** and ***B*** are the same size and are partitioned in exactly the same way, then it is natural to make the same partition of the ordinary matrix sum ***A* + *B***. In this case, each block of ***A* + *B*** is the (matrix) sum of the corresponding blocks of ***A*** and ***B***.

Theorem

If A is m×n and B is n×p, then

$$AB = Col_1(A) + Col_2(A) + ... + Col_n(A) \begin{bmatrix} Row_1(B) \\ Row_2(B) \\ \vdots \\ Row_n(B) \end{bmatrix}$$

$$= Col_1(A)Row_1(B) + Col_2(A)Row_2(B) + ... + Col_n(A)Row_n(B)$$

Proof

For each row index I and column index j, the (i,j) entry in col_k (A) row_k (B) is the product of a_{ik} from col_k (A) and b_{kj} from row_k (B).

Hence, the (i,j)-entry in the sum shown in (2) is

$$a_{i1}b_{1j} + a_{21}b_{2j} + ... + a_{n1}b_{nj}$$

$$(k = 1, 2, ..., n)$$

This sum is also the (i,j)- entry in AB by the row column rule

Example 4

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \text{ and } B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}, \text{ then}$$

$$A + B = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{bmatrix}$$

Similarly, subtraction of blocked matrices is defined.

Multiplication of a partitioned matrix by a scalar is also computed block by block.

Multiplication of Partitioned Matrices

If $$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

and if the sizes of the blocks confirm for the required operations, then

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \\ A_{31}B_{11} + A_{32}B_{21} & A_{31}B_{12} + A_{32}B_{22} \end{bmatrix}$$

It is known as **block multiplication**.

Example 5

Find the block multiplication of the following partitioned matrices:

$$A = \left[\begin{array}{ccc:cc} 1 & 2 & 1 & 0 & 2 \\ -1 & -2 & 3 & 1 & 1 \\ \hdashline 2 & 1 & -2 & 1 & 3 \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \left[\begin{array}{cc} 2 & 1 \\ 3 & 0 \\ -5 & 1 \\ \hdashline 0 & 8 \\ 0 & 2 \end{array}\right] = \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix}$$

Solution

Let

$$A_{11} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & -2 & 3 \end{bmatrix}, A_{12} = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}, A_{21} = \begin{bmatrix} 2 & 1 & -2 \end{bmatrix} \text{ and } A_{22} = \begin{bmatrix} 1 & 3 \end{bmatrix}$$

$$B_{11} = \begin{bmatrix} 2 & 1 \\ 3 & 0 \\ -5 & 1 \end{bmatrix}, B_{21} = \begin{pmatrix} 0 & 8 \\ 0 & 2 \end{pmatrix} \quad So \quad B = \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix}$$

Now

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} \\ A_{21}B_{11} + A_{22}B_{21} \end{bmatrix}$$

This is a valid formula because the sizes of the blocks are such that all of the operations can be performed:

$$A_{11}B_{11} + A_{12}B_{21} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 0 \\ -5 & 1 \end{bmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \begin{bmatrix} 0 & 8 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -23 & 14 \end{bmatrix}$$

$$A_{21}B_{11} + A_{22}B_{21} = \begin{bmatrix} 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 0 \\ -5 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 8 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 17 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 14 \end{bmatrix} = \begin{bmatrix} 17 & 14 \end{bmatrix}$$

Thus,

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} \\ A_{21}B_{11} + A_{22}B_{21} \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ -23 & 12 \\ 17 & 14 \end{bmatrix}$$

Note - We see result is same when we multiply A and B without partitions

$$AB = \begin{bmatrix} 1 & 2 & 1 & 0 & 2 \\ -1 & -2 & 3 & 1 & 1 \\ 2 & 1 & -2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 0 \\ -5 & 1 \\ 0 & 8 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ -23 & 12 \\ 17 & 14 \end{bmatrix}$$

Note - Sometimes it is more useful to find the square and cube powers of a matrix.

Example 6

The block version of the row –column rule for the product AB of the partitioned matrices

$$A = \left[\begin{array}{ccc:cc} 3 & -4 & 1 & 0 & 2 \\ -1 & 5 & -3 & 1 & 4 \\ \hdashline 2 & 0 & -2 & 1 & 6 \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \left[\begin{array}{cc} 2 & -1 \\ 3 & 0 \\ \hdashline -5 & 1 \\ 4 & -3 \\ 0 & 2 \end{array}\right] = \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix}$$

where

$$B_{11} = \begin{bmatrix} 2 & -1 \\ 3 & 0 \\ -5 & 1 \end{bmatrix} \text{ and } B_{21} = \begin{pmatrix} 4 & -3 \\ 0 & 2 \end{pmatrix}$$

So

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} \\ A_{21}B_{11} + A_{22}B_{21} \end{bmatrix}$$

This is a valid formula because the sizes of the blocks are such that all of the operations can be performed:

$$A_{11}B_{11} + A_{12}B_{21} = \begin{bmatrix} 3 & -4 & 1 \\ -1 & 5 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \\ -5 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} -11 & 2 \\ 32 & 3 \end{bmatrix}$$

$$A_{21}B_{11} + A_{22}B_{21} = \begin{bmatrix} 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \\ -5 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 6 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 18 & 5 \end{bmatrix}$$

Thus,

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} \\ A_{21}B_{11} + A_{22}B_{21} \end{bmatrix} = \begin{bmatrix} -11 & 2 \\ 32 & 3 \\ 18 & 5 \end{bmatrix}$$

This result can be confirmed by performing the computation.

$$AB = \begin{bmatrix} 3 & -4 & 1 & 0 & 2 \\ -1 & 5 & -3 & 1 & 4 \\ 2 & 0 & -2 & 1 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \\ -5 & 1 \\ 4 & -3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} -11 & 2 \\ 32 & 3 \\ 18 & 5 \end{bmatrix}$$

Example 7

Making block up of matrix

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}, \text{ evaluate } A^2 ?$$

Solution

We partition A as shown below

$$A = \left[\begin{array}{cccc:c} 2 & 0 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & 2 \\ \hdashline 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right] \quad \text{where } A_{32} = \begin{bmatrix} 0 & 2 \\ 0 & 2 \\ 0 & 2 \end{bmatrix},$$

Now

$$A^2 = \begin{pmatrix} 2I_3 & A_{32} \\ O_{23} & 2I_2 \end{pmatrix} \begin{pmatrix} 2I_3 & A_{32} \\ O_{23} & 2I_2 \end{pmatrix}$$

$$= \begin{pmatrix} 4I_3 & 4A_{32} \\ O_{23} & 4I_2 \end{pmatrix}$$

Hence

$$A^2 = \begin{bmatrix} 4 & 0 & 0 & 0 & 8 \\ 0 & 4 & 0 & 0 & 8 \\ 0 & 0 & 4 & 0 & 8 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Example 8

Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$ Evaluate A^2 ?

Solution

We partition A as shown below

$$A = \left[\begin{array}{ccc:cc:c} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ \hdashline 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ \hdashline 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right] = \begin{bmatrix} I_3 & O_{32} & A_1 \\ O & I & O \\ A_1^t & O^2 & 1_{12} \end{bmatrix} \quad \text{where } A_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Now

$$A^2 = \begin{bmatrix} I & O & A \\ O^3 & I^{32} & O^1 \\ A_1^t & O^2 & 1_{12} \end{bmatrix} \begin{bmatrix} I & O & A \\ O^3 & I^{32} & O^1 \\ A_1^t & O^2 & 1_{12} \end{bmatrix} = \begin{bmatrix} I_3 + A_1 A_1^t & O & A_1 + A_1 \\ O & I^{32} & O^1 \\ A_1^t + A_1^t & O^2 & A_1^t A_1 + 1_{11} \end{bmatrix}$$

$$I_3 + A_1 A_1^t = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}, A_1 + A_1 = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, A_1^t + A_1^t = \begin{bmatrix} 2 & 2 & 2 \end{bmatrix}$$

$$A_1^t A_1 + 1 = \begin{bmatrix} 4 \end{bmatrix}$$

Hence $A^2 = \begin{bmatrix} 2 & 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 2 & 2 & 2 & 0 & 0 & 4 \end{bmatrix}$

Toeplitz matrix

A matrix in which each descending diagonal from left to right is constant is called a **Toeplitz matrix** or **diagonal-constant matrix**

Example 9 The matrix $A = \begin{bmatrix} a & 1 & 2 & 3 \\ 4 & a & 1 & 2 \\ 5 & 4 & a & 1 \\ 6 & 5 & 4 & a \end{bmatrix}$ is a Toeplitz matrix.

Block Toeplitz matrix

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a blocked Toeplitz matrix.

A block Toeplitz matrix **B** has the form

$$B = \begin{bmatrix} B(1,1) & B(1,2) & B(1,3) & B(1,4) & B(1,5) \\ B(2,1) & B(1,1) & B(1,2) & B(1,3) & B(1,4) \\ B(3,1) & B(2,1) & B(1,1) & B(1,2) & B(1,3) \\ B(4,1) & B(3,1) & B(2,1) & B(1,1) & B(1,2) \\ B(5,1) & B(4,1) & B(3,1) & B(2,1) & B(1,1) \end{bmatrix}$$

Inverses of Partitioned Matrices

In this section, we will study about the techniques of inverse of blocked matrices.

Block Diagonal Matrices

A partitioned matrix ***A*** is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are zero, i.e.

$$A=\begin{bmatrix} D_1 & 0 & \dots & 0 \\ 0 & D_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & D_k \end{bmatrix} \qquad (1)$$

where the matrices $D_1, D_2, \ldots, D_k$ are square. It can be shown that the matrix A in (1) is invertible if and only if each matrix on the diagonal is invertible. i.e.

$$A^{-1}=\begin{bmatrix} D_1^{-1} & 0 & \dots & 0 \\ 0 & D_2^{-1} & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & D_k^{-1} \end{bmatrix}$$

Example 10 Let A be a block diagonal matrix

$$A=\left[\begin{array}{cc:cc:c} 1 & 2 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ \hdashline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 5 & 4 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

Find A^{-1}.

Solution

There are three matrices on the main diagonal; two are 2x2 matrices and one is1×1matrix.

In order to find A^{-1}, we evaluate the inverses of three matrices lie in main diagonal of A. Let $A_{11}=\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}, A_{22}=\begin{pmatrix} 1 & 1 \\ 5 & 4 \end{pmatrix}$ *and* $A_{33}=(2)$ are matrices of main diagonal of A. Then

$$A_{11}^{-1}=\frac{AdjA_{11}}{A_{11}}=\frac{\begin{pmatrix} -1 & -2 \\ -1 & 1 \end{pmatrix}}{-3}=\begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{-1}{3} \end{pmatrix}.$$

Similarly $A_{22}^{-1} = \dfrac{\begin{pmatrix} 4 & -1 \\ -5 & 1 \end{pmatrix}}{-1}$

$$A_{22}^{-1} = \begin{pmatrix} -4 & 1 \\ 5 & -1 \end{pmatrix}$$

$$\Rightarrow \quad A^{-1} = \left[\begin{array}{cc:cc:c} \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{-1}{3} & 0 & 0 & 0 \\ \hdashline 0 & 0 & -4 & 1 & 0 \\ 0 & 0 & 5 & -1 & 0 \\ \hdashline 0 & 0 & 0 & 0 & \frac{1}{2} \end{array}\right]$$

Example 11 Consider the block diagonal matrix

$$A = \left[\begin{array}{cc:cc:c} 8 & -7 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ \hdashline 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 5 & 2 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 4 \end{array}\right]$$

There are three matrices on the main diagonal – two 2x2 matrices and one 1x1 matrix.

$$\Rightarrow \quad A^{-1} = \left[\begin{array}{cc:cc:c} 1 & -7 & 0 & 0 & 0 \\ 1 & -8 & 0 & 0 & 0 \\ \hdashline 0 & 0 & 2 & -1 & 0 \\ 0 & 0 & -5 & 3 & 0 \\ \hdashline 0 & 0 & 0 & 0 & \frac{1}{4} \end{array}\right]$$

Block Upper Triangular Matrices

A partitioned square matrix ***A*** is said to be block upper triangular if the matrices on the main diagonal are square and all matrices below the main diagonal are zero; that is, the matrix is partitioned as

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1k} \\ O & A_{22} & \dots & A_{2k} \\ \vdots & \vdots & & \vdots \\ O & O & \dots & A_{kk} \end{bmatrix}$$ where the matrices $A_{11}, A_{22}, \dots, A_{kk}$ are square.

Note The definition of block lower triangular matrix is similar.

Here, we are going to introduce a formula for finding inverse of a block upper triangular matrix in the following example.

Example 12

Let A be a block upper triangular matrix of the form $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ where the orders of A_{11} and A_{22} are $p \times p$ *and* $q \times q$ *respectively.* Find A^{-1}.

Solution

Let $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ be inverse of A .i.e. $A^{-1} = B$, then

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} I_p & O \\ O & I_q \end{bmatrix}$$

$$\begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{22}B_{21} & A_{22}B_{22} \end{bmatrix} = \begin{bmatrix} I_p & O \\ O & I_q \end{bmatrix}$$

By comparing corresponding entries, we have

$$A_{11}B_{11} + A_{12}B_{21} = I_p \qquad (1)$$

$$A_{11}B_{12} + A_{12}B_{22} = O \qquad (2)$$

$$A_{22}B_{21} = O \qquad (3)$$

$$A_{22}B_{22} = I_q \qquad (4)$$

Since A_{22} is a square matrix, so by Invertible Matrix Theorem, we have $\boldsymbol{A_{22}}$ is invertible. Thus by eq.(4), $\boldsymbol{B_{22} = A_{22}^{-1}}$. Now by eq. (3), we have $B_{21} = A_{22}^{-1}O = O$. From eq.(1)

$$A_{11}B_{11} + O = I_p$$
$$\Rightarrow A_{11}B_{11} = I_p$$
$$\Rightarrow B_{11} = A_{11}^{-1}$$

Finally, form (2),

$$A_{11}B_{12} = -A_{12}B_{22} = -A_{12}A_{22}^{-1} \quad and \quad B_{12} = -A_{11}^{-1}A_{12}A_{22}^{-1}$$

Thus

$$A^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ O & A_{22}^{-1} \end{bmatrix} \qquad (5)$$

Example 13

Find A^{-1} of

$$A = \begin{bmatrix} 1 & 9 & -5 & 0 \\ 3 & 3 & 3 & -2 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

Solution

Let partition given matrix A in form

$$A = \left[\begin{array}{cc:cc} 1 & 9 & -5 & 0 \\ 3 & 3 & 3 & -2 \\ \hdashline 0 & 0 & 7 & 0 \\ 0 & 0 & 3 & 1 \end{array}\right]$$

Put $A_{11} = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix}, A_{12} = \begin{bmatrix} -5 & 0 \\ 3 & -2 \end{bmatrix}$ *and* $A_{22} = \begin{bmatrix} 7 & 0 \\ 3 & 1 \end{bmatrix}$

Thus $A_{11}^{-1} = \dfrac{AdjA}{\det(A)} = \dfrac{\begin{bmatrix} 3 & -9 \\ -3 & 1 \end{bmatrix}}{-24} = \begin{pmatrix} \frac{1}{8} & \frac{9}{24} \\ \frac{1}{8} & \frac{1}{-24} \end{pmatrix}$ *and* $A_{22}^{-1} = \begin{bmatrix} \frac{1}{7} & 0 \\ \frac{-3}{7} & 1 \end{bmatrix}$

Moreover,

$$-A_{11}^{-1}A_{12}A_{22}^{-1} = \begin{bmatrix} \frac{1}{-8} & \frac{9}{24} \\ \frac{1}{8} & \frac{1}{-24} \end{bmatrix} \begin{bmatrix} -5 & 0 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} \frac{1}{7} & 0 \\ \frac{-3}{7} & 1 \end{bmatrix} = \begin{bmatrix} \frac{-4}{7} & \frac{3}{4} \\ \frac{1}{4} & \frac{-1}{12} \end{bmatrix}$$

So by (5), we have

$$A^{-1} = \begin{bmatrix} -1 & 3 & -4 & 3 \\ -8 & 8 & -7 & 4 \\ \frac{1}{8} & \frac{-1}{24} & 1 & \frac{-1}{12} \\ 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & \frac{-3}{7} & 1 \end{bmatrix}$$

Example 14

Confirm that $A = \begin{bmatrix} 4 & 7 & -5 & 3 \\ 3 & 5 & 3 & -2 \\ 0 & 0 & 7 & 2 \\ 0 & 0 & 3 & 1 \end{bmatrix}$ is an invertible block upper triangular matrix, and then find its inverse by using formula (9).

Solution The matrix is block upper triangular because it can be partitioned into form

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1k} \\ O & A_{22} & \dots & A_{2k} \\ \vdots & \vdots & & \vdots \\ O & O & \dots & A_{kk} \end{bmatrix}$$

as

$$A = \left[\begin{array}{cc:cc} 4 & 7 & -5 & 3 \\ 3 & 5 & 3 & -2 \\ \hdashline 0 & 0 & 7 & 2 \\ 0 & 0 & 3 & 1 \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}$$

where $A_{11} = \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix}, A_{12} = \begin{bmatrix} -5 & 3 \\ 3 & -2 \end{bmatrix}, A_{22} = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}$

Now $A_{11}^{-1} = \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix}$ *and* $A_{22}^{-1} = \begin{bmatrix} 1 & -2 \\ -3 & 7 \end{bmatrix}$

Moreover, $-A_{11}^{-1}A_{12}A_{22}^{-1} = -\begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix}\begin{bmatrix} -5 & 3 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} -133 & 295 \\ 78 & -173 \end{bmatrix}$

So it follows from (9) that the inverse of A is $A^{-1} = \begin{bmatrix} -5 & 7 & -133 & 295 \\ 3 & -4 & 78 & -173 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -3 & 7 \end{bmatrix}$

NUMERICAL NOTES

1. When matrices are too large to fit in a computer's high-speed memory, partitioning permits a computer to work with only two or three sub-matrices at a time. For instance, in recent work on linear programming, a research team simplified a problem by partitioning the matrix into 837 rows and 51 columns. The problem's solution took about 4 minutes on a Cray supercomputer.

2. Some high-speed computers, particularly those with vector pipeline architecture, perform matrix calculations more efficiently when the algorithms use partitioned matrices.

3. The latest professional software for high-performance numerical linear algebra, **LAPACK**, makes intensive use of partitioned matrix calculations.

The exercises that follow give practice with matrix algebra and illustrate typical calculations found in applications.

Exercises

In exercises 1 to 3, the matrices A, B, C, X, Y, Z, and I are all n x n and satisfy the indicated equation.

1. $\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}\begin{bmatrix} I & 0 \\ X & Y \end{bmatrix}=\begin{bmatrix} 0 & I \\ Z & 0 \end{bmatrix}$

2. $\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix}\begin{bmatrix} A & 0 \\ B & C \end{bmatrix}=\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$

3. $\begin{bmatrix} X & 0 & 0 \\ & & \\ Y & 0 & I \end{bmatrix}\begin{bmatrix} A & Z \\ & \\ 0 & 0 \\ & \\ B & I \end{bmatrix}=\begin{bmatrix} I & 0 \\ & \\ 0 & I \end{bmatrix}$

4. Suppose that A_{11} is an invertible matrix. Find matrices X and Y such that the product below has the form indicated. Also compute B_{22}.

$$\begin{bmatrix} I & 0 & 0 \\ X & I & 0 \\ Y & 0 & I \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix}=\begin{bmatrix} B_{11} & B_{12} \\ 0 & B_{22} \\ 0 & B_{32} \end{bmatrix}$$

5. The inverse of $\begin{bmatrix} I & 0 & 0 \\ C & I & 0 \\ A & B & I \end{bmatrix}$ *is* $\begin{bmatrix} I & 0 & 0 \\ Z & I & 0 \\ X & Y & I \end{bmatrix}$. Find X, Y and Z.

6. Find the Inverse of matrix A.

$$A=\left[\begin{array}{cc:cc:c} 8 & 3 & 0 & 0 & 0 \\ 1 & 4 & 0 & 0 & 0 \\ \hdashline 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 5 & 1 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 8 \end{array}\right]$$

7. Show that $\begin{bmatrix} I & 0 \\ & \\ A & I \end{bmatrix}$ is invertible and find its inverse.

8. Compute $\boldsymbol{X^TX}$, when $\boldsymbol{X}$ is partitioned as $[\boldsymbol{X_1} \quad \boldsymbol{X_2}]$.

In exercises 9 and 10, determine whether block multiplication can be used to compute the product using the partitions shown. If so, compute the product by block multiplication.

9. (a) $\left[\begin{array}{cc:cc}-1 & 2 & 1 & 5\\ 0 & -3 & 4 & 2\\ \hdashline 1 & 5 & 6 & 1\end{array}\right]\left[\begin{array}{ccc}-2 & 1 & 4\\ -3 & 5 & 2\\ \hdashline 7 & -1 & 5\\ \hdashline 0 & 3 & -3\end{array}\right]$ (b) $\left[\begin{array}{ccc:c}-1 & 2 & 1 & 5\\ 0 & -3 & 4 & 2\\ \hdashline 1 & 5 & 6 & 1\end{array}\right]\left[\begin{array}{ccc}-2 & 1 & 4\\ -3 & 5 & 2\\ \hdashline 7 & -1 & 5\\ \hdashline 0 & 3 & -3\end{array}\right]$

10. (a) $\left[\begin{array}{ccc}3 & -1 & 0\\ 2 & 1 & 4\end{array}\right]\left[\begin{array}{c}-3\\ 5\end{array}\right]\left[\begin{array}{ccc}2 & -4 & 1\\ 3 & 0 & 2\\ 1 & -3 & 5\\ \hdashline 2 & 1 & 4\end{array}\right]$ (b) $\left[\begin{array}{cc}2 & -5\\ 1 & 3\\ 0 & 5\\ \hdashline 1 & 4\end{array}\right]\left[\begin{array}{cc:cc}2 & -1 & 3 & -4\\ 0 & 1 & 5 & 7\end{array}\right]$

11. Compute the product $\left[\begin{array}{cc}3 & 1\\ 2 & -4\end{array}\right]\left[\begin{array}{ccc}1 & 2 & 0\\ -1 & 6 & 2\end{array}\right]$ using the column row-rule, and check your answer by calculating the product directly.

In exercises 11 and 12, find the inverse of the block diagonal matrix A.

12. (a) $A=\left[\begin{array}{cccc}2 & 1 & 0 & 0\\ 3 & 2 & 0 & 0\\ 0 & 0 & 3 & 4\\ 0 & 0 & 1 & -1\end{array}\right]$ (b) $\left[\begin{array}{ccccc}5 & 2 & 0 & 0 & 0\\ 3 & 1 & 0 & 0 & 0\\ 0 & 0 & 5 & 0 & 0\\ 0 & 0 & 0 & 2 & 7\\ 0 & 0 & 0 & 1 & 4\end{array}\right]$

13. (a) $A=\left[\begin{array}{cccc}5 & 1 & 0 & 0\\ 4 & 1 & 0 & 0\\ 0 & 0 & 2 & -3\\ 0 & 0 & -3 & 5\end{array}\right]$ (b) $\left[\begin{array}{ccccc}2 & 0 & 0 & 0 & 0\\ 0 & 1 & 2 & 0 & 0\\ 0 & 3 & 7 & 0 & 0\\ 0 & 0 & 0 & 4 & 9\\ 0 & 0 & 0 & 1 & 2\end{array}\right]$

14. Find the inverse of the block upper triangular matrix A.

(i) $A=\left[\begin{array}{cccc}2 & 1 & 3 & -6\\ 1 & 1 & 7 & 4\\ 0 & 0 & 3 & 5\\ 0 & 0 & 2 & 3\end{array}\right]$ (ii) $A=\left[\begin{array}{cccc}-1 & -1 & 2 & 5\\ 2 & 1 & -3 & 8\\ 0 & 0 & 4 & 1\\ 0 & 0 & 7 & 2\end{array}\right]$

15. Find B_1, given that $\left[\begin{array}{cc}A_1 & B_1\\ 0 & C_1\end{array}\right]\left[\begin{array}{cc}A_2 & B_2\\ 0 & C_2\end{array}\right]=\left[\begin{array}{cc}A_3 & B_3\\ 0 & C_3\end{array}\right]$ and

$$A_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, B_2 = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, C_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$
$$A_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, C_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, B_3 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

16. Consider the partitioned linear system

$$\left[\begin{array}{cc:cc} 5 & 2 & 2 & 3 \\ 2 & 1 & -3 & 1 \\ \hdashline 1 & 0 & 4 & 1 \\ 0 & 1 & 0 & 2 \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \left[\begin{array}{c} 2 \\ 6 \\ \hdashline 0 \\ 0 \end{array}\right]$$

Solve this system by first expressing it as

$$\begin{bmatrix} A & B \\ I & D \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix} \text{ or equivalently, } \begin{aligned} Au + Bv &= b \\ u + Dv &= 0 \end{aligned}$$

next solving the second equation for u in terms of v, and then substituting in the first equation. Check your answer by solving the system directly.

17. Let $A = \begin{bmatrix} -3 & 1 & 2 \\ 1 & -4 & 5 \end{bmatrix}$ *and* $B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$.

Verify that $AB = col_1(A)row_1(B) + col_2(A)row_2(B) + col_3(A)row_3(B)$

18. Find inverse of the matrix $A = \begin{bmatrix} 4 & 7 & -5 & 3 \\ 3 & 5 & 3 & -2 \\ 0 & 0 & 7 & 2 \\ 0 & 0 & 3 & 1 \end{bmatrix}$.

Lecture 15

Matrix Factorizations

Matrix Factorization

A factorization of a matrix as a product of two or more matrices is called Matrix Factorization.

Uses of Matrix Factorization

Matrix factorizations will appear at a number of key points throughout the course. This lecture focuses on a factorization that lies at the heart of several important computer programs widely used in applications.

***LU* Factorization or LU-decomposition**

LU factorization is a matrix decomposition which writes a matrix as the product of a lower triangular matrix and an upper triangular matrix. This decomposition is used to solve systems of linear equations or calculate the determinant.

Assume A is an $m \times n$ matrix that can be row reduced to echelon form, without row interchanges. Then A can be written in the form A = LU, where L is an $m \times m$ lower triangular matrix with 1's on the diagonal and U is an $m \times n$ echelon form of A. For instance, such a factorization is called LU factorization of A. The matrix L is invertible and is called a unit lower triangular matrix.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix} \begin{bmatrix} \bullet & * & * & * & * \\ 0 & \bullet & * & * & * \\ 0 & 0 & 0 & \bullet & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

L U

***LU* factorization.**

Remarks

1) If A is the square matrix of order m, then the order of both L and U will also be m.

2) In general, not every square matrix A has an LU-decomposition, nor is an LU-decomposition unique if it exists.

Theorem **If a square matrix *A* can be reduced to row echelon form with no row interchanges, then *A* has an *LU*-decomposition.**

Note

The computational efficiency of the ***LU*** factorization depends on knowing ***L*** and ***U***. The next algorithm shows that the row reduction of ***A*** to an echelon form ***U*** amounts to an ***LU*** factorization because it produces ***L*** with essentially no extra work.

An *LU* Factorization Algorithm

Suppose ***A*** can be reduced to an echelon form ***U*** without row interchanges. Then, since row scaling is not essential, ***A*** can be reduced to ***U*** with only row replacements, adding a

multiple of one row to another row below it. In this case, there exist lower triangular elementary matrices $E_1, \ldots, E_p$ such that

$$E_p \ldots E_1 A = U \qquad (1)$$

So $A = (E_p \ldots E_1)^{-1} U = LU$

Where $L = (E_p \ldots E_1)^{-1}$ (2)

It can be shown that products and inverses of unit lower triangular matrices are also unit-lower triangular. Thus, L is unit-lower triangular.

Note that the row operations in (1), which reduce A to U, also reduce the L in (2) to I, because $E_p \ldots E_1 L = (Ep \ldots E_1)(E_p \ldots E_1)^{-1} = I$. This observation is the key to constructing L.

Procedure for finding an *LU*-decomposition

Step 1: Reduce matrix A to row echelon form U without using row interchanges, keeping track of the multipliers used to introduce the leading 1's and the multipliers used to introduce zeros below the leading 1's.

Step 2: In each position along the main diagonal of L, place the reciprocal of the multiplier that introduced the leading 1 in that position in U.

Step 3: In each position below the main diagonal of L, place the negative of the multiplier used to introduce the zero in that position in U.

Step 4: Form the decomposition $A = LU$.

Example 1 Find an *LU*-decomposition of

$$A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}$$

Solution We will reduce A to a row echelon form U and at each step we will fill in an entry of L in accordance with the four-step procedure above.

$$A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} \qquad \begin{bmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

* denotes an unknown entry of L.

$$\begin{bmatrix} \boxed{1} & -\frac{1}{3} & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} \leftarrow \text{multiplier} = \frac{1}{6} \qquad \begin{bmatrix} 6 & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ \boxed{0} & 2 & 1 \\ \boxed{0} & 8 & 5 \end{bmatrix} \begin{matrix} \leftarrow \text{multiplier} = -9 \\ \leftarrow \text{multiplier} = -3 \end{matrix} \qquad \begin{bmatrix} 6 & 0 & 0 \\ 9 & * & 0 \\ 3 & * & * \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 8 & 5 \end{bmatrix} \leftarrow multiplier = \frac{1}{2} \qquad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & * & * \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & \boxed{0} & 1 \end{bmatrix} \leftarrow multiplier = -8 \qquad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & * \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \boxed{1} \end{bmatrix} \leftarrow multiplier = 1 \qquad L = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix}$$

(No actual operation is performed here since there is already a leading 1 in the third row.)
So

$$A = LU = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

OR

Solution We will reduce ***A*** to a row echelon form ***U*** and at each step we will fill in an entry of ***L*** in accordance with the four-step procedure above.

$$A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} \qquad \begin{bmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

* denotes an unknown entry of ***L***.

$$\approx \begin{bmatrix} \frac{6}{6} & \frac{-2}{6} & \frac{0}{6} \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} \quad \frac{1}{6} R_1 \qquad \begin{bmatrix} 6 & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ 9-9(1) & -1-9(\frac{-1}{3}) & 1-9(0) \\ 3-3(1) & 7-3(\frac{-1}{3}) & 5-3(0) \end{bmatrix} \quad \begin{matrix} R_2 - 9R_1 \\ R_3 - 3R_1 \end{matrix} \quad \begin{bmatrix} 6 & 0 & 0 \\ 9 & * & 0 \\ 3 & * & * \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ 0 & 2 & 1 \\ 0 & 8 & 5 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ \frac{0}{2} & \frac{2}{2} & \frac{1}{2} \\ 0 & 8 & 5 \end{bmatrix} \frac{1}{2} R_2 \quad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & * & * \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 8 & 5 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0-8(0) & 8-8(1) & 5-8(\frac{1}{2}) \end{bmatrix} \quad R_3 - 8R_2 \quad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & * \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{-1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \boxed{1} \end{bmatrix} \leftarrow multiplier = 1 \qquad L = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix}$$

So

$$A = LU = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

Example 2 Find an ***LU*** factorization of $A = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$

Solution

$$A = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ * & 1 & 0 & 0 & 0 \\ * & * & 1 & 0 & 0 \\ * & * & * & 1 & 0 \\ * & * & * & * & 1 \end{bmatrix}$$

* denotes an unknown entry of L.

$$\begin{bmatrix} 1 & -2 & -1 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix} \leftarrow multiplier \frac{1}{2} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ * & 1 & 0 & 0 & 0 \\ * & * & 1 & 0 & 0 \\ * & * & * & 1 & 0 \\ * & * & * & * & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & -3 & -1 & 6 \\ 0 & 6 & 2 & -7 \\ 0 & -9 & -3 & 13 \end{bmatrix} \begin{matrix} \\ \\ \leftarrow multiplier -6 \\ \leftarrow multiplier -2 \\ \leftarrow multiplier -4 \\ \leftarrow multiplier 6 \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 6 & 1 & 0 & 0 & 0 \\ 2 & * & 1 & 0 & 0 \\ 4 & * & * & 1 & 0 \\ -6 & * & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & -3 & -1 & 6 \\ 0 & 6 & 2 & -7 \\ 0 & -9 & -3 & 13 \end{bmatrix} \leftarrow multiplier\ 1 \qquad \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 & 0 \\ 2 & * & 1 & 0 & 0 \\ 4 & * & * & 1 & 0 \\ -6 & * & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 10 \end{bmatrix} \begin{matrix} \\ \\ \leftarrow multiplier\ 3 \\ \leftarrow multiplier\ -6 \\ \leftarrow multiplier\ 9 \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 & 0 \\ 2 & -3 & 1 & 0 & 0 \\ 4 & 6 & * & 1 & 0 \\ -6 & -9 & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 10 \end{bmatrix} \begin{matrix} \\ \\ \leftarrow multiplier\ -\frac{1}{5} \\ \\ \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 & 0 \\ 2 & -3 & 1 & 0 & 0 \\ 4 & 6 & 0 & -5 & 0 \\ -6 & -9 & 0 & * & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ \leftarrow multiplier\ -10 \\ \\ \end{matrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 & 0 \\ 2 & -3 & 1 & 0 & 0 \\ 4 & 6 & 0 & -5 & 0 \\ -6 & -9 & 0 & 10 & 1 \end{bmatrix}$$

Thus, we have constructed the ***LU***-decomposition

$$A = LU = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 6 & 3 & 0 & 0 & 0 \\ 2 & -3 & 1 & 0 & 0 \\ 4 & 6 & 0 & -5 & 0 \\ -6 & -9 & 0 & 10 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 3 Find ***LU***–decomposition of $A = \begin{bmatrix} 6 & -2 & -4 & 4 \\ 3 & -3 & -6 & 1 \\ -12 & 8 & 21 & -8 \\ -6 & 0 & -10 & 7 \end{bmatrix}$

Solution

$$A = \begin{bmatrix} 6 & -2 & -4 & 4 \\ 3 & -3 & -6 & 1 \\ -12 & 8 & 21 & -8 \\ -6 & 0 & -10 & 7 \end{bmatrix} \qquad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}$$

* denotes an unknown entry of ***L***.

$$\begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 3 & -3 & -6 & 1 \\ -12 & 8 & 21 & -8 \\ -6 & 0 & -10 & 7 \end{bmatrix} \leftarrow \text{multiplier } \frac{1}{6} \qquad \begin{bmatrix} 6 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & -2 & -4 & -1 \\ 0 & 4 & 13 & 0 \\ 0 & -2 & -14 & 11 \end{bmatrix} \begin{matrix} \\ \leftarrow \text{multiplier } -3 \\ \leftarrow \text{multiplier } 12 \\ \leftarrow \text{multiplier } 6 \end{matrix} \qquad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ -12 & * & 1 & 0 \\ -6 & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 4 & 13 & 0 \\ 0 & -2 & -14 & 11 \end{bmatrix} \leftarrow \text{multiplier } -\frac{1}{2} \qquad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ -12 & * & 1 & 0 \\ -6 & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 5 & -2 \\ 0 & 0 & -10 & 12 \end{bmatrix} \leftarrow \text{multiplier} -4 \quad \leftarrow \text{multiplier } 2$$

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ -12 & 4 & 1 & 0 \\ -6 & -2 & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{2}{5} \\ 0 & 0 & -10 & 12 \end{bmatrix} \leftarrow \text{multiplier } \frac{1}{5}$$

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ -12 & 4 & 5 & 0 \\ -6 & -2 & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{2}{5} \\ 0 & 0 & 0 & 8 \end{bmatrix} \leftarrow \text{multiplier } 10$$

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ -12 & 4 & 5 & 0 \\ -6 & -2 & -10 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{2}{5} \\ 0 & 0 & 0 & 1 \end{bmatrix} \leftarrow \text{multiplier } \frac{1}{8}$$

$$L = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ -12 & 4 & 5 & 0 \\ -6 & -2 & -10 & 8 \end{bmatrix}$$

$$\text{Thus } A = LU = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ -12 & 4 & 5 & 0 \\ -6 & -2 & -10 & 8 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{2}{5} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 4 Find an ***LU*** factorization of $A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$

Solution

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}$$

* denotes an unknown entry of ***L***.

$$\begin{bmatrix} 1 & 2 & -\frac{1}{2} & \frac{5}{2} & -1 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} \leftarrow \text{multiplier } \frac{1}{2} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -\frac{1}{2} & \frac{5}{2} & -1 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \begin{matrix} \\ \leftarrow \text{multiplier } 4 \\ \leftarrow \text{multiplier } -2 \\ \leftarrow \text{multiplier } 6 \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 \\ -4 & 1 & 0 & 0 \\ 2 & * & 1 & 0 \\ -6 & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -\frac{1}{2} & \frac{5}{2} & -1 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} & -1 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \begin{matrix} \\ \leftarrow \text{multiplier } \frac{1}{3} \\ \\ \\ \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 \\ -4 & 3 & 0 & 0 \\ 2 & * & 1 & 0 \\ -6 & * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -\frac{1}{2} & \frac{5}{2} & -1 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} & -1 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \begin{matrix} \\ \\ \leftarrow \text{multiplier } 9 \\ \leftarrow \text{multiplier } -12 \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 0 \\ -4 & 3 & 0 & 0 \\ 2 & -9 & 1 & 0 \\ -6 & 12 & * & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 2 & -\frac{1}{2} & \frac{5}{2} & -1 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} & -1 \\ 0 & 0 & 0 & 2 & \frac{1}{4} \\ 0 & 0 & 0 & 1 & \frac{7}{4} \end{bmatrix} \leftarrow \text{multiplier } \frac{1}{4} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ -4 & 3 & 0 & 0 \\ 2 & -9 & 1 & 0 \\ -6 & 12 & 0 & 4 \end{bmatrix}$$

Thus $A = LU = \begin{bmatrix} 2 & 0 & 0 & 0 \\ -4 & 3 & 0 & 0 \\ 2 & -9 & 1 & 0 \\ -6 & 12 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & -\frac{1}{2} & \frac{5}{2} & -1 \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} & -1 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & \frac{7}{4} \end{bmatrix}$

Matrix Inversion by *LU*-Decomposition

Many of the best algorithms for inverting matrices use ***LU***-decomposition. To understand how this can be done, let ***A*** be an invertible $n \times n$ matrix, let $A^{-1} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$ be its unknown inverse partitioned into column vectors, and let $I = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix}$ be then $n \times n$ identity matrix partitioned into column vectors. The matrix equation $AA^{-1} = I$ can be expressed as

$$A\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix}$$

$$\begin{bmatrix} Ax_1 & Ax_2 & \cdots & Ax_n \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix}$$

which tells us that the unknown column vectors of A^{-1} can be obtained by solving the *n*-linear systems.

$$Ax_1 = e_1, \quad Ax_2 = e_2, \cdots, \quad Ax_n = e_n \qquad (1^*)$$

As discussed above, this can be done by finding an ***LU***-decomposition of ***A***, and then using that decomposition to solve each of the n systems in (1*).

Solving Linear System by LU-Factorization

When ***A =LU***, the equation ***Ax = b*** can be written as ***L (Ux) = b***. Writing ***y*** for ***Ux***, we can find ***x*** by solving the pair of equations; ***Ly = b*** and ***Ux = y*** (2*)

First solve ***Ly = b*** for ***y*** and then solve ***Ux = y*** for ***x***. Each equation is easy to solve because ***L*** and ***U*** are triangular.

Procedure

Step 1: Rewrite the system $\boldsymbol{A\,x = b}$ as $\boldsymbol{LU\,x = b}$ (3*)

Step 2: Define a new unknown $\boldsymbol{y}$ by letting $\boldsymbol{U\,x = y}$ (4*)
And rewrite (3*) as $\boldsymbol{L\,y = b}$

Step 3: Solve the system $\boldsymbol{L\,y = b}$ for the unknown $\boldsymbol{y}$.
Step 4: Substitute the known vector $\boldsymbol{y}$ into (4*) and solve for $\boldsymbol{x}$.

This procedure is called the method of ***LU***-Decomposition.

Although ***LU***-Decomposition converts the problem of solving the single system $\boldsymbol{A\,x = b}$ into the problem of solving the two systems, $\boldsymbol{L\,y = b}$ and $\boldsymbol{U\,x = y}$, these systems are easy to solve because their co-efficient matrices are triangular.

Example 5 Solve the given system (***Ax =b)*** by ***LU***-Decomposition

$$\begin{aligned} 2x_1 + 6x_2 + 2x_3 &= 2 \\ -3x_1 - 8x_2 \quad\;\; &= 2 \\ 4x_1 + 9x_2 + 2x_3 &= 3 \end{aligned} \qquad (1)$$

Solution We express the system (1) in matrix form:

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$\boldsymbol{A} \qquad \boldsymbol{x} \;=\; \boldsymbol{b}$$

We derive an ***LU***-decomposition of $\boldsymbol{A}$.

$$A = \begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ * & 1 & 0 \\ * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \leftarrow multiplier\ \frac{1}{2} \qquad \begin{bmatrix} 2 & 0 & 0 \\ * & 1 & 0 \\ * & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -2 \end{bmatrix} \begin{matrix} \\ \leftarrow multiplier\ 3 \\ \leftarrow multiplier\ -4 \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & * & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ \leftarrow multiplier\ 3 \\ \\ \end{matrix} \qquad \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \leftarrow multiplier\ \frac{1}{7} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix}$$

Thus
$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \qquad (2)$$

$$\boldsymbol{A} = \boldsymbol{L}\ \boldsymbol{U}$$

From (2) we can rewrite this system as

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \qquad (3)$$

$$\boldsymbol{L}\ \boldsymbol{U}\ \boldsymbol{x} = \boldsymbol{b}$$

As specified in Step 2 above, let us define y_1, y_2 and y_3 by the equation

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \qquad (4)$$

$$\boldsymbol{U}\ \boldsymbol{x} = \boldsymbol{y}$$

which allows us to rewrite (3) as

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \qquad (5)$$

$$\boldsymbol{L}\ \boldsymbol{y} = \boldsymbol{b}$$

$$\begin{aligned} 2y_1 &= 2 \\ -3y_1 + y_2 &= 2 \\ 4y_1 - 3y_2 + 7y_3 &= 3 \end{aligned}$$

or equivalently, as

This system can be solved by a procedure that is similar to back substitution, except that we solve the equations from the top down instead of from the bottom up. This procedure, called **forward substitution**, yields

$$y_1 = 1, \quad y_2 = 5, \quad y_3 = 2.$$

As indicated in Step 4 above, we substitute these values into (4), which yields the linear system

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$$

$$x_1 + 3x_2 + x_3 = 1$$

or equivalently, $x_2 + 3x_3 = 5$

$$x_3 = 2$$

Solving this system by back substitution yields $x_1 = 2$, $x_2 = -1$, $x_3 = 2$

Example 6 It can be verified that

$$\boldsymbol{A} = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \boldsymbol{LU}$$

Use this $\boldsymbol{LU}$ factorization of $\boldsymbol{A}$ to solve $\boldsymbol{Ax} = \boldsymbol{b}$, where $\boldsymbol{b} = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$

Solution The solution of $\boldsymbol{Ly} = \boldsymbol{b}$ needs only 6 multiplications and 6 additions, because the arithmetic takes place only in column 5. (The zeros below each pivot in $\boldsymbol{L}$ are created automatically by our choice of row operations.)

$$[L \quad b] = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = [I \quad y]$$

Then, for $\boldsymbol{Ux} = \boldsymbol{y}$, the "backwards" phase of row reduction requires 4 divisions, 6 multiplications, and 6 additions. (For instance, creating the zeros in column 4 of $[\boldsymbol{U} \quad \boldsymbol{y}]$ requires 1 division in row 4 and 3 multiplication – addition pairs to add multiples of row 4 to the rows above.)

$$[U \quad y] = \begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad x = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix}$$

To find $\boldsymbol{x}$ requires 28 arithmetic operations, or "flops" (floating point operations), excluding the cost of finding $\boldsymbol{L}$ and $\boldsymbol{U}$. In contrast, row reduction of $[\boldsymbol{A} \quad \boldsymbol{b}]$ to $[\boldsymbol{I} \quad \boldsymbol{x}]$ takes 62 operations.

Numerical Notes

The following operation counts apply to an $n \times n$ dense matrix $\boldsymbol{A}$ (with most entries nonzero) for n moderately large, say, $n \geq 30$.

1. Computing an ***LU*** factorization of ***A*** takes about $2n^3/3$ flops (about the same as row reducing [***A*** ***b***]), whereas finding $\boldsymbol{A^{-1}}$ requires about $2n^3$ flops.
2. Solving ***Ly*** = ***b*** and ***Ux*** = ***y*** requires about $2n^2$ flops, because $n \times n$ triangular system can be solved in about n^2 flops.
3. Multiplication of ***b*** by $\boldsymbol{A^{-1}}$ also requires about $2n^2$ flops, but the result may not be as accurate as that obtained from ***L*** and ***U*** (because of round off error when computing both $\boldsymbol{A^{-1}}$ and $\boldsymbol{A^{-1}b}$).
4. If ***A*** is sparse (with mostly zero entries), then ***L*** and ***U*** may be sparse, too, whereas $\boldsymbol{A^{-1}}$ is likely to be dense. In this case, a solution of ***Ax*** = ***b*** with an ***LU*** factorization is much faster than using $\boldsymbol{A^{-1}}$.

Example 7(Gaussian Elimination Performed as an LU-Decomposition)

In Example 5, we showed how to solve the linear system

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \qquad (6)$$

using an ***LU***-decomposition of the coefficient matrix, but we did not discuss how the factorization was derived. In the course of solving the system, we obtained the intermediate vector $\boldsymbol{y} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$ by using forward substitution to solve system (5).

We will now use the procedure discussed above to find both the ***LU***-decomposition and the vector ***y*** by row operations on the augmented matrix for (6).

$$[A|b] = \left[\begin{array}{ccc|c} 2 & 6 & 2 & 2 \\ -3 & -8 & 0 & 2 \\ 4 & 9 & 2 & 3 \end{array}\right] \qquad \begin{bmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix} = L \quad (* = \textit{unknown entries})$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ -3 & -8 & 0 & 2 \\ 4 & 9 & 2 & 3 \end{array}\right] \qquad \begin{bmatrix} 2 & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 1 & 3 & 5 \\ 0 & -3 & -2 & -1 \end{array}\right] \qquad \begin{bmatrix} 2 & 0 & 0 \\ -3 & * & 0 \\ 4 & * & * \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 7 & 14 \end{array}\right] \qquad \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & * \end{bmatrix}$$

$$\left[\begin{array}{c|c} U & y \end{array}\right] = \left[\begin{array}{cccc|c} 1 & 3 & 1 & 1 & \\ 0 & 1 & 3 & 5 & \\ 0 & 0 & 1 & 2 & \end{array}\right] \qquad \left[\begin{array}{ccc} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{array}\right] = L$$

These results agree with those in Example 5, so we have found an ***LU***-decomposition of the coefficient matrix and simultaneously have completed the forward substitution required to find ***y***.

All that remains to solve the given system is to solve the system ***Ux = y*** by back substitution. The computations were performed in Example5.

A Matrix Factorization in Electrical Engineering

Matrix factorization is intimately related to the problem of constructing an electrical network with specified properties. The following discussion gives just a glimpse of the connection between factorization and circuit design.

Suppose the box in below Figure represents some sort of electric circuit, with an input and output. Record the input voltage and current by $\begin{bmatrix} v_1 \\ i_1 \end{bmatrix}$ (with voltage ***v*** in volts and current ***i*** in amps), and record the output voltage and current by $\begin{bmatrix} v_2 \\ i_2 \end{bmatrix}$. Frequently, the transformation $\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} \rightarrow \begin{bmatrix} v_2 \\ i_2 \end{bmatrix}$ is linear. That is, there is a matrix A, called the transfer matrix, such that $\begin{bmatrix} v_2 \\ i_2 \end{bmatrix} = A \begin{bmatrix} v_1 \\ i_1 \end{bmatrix}$

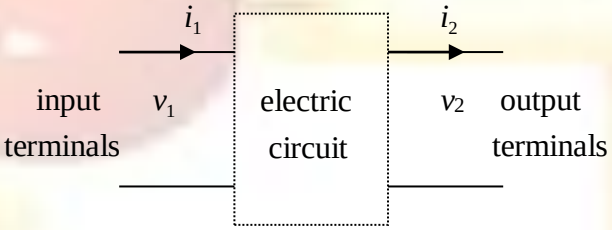


Figure A circuit with input and output terminals.

Above Figure shows a ladder network, where two circuits (there could be more) are connected in series, so that the output of one circuit becomes the input of the next circuit. The left circuit in Figure is called a series circuit, with resistance R_1 (in ohms);

The right circuit is a shunt circuit, with resistance R_2. Using Ohm's law and Kirchhoff's laws, one can show that the transfer matrices of the series and shunt circuits, respectively, are:

$$\begin{bmatrix} 1 & -R_1 \\ 0 & 1 \end{bmatrix} \quad \textit{and} \quad \begin{bmatrix} 1 & 0 \\ -1/R_2 & 1 \end{bmatrix}$$

Transfer matrix of series circuit — Transfer matrix of shunt circuit

Example 8

a) Compute the transfer matrix of the ladder network in the above Figure .

b) Design a ladder network whose transfer matrix is $\begin{bmatrix} 1 & -8 \\ -0.5 & 5 \end{bmatrix}$.

Solution

a) Let $\boldsymbol{A_1}$ and $\boldsymbol{A_2}$ be the transfer matrices of the series of the series and shunt circuits, respectively. Then an input vector $\boldsymbol{x}$ is transformed first into $\boldsymbol{A_1x}$ and then into $\boldsymbol{A_2\,(A_1x)}$. The series connection of the circuits corresponds to composition of linear transformations; and the transfer matrix of the ladder network is (note the order)

$$A_2A_1 = \begin{bmatrix} 1 & 0 \\ -1/R_2 & 1 \end{bmatrix}\begin{bmatrix} 1 & -R_1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -R_1 \\ -1/R_2 & 1+R_1/R_2 \end{bmatrix} \quad (6)$$

b) We seek to factor the matrix $\begin{bmatrix} 1 & -8 \\ -0.5 & 5 \end{bmatrix}$ into the product of transfer matrices, such as in (6). So we look for R_1 and R_2 to satisfy

$$\begin{bmatrix} 1 & -R_1 \\ -1/R_2 & 1+R_1/R_2 \end{bmatrix} = \begin{bmatrix} 1 & -8 \\ -0.5 & 5 \end{bmatrix}$$

From the (1, 2) – entries, R_1 = 8 ohms, and from the (2, 1) – entries, $1/R_2$ = 0.5 ohm and R_2 = 1/0.5 = 2 ohms. With these values, the network has the desired transfer matrix.

Note

A network transfer matrix summarizes the input-output behavior ("Design specifications") of the network without reference to the interior circuits. To physically build a network with specified properties, an engineer first determines if such a network can be constructed (or realized). Then the engineer tries to factor the transfer matrix into matrices corresponding to smaller circuits that perhaps are already manufactured and ready for assembly. In the common case of alternating current, the entries in the transfer matrix are usually rational complex-valued functions. A standard problem is to find a minimal realization that uses the smallest number of electrical components.

Exercises

Find an LU factorization of the matrices in exercises 1 to 8.

1. $\begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}$

2. $\begin{bmatrix} 3 & -1 & 2 \\ -3 & -2 & 10 \\ 9 & -5 & 6 \end{bmatrix}$

3. $\begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ -1 & 7 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 3 & -5 & -3 \\ -1 & -5 & 8 & 4 \\ 4 & 2 & -5 & -7 \\ -2 & -4 & 7 & 5 \end{bmatrix}$

5. $\begin{bmatrix} 2 & -4 & 4 & -2 \\ 6 & -9 & 7 & -3 \\ -1 & -4 & 8 & 0 \end{bmatrix}$

6. $\begin{bmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 4 & -1 & 5 \\ 3 & 7 & -2 & 9 \\ -2 & -3 & 1 & -4 \\ -1 & 6 & -1 & 7 \end{bmatrix}$

8. $\begin{bmatrix} 2 & -4 & -2 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$

Solve the equation $\boldsymbol{Ax = b}$ by using $\boldsymbol{LU}$-factorization.

9. $A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}, b = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}$

10. $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}, b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$

11. $A = \begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$

12. $A = \begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}, b = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$

13. $A = \begin{bmatrix} 1 & -2 & -4 & -3 \\ 2 & -7 & -7 & -6 \\ -1 & 2 & 6 & 4 \\ -4 & -1 & 9 & 8 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 7 \\ 0 \\ 3 \end{bmatrix}$

14. $A = \begin{bmatrix} 1 & 3 & 4 & 0 \\ -3 & -6 & -7 & 2 \\ 3 & 3 & 0 & -4 \\ -5 & -3 & 2 & 9 \end{bmatrix}, b = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 2 \end{bmatrix}$

Lecture 16

Iterative Solutions of Linear Systems

Consistent linear systems are solved in one of two ways by direct calculation (matrix factorization) or by an iterative procedure that generates a sequence of vectors that approach the exact solution. When the coefficient matrix is large and sparse (with a high proportion of zero entries), iterative algorithms can be more rapid than direct methods and can require less computer memory. Also, an iterative process may be stopped as soon as an approximate solution is sufficiently accurate for practical work.

General Framework for an Iterative Solution of *Ax = b*

Throughout the section, ***A*** is an invertible matrix. The goal of an iterative algorithm is to produce a sequence of vectors,

$$x^{(0)}, x^{(1)}, \ldots, x^{(k)}, \ldots$$

that converges to the unique solution say x^* of $\boldsymbol{Ax = b}$, in the sense that the entries in $x^{(k)}$ are as close as desired to the corresponding entries in x^* for all k sufficiently large.

To describe a recursion algorithm that produces $x^{(k+1)}$ from $x^{(k)}$, we write $\boldsymbol{A = M - N}$ for suitable matrices ***M*** and ***N,*** and then we rewrite the equation $\boldsymbol{Ax = b}$ as $\boldsymbol{Mx - Nx = b}$ and

$$Mx = Nx + b$$

If a sequence $\{ x^{(k)} \}$ satisfies

$$\boxed{Mx^{(k+1)} = Nx^{(k)} + b \qquad (k = 0,1,\ldots)} \qquad (1)$$

and if the sequence converges to some vector x^*, then it can be shown that $Ax^* = b$.[The vector on the left in (1) approaches Mx^*, while the vector on the right in (1) approaches $Nx^* + b$. This implies that $Mx^* = Nx^* + b$ and $Ax^* = b$.

For $x^{(k+1)}$ to be uniquely specified in (1), ***M*** must be invertible. Also, ***M*** should be chosen so that $x^{(k+1)}$ is easy to calculate. There are two iterative methods below to illustrate two simple choices for ***M***.

1) Jacobi's Method

This method assumes that the diagonal entries of ***A*** are all nonzero.
Choosing ***M*** as the diagonal matrix formed from the diagonal entries of ***A***. So next
N = M – A,

$$\therefore (1) \Rightarrow Mx^{(k+1)} = (M - A)x^{(k)} + b \qquad (k = 0,1,\ldots)$$

For simplicity, we take the zero vector as $x^{(0)}$ as the initial approximation.

Example 1

Apply Jacobi's method to the system

$$\begin{aligned} 10x_1 + x_2 - x_3 &= 18 \\ x_1 + 15x_2 + x_3 &= -12 \\ -x_1 + x_2 + 20x_3 &= 17 \end{aligned} \qquad (2)$$

Take $\boldsymbol{x}^{(0)} = (0, 0, 0)$ as an initial approximation to the solution, and use six iterations (that is, compute $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(6)}$).

Solution

For some $\boldsymbol{k}$, let $\boldsymbol{x}^{(k)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = (x_1, x_2, x_3)$ and $\boldsymbol{x}^{(k+1)} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = (y_1, y_2, y_3)$

Firstly we will construct $\boldsymbol{M}$ and $\boldsymbol{N}$ from $\boldsymbol{A}$.

Here

$$A = \begin{bmatrix} 10 & 1 & -1 \\ 1 & 15 & 1 \\ -1 & 1 & 20 \end{bmatrix}$$

Its diagonal entries will give

$$M = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 20 \end{bmatrix} \text{ and}$$

$$N = M - A = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 20 \end{bmatrix} - \begin{bmatrix} 10 & 1 & -1 \\ 1 & 15 & 1 \\ -1 & 1 & 20 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

Now the recursion: $Mx^{(k+1)} = (M - A)x^{(k)} + b \qquad (here\ k = 0,1,\ldots 6)$

implies

$$\begin{bmatrix} 10 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 20 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 18 \\ -12 \\ 17 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 10y_1 \\ 15y_2 \\ 20y_3 \end{bmatrix} = \begin{bmatrix} 0x_1 - 1x_2 + 1x_3 \\ -1x_1 + 0x_2 - x_3 \\ 1x_1 - 1x_2 + 0x_3 \end{bmatrix} + \begin{bmatrix} 18 \\ -12 \\ 17 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 10y_1 \\ 15y_2 \\ 20y_3 \end{bmatrix} = \begin{bmatrix} 0x_1 - 1x_2 + 1x_3 + 18 \\ -1x_1 + 0x_2 - x_3 - 12 \\ 1x_1 - 1x_2 + 0x_3 + 17 \end{bmatrix}$$

Comparing the corresponding entries on both sides, we have

$$10y_1 = -x_2 + x_3 + 18$$

$$15y_2 = -x_1 - x_3 - 12$$
$$20y_3 = x_1 - x_2 + 17$$

And

$$y_1 = (-x_2 + x_3 + 18)/10$$
$$y_2 = (-x_1 - x_3 - 12)/15 \qquad (3)$$
$$y_3 = (x_1 - x_2 + 17)/20$$

1st Iteration

For $\boldsymbol{k} = 0$, put $\boldsymbol{x}^{(0)} = (x_1, x_2, x_3) = (0, 0, 0)$ in (3) and compute

$$\boldsymbol{x}^{(1)} = (y_1, y_2, y_3) = (18/10, -12/15, 17/20) = (1.8, -0.8, 0.85)$$

2nd Iteration

For k = 1, put $\boldsymbol{x}^{(1)} = (1.8, -.8, .85)$

$$y_1 = [-(-0.8) + (0.85) + 18]/10 = 1.965$$
$$y_2 = [-(1.8) - (0.85) - 12]/15 = -0.9767$$
$$y_3 = [(1.8) - (-0.8) + 17]/20 = 0.98$$

Thus, $\boldsymbol{x}^{(2)} = (1.965, -.9767, .98)$.

The entries in $\boldsymbol{x}^{(2)}$ are used on the right in (3) to compute the entries in $\boldsymbol{x}^{(3)}$, and so on. Here are the results, with calculations using **MATLAB** and results reported to four decimal places:

$$\begin{array}{ccccccc} x^{(0)} & x^{(1)} & x^{(2)} & x^{(3)} & x^{(4)} & x^{(5)} & x^{(6)} \\ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 1.8 \\ -.8 \\ .85 \end{bmatrix} & \begin{bmatrix} 1.965 \\ -.9767 \\ .98 \end{bmatrix} & \begin{bmatrix} 1.9957 \\ -.9963 \\ .9971 \end{bmatrix} & \begin{bmatrix} 1.9993 \\ -.9995 \\ .9996 \end{bmatrix} & \begin{bmatrix} 1.9999 \\ -.9999 \\ .9999 \end{bmatrix} & \begin{bmatrix} 2.0000 \\ -1.0000 \\ 1.0000 \end{bmatrix} \end{array}$$

If we decide to stop when the entries in $x^{(k)}$ and $x^{(k-1)}$ differ by less than .001, then we need five iterations ($k = 5$).

Alternative Approach

If we express the above system as

$$10x_1 + x_2 - x_3 = 18 \Rightarrow x_1 = \frac{18 - x_2 + x_3}{10}$$

$$x_1 + 15x_2 + x_3 = -12 \Rightarrow x_2 = \frac{-12 - x_1 - x_3}{15}$$

$$-x_1 + x_2 + 20x_3 = 17 \Rightarrow x_3 = \frac{17 + x_1 - x_2}{20}$$

$\therefore$ the equivalent system is

$$x_1 = \frac{18 - x_2 + x_3}{10}$$
$$x_2 = \frac{-12 - x_1 - x_3}{15}$$
$$x_3 = \frac{17 + x_1 - x_2}{20}$$

Now put $(x_1,x_2,x_3)=(0,0,0)= x^{(0)}$ in the RHS to have

$x_1=(18-0+0)/10 = 1.80$

$x_2= (-12-0-0)/15 = -0.80$

$x_3 = (17+0-0)/20 = 0.85$

Which gives $x^{(1)} = (1.80,-0.80,0.85)$ ----put this again on RHS of the equivalent system to get

$x_1=(18+0.80+0.85)/10 =1.965$

$x_2= (-12-1.80-0.85)/15 = -0.9767$

$x_3= (17+1.80+0.80)/20 = 0.98$

So in the similar fashion, we can get the next approximate solutions: $x^{(3)}, x^{(4)}, x^{(5)}$ and $x^{(6)}$ Next example will be solved by following this approach.

Example 2

Use Jacobi iteration to approximate the solution of the system

$$20\,x_1 + x_2 - x_3 = 17$$
$$x_1 - 10x_2 + x_3 = 13$$
$$-x_1 + x_2 + 10x_3 = 18$$

Stop the process when the entries in two successive iterations are the same when rounded to four decimal places.

Solution

As required for Jacobi iteration, we begin by solving the first equation for x_1, the second for x_2, and the third for x_3. This yields

$$x_1 = \frac{17}{20} - \frac{1}{20}x_2 + \frac{1}{20}x_3 = 0.85 - 0.05x_2 + 0.05x_3$$
$$x_2 = -\frac{13}{10} + \frac{1}{10}x_1 + \frac{1}{10}x_3 = -1.3 + 0.1x_1 + 0.1x_3 \qquad (4)$$
$$x_3 = \frac{18}{10} + \frac{1}{10}x_1 - \frac{1}{10}x_2 = 1.8 + 0.1x_1 - 0.1x_2$$

which we can write in matrix form as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & -0.05 & 0.05 \\ 0.1 & 0 & 0.1 \\ 0.1 & -0.1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0.85 \\ -1.3 \\ 1.8 \end{bmatrix} \qquad (5)$$

Since we have no special information about the solution, we will take the initial approximation to be $x_1 = x_2 = x_3 = 0$. To obtain the first iterate, we substitute these values into the right side of (5). This yields

$$y_1 = \begin{bmatrix} 0.85 \\ -1.3 \\ 1.8 \end{bmatrix}$$

To obtain the second iterate, we substitute the entries of y_1 into the right side of (5). This yields

$$y_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & -0.05 & 0.05 \\ 0.1 & 0 & 0.1 \\ 0.1 & -0.1 & 0 \end{bmatrix} \begin{bmatrix} 0.85 \\ -1.3 \\ 1.8 \end{bmatrix} + \begin{bmatrix} 0.85 \\ -1.3 \\ 1.8 \end{bmatrix} = \begin{bmatrix} 1.005 \\ -1.035 \\ 2.015 \end{bmatrix}$$

Repeating this process until two successive iterations match to four decimal places yields the results in the following table:

	y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7
x_1	0	0.8500	1.0050	1.0025	1.0001	1.0000	1.0000	1.0000
x_2	0	-1.3000	-1.0350	-0.9980	-0.9994	-1.0000	-1.0000	-1.0000
x_3	0	1.8000	2.0150	2.0040	2.0000	1.9999	2.0000	2.0000

The Gauss-Seidel Method

This method uses the recursion (1) with M the lower triangular part of A. That is, M has the same entries as A on the diagonal and below, and M has zeros above the diagonal. See Fig. 1. As in Jacobi's method, the diagonal entries of A must be nonzero in order for M to be invertible.

$$A = \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} \qquad M = \begin{bmatrix} * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 \\ * & * & * & 0 & 0 \\ * & * & * & * & 0 \\ * & * & * & * & * \end{bmatrix}$$

Figure 01:The Lower Triangular Part of A

Example 3

Apply the Gauss – Seidel method to the system in Example 1 with $x^{(0)} = (0,0,0)$ and six iterations.

$$\begin{aligned} 10x_1 + x_2 - x_3 &= 18 \\ x_1 + 15x_2 + x_3 &= -12 \\ -x_1 + x_2 + 20x_3 &= 17 \end{aligned} \qquad (6)$$

Solution

For some k, let $x^{(k)} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = (x_1, x_2, x_3)$ and $x^{(k+1)} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = (y_1, y_2, y_3)$

Again, firstly we construct matrices M and N from the coefficient matrix A.

Here $A = \begin{bmatrix} 10 & 1 & -1 \\ 1 & 15 & 1 \\ -1 & 1 & 20 \end{bmatrix}$

Since matrix M is constructed by

1) taking the values along the diagonal and below the diagonal of coefficient matrix A.
2) putting the zeros above the diagonal at upper trianular position.

So

$$M = \begin{bmatrix} 10 & 0 & 0 \\ 1 & 15 & 0 \\ -1 & 1 & 20 \end{bmatrix}$$

Now,

$$N = M - A = \begin{bmatrix} 10 & 0 & 0 \\ 1 & 15 & 0 \\ -1 & 1 & 20 \end{bmatrix} - \begin{bmatrix} 10 & 1 & -1 \\ 1 & 15 & 1 \\ -1 & 1 & 20 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Now the recursion: $Mx^{(k+1)} = (M - A)x^{(k)} + b \qquad (here\ k = 0,1,...6)$

implies

$$\begin{bmatrix} 10 & 0 & 0 \\ 1 & 15 & 0 \\ -1 & 1 & 20 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 18 \\ -12 \\ 17 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 10y_1 & 0 & 0 \\ y_1 & 15y_2 & 0 \\ -y_1 & y_2 & 20y_3 \end{bmatrix} = \begin{bmatrix} -x_2 + x_3 \\ -x_3 \\ 0 \end{bmatrix} + \begin{bmatrix} 18 \\ -12 \\ 17 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 10y_1 & 0 & 0 \\ y_1 & 15y_2 & 0 \\ -y_1 & y_2 & 20y_3 \end{bmatrix} = \begin{bmatrix} -x_2 + x_3 + 18 \\ -x_3 - 12 \\ 0 + 17 \end{bmatrix}$$

Comparing the corresponding entries on both sides, we have

$$10y_1 = -x_2 + x_3 + 18$$
$$y_1 + 15y_2 = -x_3 - 12$$
$$-y_1 + y_2 + 20y_3 = 17$$

This further implies as

$$\left.\begin{array}{l} 10y_1 = -x_1 - x_3 + 18 \Rightarrow y_1 = (-x_2 + x_3 + 18)/10 \\ y_1 + 15y_2 = -x_2 - 12 \Rightarrow y_2 = (-y_1 - x_2 - 12)/15 \\ -y_1 + y_2 + 20y_3 = 17 \Rightarrow y_3 = (y_1 - y_2 + 17)/20 \end{array}\right\} \text{---------- (7)}$$

Another way to view (7) is to solve each equation in (6) for x_1, x_2, x_3, respectively and regard the highlighted x's as the values:

$$x_1 = (-x_2 + x_3 + 18)/10$$
$$x_2 = (-x_1 - x_3 - 12)/15$$
$$x_3 = (x_1 - x_2 + 17)/20 \qquad (8)$$

Use the first equation to calculate the new x_1 [called y_1 in (7)] from x_2 and x_3. Then, use this new x_1 along with x_3 in the second equation to compute the new x_2. Finally, in the third equation, use the new values for x_1 and x_2 to compute x_3. In this way, the latest information about the variables is used to compute new values. [A computer program would use statements corresponding to the equations in (8).]

From $x^{(0)} = (0, 0, 0)$, we obtain

$$x_1 = [-(0) + (0) + 18]/10 = 1.8$$
$$x_2 = [-(1.8) - (0) - 12]/15 = -.92$$
$$x_3 = [+(1.8) - (-.92) + 17]/20 = .986$$

Thus, $x^{(1)} = (1.8, -.92, .986)$. The entries in $x^{(1)}$ are used in (8) to produce $x^{(2)}$ and so on. Here are the **MATLAB** calculations reported to four decimal places:

$$\begin{array}{ccccccc} x^{(0)} & x^{(1)} & x^{(2)} & x^{(3)} & x^{(4)} & x^{(5)} & x^{(6)} \\ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 1.8 \\ -.92 \\ .986 \end{bmatrix} & \begin{bmatrix} 1.9906 \\ -.9984 \\ .9995 \end{bmatrix} & \begin{bmatrix} 1.9998 \\ -.9999 \\ 1.0000 \end{bmatrix} & \begin{bmatrix} 2.0000 \\ -1.0000 \\ 1.0000 \end{bmatrix} & \begin{bmatrix} 2.0000 \\ -1.0000 \\ 1.0000 \end{bmatrix} & \begin{bmatrix} 2.0000 \\ -1.0000 \\ 1.0000 \end{bmatrix} \end{array}$$

Observe that when k is 4, the entries in $x^{(4)}$ and $x^{(k-1)}$ differ by less than .001. The values in $x^{(6)}$ in this case happen to be accurate to eight decimal places.

Alternative Approach

If we express the above system as

$10x_1 + x_2 - x_3 = 18 \Rightarrow x_1 = (-x_2 + x_3 + 18)/10$ ------ (a)

$x_1 + 15x_2 + x_3 = -12 \Rightarrow x_2 = (-x_1 - x_3 - 12)/15$ ----- (b)

$-x_1 + x_2 + 20x_3 = 17 \Rightarrow x_3 = (x_1 - x_2 + 17)/20$ ------(c)

Ist Iteration

Put $x_2 = x_3 = 0$ in (a)

$x_1 = 18/10 = 1.80$

Put $x_1 = 1.80$ and $x_3 = 0$ in (b)

$x_2 = (-1.80 - 0 - 12)/15 = -0.92$

Put $x_1 = 1.80$, $x_2 = -0.92$ in (c)

$x_3 = (1.80 + 0.92 + 17)/20 = 0.9863$

So, $\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} 1.8 \\ -.92 \\ .986 \end{bmatrix} = x^{(1)}$

2nd iteration

Put $x_2 = -0.92$, x3=0.9863 in (a)

$x_1 = (0.92 + 0.9863 + 18)/10 = 1.9906$

Put $x_1 = 1.9906$(from 2nd iteration) and $x_3 = 0.9863$(from 1st iteration) in (b)

$x_2 = (-1.9906 - 0.9863 - 12)/15 = -0.9984$

Put $x_1 = 1.9906$, $x2 = -0.9984$(both from 2nd iteration) in (c)

$x_3 = (1.9906 + 0.9984 + 17)/20 = 0.9995$

So, $\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} 1.9906 \\ -.9984 \\ .9995 \end{bmatrix} = x^{(2)}$

So in the similar fashion, we can get the next approximate solutions: $x^{(3)}, x^{(4)}, x^{(5)}$ and $x^{(6)}$
Next example will be solved by following this approach.

Example 4

Use Gauss-Seidel to approximate the solution of the linear system in example 2 to four decimal places.

Solution

As before, we will take $x_1 = x_2 = x_3 = 0$ as the initial approximation. First, we will substitute $x_2 = 0$ and $x_3 = 0$ into the right side of the first equation of (4) to obtain the new x_1. Then, we will substitute $x_3 = 0$ and the new x_1 into the right side of the second equation to obtain the new x_2, and finally, we will substitute the new x_1 and new x_2 into the right side of the third equation to obtain the new x_3. The computations are as follows:

$$x_1 = 0.85 - (0.05)(0) + (0.05)(0) = 0.85$$
$$x_2 = -1.3 + (0.1)(0.85) + (0.1)(0) = -1.215$$
$$x_3 = 1.8 + (0.1)(0.85) - (0.1)(-1.215) = 2.0065$$

Thus, the first Gauss-Seidel iterate is

$$y_1 = \begin{bmatrix} 0.8500 \\ -1.2150 \\ 2.0065 \end{bmatrix}$$

Similarly, the computations for second iterate are

$$x_1 = 0.85 - (0.05)(-1.215) + (0.05)(2.0065) = 1.011075$$
$$x_2 = -1.3 + (0.1)(1.011075) + (0.1)(2.0065) = -0.9982425$$
$$x_3 = 1.8 + (0.1)(1.011075) - (0.1)(-0.9982425) = 2.00093175$$

Thus, the second Gauss-Seidel iterate to four decimal places is

$$y_2 \approx \begin{bmatrix} 1.0111 \\ -0.9982 \\ 2.0009 \end{bmatrix}$$

The following table shows the first four Gauss-Seidel iterates to four decimal places. Comparing both tables, we see that the Gauss-Seidel method produced the solution to four decimal places in four iterations whereas, the Jacobi method required six.

	y_0	y_1	y_2	y_3	y_4
x_1	0	0.8500	1.0111	1.0000	1.0000
x_2	0	-1.2150	-0.9982	-0.9999	-1.0000
x_3	0	2.0065	2.0009	2.0000	2.0000

Comparison of Jacobi's and Gauss-Seidel method

There exist examples where Jacobi's method is faster than the Gauss-Seidel method, but usually a Gauss-Seidel sequence converges faster (means to say iterative solution approaches to the unique solution), as in Example 2. (If parallel processing is available, Jacobi might be faster because the entries in $x^{(k)}$ can be computed simultaneously.) There are also examples where one or both methods fail to produce a convergent sequence, and other examples where a sequence is convergent, but converges too slowly for practical use.

Condition for the Convergence of both Iterative Mthods

Fortunately, there is a simple condition that guarantees (but is not essential for) the convergence of both Jacobi and Gauss-Seidel sequences. This condition is often satisfied, for instance, in large-scale systems that can occur during numerical solutions of partial differential equations (such as Laplace's equation for steady-state heat flow).

*An $n \times n$ matrix A is said to be **strictly diagonally dominant** if the absolute value of each diagonal entry exceeds the sum of the absolute values of the other entries in the same row.*

In this case it can be shown that A is invertible and that both the Jacobi and Gauss-Seidel sequences converge to the unique solution of $Ax = b$, for any initial $x^{(0)}$. (The speed of the convergence depends on how much the diagonal entries dominate the corresponding row sums.)

The coefficient matrices in Examples 1 and 2 are strictly diagonally dominant, but the following matrix is not. Examine each row:

$$\begin{bmatrix} -6 & 2 & -3 \\ 1 & 4 & -2 \\ 3 & -5 & 8 \end{bmatrix} \qquad \begin{aligned} |-6| &> |2| + |-3| \\ |4| &> |1| + |-2| \\ |8| &= |3| + |-5| \end{aligned}$$

The problem lies in the third row, because $8 \left| \right|$ is not *larger* than the sum of the magnitudes of the other entries.

Note

The practice problem below suggests a TRICK(rearrangement of the system of equations) that sometimes works when a system is not strictly diagonally dominant.

Example 5

Show that the Gauss-Seidel method will produce a sequence converging to the solution of the following systems, provided the equations are arranged properly:

$$\begin{aligned} x_1 - 3x_2 + x_3 &= -2 - \\ 6x_1 + 4x_2 + 11x_3 &= 1 \\ 5x_1 - 2x_2 - 2x_3 &= 9 \end{aligned}$$

Solution

The system is not strictly diagonally dominant, as for the 1st row

$$|\textit{coefficient of } x_1| < |\textit{coefficient of } x_2| + |\textit{coefficient of } x_3|$$

$$\textit{or} \quad |1| < |-3| + |1|$$

so neither Jacobi nor Gauss- Seidel is guaranteed to work. In fact, both iterative methods produce sequences that fail to converge, even though the system has the unique solution $x_1 = 3, x_2 = 2, x_3 = 1$. However, the equations can be rearranged as

$$\begin{aligned} 5x_1 - 2x_2 - 2x_3 &= 9 \\ x_1 - 3x_2 + x_3 &= -2 \\ -6x_1 + 4x_2 + 11x_3 &= 1 \end{aligned}$$

So,

for 1st equation (row);

$$|\textit{coefficient of } x_1| > |\textit{coefficient of } x_2| + |\textit{coefficient of } x_3|$$

$$\textit{or} \quad |5| > |-2| + |-2|$$

for 2nd equation(row);

$$|\textit{coefficient of } x_2| > |\textit{coefficient of } x_1| + |\textit{coefficient of } x_3|$$

$$\textit{or} \quad |-3| > |1| + |1|$$

for 3rd equation(row);

$$|\textit{coefficient of } x_3| > |\textit{coefficient of } x_1| + |\textit{coefficient of } x_2|$$

$$\textit{or} \quad |11| > |-6| + |4|$$

Now the coefficient matrix is strictly diagonally dominant, so we know Gauss-Seidel works with any initial vector. In fact, if $x^{(0)} = 0$, then $x^{(8)} = (2.9987, 1.9992, .9996)$.

Exercises

Solve the system in exercise 1 to 3 using Jacobi's method, with $x^{(0)} = 0$ and three iterations. Repeat the iterations until two successive approximations agree within a tolerance of .001 in each entry.

1. $$\begin{aligned} 4x_1 + x_2 &= 7 \\ -x_1 + 5x_2 &= -7 \end{aligned}$$

2. $$\begin{aligned} 10x_1 - x_2 &= 25 \\ x_1 + 8x_2 &= 43 \end{aligned}$$

3. $$\begin{aligned} 3x_1 + x_2 &= 11 \\ -x_1 - 5x_2 + 2x_3 &= 15 \\ 3x_2 + 7x_3 &= 17 \end{aligned}$$

4. $$\begin{aligned} 50x_1 - x_2 &= 149 \\ x_1 - 100x_2 + 2x_3 &= -101 \\ 2x_2 + 50x_3 &= -98 \end{aligned}$$

In exercises 5 to 8, use the Gauss Seidel method, with $x^{(0)} = 0$ and two iterations. Compare the number of iterations needed by Gauss Seidel and Jacobi to make two successive approximations agree within a tolerance of .001.

5. The system in exercise 1

6. The system in exercise 2

7. The system in exercise 3

8. The system in exercise 4

Determine which of the matrices in exercises 9 and 10 are strictly diagonally dominant.

9. (a) $\begin{bmatrix} 5 & 4 \\ 4 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 9 & -5 & 2 \\ 5 & -8 & -1 \\ -2 & 1 & 4 \end{bmatrix}$

10. (a) $\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 5 & 3 & 1 \\ 3 & 6 & -4 \\ 1 & -4 & 7 \end{bmatrix}$

Show that the Gauss Seidel method will produce a sequence converging to the solution of the following system, provided the equations are arranged properly:

11. $$\begin{aligned} x_1 - 3x_2 + x_3 &= -2 \\ -6x_1 + 4x_2 + 11x_3 &= 1 \\ 5x_1 - 2x_2 - 2x_3 &= 9 \end{aligned}$$

12. $$\begin{aligned} -x_1 + 4x_2 - x_3 &= 3 \\ 4x_1 - x_2 &= 10 \\ -x_2 + 4x_3 &= 6 \end{aligned}$$

Lecture 17

Introduction to Determinant

In algebra, the **determinant** is a special number associated with any square matrix. As we have studied in earlier classes, that the determinant of 2 x 2 matrix is defined as the product of the entries on the main diagonal minus the product of the entries off the main diagonal. The determinant of a matrix A is denoted by det (***A***) or |A|

For example: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Then det (***A***) = *ad-bc*.

or |A| = ad – bc

Example Find the determinant of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1\times 4 - 2\times 3 = 4 - 6 = -2$$

To extend the definition of the det(***A***) to matrices of higher order, we will use subscripted entries for ***A***.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$\det(\boldsymbol{A}) = \begin{vmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{vmatrix} = a_{11}b_{22} - a_{12}b_{21}$$

This is called a 2x2 determinant.

The determinant of a 3x3 matrix is also called a 3x3 determinant is defined by the following formula.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \qquad (1)$$

For finding the determinant of the 3x3 matrix, we look at the following diagram:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{matrix}$$

We write 1st and 2nd columns again beside the determinant. The first arrow goes from a_{11} to a_{33}, which gives us product: $a_{11}a_{22}a_{33}$. The second arrow goes from a_{12} to a_{31}, which gives us product: $a_{12}a_{23}a_{31}$. The third arrow goes from a_{13} to a_{32}, which gives us the product: $a_{13}a_{21}a_{32}$. These values are taken with positive signs.

The same method is used for the next three arrows that go from right to left downwards, but these products are taken as negative signs.

$$= a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$$

Example 2 Find the determinant of the matrix $A=\begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{bmatrix}$

$$detA=\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix}$$

$$=\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix}\begin{matrix} 1 & 2 \\ -4 & 5 \\ 7 & -8 \end{matrix}$$

$$= 1\times5\times9 \;+\; 2\times6\times7 \;+\; 3\times(-4)\times(-8) - 3\times5\times7 - 1\times6\times(-8) - 2\times(-4)\times9$$

$$= 45+84+96-105+48+72$$

$$= 240$$

We saw earlier that a 2×2 matrix is invertible if and only if its determinant is nonzero. In simple words, a matrix has its inverse if its determinant is nonzero. To extend this useful fact to larger matrices, we need a definition for the determinant of the $n\times n$ matrix. We can discover the definition for the 3×3 case by watching what happens when an invertible 3×3 matrix $\boldsymbol{A}$ is row reduced.

Gauss' algorithm for evaluation of determinants

1) Firstly, we apply it for 2×2 matrix say

$$A=\begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$$

$R' \to R \;-2R$ (Multiplying 1st row by 2 and then subtracting from 2nd row)

$$\begin{bmatrix} 2 & 3 \\ 4-2(2) & 3-2(3) \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & -3 \end{bmatrix}$$

Now the determinant of this upper triangular matrix is the product of its entries on main diagonal that is

$Det(A) = 2(-3) - 0 \times 3 \;\; = -6 - 0 \;\; = -6$

2) For 3×3 matrix say

$$B = \begin{bmatrix} -2 & 2 & -3 \\ -1 & 1 & 3 \\ 2 & 0 & -1 \end{bmatrix}$$

By $R_1' \to R_{21}$ (Interchanging of 1st and 2nd rows)

$$\begin{bmatrix} -1 & 1 & 3 \\ -2 & 2 & -3 \\ 2 & 0 & -1 \end{bmatrix}$$

$R_2' \to R_2 - 2R_1$ (Multiplying 1st row by '-2' and then adding in the 2nd row)

$R_3' \to R_3 + 2R_1$ (Multiplying 1st row by '2' and then adding in the 3rd row)

$$\begin{bmatrix} -1 & 1 & 3 \\ 0 & 0 & -9 \\ 0 & 2 & 5 \end{bmatrix}$$

By $R_2' \to R_{32}$ (Interchanging of 2nd and 3rd rows)

$$\begin{bmatrix} -1 & 1 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -9 \end{bmatrix}$$

Now the determinant of this upper triangular matrix is the product of its entries on main diagonal and that is

$Det(B) = (-1) \cdot 2 \cdot (-9) = 18$

So in general,

For a 1×1 matrix

say, $A = [a_{ij}]$ - we define $\det A = a_{11}$.

For 2×2 matrix

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

By $R_2' \to R_2 - \left(\frac{a_{21}}{a_{11}}\right) R_1$ provided that $a_{11} \neq 0$

$$\begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{22} - \frac{a_{21}}{a_{11}} a_{12} \end{bmatrix}$$

$\therefore \Delta = \det A =$ product of the diagonal entries

$$= a_{11}\left(a_{22} - \frac{a_{21}}{a_{11}}a_{12}\right) = a_{11}a_{22} - a_{12}a_{21}$$

For 3×3 matrix say

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

By $R_2' \to R_2 - \left(\frac{a_{21}}{a_{11}}\right)R_1$, $R_3' \to R_3 - \left(\frac{a_{31}}{a_{11}}\right)R_1$ provided that $a_{11} \neq 0$

$$\sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & \frac{a_{23}a_{11} - a_{13}a_{21}}{a_{11}} \\ 0 & \frac{a_{32}a_{11} - a_{12}a_{31}}{a_{11}} & \frac{a_{11}a_{33} - a_{13}a_{31}}{a_{11}} \end{bmatrix}$$

By $R_3' \to R_3 - \left(\frac{\frac{a_{32}a_{11} - a_{12}a_{31}}{a_{11}}}{\frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}}}\right)R_2$ provided that $\frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} \neq 0$

$$\sim \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & \frac{a_{23}a_{11} - a_{13}a_{21}}{a_{11}} \\ 0 & 0 & \frac{a_{11}a_{33} - a_{13}a_{31}}{a_{11}} - \left(\frac{a_{23}a_{11} - a_{13}a_{21}}{a_{11}}\right)\left(\frac{\frac{a_{32}a_{11} - a_{12}a_{31}}{a_{11}}}{\frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}}}\right) \end{bmatrix} \quad \because a_{11} \neq 0$$

Which is in echelon form.Now,

$\Delta = \det A =$ product of the diagonal entries

$$= a_{11}\left|\frac{a_{22}a_{11}-a_{12}a_{21}}{a_{11}}\right|\left|\frac{a_{11}a_{33}-a_{13}a_{31}}{a_{11}}\right| - \left|\frac{a_{23}a_{11}-a_{13}a_{21}}{a_{11}}\right|\left|\frac{a_{32}a_{11}-a_{12}a_{31}}{a_{22}a_{11}-a_{12}a_{21}}\right|$$

$$= (a_{22}a_{11}-a_{12}a_{21})\left|\frac{a_{11}a_{33}-a_{13}a_{31}}{a_{11}}\right| - (a_{22}a_{11}-a_{12}a_{21})\left|\frac{a_{23}a_{11}-a_{13}a_{21}}{a_{11}}\right|\left|\frac{a_{32}a_{11}-a_{12}a_{31}}{a_{22}a_{11}-a_{12}a_{21}}\right|$$

$$= \frac{1}{a_{11}}\{(a_{22}a_{11}-a_{12}a_{21})(a_{11}a_{33}-a_{13}a_{31}) - (a_{23}a_{11}-a_{13}a_{21})(a_{32}a_{11}-a_{12}a_{31})\}$$

$$= \frac{1}{a_{11}}\{a^2_{11}a_{22}a_{33} - a_{11}a_{22}a_{13}a_{31} - a_{11}a_{12}a_{21}a_{33} + a_{12}a_{21}a_{13}a_{31} - a_{23}a_{11}^2a_{32} + a_{23}a_{11}a_{12}a_{31} + a_{13}a_{21}a_{32}a_{11} - a_{12}a_{21}a_{13}a_{31}\}$$

$$= \frac{1}{a_{11}}\{a^2_{11}a_{22}a_{33} - a_{11}a_{22}a_{13}a_{31} - a_{11}a_{12}a_{21}a_{33} - a_{23}a_{11}^2a_{32} + a_{23}a_{11}a_{12}a_{31} + a_{13}a_{21}a_{32}a_{11}\}$$

$$= \frac{a_{11}}{a_{11}}\{a_{11}a_{22}a_{33} - a_{22}a_{13}a_{31} - a_{12}a_{21}a_{33} - a_{23}a_{11}a_{32} + a_{23}a_{12}a_{31} + a_{13}a_{21}a_{32}\}$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31}$$

□

Since ***A*** is invertible, Δ must be nonzero. *The converse is true as well*.

To generalize the definition of the determinant to larger matrices, we will use 2×2 determinants to rewrite the 3×3 determinant Δ described above. Since the terms in Δ can be grouped as:

$$\Delta = (a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32}) - (a_{12}a_{23}a_{31} - a_{12}a_{21}a_{33}) + (a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31})$$

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}),$$

$$\Delta = a_{11}\cdot\det\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{bmatrix} - a_{12}\cdot\det\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}\end{bmatrix} + a_{13}\cdot\det\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32}\end{bmatrix}$$

$$\Delta = a_{11}\cdot\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}\end{vmatrix} + a_{13}\cdot\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32}\end{vmatrix}$$

For brevity, we write $\Delta = a_{11}\cdot\det A_{11} - a_{12}\cdot\det A_{12} + a_{13}\cdot\det A_{13}$ (3)

$$\det(A_{11}) = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33}\end{vmatrix}, \quad \det(A_{12}) = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33}\end{vmatrix} \quad \text{and} \quad \det(A_{13}) = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32}\end{vmatrix}$$

where

A₁₁ is obtained from ***A*** by deleting the first row and first column.
A₁₂ is obtained from ***A*** by deleting the first row and second column.
A₁₃ is obtained from ***A*** by deleting the first row and third column.
So in general, for any square matrix ***A***, let ***Aᵢⱼ*** denote the sub-matrix formed by deleting the *ith* row and *jth* column of ***A***.

Let's understand it with the help of an example.

Example3

Find the determinant of the matrix $A=\begin{bmatrix}1 & 4 & 3\\ 5 & 2 & 4\\ 3 & 6 & 3\end{bmatrix}$

Solution Given $A=\begin{bmatrix}1 & 4 & 3\\ 5 & 2 & 4\\ 3 & 6 & 3\end{bmatrix}$

$$|A|=\begin{vmatrix}1 & 4 & 3\\ 5 & 2 & 4\\ 3 & 6 & 3\end{vmatrix}$$

$$=1\begin{vmatrix}2 & 4\\ 6 & 3\end{vmatrix}-4\begin{vmatrix}5 & 4\\ 3 & 3\end{vmatrix}+3\begin{vmatrix}5 & 2\\ 3 & 6\end{vmatrix}$$

$$=1(2\times 3-4\times 6)-4(5\times 3-4\times 3)+3(5\times 6-2\times 3)$$

$$=1(6-24)-4(15-12)+3(30-6)$$

$$=1(-18)-4(3)+3(24)$$

$$=-18-12+72$$

$$=42$$

For instance, if $A=\begin{bmatrix}1 & -2 & 5 & 0\\ 2 & 0 & 4 & -1\\ 3 & 1 & 0 & 7\\ 0 & 4 & -2 & 0\end{bmatrix}$

then A_{32} is obtained by crossing out row 3 and column 2,

$$\begin{bmatrix}1 & -2 & 5 & 0\\ 2 & 0 & 4 & -1\\ \hline 3 & 1 & 0 & 7\\ 0 & 4 & -2 & 0\end{bmatrix}$$

so that $A_{32}=\begin{bmatrix}1 & 5 & 0\\ 2 & 4 & -1\\ 0 & -2 & 0\end{bmatrix}$

We can now give a recursive definition of a determinant.

When $n = 3$, det $\boldsymbol{A}$ is defined using determinants of the 2×2 submatrices A_{1j}.

When $n = 4$, det $\boldsymbol{A}$ uses determinants of the 3×3 submatrices A_{1j}

In general, an nxn determinant is defined by determinants of $(n-1)\times(n-1)$ sub matrices.

Definition

For $n \geq 2$, the **determinant** of $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \times (\det A_{1j})$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \cdots, a_{1n}$ are from the first row of ***A***.

Here for a_{ij},

$i = 1, 2, 3, \square, n \quad (1 \leq i \leq n)$

$j = 1, 2, 3, \square, n \quad (1 \leq j \leq n)$

In symbols, $\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + ... + (-1)^{1+n} a_{1n} \det A_{1n} = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A_{1j}$

Example 4

Compute the determinant of $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$

Solution

Here A is $n \times n = 3 \times 3$ matrix such that

$i = 1, 2, 3$

$j = 1, 2, 3$

$\because \det(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A_{1j}$ and here $j = 1, 2, 3$

$\therefore \det(A) = \sum_{j=1}^{3} (-1)^{1+j} a_{1j} \det A_{1j} = (-1)^{1+1} a_{11} \det A_{11} + (-1)^{1+2} a_{12} \det A_{12} + (-1)^{1+3} a_{13} \det A_{13}$

$= a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$

$$\det A = 1.\det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} - 5.\det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + 0.\det \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix}$$

$$\det A = 1.\begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5.\begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0.\begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix}$$

$= 1\,[4(0) - (-1)(-2)] - 5\,[2(0) - 0(-1)] + 0[2(-2) - 4(0)]$

$= 1(0 - 2) - 5(0 - 0) + 0(-4 - 0) = -2$

Minor of an element

If ***A*** is a square matrix, then the **Minor** of entry a_{ij} (called the ijth minor of ***A***) is denoted by $\boldsymbol{M_{ij}}$ and is defined to be the determinant of the sub matrix that remains when the *ith* row and *jth* column of ***A*** are deleted.

In the above example, Minors are as follows:

$$M_{11}=\begin{vmatrix}4 & -1\\ -2 & 0\end{vmatrix},\qquad M_{12}=\begin{vmatrix}2 & -1\\ 0 & 0\end{vmatrix},\qquad M_{13}=\begin{vmatrix}2 & 4\\ 0 & -2\end{vmatrix}$$

Cofactor of an element

The number $C_{ij}=(-1)^{i+j}M_{ij}$ is called the cofactor of entry a_{ij}(or the *ijth* cofactor of A). When the + or – sign is attached to the Minor, then Minor becomes a cofactor.

In the above example, following are the Cofactors:

$$C_{11}=(-1)^{1+1}M_{11},\qquad C_{12}=(-1)^{1+2}M_{12},\qquad C_{13}=(-1)^{1+3}M_{13}$$

$$C_{11}=(-1)^{1+1}\begin{vmatrix}4 & -1\\ -2 & 0\end{vmatrix},\qquad C_{12}=(-1)^{1+2}\begin{vmatrix}2 & -1\\ 0 & 0\end{vmatrix},\qquad C_{13}=(-1)^{1+3}\begin{vmatrix}2 & 4\\ 0 & -2\end{vmatrix}$$

Example 5 Find the minor and the cofactor of the matrix $A=\begin{bmatrix}3 & 1 & -4\\ 2 & 5 & 6\\ 1 & 4 & 8\end{bmatrix}$

Solution Here $A=\begin{bmatrix}3 & 1 & -4\\ 2 & 5 & 6\\ 1 & 4 & 8\end{bmatrix}$

The minor of entry a_{11} is

$$M_{11}=\begin{vmatrix}3 & 1 & -4\\ 2 & 5 & 6\\ 1 & 4 & 8\end{vmatrix}=\begin{vmatrix}5 & 6\\ 4 & 8\end{vmatrix}=5\times 8-6\times 4=40-24=16$$

and the corresponding cofactor is

$$C_{11}=(-1)^{1+1}M_{11}=M_{11}=16$$

The minor of entry a_{32} is

$$M_{32}=\begin{vmatrix}3 & 1 & -4\\ 2 & 5 & 6\\ 1 & 4 & 8\end{vmatrix}=\begin{vmatrix}3 & -4\\ 2 & 6\end{vmatrix}=26$$

and the corresponding cofactor is

$$C_{32}=(-1)^{3+2}M_{32}=-M_{32}=-\begin{vmatrix}3 & -4\\ 2 & 6\end{vmatrix}=-26$$

Alternate Definition

Given $A = [a_{ij}]$, the (i, j)-cofactor of ***A*** is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij} \qquad (4)$$

Then $\det A = a_{11}C_{11} + a_{12}C_{12} + \ldots\ldots + a_{1n}C_{1n}$

This formula is called the *cofactor expansion across the first row* of ***A***.

Example 6 Expand a 3x3 determinant using cofactor concept $A = \begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix}$

Solution Using cofactor expansion along the first column;

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix} = (1)(-1)^{1+1}\begin{vmatrix} 5 & 6 \\ -8 & 9 \end{vmatrix} + (-4)(-1)^{2+1}\begin{vmatrix} 2 & 3 \\ -8 & 9 \end{vmatrix} + (7)(-1)^{3+1}\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

Nowif we compareit with the formula (4),

$$= 1C_{11} + (-4)C_{21} + 7C_{31}$$

$$= (1)(-1)^2\begin{vmatrix} 5 & 6 \\ -8 & 9 \end{vmatrix} + (-4)(-1)^3\begin{vmatrix} 2 & 3 \\ -8 & 9 \end{vmatrix} + (7)(-1)^4\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

$$= (1)(1)\begin{vmatrix} 5 & 6 \\ -8 & 9 \end{vmatrix} + (-4)(-1)\begin{vmatrix} 2 & 3 \\ -8 & 9 \end{vmatrix} + (7)(1)\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

$$= 1\begin{vmatrix} 5 & 6 \\ -8 & 9 \end{vmatrix} + 4\begin{vmatrix} 2 & 3 \\ -8 & 9 \end{vmatrix} + 7\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

$$= 1(45 - (-48)) + 4(18 - (-24)) + 7(12 - 15)$$

$$= 1(45 + 48) + 4(18 + 24) + 7(12 - 15)$$

$$= (1)(93) + (4)(42) + (7)(-3) = 240$$

Using cofactor expansion along the second column,

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix} = (2)(-1)^{1+2}\begin{vmatrix} -4 & 6 \\ 7 & 9 \end{vmatrix} + (5)(-1)^{2+2}\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + (-8)(-1)^{3+2}\begin{vmatrix} 1 & 3 \\ -4 & 6 \end{vmatrix}$$

$$= (2)(-1)^{3}\begin{vmatrix} -4 & 6 \\ 7 & 9 \end{vmatrix} + (5)(-1)^{4}\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + (-8)(-1)^{5}\begin{vmatrix} 1 & 3 \\ -4 & 6 \end{vmatrix}$$

$$= (2)(-1)\begin{vmatrix} -4 & 6 \\ 7 & 9 \end{vmatrix} + (5)(1)\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + (-8)(-1)\begin{vmatrix} 1 & 3 \\ -4 & 6 \end{vmatrix}$$

$$= -2\begin{vmatrix} -4 & 6 \\ 7 & 9 \end{vmatrix} + 5\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + 8\begin{vmatrix} 1 & 3 \\ -4 & 6 \end{vmatrix}$$

$$= -2(-36-42) + 5(9-21) + 8(6-(-12))$$

$$= (-2)(-78) + (5)(-12) + (8)(18) = 240$$

Theorem 1 The determinant of an $n \times n$ matrix $\boldsymbol{A}$ can be computed by a cofactor expansion across any row or down any column. The expansion across the ith row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

The cofactor expansion down the jth column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

The plus or minus sign in the (i, j)-cofactor depends on the position of a_{ij} in the matrix, regardless of the sign of a_{ij} itself. The factor $(-1)^{i+j}$ determines the following checkerboard pattern of signs:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

Example 7 Use a cofactor expansion across the third row to compute det ***A***, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Solution Compute $\det A = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$

$$= (-1)^{3+1} a_{31} \det A_{31} + (-1)^{3+2} a_{32} \det A_{32} + (-1)^{3+3} a_{33} \det A_{33}$$

$$= 0\begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2)\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix}$$

$$= 0 + 2(-1) + 0 = -2$$

Theorem 1 is helpful for computing the determinant of a matrix that contains many zeros. For example, if a row is mostly zeros, then the cofactor expansion across that row has many terms that are zero, and the cofactors in those terms need not be calculated. The same approach works with a column that contains many zeros.

Example 8 Evaluate the determinant of $A = \begin{bmatrix} 2 & 0 & 0 & 5 \\ -1 & 2 & 4 & 1 \\ 3 & 0 & 0 & 3 \\ 8 & 6 & 0 & 0 \end{bmatrix}$

Solution $\det(A) = \begin{vmatrix} 2 & 0 & 0 & 5 \\ -1 & 2 & 4 & 1 \\ 3 & 0 & 0 & 3 \\ 8 & 6 & 0 & 0 \end{vmatrix}$

Expand from third column

$$\det(A) = 0\times C_{13} + 4\times C_{23} + 0\times C_{33} + 0\times C_{43}$$

$$= 0 + 4\times C_{23} + 0 + 0$$

$$= 4\times C_{23}$$

$$= 4\times(-1)^{2+3}\begin{vmatrix} 2 & 0 & 5 \\ 3 & 0 & 3 \\ 8 & 6 & 0 \end{vmatrix}$$

Expand from second column

$$= -4\left(0 + 0 + (-6)\begin{vmatrix} 2 & 5 \\ 3 & 3 \end{vmatrix}\right)$$

$$= (-4)\ (-6)\begin{vmatrix} 2 & 5 \\ 3 & 3 \end{vmatrix}$$

$$= -216$$

Example 9 Show that the value of the determinant is independent of θ

$$A = \begin{vmatrix} \sin\theta & \cos\theta & 0 \\ -\cos\theta & \sin\theta & 0 \\ \cos\theta - \sin\theta & \sin\theta + \cos\theta & 1 \end{vmatrix}$$

Solution Consider $A = \begin{vmatrix} \sin\theta & \cos\theta & 0 \\ -\cos\theta & \sin\theta & 0 \\ \cos\theta - \sin\theta & \sin\theta + \cos\theta & 1 \end{vmatrix}$

Expand the given determinant from 3rd column we have

$$= 0 - 0 + (-1)^{3+3}[\sin^2\theta + \cos^2\theta] = 1$$

Example 10 Compute det A, where $A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$

Solution The cofactor expansion down the first column of A has all terms equal to zero except the first.

Thus $$\det A = 3\begin{vmatrix} 2 & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{vmatrix} - 0.C_{21} + 0.C_{31} - 0.C_{41} + 0.C_{51}$$

Henceforth, we will omit the zero terms in the cofactor expansion.

Next, expand this 4×4 determinant down the first column, in order to take advantage of the zeros there.

We have $$\det A = 3 \times 2\begin{vmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{vmatrix}$$

This 3×3 determinant was computed above and found to equal -2.

Hence, det $A = 3 \times 2 \times (-2) = -12$.

The matrix in this example was nearly triangular. The method in that example is easily adapted to prove the following theorem.

Triangular Matrix

A triangular matrix is a special kind of m x n matrix where the entries either below or above the main diagonal are zero.

$\begin{bmatrix} 1 & 4 & 2 \\ 0 & 3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ is upper triangular and $25 \times 25 \begin{bmatrix} 1 & 0 & 0 \\ 2 & 8 & 0 \\ 4 & 9 & 7 \end{bmatrix}$ is lower triangular matrices.

Determinants of Triangular Matrices

Determinants of the triangular matrices are also easy to evaluate regardless of size.

Theorem If ***A*** is triangular matrix, then det (***A***) is the product of the entries on the main diagonal.

Consider a 4×4 lower triangular matrix.

$$A=\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

Keeping in mind that an elementary product must have exactly one factor from each row and one factor from each column, the only elementary product that does not have one of the six zeros as a factor is $(a_{11}a_{22}a_{33}a_{44})$. The column indices of this elementary product are in natural order, so the associated signed elementary product takes a +.

Thus, $\det(A) = a_{11}\times a_{22}\times a_{33}\times a_{44}$

Example 11

$$\begin{vmatrix} -2 & 5 & 7 \\ 0 & 3 & 8 \\ 0 & 0 & 5 \end{vmatrix} = (-2)(3)(5) = -30$$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 4 & 9 & 0 & 0 \\ -7 & 6 & -1 & 0 \\ 3 & 8 & -5 & -2 \end{vmatrix} = (1)(9)(-1)(-2) = 18$$

$$\begin{vmatrix} 1 & 2 & 7 & -3 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 3 \end{vmatrix} = (1)(1)(2)(3) = 6$$

The strategy in the above Example of looking for zeros works extremely well when an entire row or column consists of zeros. In such a case, the cofactor expansion along such a row or column is a sum of zeros. So, the determinant is zero. Unfortunately, most cofactor expansions are not so quickly evaluated.

Numerical Note By today's standards, a 25× 25 matrix is small. Yet it would be impossible to calculate a 25×25 determinant by cofactor expansion. In general, a cofactor expansion requires over *n!* multiplications, and $25! \approx 1.5x10^{25}$.

If a supercomputer could make one trillion multiplications per second, it would have to run for over 500,000 years to compute a 25×25 determinant by this method. Fortunately, there are faster methods, as we'll soon discover.

Example 12 Compute $\begin{vmatrix} 5 & -7 & 2 & 2 \\ 0 & 3 & 0 & -4 \\ -5 & -8 & 0 & 3 \\ 0 & 5 & 0 & -6 \end{vmatrix}$

Solution Take advantage of the zeros. Begin with a cofactor expansion down the third column to obtain a 3×3 matrix, which may be evaluated by an expansion down its first column,

$$\begin{vmatrix} 5 & -7 & 2 & 2 \\ 0 & 3 & 0 & -4 \\ -5 & -8 & 0 & 3 \\ 0 & 5 & 0 & -6 \end{vmatrix} = (-1)^{1+3}2\begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix}$$

$$= 2\cdot(-1)^{2+1}(-5)\begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix} = 20$$

The –1 in the next-to-last calculation came from the position of the –5 in the 3×3 determinant.

Exercises

Compute the determinants in exercises 1 to 6 by cofactor expansions. At each step, choose a row or column that involves the least amount of computation.

1. $\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$

2. $\begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix}$

3. $\begin{vmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{vmatrix}$

4. $\begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{vmatrix}$

5. $\begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix}$

6. $\begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix}$

Use the method of Example 2 to compute the determinants in exercises 7 and 8. In exercises 9 to 11, compute the determinant of elementary matrix. In exercises 12 and 13, verify that det $\boldsymbol{EA}$ = (det $\boldsymbol{E}$) . (det $\boldsymbol{A}$), where $\boldsymbol{E}$ is the elementary matrix and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

7. $\begin{vmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{vmatrix}$

8. $\begin{vmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{vmatrix}$

9. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & k & 1 \end{vmatrix}$

10. $\begin{vmatrix} k & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$

11. $\begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$

12. $\begin{vmatrix} 1 & k \\ 0 & 1 \end{vmatrix}$

13. $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$

14. Let $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$. Write $5\boldsymbol{A}$. Is det $5\boldsymbol{A}$ = 5 det $\boldsymbol{A}$?

15. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and k be a scalar. Find a formula that relates det ($k\boldsymbol{A}$)to k and det $\boldsymbol{A}$.

Lecture 18

Properties of Determinants

In this lecture, we will study the properties of the determinants. Some of them have already been discussed and you will be familiar with these. These properties become helpful, while computing the values of the determinants. The secret of determinants lies in how they change when row or column operations are performed.

Theorem 3 (Row Operations)**:** Let ***A*** be a square matrix.

a. If a multiple of one row of *A* is added to another row, the resulting determinant will remain same.
b. If two rows of ***A*** are interchanged to produce ***B***, then *det* ***B*** = –*det* ***A***.
c. If one row of ***A*** is multiplied by *k* to produce ***B***, then *det* ***B*** = *k* . *det* ***A***.

The following examples show how to use Theorem 3 to find determinants efficiently.

a. If a multiple of one row of *A* is added to another row, the resulting determinant will remain same.

Example

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Multiplying 2nd row by non – zero scalar say 'k 'as

$ka_{21} \quad ka_{22} \quad ka_{23}$ *– – – –adding this in 1st row then 'A' becomes*

$$= \begin{vmatrix} a_{11}+ka_{21} & a_{12}+ka_{22} & a_{13}+ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \qquad R_1' \to R_1 + kR_2$$

If each element of any row(column) can be expressed as sum of two elements then the resulting determinant can be expressed as sum of two determinants, so in this case

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} ka_{21} & ka_{22} & ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + k \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

By using property (c) of above theorem 3.

If any two rows or columns in a determinant are identical then value of this determinant is zero. So in this case $R_1 \equiv R_2$

$$\therefore \quad \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + k(0)$$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = A$$

b. If two rows of A are interchanged to produce B, then $det\ B = -det\ A$.

Example 1

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 1 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$\text{Now, } \det A = \begin{vmatrix} 1 & 2 & 3 \\ 5 & 1 & 1 \\ 0 & 8 & 9 \end{vmatrix} = 1(9-8) - 2(45-0) + 3(40-0) = 1 - 90 + 120 = 31$$

Now interchange column 1st with 2nd we get a new matrix, $B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 5 & 1 \\ 8 & 0 & 9 \end{bmatrix}$

$$\det B = \begin{vmatrix} 2 & 1 & 3 \\ 1 & 5 & 1 \\ 8 & 0 & 9 \end{vmatrix} = 2(45-0) - 1(9-8) + 3(0-40) = 90 - 1 - 120 = -31$$

c. If one row of A is multiplied by k to produce B, then $det\ B = k\ .\ det\ A$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 0 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$|A| = 1(0-8) - 2(45-0) + 3(40-0)$$

$$= -8 - 90 + 120 = 22$$

Multiplying R_1 by k, we get say

$$B = \begin{bmatrix} 1k & 2k & 3k \\ 5 & 0 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$|B| = k(40-0) - 2k(45-0) + 3k(40-0)$$
$$= 40k - 90k + 120k = 22k$$
$$= k|A|$$

Example 2

Evaluate $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$

Solution

$$\det A = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -7 \\ 0 & -2 & -8 & -10 \\ 0 & -7 & -10 & -13 \end{vmatrix} \text{ by } R_2' \to R_2 + (-2)R_1, R_3' \to R_3 + (-3)R_1, R_4' \to R_4 + (-4)R_1$$

$$= \begin{vmatrix} -1 & -2 & -7 \\ -2 & -8 & -10 \\ -7 & -10 & -13 \end{vmatrix} \text{ expanding from Ist column}$$

$$= (-1)(-2)(-1)\begin{vmatrix} 1 & 2 & 7 \\ 1 & 4 & 5 \\ 7 & 10 & 13 \end{vmatrix} \text{ taking } (-1),(-2) \text{ and } (-1) \text{ common from1st, 2nd, 3rd rows}$$

$$= (-2)\begin{vmatrix} 1 & 2 & 7 \\ 0 & 2 & -2 \\ 0 & -4 & -36 \end{vmatrix} \text{ by } R_2' \to R_2 + (-1)R_1, R_3' \to R_3 + (-7)R_1$$

$$= (-2)\begin{vmatrix} 2 & -2 \\ -4 & -36 \end{vmatrix} \text{ expanding by1st column}$$

$$= (-2)(2)(-4)\begin{vmatrix} 1 & -1 \\ 1 & 9 \end{vmatrix} \text{ taking 2 and (-4) common from1st and 2nd rows respectively.}$$

$$= 16\begin{vmatrix} 1 & -1 \\ 0 & 10 \end{vmatrix} \text{ by } R_2 + (-1)R_1$$
$$= 160$$

Example 3 Evaluate the determinant of the matrix $A = \begin{bmatrix} 4 & 2 & 5 & 10 \\ 1 & 1 & 6 & 3 \\ 7 & 3 & 0 & 5 \\ 0 & 2 & 5 & 8 \end{bmatrix}$

Solution

$$\det A = \begin{vmatrix} 4 & 2 & 5 & 10 \\ 1 & 1 & 6 & 3 \\ 7 & 3 & 0 & 5 \\ 0 & 2 & 5 & 8 \end{vmatrix}$$

$$= -\begin{vmatrix} 1 & 1 & 6 & 3 \\ 4 & 2 & 5 & 10 \\ 7 & 3 & 0 & 5 \\ 0 & 2 & 5 & 8 \end{vmatrix}$$ interchanging R_1 and R_2 (R_{12}')

$$= -\begin{vmatrix} 1 & 1 & 6 & 3 \\ 0 & -2 & -19 & -2 \\ 0 & -4 & -42 & -16 \\ 0 & 2 & 5 & 8 \end{vmatrix}$$ By $R_2' \to R_2 + (-4)R_1, R_3' \to R_3 + (-7)R_1$

$$= -\begin{vmatrix} -2 & -19 & -2 \\ -4 & -42 & -16 \\ 2 & 5 & 8 \end{vmatrix}$$ expanding from 1st column

$$= (-1)^3 \begin{vmatrix} 2 & 19 & 2 \\ 4 & 42 & 16 \\ 2 & 5 & 8 \end{vmatrix}$$ taking (-1) as a common factor from R_1 and R_2

$$= -\begin{vmatrix} 2 & 19 & 2 \\ 4 & 42 & 16 \\ 2 & 5 & 8 \end{vmatrix}$$

$$= -2\begin{vmatrix} 1 & 19 & 2 \\ 2 & 42 & 16 \\ 1 & 5 & 8 \end{vmatrix}$$

$$= (-2)\begin{vmatrix} 1 & 19 & 2 \\ 0 & 4 & 12 \\ 0 & -14 & 6 \end{vmatrix}$$ By $R_2' \to R_2 + (-2)R_1, R_3' \to R_3 + (-1)R_1$

$$= (-2)\begin{vmatrix} 1 & 19 & 2 \\ 0 & 4 & 12 \\ 0 & -14 & 6 \end{vmatrix} \quad R_2 + (-2)R_1, R_3 + (-1)R_1$$

$$= -2\begin{vmatrix} 4 & 12 \\ -14 & 6 \end{vmatrix} \text{ expand from Ist column}$$

$$= -2(24+168) = -384$$

Example 4 Without expansion, show that $\begin{vmatrix} x & a+x & b+c \\ x & b+x & c+a \\ x & c+x & a+b \end{vmatrix} = 0$

Solution

$$\begin{vmatrix} x & a+x & b+c \\ x & b+x & c+a \\ x & c+x & a+b \end{vmatrix}$$

$$= \begin{vmatrix} x & a+x-x & b+c \\ x & b+x-x & c+a \\ x & c+x-x & a+b \end{vmatrix} \quad By\ C_2' \to C_2 - C_1$$

$$= \begin{vmatrix} x & a & b+c \\ x & b & c+a \\ x & c & a+b \end{vmatrix}$$

Taking 'x' common from C_1

$$= x\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$$

$$= x\begin{vmatrix} 1 & a+b+c & b+c \\ 1 & b+c+a & c+a \\ 1 & c+a+b & a+b \end{vmatrix} \quad By\ C_2' \to C_2 + C_3$$

Now taking (a+b+c) common form C_2

$$= x(a+b+c)\begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & c+a \\ 1 & 1 & a+b \end{vmatrix}$$

$= 0$ as column Ist and 2nd are identical ($C_1 \equiv C_2$). So its value will be zero.

Example 5 Evaluate $A=\begin{vmatrix} 2 & 3 & 1 & 0 & 1 \\ 1 & 1 & 3 & 1 & 2 \\ 2 & 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 1 & 2 \\ 4 & 1 & 1 & 0 & 0 \end{vmatrix}$

Solution Interchanging R_1 and R_2, we get

$$A=-\begin{vmatrix} 1 & 1 & 3 & 1 & 2 \\ 2 & 3 & 1 & 0 & 1 \\ 2 & 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 1 & 2 \\ 4 & 1 & 1 & 0 & 0 \end{vmatrix}$$

$$R_2' \to R_2 - 2R_1, R_3' \to R_3 - 2R_1, R_4' \to R_4 - 3R_1, R_5' \to R_5 - 4R_1$$

$$=-\begin{vmatrix} 1 & 1 & 3 & 1 & 2 \\ 0 & 1 & -5 & -2 & -3 \\ 0 & -1 & -4 & 1 & 0 \\ 0 & -1 & -8 & -2 & -4 \\ 0 & -3 & -11 & -4 & -8 \end{vmatrix}$$

expand from C_1

$$=-\begin{vmatrix} 1 & -5 & -2 & -3 \\ -1 & -4 & 1 & 0 \\ -1 & -8 & -2 & -4 \\ -3 & -11 & -4 & -8 \end{vmatrix}$$

$$R_2' \to R_2 + R_1, R_3' \to R_3 + R_1, R_4' \to R_4 + 3R_1$$

$$=-\begin{vmatrix} 1 & -5 & -2 & -3 \\ 0 & -9 & -1 & -3 \\ 0 & -13 & -4 & -7 \\ 0 & -26 & -10 & -17 \end{vmatrix}$$

expand from C_1

$$=-\begin{vmatrix} -9 & -1 & -3 \\ -13 & -4 & -7 \\ -26 & -10 & -17 \end{vmatrix}$$

taking (-1) common from Ist, 2nd and 3rd row

$$= \begin{vmatrix} 9 & 1 & 3 \\ 13 & 4 & 7 \\ 26 & 10 & 17 \end{vmatrix}$$

interchange Ist and 2nd Column($C_1' _2$)

$$= - \begin{vmatrix} 1 & 9 & 3 \\ 4 & 13 & 7 \\ 10 & 26 & 17 \end{vmatrix}$$

$C_2' \to C_2 - 9C_1, C_3' \to C_3 - 3C_1$

$$= - \begin{vmatrix} 1 & 0 & 0 \\ 4 & -23 & -5 \\ 10 & -64 & -13 \end{vmatrix}$$

expand from Ist row

$$= - \begin{vmatrix} -23 & -5 \\ -64 & -13 \end{vmatrix} = -(299 - 320) = 21$$

An Algorithm to evaluate the determinant

Algorithm means a sequence of a finite number of steps to get a desired result. The word Algorithm comes from the famous Muslim mathematician AL-Khwarizmi who invented the word algebra.

The step-by-step evaluation of det(***A***) of order ***n*** is obtained as follows:

Step 1: By an interchange of rows of ***A*** (and taking the resulting sign into account) bring a non zero entry to (1,1) the position (unless all the entries in the first column are zero in which case det ***A***=0).

Step 2: By adding suitable multiples of the first row to all the other rows, reduce the (n-1) entries, except (1,1) in the first column, to 0. Expand det(***A***) by its first column. Repeat this process or continue the following steps.

Step 3: Repeat step 1 and step 2 with the last remaining rows concentrating on the second column.

Step 4: Repeat step 1,step2 and step 3 with the remaining (n-2) rows, (n-3) rows and so on, until a triangular matrix is obtained.

Step5: Multiply all the diagonal entries of the resulting triangular matrix and then multiply it by its sign to get det(A)

Example 6 Compute det ***A***, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$.

Solution *The strategy is to reduce **A** to echelon form and then to use the fact that the determinant of a triangular matrix is the product of the diagonal entries.* The first two row replacements in column 1 do not change the determinant:

$$\det A = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix} \quad By\ R_2' \to R_2 + 2R_1, R_3' \to R_1 + R_3$$

An interchange of rows 2 and 3 (R'_{23}), it reverses the sign of the determinant, so

$$\det A = -\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

Example 7 Compute det ***A***, where

$$A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}.$$

Solution Taking'2' common from 1st row

$$\det A = 2\begin{vmatrix} 1 & -4 & 3 & 4 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{vmatrix}$$

$$\det A = 2\begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & -3 & 2 \end{vmatrix} \quad By\ R_2' \to R_2 - 3R_1, R_3' \to R_3 + 3R_1, R_4' \to R_4 - R_1$$

$$\det A = 2\begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} \quad (By\ R_4' \to R_4 - \tfrac{1}{2}R_3)$$

$$= 2.\{(1)(3)(-6)(1)\} = -36$$

Example 8 Show that $\begin{vmatrix} x & 2 & 2 & 2 \\ 2 & x & 2 & 2 \\ 2 & 2 & x & 2 \\ 2 & 2 & 2 & x \end{vmatrix} = (x+6)(x-2)^3$

Solution

$$\begin{vmatrix} x & 2 & 2 & 2 \\ 2 & x & 2 & 2 \\ 2 & 2 & x & 2 \\ 2 & 2 & 2 & x \end{vmatrix}$$

$$= \begin{vmatrix} x+6 & 2 & 2 & 2 \\ x+6 & x & 2 & 2 \\ x+6 & 2 & x & 2 \\ x+6 & 2 & 2 & x \end{vmatrix} \quad By\ \ C_1' \to C_1 + (C_2 + C_3 + C_4)$$

Taking (x+6) common from 1st column

$$= (x+6)\begin{vmatrix} 1 & 2 & 2 & 2 \\ 1 & x & 2 & 2 \\ 1 & 2 & x & 2 \\ 1 & 2 & 2 & x \end{vmatrix}$$

$$= (x+6)\begin{vmatrix} 1 & 2 & 2 & 2 \\ 0 & x-2 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 0 & 0 & 0 & x-2 \end{vmatrix} \quad By\ R_2' \to R_2 - R_1, R_3' \to R_3 - R_1, R_4' \to R_4 - R_1$$

And this is the triangular matrix and its determinant is the prodcut of main diagonal's entries.

$$= (x+6)(x-2)^3$$

Example 9 Compute det ***A***, where $A=\begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$.

Solution $A=\begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

$$\det A=\begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix} R_3' \to R_3+2R_1$$

$= \mathbf{0}$ as $R_2 \equiv R_3$

Example10 Compute det ***A***, where

$$A=\begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$$

Solution

$$A=\begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$$

$$=\begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{bmatrix} R_4' \to R_4+R_2$$

$$=(-1)\begin{bmatrix} 2 & 1 & 2 & -1 \\ 0 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{bmatrix} \quad By\ R_{12}'$$

Expanding from 1st row and 1st column

$$= -2\begin{vmatrix} 5 & -7 & 3 \\ 3 & 6 & 2 \\ 0 & -3 & 1 \end{vmatrix}$$

$$= (-2)\{5(6+6) - (-7)(3-0) + 3(-9-0)\}$$

$$= 54$$

Remarks

Suppose that a square matrix ***A*** has been reduced to an echelon form ***U*** by row replacements and row interchanges.

If there are r interchanges, then $\det(A) = (-1)^r \det(U)$

Furthermore, all of the pivots are still visible in ***U*** (because they have not been scaled to ones). If ***A*** is invertible, then the pivots in ***U*** are on the diagonal (since ***A*** is row equivalent to the identity matrix). In this case, *det* ***U*** is the product of the pivots. If ***A*** is not invertible, then ***U*** has a row of zero and *det* ***U*** = *0*.

$$U = \begin{bmatrix} \bullet & * & * & * \\ 0 & \bullet & * & * \\ 0 & 0 & \bullet & * \\ 0 & 0 & 0 & \bullet \end{bmatrix} \qquad U = \begin{bmatrix} \bullet & * & * & * \\ 0 & \bullet & * & * \\ 0 & 0 & \bullet & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\det U \neq 0 \qquad \det U = 0$$

Thus we have the following formula

$$\det A = \begin{cases} (-1)^r .(\textit{Product of pivots in } U) & \textit{When } A \textit{ is invertible} \\ 0 & \textit{When } A \textit{ is not invertible} \end{cases} \qquad (1)$$

Example

Case-01 For 2×2 invertible matrix

Reducing given 2×2 invertible matrix into Echelon form as follows;

$$A = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$$

By interchanging 1st and 2nd rows(R_{12}')

$\Rightarrow \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$ $\because$ one replacement of rows has occurred, $\therefore r = 1$

$\Rightarrow \begin{bmatrix} 3 & 2 \\ 0 & \frac{7}{3} \end{bmatrix}$ By $R_2' \to R_2 - \frac{4}{3}R_1$, we have desired row-echelon form $U = \begin{bmatrix} 3 & 2 \\ 0 & \frac{7}{3} \end{bmatrix}$.

Thus using the above formula as follows;

$$\det A = (-1)^r .(Product\ of\ \ pivots\ in U) = (-1)^1(3\cdot \frac{7}{3}) = -7$$

Case-02 For 2×2 non-invertible matrix

In this case say;

$$A = \begin{bmatrix} 4 & 5 \\ 8 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 \\ 0 & 0 \end{bmatrix} \quad By\ R_2' \to R_2 - 2R_1, \text{desired row-echelon form is } U = \begin{bmatrix} 4 & 5 \\ 0 & 0 \end{bmatrix}$$

Here no interchange of rows has occurred. So, $r = 0$ and

$$\therefore \det A = (-1)^r .(Product\ of\ \ pivots\ in U) = (-1)^0(4\cdot 0) = 0$$

Theorem 5 If A is an $n\times n$ matrix, then *det* $\boldsymbol{A^T}$ = *det* $\boldsymbol{A}$.

Example 11 If $A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 3 \end{bmatrix}$, find det($\boldsymbol{A}$) and det ($\boldsymbol{A^T}$)

$$\det A = \begin{vmatrix} 1 & 4 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 3 \end{vmatrix} = 1(3-2) - 4(6-6) + 1(2-3) = 1 - 0 - 1 = 0$$

Now

$$A^t = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\det A^t = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 1(3-2) - 2(12-1) + 3(8-1) = 1 - 22 + 21 = 0$$

Remark

Column operations are useful for both theoretical purposes and hand computations. However, for simplicity we'll perform only row operations in numerical calculations.

Theorem 6 (Multiplicative Property)

If $\boldsymbol{A}$ and $\boldsymbol{B}$ are $n\times n$ matrices, then $\det(AB) = (\det A)(\det B)$.

Example 12 Verify Theorem 6 for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ *and* $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

Solution $AB = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$

and $\det AB = 25.13 - 20.14 = 325 - 280 = 45$

Since *det* $\boldsymbol{A}$ = *9* and *det* $\boldsymbol{B}$ = *5*, $(\det A)(\det B) = 9.5 = 45 = \det AB$

Remark

$\det(\boldsymbol{A} + \boldsymbol{B}) \neq \det \boldsymbol{A} + \det \boldsymbol{B}$, in general.

For example,

If $A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & -3 \\ -1 & 5 \end{bmatrix}$. Then

$$A + B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \det(A + B) = 0$$

$$\det A + \det B = \begin{vmatrix} 2 & 3 \\ 1 & -5 \end{vmatrix} + \begin{vmatrix} -2 & -3 \\ -1 & 5 \end{vmatrix} = (-10 - 3) + (-10 - 3) = -26 \neq \det(A + B)$$

Exercise

Find the determinants in exercises 1 to 6 by row reduction to echelon form.

1. $\begin{vmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{vmatrix}$

2. $\begin{vmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 2 & 5 & 4 & -3 \\ -3 & -7 & -5 & 2 \end{vmatrix}$

3. $\begin{vmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & -2 & 3 \end{vmatrix}$

4. $\begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{vmatrix}$

5. $\begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix}$

6. $\begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix}$

Combine the methods of row reduction and cofactor expansion to compute the determinants in exercises 7 and 8.

7. $\begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{vmatrix}$

8. $\begin{vmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{vmatrix}$

9. Use determinant to find out whether the matrix is invertible $\begin{bmatrix} 2 & 0 & 0 & 8 \\ 1 & -7 & -5 & 0 \\ 3 & 8 & 6 & 0 \\ 0 & 7 & 5 & 4 \end{bmatrix}$

10. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 3×3 matrices, with det $\boldsymbol{A}$ = 4 and det $\boldsymbol{B}$ = -3. Use properties of determinants to compute

(a) det $\boldsymbol{AB}$ (b) det $7\boldsymbol{A}$ (c) det $\boldsymbol{B^T}$ (d) det $\boldsymbol{A^T}$
(e) det $\boldsymbol{A^TA}$

11 Show that

(a) $\begin{vmatrix} a_1 & b_1 & a_1+b_1+c_1 \\ a_2 & b_2 & a_2+b_2+c_2 \\ a_3 & b_3 & a_3+b_3+c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(b) $\begin{vmatrix} a_1+b_1 & a_1-b_1 & c_1 \\ a_2+b_2 & a_2-b_2 & c_2 \\ a_3+b_3 & a_3-b_3 & c_3 \end{vmatrix} = -2\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

12 Show that

(a) $\begin{vmatrix} a_1+b_1t & a_2+b_2t & a_3+b_3t \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (1-t^2)\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

(b) $\begin{vmatrix} a_1 & b_1+ta_1 & c_1+rb_1+sa_1 \\ a_2 & b_2+ta_2 & c_2+rb_2+sa_2 \\ a_3 & b_3+ta_3 & c_3+rb_3+sa_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

13. Show that $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (y-x)(z-x)(z-y)$

Lecture 19

Cramer's Rule, Volume, and Linear Transformations

In this lecture, we shall apply the theory discussed in the last two lectures to obtain important theoretical formulae and a geometric interpretation of the determinant.

Cramer's Rule Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $\boldsymbol{Ax} = \boldsymbol{b}$ is affected by changes in the entries of $\boldsymbol{b}$. However, the formula is inefficient for hand calculations, except for 2×2 or perhaps 3×3 matrices.

Theorem 1 (Crammer's Rule) Let $\boldsymbol{A}$ be an invertible $n\times n$ matrix. For any $\boldsymbol{b}$ in $\boldsymbol{R}^n$, the unique solution x of $\boldsymbol{Ax} = \boldsymbol{b}$ has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \qquad i = 1, 2, ..., n \qquad (1)$$

Example 1 Use Cramer's rule to solve the system

$$3x_1 - 2x_2 = 6$$
$$-5x_1 + 4x_2 = 8$$

Solution Write the system in matrix form, $\boldsymbol{Ax} = \boldsymbol{b}$

$$\begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

where

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \& b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\det A = \begin{vmatrix} 3 & -2 \\ -5 & 4 \end{vmatrix} = 12 - 10 = 2$$

$$A_1(b) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, A_2(b) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since $\det \boldsymbol{A} = 2$, the system has a unique solution. By Cramer's rule,

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{24+16}{2} = 20$$
$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{24+30}{2} = 27$$

Example 2 Consider the following system in which s is an unspecified parameter. Determine the values of s for which the system has a unique solution and use Cramer's rule to describe the solution.

$$3sx_1 - 2x_2 = 4$$
$$-6x_1 + sx_2 = 1$$

Solution Here

$$A = \begin{bmatrix} 3s & -2 \\ -6 & s \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad A_1(b) = \begin{bmatrix} 4 & -2 \\ 1 & s \end{bmatrix}, \quad A_2(b) = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}$$

Since $\det A = 3s^2 - 12 = 3(s+2)(s-2)$

the system has a unique solution when

$\det A \neq 0$

$\Rightarrow 3(s+2)(s-2) \neq 0$

$\Rightarrow s^2 - 4 \neq 0$

$\Rightarrow s \neq \pm 2$

For such an s, the solution is (x_1, x_2), where

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{4s+2}{3(s+2)(s-2)}, \qquad s \neq \pm 2$$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{3s+24}{3(s+2)(s-2)} = \frac{s+8}{(s+2)(s-2)}, \qquad s \neq \pm 2$$

Example 3 Solve, by Cramer's Rule, the system of equations

$$2x_1 - x_2 + 3x_3 = 1$$
$$x_1 + 2x_2 - x_3 = 2$$
$$3x_1 + 2x_2 + 2x_3 = 3$$

Solution Here $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 3 & 2 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $A_1 = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 2 & -1 \\ 3 & 2 & 2 \end{bmatrix}$

$$A_2 = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & -1 \\ 3 & 3 & 2 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

$$D = \det A = 2\cdot 6 - 1\cdot(-5) + 3(-4) = 5$$
$$D_1 = \det A_1 b = 1\cdot 6 + 2\cdot 8 + 3(-5) = 7$$
$$D_2 = \det A_2 b = 2\cdot(7) - 1\cdot(5) + 3(-3) = 0$$
$$D_3 = \det A_3 b = 2\cdot(2) + 1\cdot(-3) + 1(-4) = -3$$
$$So\ x_1 = \frac{D_1}{D} = \frac{7}{5},\quad x_2 = \frac{D_2}{D} = 0,\quad x_3 = \frac{D_3}{D} = -\frac{3}{5}$$

Example 4 Use Cramer's Rule to solve.

$$x_1 + \quad 2x_3 = 6$$
$$-3x_1 + 4x_2 + 6x_3 = 30$$
$$-\ x_1 - 2x_2 + 3x_3 = 8$$

Solution

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 30 \\ 8 \end{bmatrix}$$

$$\therefore A_1 = \begin{bmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{bmatrix}, A_2 = \begin{bmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & -8 & 3 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{bmatrix}$$

Therefore,

$$x_1 = \frac{\det(A_1 b)}{\det(A)} = \frac{-40}{44} = \frac{-10}{11}, \qquad x_2 = \frac{\det(A_2 b)}{\det(A)} = \frac{72}{44} = \frac{18}{11}$$
$$x_3 = \frac{\det(A_3 b)}{\det(A)} = \frac{152}{44} = \frac{38}{11}$$

Note For any $n \times n$ matrix ***A*** and any ***b*** in $\boldsymbol{R^n}$, *let* $\boldsymbol{A_i(b)}$ be the matrix obtained from ***A*** by replacing ***i***th column by the vector ***b***.

$$A_i(b) = \begin{bmatrix} a_1 & \ldots & b & \ldots & a_n \end{bmatrix}$$
$$\uparrow$$
$$i\text{th column}$$

Formula for A^{-1}

Cramer's rule leads easily to a general formula for the inverse of $n \times n$ matrix ***A***. The ***j****th* column of $\boldsymbol{A^{-1}}$ is a vector ***x*** that satisfies $\boldsymbol{Ax = e_j}$
where $\boldsymbol{e_j}$ is the ***j****th* column of the identity matrix, and the ***i****th* entry of ***x*** is the (i, j)-entry of $\boldsymbol{A^{-1}}$**.** By Cramer's rule,

$$\left\{(i,j) - entry\ of\ A^{-1}\right\} = x_{ij} = \frac{\det A_i(e_j)}{\det A} \qquad (2)$$

Recall that A_{ji} denotes the submatrix of A formed by deleting row j and column i. A cofactor expansion down column i of $A_i(e_j)$ shows that

$$\det A_{i}(e_{j}) = (-1)^{i+j} \det A_{ji} = C_{ji} \qquad (3)$$

where C_{ji} is a cofactor of A.

By (2), the (i, j)-entry of A^{-1} is the cofactor C_{ji} divided by $\det A$.

[Note that the subscripts on C_{ji} are the reverse of (i, j).] Thus

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix} \qquad (4)$$

The matrix of cofactors on the right side of (4) is called the **adjugate** (or **classical adjoint**) of A, denoted by adj A. (The term adjoint also has another meaning in advance texts on linear transformations.) The next theorem simply restates (4).

Theorem 2 (An Inverse Formula)

Let A be an invertible $n \times n$ matrix, then $A^{-1} = \frac{1}{\det A} adj\ A$

Example

For the matrrix say

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix} \Rightarrow \det A = 10 - (-3) = 13$$

$\Rightarrow A^{-1}$ will also be a 2×2 matrix

As

$Aji =$ submatrix of A formed by deleting row j and column i

So in this case

$A_{11} =$ submatrix of A formed by deleting row **1** and column **1** $= [5]$

$A_{12} =$ submatrix of A formed by deleting row **1** and column **2** $= [-1]$

$A_{21} =$ submatrix of A formed by deleting row **2** and column **1** $= [3]$

$A_{22} =$ submatrix of A formed by deleting row **2** and column **2** $= [2]$

and

$$\det A_{i}(e_{j}) = (-1)^{i+j} \det(A_{ji}) = C_{ji}$$

where e_j is the jth column of identity matrix $I_{n \times n}$

So in this case

$$C_{11} = \det A_{11}(e_1) = (-1)^{1+1} \det A_{11} = (+1)\det[5] = 5$$
$$C_{12} = \det A_{12}(e_1) = (-1)^{1+2} \det A_{12} = (-1)\det[-1] = (-1)(-1) = 1$$
$$C_{21} = \det A_1(e_2) = (-1)^{2+1} \det A_{21} = (-1)\det[3] = -3$$
$$C_{22} = \det A_2(e_2) = (-1)^{2+2} \det A_{22} = (+1)\det[2] = 2$$

By Cramer's rule,

$$\{(i,j)-entry\ of\ A^{-1}\} = x_{ij} = \frac{\det A_i(e_j)}{\det A} = \frac{C_{ji}}{\det A}$$

So for the current matrix;

$$\{(1,1)-entry\ of\ A^{-1}\} = x_{11} = \frac{\det A_1(e_1)}{\det A} = \frac{C_{11}}{\det A} = \frac{5}{13}$$
$$\{(1,2)-entry\ of\ A^{-1}\} = x_{12} = \frac{\det A_1(e_2)}{\det A} = \frac{C_{21}}{\det A} = \frac{-3}{13}$$
$$\{(2,1)-entry\ of\ A^{-1}\} = x_{21} = \frac{\det A_2(e_1)}{\det A} = \frac{C_{12}}{\det A} = \frac{1}{13}$$
$$\{(2,2)-entry\ of\ A^{-1}\} = x_{22} = \frac{\det A_2(e_2)}{\det A} = \frac{C_{22}}{\det A} = \frac{2}{13}$$

Hence by using equation # 4, we get

$$A^{-1} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} \frac{C_{11}}{\det A} & \frac{C_{21}}{\det A} \\ \frac{C_{12}}{\det A} & \frac{C_{22}}{\det A} \end{bmatrix} = \frac{1}{\det A}\begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} = \begin{bmatrix} \frac{5}{13} & \frac{-3}{13} \\ \frac{1}{13} & \frac{2}{13} \end{bmatrix}$$

Example 5 Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$.

Solution The nine cofactors are

$$C_{11} = +\begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, \quad C_{12} = -\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, \quad C_{13} = +\begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$C_{21} = -\begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, \quad C_{22} = +\begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, \quad C_{23} = -\begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$$

$$C_{31} = +\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4, \quad C_{32} = -\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, \quad C_{33} = +\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

The adjoint matrix is the transpose of the matrix of cofactors. [For instance, C_{12} goes in the (*2, 1*) position.] Thus

$$adjA = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

We could compute det $\boldsymbol{A}$ directly, but the following computation provides a check on the calculations above and produces det $\boldsymbol{A}$:

$$(adjA).A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14I$$

Since (adj $\boldsymbol{A}$) $\boldsymbol{A}$ = 14 $\boldsymbol{I}$, Theorem 2 shows that det $\boldsymbol{A}$ = 14 and

$$A^{-1} = \frac{1}{14}\begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} = \begin{bmatrix} -2/14 & 14/14 & 4/14 \\ 3/14 & -7/14 & 1/14 \\ 5/14 & -7/14 & -3/14 \end{bmatrix}$$

Determinants as Area or Volume

In the next application, we verify the geometric interpretation of determinants and we assume here that the usual Euclidean concepts of length, area, and volume are already understood for $\boldsymbol{R^2}$ and $\boldsymbol{R^3}$.

Theorem 3 If $\boldsymbol{A}$ is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If $\boldsymbol{A}$ is a 3×3 matrix, the volume of the parallelepiped determined by the columns of $\boldsymbol{A}$ is $|\det A|$.

Example 6 Calculate the area of the parallelogram determined by the points (-2, -2), (0, 3), (4, -1) and (6, 4).

Solution

Let A(-2,-2), B(0,3), C(4,-1) and D(6,4). Fixing one point say A(-2,-2) and find the adjacent lengths of parallelogram which are given by the column vectors as follows**;**

$$AB = \begin{bmatrix} 0-(-2) \\ 3-(-2) \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$AC = \begin{bmatrix} 4-(-2) \\ -1-(-2) \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

So the area of parallelogram ABCD determined by above column vectors

$$= \left|\det \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}\right| = |2-30| = |-28| = 28$$

Now we translate the parallelogram ABCD to one having the origin as a vertex. For which we subtract the vertex (-2, -2) from each of the four vertices. The new parallelogram has the vertices say

A′=(-2-(-2),-2-(-2))=(0, 0)

B′=(0-(-2),3-(-2))=(2, 5)

C′=(4-(-2),-1-(-2))=(6,1)

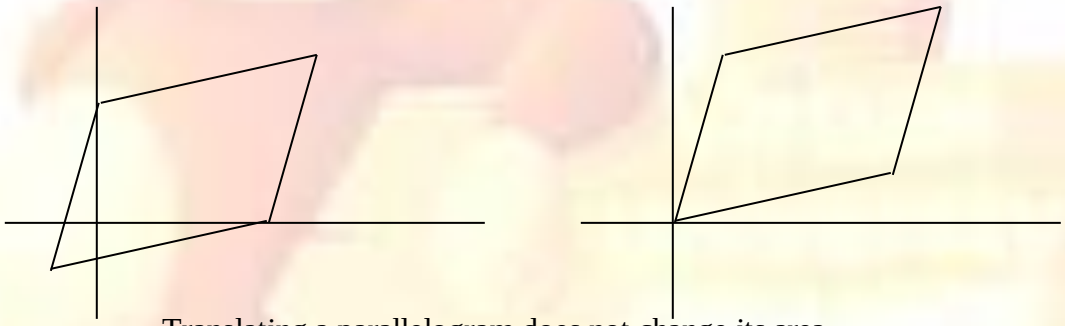
D′=(6-(-2),4-(-2))=(8, 6)

And fixing $A'(0,0)$ in this case, so

$$A'B' = \begin{bmatrix} 2-0 \\ 5-0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$A'C' = \begin{bmatrix} 6-0 \\ 5-0 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

See Fig below. The area of this parallelogram is also determined by the above columns vectors = $\left|\det \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}\right| = |2-30| = |-28| = 28$



Translating a parallelogram does not change its area

Linear Transformations

Determinants can be used to describe an important geometric property of linear transformations in the plane and in $\boldsymbol{R^3}$. If ***T*** is a linear transformation and ***S*** is a set in the domain of ***T***, let ***T* (*S*)** denote the set of images of points in ***S***. We are interested in how the area (or volume) of ***T* (*S*)** compares with the area (or volume) of the original set ***S***. For convenience, when ***S*** is a region bounded by a parallelogram, we also refer to ***S*** as a parallelogram.

Theorem 4 Let $T: R^2 \to R^2$ be the linear transformation determined by a 2×2 matrix ***A***. If ***S*** is a parallelogram in $\boldsymbol{R^2}$, then

{area of ***T* (*S*)**} = |det***A***|. {area of ***S***}

If ***T*** is determined by a 3 x 3 matrix ***A***, and if ***S*** is a parallelepiped in $\boldsymbol{R^3}$, then

{volume of ***T* (*S*)**} = |det***A***|. {volume of ***S***}

Example 7 Let *a* and *b* be positive numbers. Find the area of the region *E* bounded by the ellipse whose equation is $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$.

Solution We claim that E is the image of the unit disk D under the linear transformation **A:D→E** determined by the matrix $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, given as

$A\boldsymbol{u} = \boldsymbol{x}$ where $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in D$, $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in E$.

$$\Rightarrow \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Now $A\boldsymbol{u} = \boldsymbol{x} \Rightarrow \begin{bmatrix} au_1 \\ bu_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ then

$$\Rightarrow au_1 = x_1 \quad and \quad bu_2 = x_2$$

$$\Rightarrow u_1 = \frac{x_1}{a} \text{ and } u_2 = \frac{x_2}{b}$$

Since $u \in D$ (in the circular disk),it follows that the distance of $\boldsymbol{u}$ from origin will be less than unity i-e

$$\left(u_1^2 - 0\right) + \left(u_2^2 - 0\right) \le 1$$

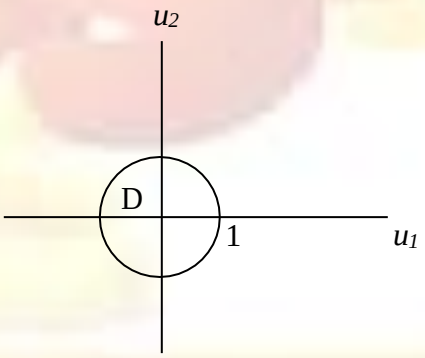
$$\Rightarrow \left(\frac{x_1}{a}\right)^2 + \left(\frac{x_2}{b}\right)^2 \le 1 \qquad \because u_1 = \frac{x_1}{a}, u_2 = \frac{x_2}{b}$$

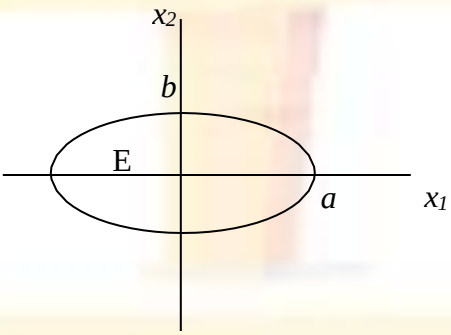
Hence by the generalization of theorem 4,

{area of ellipse} = {area of $\boldsymbol{A(D)}$} (*here* $T \equiv A$)

= |det $\boldsymbol{A}$|. {area of $\boldsymbol{D}$}

= $ab.\pi\,(1)^2 = \pi$ ab





Example 8 Let ***S*** be the parallelogram determined by the vectors $b_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$. Compute the area of image of ***S*** under the mapping $x \to Ax$.

Solution The area of ***S*** is $\left|\det \begin{bmatrix} 1 & 5 \\ 3 & 1 \end{bmatrix}\right| = 14$, and det ***A*** = 2. By theorem 4, the area of image of ***S*** under the mapping $x \to Ax$ is |det ***A***|. {area of ***S***} = 2.14 = 28

Exercises

Use Cramer's Rule to compute the solutions of the systems in exercises 1 and 2.

1. $2x_1 + x_2 = 7$
   $-3x_1 + x_3 = -8$
   $x_2 + 2x_3 = -3$

2. $2x_1 + x_2 + x_3 = 4$
   $-x_1 + 2x_3 = 2$
   $3x_1 + x_2 + 3x_3 = -2$

In exercises 3-6, determine the values of the parameter s for which the system has a unique solution, and describe the solution.

3. $6sx_1 + 4x_2 = 5$
   $9x_1 + 2sx_2 = -2$

4. $3sx_1 - 5x_2 = 3$
   $9x_1 + 5sx_2 = 2$

5. $sx_1 - 2sx_2 = -1$
   $3x_1 + 6sx_2 = 4$

6. $2sx_1 + x_2 = 1$
   $3sx_1 + 6sx_2 = 2$

In exercises 7 and 8, compute the adjoint of the given matrix, and then find the inverse of the matrix.

7. $\begin{bmatrix} 3 & 5 & 4 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

8. $\begin{bmatrix} 3 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 3 & 2 \end{bmatrix}$

In exercises 9 and 10, find the area of the parallelogram whose vertices are listed.

9. *(0, 0), (5, 2), (6, 4), (11, 6)*

10. *(-1, 0), (0, 5), (1, -4), (2, 1)*

11. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at *(1, 0, -2), (1, 2, 4), (7, 1, 0)*.

12. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at *(1, 4, 0), (-2, -5, 2), (-1, 2, -1)*.

13. Let $\boldsymbol{S}$ be the parallelogram determined by the vectors $\boldsymbol{b_1} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\boldsymbol{b_2} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and let $\boldsymbol{A} = \begin{bmatrix} 6 & -2 \\ -3 & 2 \end{bmatrix}$. Compute the area of the image of $\boldsymbol{S}$ under the mapping $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$.

14. Let $\boldsymbol{S}$ be the parallelogram determined by the vectors $\boldsymbol{b_1} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$ and $\boldsymbol{b_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and let $\boldsymbol{A} = \begin{bmatrix} 7 & 2 \\ 1 & 1 \end{bmatrix}$. Compute the area of the image of $\boldsymbol{S}$ under the mapping $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$.

15. Let $\boldsymbol{T}: \boldsymbol{R^3} \rightarrow \boldsymbol{R^3}$ be the linear transformation determined by the matrix $\boldsymbol{A} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$, where a, b, c are positive numbers. Let $\boldsymbol{S}$ be the unit ball, whose bounding surface has the equation $x_1^2 + x_2^2 + x_3^2 = 1$.

a. Show that $\boldsymbol{T}(\boldsymbol{S})$ is bounded by the ellipsoid with the equation $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} = 1$.

b. Use the fact that the volume of the unit ball is $4\pi/3$ to determine the volume of the region bounded by the ellipsoid in part (a).

Lecture 20

Vector Spaces and Subspaces

Case Example

The space shuttle's control systems are absolutely critical for flight. Because the shuttle is an unstable airframe, it requires constant computer monitoring during atmospheric flight. The flight control system sends a stream of commands to aerodynamic control surfaces.

Mathematically, the input and output signals to an engineering system are functions. It is important in applications that these functions can be added, and multiplied by scalars. These two operations on functions have algebraic properties that are completely analogous to the operation of adding vectors in $\boldsymbol{R^n}$ and multiplying a vector by a scalar, as we shall see in the lectures 20 and 27. For this reason, the set of all possible inputs (functions) is called a *vector space.* The mathematical foundation for systems engineering rests on vector spaces of functions, and we need to extend the theory of vectors in $\boldsymbol{R^n}$ to include such functions. Later on, we will see how other vector spaces arise in engineering, physics, and statistics.

Definition Let $\boldsymbol{V}$ be an arbitrary nonempty set of objects on which two operations are defined, addition and multiplication by scalars (numbers). If the following axioms are satisfied by all objects $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in $\boldsymbol{V}$ and all scalars k and l, then we call $\boldsymbol{V}$ a vector space.

Axioms of Vector Space

1. Closure Property For any two vectors $\boldsymbol{u}$ & $\boldsymbol{v} \in \boldsymbol{V}$, implies $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{V}$

2. Commutative Property For any two vectors $\boldsymbol{u}$ & $\boldsymbol{v} \in \boldsymbol{V}$, implies $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$

3. Associative Property For any three vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \boldsymbol{V}$, $\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$

4. Additive Identity For any vector $\boldsymbol{u} \in \boldsymbol{V}$, there exist a zero vector $\boldsymbol{0}$ such that
$\boldsymbol{0} + \boldsymbol{u} = \boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$

5. Additive Inverse For each vector $\boldsymbol{u} \in \boldsymbol{V}$, there exist a vector $-\boldsymbol{u}$ in $\boldsymbol{V}$ such that
$-\boldsymbol{u} + \boldsymbol{u} = \boldsymbol{0} = \boldsymbol{u} + (-\boldsymbol{u})$

6. Scalar Multiplication For any scalar k and a vector $\boldsymbol{u} \in \boldsymbol{V}$ implies k $\boldsymbol{u} \in \boldsymbol{V}$

7. Distributive Law For any scalar k if $\boldsymbol{u}$ & $\boldsymbol{v} \in \boldsymbol{V}$, then $k(\boldsymbol{u} + \boldsymbol{v}) = k\,\boldsymbol{u} + k\,\boldsymbol{v}$

8. For scalars m, n and for any vector $\boldsymbol{u} \in \boldsymbol{V}$, $(m + n)\,\boldsymbol{u} = m\,\boldsymbol{u} + n\,\boldsymbol{u}$

9. For scalars m, n and for any vector $\boldsymbol{u} \in \boldsymbol{V}$, $m\,(n\,\boldsymbol{u}) = (m\,n)\,\boldsymbol{u} = n\,(m\,\boldsymbol{u})$

10. For any vector $\boldsymbol{u} \in \boldsymbol{V}$, $1\boldsymbol{u} = \boldsymbol{u}$ where 1 is the multiplicative identity of real numbers.

Examples of vector spaces The following examples will specify a non empty set $\boldsymbol{V}$ and two operations: addition and scalar multiplication; then we shall verify that the ten vector space axioms are satisfied.

Example 1 Show that the set of all ordered n-tuple $\boldsymbol{R^n}$ is a vector space under the standard operations of addition and scalar multiplication.

Solution

(i) **Closure Property:**

Suppose that $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \boldsymbol{R^n}$

Then by definition, $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)$

$= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n) \in \boldsymbol{R^n}$ (By closure property)

Therefore, $\boldsymbol{R^n}$ is closed under addition.

(ii) **Commutative Property**

Suppose that $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \boldsymbol{R^n}$

Now $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)$

$= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$ (By closure property)

$= (v_1 + u_1, v_2 + u_2, \ldots, v_n + u_n)$ (By commutative law of real numbers)

$= (v_1, v_2, \ldots, v_n) + (u_1, u_2, \ldots, u_n)$ (By closure property)

$= \boldsymbol{v} + \boldsymbol{u}$

Therefore, $\boldsymbol{R^n}$ is commutative under addition.

(iii) **Associative Property**

Suppose that $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$, $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ *and* $\boldsymbol{w} = (w_1, w_2, \ldots, w_n) \in \boldsymbol{R^n}$

Now $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = [(u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)] + (w_1, w_2, \ldots, w_n)$

$= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n) + (w_1, w_2, \ldots, w_n)$ (By closure property)

$= ((u_1 + v_1) + w_1, (u_2 + v_2) + w_2, \ldots, (u_n + v_n) + w_n))$ (By closure property)

$= (u_1 + (v_1 + w_1), u_2 + (v_2 + w_2), \ldots, u_n + (v_n + w_n))$ (By associative law of real numbers)

$= (u_1, u_2, \ldots, u_n) + (v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n)$ (By closure property)

$= (u_1, u_2, \ldots, u_n) + [(v_1, v_2, \ldots, v_n) + (w_1, w_2, \ldots, w_n)]$ (By closure property)

$= \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$

Hence $\boldsymbol{R^n}$ is associative under addition.

(iv) **Additive Identity**

Suppose $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R^n}$. There exists $\boldsymbol{0} = (0, 0, \ldots, 0) \in \boldsymbol{R^n}$ such that
$\boldsymbol{0} + \boldsymbol{u} = (0, 0, \ldots, 0) + (u_1, u_2, \ldots, u_n)$

$= (0 + u_1, 0 + u_2, \ldots, 0 + u_n)$ (By closure property)

$= (u_1, u_2, \ldots, u_n) = \boldsymbol{u}$ (Existence of identity of real numbers)

Similarly, $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$

Hence $\boldsymbol{0} = (0, 0, \ldots, 0)$ is the additive identity for $\boldsymbol{R^n}$.

(v) **Additive Inverse**

Suppose $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R^n}$. There exists $-\boldsymbol{u} = (-u_1, -u_2, \ldots, -u_n) \in \boldsymbol{R^n}$

Such that $\boldsymbol{u} + (-\boldsymbol{u}) = (u_1, u_2, \ldots, u_n) + (-u_1, -u_2, \ldots, -u_n)$

$= (u_1 + (-u_1), u_2 + (-u_2), \ldots, u_n + (-u_n))$ (By closure property)

$= (0, 0, \ldots, 0) = \boldsymbol{0}$

Similarly, $(-\boldsymbol{u}) + \boldsymbol{u} = \boldsymbol{0}$

Hence the inverse of each element of $\boldsymbol{R^n}$ exists in $\boldsymbol{R^n}$.

(vi) **Scalar Multiplication**

If k is any scalar and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R^n}$.

Then by definition, $k\,\boldsymbol{u} = k\,(u_1, u_2, \ldots, u_n) = (k\,u_1, k\,u_2, \ldots, k\,u_n) \in \boldsymbol{R^n}$
(By closure property)

(vii) **Distributive Law**

Suppose k is any scalar and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$, $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \boldsymbol{R^n}$

Now $k\,(\boldsymbol{u} + \boldsymbol{v}) = k\,[(u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)]$

$= k(u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$ (By closure property)

$= (k(u_1 + v_1), k(u_2 + v_2), \ldots, k(u_n + v_n))$ (By scalar multiplication)

$= (k u_1 + k v_1, k u_2 + k v_2, \ldots, k u_n + k v_n)$ (By Distributive Law)

$= (k u_1, k u_2, \ldots, k u_n) + (k v_1, k v_2, \ldots, k v_n)$ (By closure property)

$= k(u_1, u_2, \ldots, u_n) + k(v_1, v_2, \ldots, v_n)$ (By scalar multiplication)

$= k\,\boldsymbol{u} + k\,\boldsymbol{v}$

(viii) Suppose k and l be any scalars and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R}^n$

Then $(k + l)\,\boldsymbol{u} = (k + l)(u_1, u_2, \ldots, u_n)$

$= ((k + l)u_1, (k + l)u_2, \ldots, (k + l)u_n)$ (By scalar multiplication)

$= (k u_1 + l u_1, k u_2 + l u_2, \ldots, k u_n + l u_n)$ (By Distributive Law)

$= (k u_1, k u_2, \ldots, k u_n) + (l u_1, l u_2, \ldots, l u_n)$ (By closure property)

$= k(u_1, u_2, \ldots, u_n) + l(u_1, u_2, \ldots, u_n)$ (By scalar multiplication)

$= k\,\boldsymbol{u} + l\,\boldsymbol{u}$

(ix) Suppose k and l be any scalars and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R}^n$

Then $k(l\,\boldsymbol{u}) = k[l(u_1, u_2, \ldots, u_n)]$

$= k(l u_1, l u_2, \ldots, l u_n)$ (By scalar multiplication)

$= (k(l u_1), k(l u_2), \ldots, k(l u_n))$ (By scalar multiplication)

$= ((k l)u_1, (k l)u_2, \ldots, (k l)u_n)$ (By associative law)

$= (k l)(u_1, u_2, \ldots, u_n)$ (By scalar multiplication)

$= (k l)\,\boldsymbol{u}$

(x) Suppose $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R}^n$

Then $1\,\boldsymbol{u} = 1(u_1, u_2, \ldots, u_n)$

$= (1u_1, 1u_2, \ldots, 1u_n)$ (By scalar multiplication)

$= (u_1, u_2, \ldots, u_n) = \boldsymbol{u}$ (Existence of identity in scalrs)

Hence, $\boldsymbol{R^n}$ is the real vector space with the standard operations of addition and scalar multiplication.

Note The three most important special cases of $\boldsymbol{R^n}$ are $\boldsymbol{R}$ (the real numbers), $\boldsymbol{R^2}$ (the vectors in the plane), and $\boldsymbol{R^3}$ (the vectors in 3-space).

Example 2 Show that the set $\boldsymbol{V}$ of all 2x2 matrices with real entries is a vector space if vector addition is defined to be matrix addition and vector scalar multiplication is defined to be matrix scalar multiplication.

Solution Suppose that $\boldsymbol{u} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} \in \boldsymbol{V}$

and k and l be two any scalars.

(i) **Closure property** To prove axiom (i), we must show that $\boldsymbol{u} + \boldsymbol{v}$ is an object in $\boldsymbol{V}$: that is , we must show that $\boldsymbol{u} + \boldsymbol{v}$ is a 2x2 matrix. But this is clear from the definition of matrix addition, since $\boldsymbol{u}+\boldsymbol{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix}$

(By closure property)

(ii) **Commutative property** Now it is very easy to verify the Axiom (ii)

$\boldsymbol{u}+\boldsymbol{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix}$ (By closure property)

$= \begin{bmatrix} v_{11}+u_{11} & v_{12}+u_{12} \\ v_{21}+u_{21} & v_{22}+u_{22} \end{bmatrix}$ (Commutative property of real numbers)

$= \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} + \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \boldsymbol{v}+\boldsymbol{u}$

(iii) **Associative property** $(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w} = \left(\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \right) + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$

$= \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix} + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$ (By closure property)

$= \begin{bmatrix} (u_{11}+v_{11})+w_{11} & (u_{12}+v_{12})+w_{12} \\ (u_{21}+v_{21})+w_{21} & (u_{22}+v_{22})+w_{22} \end{bmatrix}$

$= \begin{bmatrix} u_{11}+(v_{11}+w_{11}) & u_{12}+(v_{12}+w_{12}) \\ u_{21}+(v_{21}+w_{21}) & u_{22}+(v_{22}+w_{22}) \end{bmatrix}$ (By associative property of real numbers)

$$= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11}+w_{11} & v_{12}+w_{12} \\ v_{21}+w_{21} & v_{22}+w_{22} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \left(\begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} \right) = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$$

Therefore, $\boldsymbol{V}$ is associative under '+'.

(iv) **Additive Identity** Now to prove the axiom (iv), we must find an object $\boldsymbol{0}$ in $\boldsymbol{V}$ such that $\boldsymbol{0} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{0} = \boldsymbol{v}$ for all $\boldsymbol{u}$ in $\boldsymbol{V}$. This can be done by defining $\boldsymbol{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

$$\boldsymbol{0} + \boldsymbol{u} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} 0+u_{11} & 0+u_{12} \\ 0+u_{21} & 0+u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \boldsymbol{u}$$

and similarly $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$.

(v) **Additive Inverse** Now to prove the axiom (v) we must show that each object $\boldsymbol{u}$ in $\boldsymbol{V}$ has a negative $-\boldsymbol{u}$ such that $\boldsymbol{u} + (-\boldsymbol{u}) = \boldsymbol{0} = (-\boldsymbol{u}) + \boldsymbol{0}$. Defining the negative of $\boldsymbol{u}$ to be

$$-\boldsymbol{u} = \begin{bmatrix} -u_{11} & -u_{12} \\ -u_{21} & -u_{22} \end{bmatrix}.$$

$$\boldsymbol{u} + (-\boldsymbol{u}) = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} -u_{11} & -u_{12} \\ -u_{21} & -u_{22} \end{bmatrix} = \begin{bmatrix} u_{11}+(-u_{11}) & u_{12}+(-u_{12}) \\ u_{21}+(-u_{21}) & u_{22}+(-u_{22}) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \boldsymbol{0}$$

Similarly, $(-\boldsymbol{u}) + \boldsymbol{u} = \boldsymbol{0}$

(vi) **Scalar Multiplication**

Axiom (vi) also holds because for any real number k we have

$$k\boldsymbol{u} = k \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix}$$ (By closure property)

so that k $\boldsymbol{u}$ is a 2x2 matrix and consequently is an object in $\boldsymbol{V}$.

(vii) **Distributive Law**

$$k(\boldsymbol{u} + \boldsymbol{v}) = k \left(\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \right)$$

$$= k\begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix} = \begin{bmatrix} k(u_{11}+v_{11}) & k(u_{12}+v_{12}) \\ k(u_{21}+v_{21}) & k(u_{22}+v_{22}) \end{bmatrix}$$

$$= \begin{bmatrix} ku_{11}+kv_{11} & ku_{12}+kv_{12} \\ ku_{21}+kv_{21} & ku_{22}+kv_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix} + \begin{bmatrix} kv_{11} & kv_{12} \\ kv_{21} & kv_{22} \end{bmatrix}$$

$$= k\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + k\begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = k\boldsymbol{u} + k\boldsymbol{v}$$

(viii) $(k+l)\boldsymbol{u} = (k+l)\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} (k+l)u_{11} & (k+l)u_{12} \\ (k+l)u_{21} & (k+l)u_{22} \end{bmatrix}$

$$= \begin{bmatrix} ku_{11}+lu_{11} & ku_{12}+lu_{12} \\ ku_{21}+lu_{21} & ku_{22}+lu_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix} + \begin{bmatrix} lu_{11} & lu_{12} \\ lu_{21} & lu_{22} \end{bmatrix}$$

$$= k\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + l\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = k\boldsymbol{u} + l\boldsymbol{u}$$

(ix) $k(l\boldsymbol{u}) = k\left(l\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}\right) = k\begin{bmatrix} lu_{11} & lu_{12} \\ lu_{21} & lu_{22} \end{bmatrix}$

$$= \begin{bmatrix} k(lu_{11}) & k(lu_{12}) \\ k(lu_{21}) & k(lu_{22}) \end{bmatrix} = \begin{bmatrix} (kl)u_{11} & (kl)u_{12} \\ (kl)u_{21} & (kl)u_{22} \end{bmatrix} = (kl)\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = (kl)\boldsymbol{u}$$

(x) Finally axiom (x) is a simple computation

$$1\boldsymbol{u} = 1\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} 1u_{11} & 1u_{12} \\ 1u_{21} & 1u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \boldsymbol{u}$$

Hence the set of all 2x2 matrices with real entries is vector space under matrix addition and matrix scalar multiplication.

Note Example 2 is a special case of a more general class of vector spaces. The arguments in that example can be adapted to show that a set $\boldsymbol{V}$ of all $\boldsymbol{m} \times \boldsymbol{n}$ matrices with real entries, together with the operations of matrix addition and scalar multiplication, is a vector space.

Example 3 Let V be the set of all real-valued functions defined on the entire real line $(-\infty,\infty)$. If $f, g \in V$, then $f + g$ is a function defined by

$$(f+g)(x) = f(x) + g(x), \text{ for all } x \in R.$$

The product of a scalar $a \in R$ and a function f in V is defined by

$$(a f)(x) = a f(x), \text{ for all } x \in R.$$

Solution

(i) **Closure Property** If $f, g \in V$, then by definition
$(f+g)(x) = f(x) + g(x) \in V$. Therefore, V is closed under addition.

(ii) **Commutative Property** If f and g are in V, then for all $x \in R$

$(f + g)(x) = f(x) + g(x)$ (By definition)

$= g(x) + f(x)$ (By commutative property)

$= (g + f)(x)$ (By definition)

So that $f + g = g + f$

(iii) **Associative Property** If f, g and h are in V, then for all $x \in R$

$((f + g) + h)(x) = (f + g)(x) + h(x)$ (By definition)

$= (f(x) + g(x)) + h(x)$ (By definition)

$= f(x) + (g(x) + h(x))$ (By associative property)

$= f(x) + (g + h)(x)$ (By definition)

$= (f + (g + h))(x)$

And so $(f + g) + h = f + (g + h)$

(iv) **Additive Identity** The additive identity of V is the zero function defined by
$0(x) = 0$, for all $x \in R$ because $(0 + f)(x) = 0(x) + f(x)$ (By definition)

$= 0 + f(x) = f(x)$ (Existence of identity)

i.e. $0 + f = f$. Similarly, $f + 0 = f$.

(v) **Additive Inverse** The additive inverse of a function f in V is $(-1) f = -f \in V$ because

$(f + (-f))(x) = f(x) + (-f)(x)$ (By definition)

$= f(x) - f(x)$ (By definition)

$= 0$ (Existence of inverse)

i.e. $\boldsymbol{f} + (-\boldsymbol{f}) = \boldsymbol{0}$. Similarly, $(-\boldsymbol{f}) + \boldsymbol{f} = \boldsymbol{0}$.

(vi) **Scalar Multiplication** If $\boldsymbol{f}$ is in $\boldsymbol{V}$ and a is in R, then by definition $(a\,\boldsymbol{f})(x) = a\,\boldsymbol{f}(x) \in \boldsymbol{V}$.

(vii) **Distributive Law** If $\boldsymbol{f}, \boldsymbol{g}$ are in $\boldsymbol{V}$ and $a \in R$, then
$(a(\boldsymbol{f} + \boldsymbol{g}))(x) = a(\boldsymbol{f} + \boldsymbol{g})(x) = a(\boldsymbol{f}(x) + \boldsymbol{g}(x)) = a\,\boldsymbol{f}(x) + a\,\boldsymbol{g}(x)$

$= (a\,\boldsymbol{f})(x) + (a\,\boldsymbol{g})(x) = (a\,\boldsymbol{f} + a\,\boldsymbol{g})(x)$ and, therefore, $a(\boldsymbol{f} + \boldsymbol{g}) = a\,\boldsymbol{f} + a\,\boldsymbol{g}$

(viii) Let a, b in R and $\boldsymbol{f} \in \boldsymbol{V}$, then

$((a + b)\,\boldsymbol{f})(x) = (a + b)\,\boldsymbol{f}(x) = a\,\boldsymbol{f}(x) + b\,\boldsymbol{f}(x) = (a\,\boldsymbol{f})(x) + (b\,\boldsymbol{f})(x) = (a\,\boldsymbol{f} + b\,\boldsymbol{f})(x)$

Thus $(a + b)\,\boldsymbol{f} = a\,\boldsymbol{f} + b\,\boldsymbol{f}$

(ix) $a(b\,\boldsymbol{f})(x) = a(b\,\boldsymbol{f}(x)) = (a\,b)\,\boldsymbol{f}(x)$ showing that $a(b\,\boldsymbol{f}) = (a\,b)\,\boldsymbol{f}$

(x) $(1.\boldsymbol{f})(x) = 1\,\boldsymbol{f}(x) = \boldsymbol{f}(x)$ (Existence of identity)

And so $1.\boldsymbol{f} = \boldsymbol{f}$

Hence $\boldsymbol{V}$ is a real vector space.

Example 4 If $\boldsymbol{p}(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$

and $\boldsymbol{q}(x) = b_0 + b_1 x + b_2 x^2 + \ldots + b_n x^n$

We define

$$\boldsymbol{p}(x) + \boldsymbol{q}(x) = (a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n) + (b_0 + b_1 x + b_2 x^2 + \ldots + b_n x^n)$$

$$= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \ldots + (a_n + b_n)x^n$$ and for any scalar k,

$$k\boldsymbol{p}(x) = k(a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n) = ka_0 + ka_1 x + ka_2 x^2 + \ldots + k\,a_n x^n$$

Clearly the given polynomial is a vector space under the addition and scalar multiplication.

Example 5 **(The Zero Vector Space)** Let $\boldsymbol{V}$ consists of a single object, which we define by $\boldsymbol{0}$ and $\boldsymbol{0} + \boldsymbol{0} = \boldsymbol{0}$ and k $\boldsymbol{0} = \boldsymbol{0}$ for all scalars k. It is easy to check that all the vector space axioms are satisfied. We call $\boldsymbol{V} = \{\boldsymbol{0}\}$ as the zero vector space.

Example 6 (Every plane through the origin is a vector space)

Let $\boldsymbol{V}$ be any plane through the origin in $\boldsymbol{R^3}$. We shall show that the points in $\boldsymbol{V}$ form a vector space under a standard addition and scalar multiplication operations for vectors in $\boldsymbol{R^3}$.

From example1, we know that $\boldsymbol{R^3}$ itself is a vector space under these operations. Thus, Axioms 2, 3, 7, 8, 9 and 10 hold for all points in $\boldsymbol{R^3}$ and consequently for all points in the plane $\boldsymbol{V}$. We therefore need only show that Axioms 1, 4, 5 and 6 are satisfied.

Since the plane is passing through the origin, it has an equation of the form

$$a\,x + b\,y + c\,z = 0 \qquad (1)$$

Thus, if $\boldsymbol{u} = (u_1, u_2, u_3)$ and $\boldsymbol{v} = (v_1, v_2, v_3)$ are points in $\boldsymbol{V}$, then

$$a\,u_1 + b\,u_2 + c\,u_3 = 0 \text{ and } a\,v_1 + b\,v_2 + c\,v_3 = 0.$$

Adding these equations gives $a\,(u_1 + v_1) + b\,(u_2 + v_2) + c\,(u_3 + v_3) = 0$

This equality tell us that the coordinates of the point

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3)$$

satisfies (1); thus, $\boldsymbol{u} + \boldsymbol{v}$ lies in plane $\boldsymbol{V}$. This proves that the Axiom 1 is satisfied.

There exists $\boldsymbol{0} = (0, 0, 0)$ such that $a\,(0) + b\,(0) + c\,(0) = 0$. Therefore, Axiom 4 is satisfied.

Multiplying $a\,u_1 + b\,u_2 + c\,u_3 = 0$ through by k gives

$$a\,(ku_1) + b\,(ku_2) + c\,(ku_3) = 0$$

Thus, $(ku_1, k\,u_2, k\,u_3) = k\,(u_1, u_2, u_3) = k\,\boldsymbol{u} \in \boldsymbol{V}$. Hence, Axiom 6 is satisfied.

We shall prove the axiom 5 is satisfied. Multiplying $a\,u_1 + b\,u_2 + c\,u_3 = 0$ through by -1 gives a $(-1u_1) + b\,(-1u_2) + c\,(-1u_3) = 0$

Thus, $(-u_1, -u_2, -u_3) = -\,(u_1, u_2, u_3) = -\boldsymbol{u} \in \mathbf{V}$. This establishes Axiom 5.

Example 7 (A set that is not a vector space)

Let $\boldsymbol{V}=\boldsymbol{R^2}$ and define addition and scalar multiplication operation as follows. If $\boldsymbol{u} = (u_1, u_2)$ and $\boldsymbol{v} = (v_1, v_2)$ then define

$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2)$ and if k is any real number then define $k\boldsymbol{u} = (ku_1, 0)$.

For any vector $\boldsymbol{u} \in \boldsymbol{V}$, $1\boldsymbol{u} = 1(u_1, u_2) = (1\,u_1, 0) = (u_1, 0) \neq \boldsymbol{u}$ where 1 is the multiplicative identity of real numbers. Therefore, the axiom 10 is not satisfied.

Hence, $\boldsymbol{V}=\boldsymbol{R^2}$ is not a vector space.

Theorem 1 Let $\boldsymbol{V}$ be a vector space, $\boldsymbol{u}$ a vector in $\boldsymbol{V}$, and k is a scalar, then

(i) $0\boldsymbol{u} = \boldsymbol{0}$ (ii) $k\boldsymbol{0} = \boldsymbol{0}$

(iii) $(-1)\,\boldsymbol{u} = -\boldsymbol{u}$ (iv) If $k\,\boldsymbol{u} = \boldsymbol{0}$ then $k = 0$ or $\boldsymbol{u} = \boldsymbol{0}$

Definition A subset $\boldsymbol{W}$ of a vector space $\boldsymbol{V}$ is called a subspace of $\boldsymbol{V}$ if $\boldsymbol{W}$ itself a vector space under the addition and scalar multiplication defined on $\boldsymbol{V}$.

Note If $\boldsymbol{W}$ is a part of a larger set $\boldsymbol{V}$ that is already known to be a vector space, then certain axioms need not be verified for $\boldsymbol{W}$ because they are "inherited" from $\boldsymbol{V}$. For example, there is no need to check that $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ (Axiom 2) for $\boldsymbol{W}$ because this holds for all vectors in $\boldsymbol{V}$ and consequently for all vectors in $\boldsymbol{W}$. Other Axioms are inherited by $\boldsymbol{W}$ from $\boldsymbol{V}$ are 3, 7, 8, 9, and 10. Thus, to show that a set $\boldsymbol{W}$ is a subspace of a vector space $\boldsymbol{V}$, we need only verify Axioms 1, 4, 5 and 6. The following theorem shows that even Axioms 4 and 5 can be omitted.

Theorem 2 If $\boldsymbol{W}$ is a set of one or more vectors from a vector space $\boldsymbol{V}$, then $\boldsymbol{W}$ is subspace of $\boldsymbol{V}$ if and only if the following conditions hold.

(a) If $\boldsymbol{u}$ and $\boldsymbol{v}$ are vectors in $\boldsymbol{W}$, then $\boldsymbol{u} + \boldsymbol{v}$ is in $\boldsymbol{W}$
(b) If k is any scalar and $\boldsymbol{u}$ is any vector in $\boldsymbol{W}$, then $k\,\boldsymbol{u}$ is in $\boldsymbol{W}$.

Proof If $\boldsymbol{W}$ is a subspace of $\boldsymbol{V}$, then all the vector space axioms are satisfied; in particular, Axioms 1 and 6 hold. But these are precisely conditions (a) and (b).

Conversely, assume conditions (a) and (b) hold. Since these conditions are vector space Axioms 1 and 6, we need only show that $\boldsymbol{W}$ satisfies the remaining 8 axioms. The vectors in $\boldsymbol{W}$ automatically satisfy axioms 2, 3, 7, 8, 9, and 10 since they are satisfied by all vectors in $\boldsymbol{V}$. Therefore, to complete the proof, we need only to verify that vectors in $\boldsymbol{W}$ satisfy axioms 4 and 5.

Let $\boldsymbol{u}$ be any vector in $\boldsymbol{W}$. By condition (b), $k\,\boldsymbol{u}$ is in $\boldsymbol{W}$ for every scalar k. Setting $k = 0$, it follows from theorem 1 that $0\,\boldsymbol{u} = \boldsymbol{0}$ is in $\boldsymbol{W}$, and setting $k = -1$, it follows that $(-1)\,\boldsymbol{u} = -\boldsymbol{u}$ is in $\boldsymbol{W}$. □

Remark

(1) The theorem states that $\boldsymbol{W}$ is a subspace of $\boldsymbol{V}$ if and only if $\boldsymbol{W}$ is closed under addition and closed under scalar multiplication.

(2) Every vector space has at least two subspaces, itself and the subspace $\{\boldsymbol{0}\}$ consisting only of the zero vector. Thus the subspace $\{\boldsymbol{0}\}$ is called the zero subspace.

Example 8 Let $\boldsymbol{W}$ be the subset of $\boldsymbol{R^3}$ consisting of the all the vectors of the form $(a, b, 0)$, where a and b are real numbers. To check if $\boldsymbol{W}$ is subspace of $\boldsymbol{R^3}$, we first see that axiom 1 and 6 of a vector space holds.

Let $\boldsymbol{u} = (a_1, b_1, 0)$ *and* $\boldsymbol{v} = (a_2, b_2, 0)$ be vectors in $\boldsymbol{W}$ then $\boldsymbol{u} + \boldsymbol{v} = (a_1, b_1, 0) + (a_2, b_2, 0) = (a_1 + a_2, b_1 + b_2, 0)$ is in $\boldsymbol{W}$. Since the third component is zero. Also c is scalar, and then $c\boldsymbol{u} = c(a_1, b_1, 0) = (ca_1, cb_1, 0)$ is in $\boldsymbol{W}$. Therefore the Ist and 6th axioms of the vector space holds. We can also verify the other axioms of vector space. Hence $\boldsymbol{W}$ is a subspace.

Example 9 Consider the set $\boldsymbol{W}$ consisting of all 2x3 matrices of the form $\begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix}$, Where a, b, c and d are arbitrary real numbers. Show that the $\boldsymbol{W}$ is a subspace $\boldsymbol{M_{2x3}}$.

Solution Consider $\boldsymbol{u} = \begin{bmatrix} a_1 & b_1 & 0 \\ 0 & c_1 & d_1 \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} a_2 & b_2 & 0 \\ 0 & c_2 & d_2 \end{bmatrix}$ in $\boldsymbol{W}$

Then $\boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} a_1 & b_1 & 0 \\ 0 & c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 & 0 \\ 0 & c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 & 0 \\ 0 & c_1 + c_2 & d_1 + d_2 \end{bmatrix}$ is in $\boldsymbol{W}$.

So that the (a) part of the theorem is satisfied. Also k is a scalar, and then $k\boldsymbol{u} = \begin{bmatrix} ka_1 & kb_1 & 0 \\ 0 & kc_1 & kd_1 \end{bmatrix}$ is in $\boldsymbol{W}$. So the (b) part of the above theorem is also satisfied.

Hence $\boldsymbol{W}$ is a subspace of $\boldsymbol{M_{2x3}}$.

Note Let $\boldsymbol{V}$ is a vector space then every subset of $\boldsymbol{V}$ is not necessary a subspace of $\boldsymbol{V}$. For example, let $\boldsymbol{V} = \boldsymbol{R^2}$ then any line in $\boldsymbol{R^2}$ not passing through origin is not a subspace of $\boldsymbol{R^2}$. Similarly, a plane in $\boldsymbol{R^3}$ not passing through the origin is not a subspace of $\boldsymbol{R^3}$.

Example 10 Let $\boldsymbol{W}$ be the subset of $\boldsymbol{R^3}$ consisting of all vectors of the form $(a, b, 1)$, where a, b are any real numbers. To check whether property (a) and (b) of the above theorem holds. Let $\boldsymbol{u} = (a_1, b_1, 1)$ *and* $\boldsymbol{v} = (a_2, b_2, 1)$ be vectors in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (a_1, b_1, 1) + (a_2, b_2, 1) = (a_1 + a_2, b_1 + b_2, 1 + 1)$ which is not in $\boldsymbol{W}$ because the third component 2 is not 1. As the Ist property does not hold therefore, the given set of vectors is not a vector space.

Example 11 Which of the following are subspaces of $\boldsymbol{R^3}$

(i) All vectors of the form $(a, 0, 0)$

(ii) All vectors of the form $(a, 1, 1)$

(iii) All vectors of the form (a, b, c), where $b = a + c$

(iv) All vectors of the form (a, b, c), where $b = a + c + 1$

Solution Let $\boldsymbol{W}$ is the set of all vectors of the form $(a, 0, 0)$.

(i) Suppose $\boldsymbol{u} = (u_1, 0, 0)$ and $\boldsymbol{v} = (v_1, 0, 0)$ are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, 0, 0) + (v_1, 0, 0) = (u_1 + v_1, 0, 0)$ which is of the form $(a, 0, 0)$.

Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If k is any scalar and $\boldsymbol{u} = (u_1, 0, 0)$ is any vector in $\boldsymbol{W}$, then $k\,\boldsymbol{u} = k\,(u_1, 0, 0) = (k\,u_1, 0, 0)$
which is of the form $(a, 0, 0)$. Therefore, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Hence $\boldsymbol{W}$ is the subspace of $\boldsymbol{R^3}$.

(ii) Let $\boldsymbol{W}$ is the set of all vectors of the form $(a, 1, 1)$.

Suppose $\boldsymbol{u} = (u_1, 1, 1)$ and $\boldsymbol{v} = (v_1, 1, 1)$ are in $\boldsymbol{W}$. Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, 1, 1) + (v_1, 1, 1) = (u_1 + v_1, 2, 2)$ which is not of the form $(a, 1, 1)$. Therefore, $\boldsymbol{u} + \boldsymbol{v} \notin \boldsymbol{W}$. Hence $\boldsymbol{W}$ is not the subspace of $\boldsymbol{R^3}$.

(iii) Suppose $\boldsymbol{W}$ is the set of all vectors of the form (a, b, c), where $b = a + c$

Suppose $\boldsymbol{u} = (u_1, u_1 + u_3, u_3)$ and $\boldsymbol{v} = (v_1, v_1 + v_3, v_3)$ are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_1 + u_3, u_3) + (v_1, v_1 + v_3, v_3)$
$= (u_1 + v_1, u_1 + u_3 + v_1 + v_3, u_3 + v_3)$
$= (u_1 + v_1, (u_1 + v_1) + (u_3 + v_3), u_3 + v_3)$, which is of the form $(a, a + c, c)$.

Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If k is any scalar and $\boldsymbol{u} = (u_1, u_1 + u_3, u_3)$ is any vector in $\boldsymbol{W}$, then
$k\,\boldsymbol{u} = k\,(u_1, u_1 + u_3, u_3) = (k\,u_1, k\,(u_1 + u_3), k\,u_3)$ (By definition)
$= (k\,u_1, k\,u_1 + k\,u_3, k\,u_3)$ (By Distributive Law)
Which is of the form $(a, a + c, c)$. Therefore, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Hence $\boldsymbol{W}$ is the subspace of $\boldsymbol{R^3}$.

(iv) Let $\boldsymbol{W}$ is the set of all vectors of the form (a, b, c), where $b = a + c + 1$
Suppose $\boldsymbol{u} = (u_1, u_1 + u_3 + 1, u_3)$ and $\boldsymbol{v} = (v_1, v_1 + v_3 + 1, v_3)$ are in $\boldsymbol{W}$.
Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_1 + u_3 + 1, u_3) + (v_1, v_1 + v_3 + 1, v_3)$
$= (u_1 + v_1, u_1 + u_3 + 1 + v_1 + v_3 + 1, u_3 + v_3)$
$= (u_1 + v_1, (u_1 + v_1) + (u_3 + v_3) + 2, u_3 + v_3)$
Which is not of the form $(a, a + c + 1, c)$. Therefore, $\boldsymbol{u} + \boldsymbol{v} \notin \boldsymbol{W}$. Hence $\boldsymbol{W}$ is not the subspace of $\boldsymbol{R^3}$.

Example 12 Determine which of the following are subspaces of $\boldsymbol{P_3}$.
(i) All polynomials $a_0 + a_1\,\boldsymbol{x} + a_2\,\boldsymbol{x^2} + a_3\,\boldsymbol{x^3}$ for which $a_0 = 0$
(ii) All polynomials $a_0 + a_1\,\boldsymbol{x} + a_2\,\boldsymbol{x^2} + a_3\,\boldsymbol{x^3}$ for which $a_0 + a_1 + a_2 + a_3 = 0$
(iii) All polynomials $a_0 + a_1\,\boldsymbol{x} + a_2\,\boldsymbol{x^2} + a_3\,\boldsymbol{x^3}$ for which a_0, a_1, a_2, and a_3 are integers
(iv) All polynomials of the form $a_0 + a_1\,\boldsymbol{x}$, where a_0 and a_1 are real numbers.

Solution (i) Let W is the set of all polynomials $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ for which $a_0 = 0$.
Suppose that $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 = 0$) and $v = b_0 + b_1 x + b_2 x^2 + b_3 x^3$ (where $b_0 = 0$) are in W. Then $u + v = (c_0 + c_1x + c_2x^2 + c_3x^3) + (b_0 + b_1x + b_2x^2 + b_3x^3) = (c_0 + b_0) + (c_1 + b_1)x + (c_2 + b_2)x^2 + (c_3 + b_3) x^3$, where $c_0 + b_0 = 0$.
Therefore, $u + v \in W$

If k is any scalar and $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 = 0$) is any vector in W.
Then $k\,u = k\,(c_0 + c_1x + c_2x^2 + c_3x^3) = (kc_0) + (kc_1)\,x + (kc_2)\,x^2 + (kc_3)\,x^3$ where $k\,c_0 = 0$. Therefore, $k\,u \in W$. Hence W is the subspace of P_3.

(ii) Let W is the set of all polynomials $a_0 + a_1x + a_2x^2 + a_3x^3$ for which $a_0+a_1+a_2+a_3 = 0$.
Suppose that $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 + c_1 + c_2 + c_3 = 0$) and $v = b_0 + b_1 x + b_2 x^2 + b_3 x^3$ (where $b_0 + b_1 + b_2 + b_3 = 0$) are in W.

Now
$u + v = (c_0 + c_1 x + c_2 x^2 + c_3 x^3) + (b_0 + b_1 x + b_2 x^2 + b_3 x^3)$

$= (c_0 + b_0) + (c_1 + b_1) x + (c_2 + b_2) x^2 + (c_3 + b_3) x^3$

Where $(c_0+b_0) + (c_1+b_1) + (c_2+b_2) + (c_3+b_3) = (c_0+c_1+c_2+c_3) + (b_0+b_1+b_2+b_3) = 0 + 0 = 0$. Therefore, $u + v \in W$

If k is any scalar and $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 + c_1 + c_2 + c_3 = 0$) is any vector in W. Then $k\,u = k\,(c_0 + c_1x + c_2x^2 + c_3x^3) = (kc_0) + (kc_1)\,x + (kc_2)\,x^2 + (kc_3)\,x^3$
Where $(k\,c_0) + (k\,c_1) + (k\,c_2) + (k\,c_3) = k(c_0 + c_1 + c_2 + c_3) = k.0 = 0$

Therefore, $k\,u \in W$. Hence W is the subspace of P_3.

(iii) Let W is the set of all polynomials $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ for which a_0, a_1, a_2, and a_3 are integers.
Suppose that the vectors $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where c_0, c_1, c_2, and c_3 are integers) and $v = b_0 + b_1 x + b_2 x^2 + b_3 x^3$ (where b_0, b_1, b_2, and b_3 are integers) are in W.
Now
$u + v = (c_0 + c_1 x + c_2 x^2 + c_3 x^3) + (b_0 + b_1 x + b_2 x^2 + b_3 x^3)$
$= (c_0 + b_0) + (c_1 + b_1) x + (c_2 + b_2) x^2 + (c_3 + b_3) x^3$, where
$(c_0 + b_0)$, $(c_1 + b_1)$, $(c_2 + b_2)$, and $(c_3 + b_3)$ are integers (integers are closed under addition). Therefore, $u + v \in W$

If k is any scalar and $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where c_0, c_1, c_2, and c_3 are integers) is any vector in W. Then $k\,u = k\,(c_0 + c_1x + c_2x^2 + c_3x^3 = (kc_0) + (kc_1)\,x + (kc_2)\,x^2 + (kc_3)\,x^3$, where $(k\,c_0)$, $(k\,c_1)$, $(k\,c_2)$, and $(k\,c_3)$ are not integers (product of real number and integer). Therefore, $k\,u \notin W$. Hence, W is not the subspace of P_3.

(iv) Let $\boldsymbol{W}$ is the set of all polynomials of the form $a_0 + a_1 \boldsymbol{x}$, where a_0 and a_1 are real numbers. Suppose that $\boldsymbol{u} = c_0 + c_1 \boldsymbol{x}$ (where c_0 and c_1 are real numbers) and $\boldsymbol{v} = b_0 + b_1 \boldsymbol{x}$ (where b_0 and b_1 are real numbers) are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (c_0 + c_1 \boldsymbol{x}) + (b_0 + b_1 \boldsymbol{x}) = (c_0 + b_0) + (c_1 + b_1) \boldsymbol{x}$
Where $(c_0 + b_0)$ and $(c_1 + b_1)$ are real numbers.
Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If k is any scalar and $\boldsymbol{u} = c_0 + c_1 \boldsymbol{x}$ (where c_0 and c_1 are real numbers) is any vector in $\boldsymbol{W}$.
Then $k\,\boldsymbol{u} = k(c_0 + c_1 \boldsymbol{x}) = (k\,c_0) + (k\,c_1) \boldsymbol{x}$
Where $(k\,c_0)$ and $(k\,c_1)$ are real numbers.
Therefore, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Hence $\boldsymbol{W}$ is the subspace of $\boldsymbol{P_3}$.

Example 13 Determine which of the following are subspaces of $\boldsymbol{M_{22}}$.

(i) All matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b + c + d = 0$

(ii) All 2 x 2 matrices $\boldsymbol{A}$ such that det $(\boldsymbol{A}) = 0$

(iii) All the matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$

Solution Let $\boldsymbol{W}$ is the set of all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b + c + d = 0$.

(i) Suppose $\boldsymbol{u} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ (where $e + f + g + h = 0$) and $\boldsymbol{v} = \begin{bmatrix} l & m \\ n & p \end{bmatrix}$
(Where $l + m + n + p = 0$) are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} e & f \\ g & h \end{bmatrix} + \begin{bmatrix} l & m \\ n & p \end{bmatrix} = \begin{bmatrix} e+l & f+m \\ g+n & h+p \end{bmatrix}$ (By definition)

Where $(e + l) + (f + m) + (g + n) + (h + p)$
$= (e + f + g + h) + (l + m + n + p) = 0 + 0 = 0$
Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If k is any scalar and $\boldsymbol{u} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ (where $e + f + g + h = 0$) is any vector in $\boldsymbol{W}$.

Then $k\,\boldsymbol{u} = k\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ke & kf \\ kg & kh \end{bmatrix}$ (by definition)

Where $k\,e + k\,f + k\,g + k\,h = k\,(e + f + g + h) = k\,0 = 0$
Hence, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Therefore, $\boldsymbol{W}$ is subspace of $\boldsymbol{M_{22}}$.

(ii) Let $\boldsymbol{W}$ is the set of all 2 x 2 matrices $\boldsymbol{A}$ such that det $(\boldsymbol{A}) = 0$

Suppose $\boldsymbol{u} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ (Where det $(\boldsymbol{u}) = e\,h - f\,g = 0$) and $\boldsymbol{v} = \begin{bmatrix} l & m \\ n & p \end{bmatrix}$

(Where det $(v) = l\,p - m\,n = 0$) are in W.

Then $u + v = \begin{bmatrix} e & f \\ g & h \end{bmatrix} + \begin{bmatrix} l & m \\ n & p \end{bmatrix} = \begin{bmatrix} e+l & f+m \\ g+n & h+p \end{bmatrix}$ (By definition)

Where det $(u + v) = (e + l)(h + p) - (f + m)(g + n)$

$= e\,h + e\,p + l\,h + l\,p - f\,g - f\,n - m\,g - m\,n$

$= (e\,h - f\,g) + (l\,p - m\,n) + e\,p + l\,h - f\,n - m\,g = e\,p + l\,h - f\,n - m\,g \neq 0$

Therefore, $u + v \notin W$. Therefore, W is not subspace of M_{22}.

(iii) Let W is the set of all matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$

Suppose $u = \begin{bmatrix} e & f \\ 0 & g \end{bmatrix}$ and $v = \begin{bmatrix} l & m \\ 0 & n \end{bmatrix}$ are in W.

Then $u + v = \begin{bmatrix} e & f \\ 0 & g \end{bmatrix} + \begin{bmatrix} l & m \\ 0 & n \end{bmatrix} = \begin{bmatrix} e+l & f+m \\ 0 & g+n \end{bmatrix}$ (By definition)

Which is of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$. Therefore, $u + v \in W$

If k is any scalar and $u = \begin{bmatrix} e & f \\ 0 & g \end{bmatrix}$ is any vector in W.

Then $k\,u = k\begin{bmatrix} e & f \\ 0 & g \end{bmatrix} = \begin{bmatrix} ke & kf \\ 0 & kg \end{bmatrix}$ (By definition)

Which is of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$. Hence, $k\,u \in W$

Therefore, W is subspace of M_{22}.

Example 14 Determine which of the following are subspaces of the space $F(-\infty,\infty)$.

(i) All f such that $f(x) \leq 0$ for all x (ii) all f such that $f(0) = 0$

(iii) All f such that $f(0) = 2$ (iv) all constant functions

(v) All f of the form $k_1 + k_2 \sin x$, where k_1 and k_2 are real numbers

(vi) All everywhere differentiable functions that satisfy $f' + 2f = 0$.

Solution **(i)** Let W is the set of all f such that $f(x) \leq 0$ for all x.

Suppose g and h are the vectors in W. Then $g(x) \leq 0$ for all x and $h(x) \leq 0$ for all x.

Now $(g + h)(x) = g(x) + h(x) \leq 0$. Therefore, $g + h \in W$

If k is any scalar and g is any vector in W. Then $g(x) \leq 0$ for all x

Now $(k\,g)(x) = k\,g(x)$, which is greater than 0 for negative real values of k.

$\therefore kg \notin W \quad \forall\ k < 0$.

Hence W is not the subspace of $F(-\infty,\infty)$.

(ii) Let W is the set of all f such that $f(0) = 0$.
Suppose g and h are the vectors in W. Then $g(0) = 0$ and $h(0) = 0$
Now $(g + h)(0) = g(0) + h(0) = 0 + 0 = 0$. Therefore, $g + h \in W$
If k is any scalar and g is any vector in W. Then $g(0) = 0$
Now $(k g)(0) = k\, g(0) = k\, 0 = 0$. $\therefore kg \in W$. Hence W is the subspace of $F(-\infty,\infty)$.

(iii) Let W is the set of all f such that $f(0) = 2$
Suppose g and h are the vectors in W. Then $g(0) = 2$ and $h(0) = 2$
Now $(g + h)(0) = g(0) + h(0) = 2 + 2 \neq 2 \therefore kg \notin W$. Hence W is not the subspace of $F(-\infty,\infty)$.

(iv) Let W is the set of all constant functions. Suppose g and h are the vectors in W.
Then $g(x) = a$ and $h(x) = b$, where a and b are constants.
Now $(g + h)(x) = g(x) + h(x) = a + b$, which is constant. Therefore, $g + h \in W$
If k is any scalar and g is any vector in W. Then $g(x) = a$, where a is any constant.
Now $(k g)(x) = k\, g(x) = k\, a$, which is a constant. $\therefore kg \in W$. Hence W is the subspace of $F(-\infty,\infty)$.

(v) Let W is the set of all f of the form $k_1 + k_2 \sin x$, where k_1 and k_2 are real numbers
Suppose g and h are the vectors in W. Then $g(x) = m_1 + m_2 \sin x$ and $h(x) = n_1 + n_2 \sin x$, where m_1, m_2, $n_1$and n_2 are real numbers.

Now $(g + h)(x) = g(x) + h(x) = [m_1 + m_2 \sin x] + [n_1 + n_2 \sin x] = (m_1 + n_1) + (m_2 + n_2) \sin x$
Which is of the form $k_1 + k_2 \sin x$. Therefore, $g + h \in W$

If k is any scalar and g is any vector in W. Then $g(x) = m_1 + m_2 \sin x$, where m_1 and m_2 are any real numbers.

Now $(k g)(x) = k\, g(x) = k\,[m_1 + m_2 \sin x] = (k\, m_1) + (k\, m_2) \sin x$
Which is of the form $k_1 + k_2 \sin x$. $\therefore kg \in W$. Hence W is the subspace of $F(-\infty,\infty)$.

(vi) Let W is the set of all everywhere differentiable functions that satisfy $f' + 2f = 0$.
Suppose g and h are the vectors in W. Then $g' + 2g = 0$ and $h' + 2h = 0$
Now $(g + h)' + 2(g + h) = g' + h' + 2(g + h) = (g' + 2g) + (h' + 2h) = 0 + 0 = 0$
Therefore, $g + h \in W$
If k is any scalar and g is any vector in W. Then $g' + 2g = 0$
Now $(kg)' + 2(kg) = kg' + 2kg = k(g' + 2g) = k.0 = 0$
$\therefore kg \in W$. Hence W is the subspace of $F(-\infty,\infty)$.

Remark Let n be a non-negative integer, and let $\boldsymbol{P_n}$ be the set of real valued function of the form $\boldsymbol{p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n}$ where $a_0, a_1, a_2, \ldots, a_n$ are real numbers, then $\boldsymbol{P_n}$ is a subspace $\boldsymbol{F(-\infty, \infty)}$.

Example 15 Show that the invertible n x n matrices do not form a subspace of $\boldsymbol{M_{n\,x\,n}}$.

Solution Let $\boldsymbol{W}$ is the set of invertible matrices in $\boldsymbol{M_{n\,x\,n}}$. This set fails to be a subspace on both counts- it is closed under neither scalar multiplication nor addition.

For example consider invertible matrices $\boldsymbol{W} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, \boldsymbol{V} = \begin{bmatrix} -1 & 2 \\ -2 & 5 \end{bmatrix}$ in $\boldsymbol{M_{n\,x\,n}}$.

The matrix $\boldsymbol{0.U}$ is a 2x2 zero matrix, hence is not invertible; and the matrix $\boldsymbol{U + V}$ has a column of zeros, hence is not invertible.

Theorem If $\boldsymbol{Ax = 0}$ is a homogeneous linear system of m equations in n unknowns, then the set of solution vectors is a subspace of $\boldsymbol{R^n}$.

Example 16 Consider the linear systems

(a) $\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ *(b)* $\begin{bmatrix} 1 & -2 & 3 \\ -3 & 7 & -8 \\ -2 & 4 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & -2 & 3 \\ -3 & 7 & -8 \\ 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ *(d)* $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Each of the systems has three unknowns, so the solutions form subspaces of $\boldsymbol{R^3}$. Geometrically, this means that each solution space must be a line through origin, a plane through origin, the origin only, or all of $\boldsymbol{R^3}$.

Solution *(a)* The solutions are $x = 2s - 3t, y = s, z = t$. From which it follows that $x = 2y - 3z$ *or* $x - 2y + 3z = 0$.
This is the equation of the plane through the origin with $\boldsymbol{n} = (1, -2, 3)$ as a normal vector.

(b) The solutions are $x = -5t, y = -t, z = t$, which are parametric equations for the line through the origin parallel to the vector $\boldsymbol{v} = (-5, -1, 1)$.

(c) The solution is $x = 0, y = 0, z = 0$ so the solution space is the origin only, that is $\{\boldsymbol{0}\}$.

(d) The solutions are $x = r, y = s, z = t$. where r, s and t have arbitrary values, so the solution space is all $\boldsymbol{R^3}$.

A Subspace Spanned by a Set: The next example illustrates one of the most common ways of describing a subspace. We know that the term linear combination refers to any

sum of scalar multiples of vectors, and Span $\{v_1, \ldots, v_p\}$ denotes the set of all vectors that can be written as linear combinations of $v_1, \ldots, v_p$.

Example 17 Given v_1 and v_2 in a vector space V, let H = Span $\{v_1, v_2\}$. Show that H is a subspace of V.

Solution The zero vector is in H, since $0 = 0v_1 + 0v_2$. To show that H is closed under vector addition, take two arbitrary vectors in H, say,

$$u = s_1v_1 + s_2v_2 \quad \text{and} \quad w = t_1v_1 + t_2v_2$$

By Axioms 2, 3 and 8 for the vector space V.

$$u + w = (s_1v_1 + s_2v_2) + (t_1v_1 + t_2v_2)$$
$$= (s_1 + t_1)v_1 + (s_2 + t_2)v_2$$

So $u + w$ is in H. Furthermore, if c is any scalar, then by Axioms 7 and 9,

$$cu = c(s_1v_1 + s_2v_2) = (cs_1)v_1 + (cs_2)v_2$$

Which shows that cu is in H and H is closed under scalar multiplication.

Thus H is a subspace of V. □

Later on we will prove that every nonzero subspace of R^3, other than R^3 itself, is either Span $\{v_1, v_2\}$ for some linearly independent v_1 and v_2 or Span $\{v\}$ for $v \neq 0$. In the first case the subspace is a plane through the origin and in the second case a line through the origin. (See Figure below) It is helpful to keep these geometric pictures in mind, even for an abstract vector space.

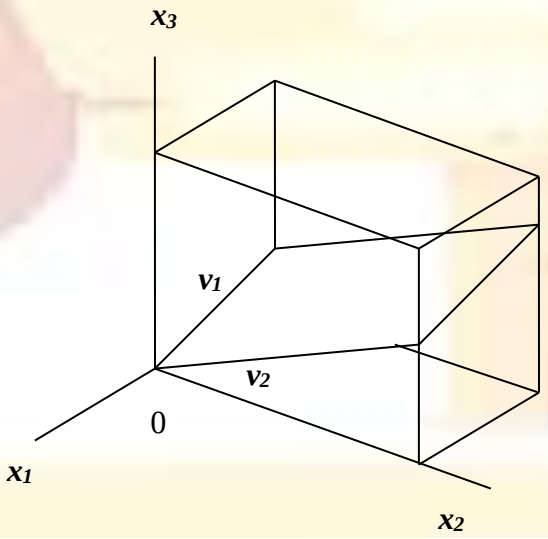


Figure 9 – An example of a subspace

The argument in Example 17 can easily be generalized to prove the following theorem.

Theorem 3 If $v_1, \ldots, v_p$ are in a vector space V, then Span $\{v_1, \ldots, v_p\}$ is a subspace of V.

We call Span $\{v_1, \ldots, v_p\}$ the subspace spanned (or generated) by $\{v_1, \ldots, v_p\}$. Given any subspace H of V, a spanning (or generating) set for H is a set $\{v_1, \ldots, v_p\}$ in H such that H = Span $\{v_1, \ldots, v_p\}$.

Proof

The zero vector is in H, since $0 = 0v_1 + 0v_2 + \ldots + 0v_n = \sum_{j=0}^{j=n} 0V_j = 0\left(\sum_{j=0}^{j=n} V_j\right) = 0$

To show that H is closed under vector addition, take two arbitrary vectors in H, say,

$$u = s_1v_1 + s_2v_2 + \ldots + t_nv_n = \sum_{i=0}^{i=n} s_iV_i$$

and

$$w = t_1v_1 + t_2v_2 \ldots + t_nv_n = \sum_{k=0}^{k=n} t_kV_k$$

By Axioms 2, 3 and 8 for the vector space V.

$$u + w = \sum_{i=0}^{i=n} s_iV_i + \sum_{k=0}^{k=n} t_kV_k = (s_1v_1 + s_2v_2 + \ldots + s_nv_n) + (t_1v_1 + t_2v_2 \ldots + t_nv_n)$$

$$= (s_1 + t_1)v_1 + (s_2 + t_2)v_2 + \ldots + (s_n + t_n)v_n = \sum_{p=0}^{p=n} (s_p + t_p)V_p$$

So $u + w$ is in H. Furthermore, if c is any scalar, then by Axioms 7 and 9,

$$cu = c(s_1v_1 + s_2v_2 + \ldots + s_nv_n) = (cs_1)v_1 + (cs_2)v_2 + \ldots + (cs_n)v_n = \sum_{r=0}^{r=n} cs_rV_r .$$

Which shows that cu is in H and H is closed under scalar multiplication.

Thus H is a subspace of V. □

Example 18 Let H be the set of all vectors of the form $(a - 3b, b - a, a, b)$, where a and b are arbitrary scalars. That is, let $H = \{(a - 3b, b - a, a, b): a \text{ and } b \text{ in } R\}$. Show that H is a subspace of R^4.

Solution Write the vectors in H as column vectors. Then an arbitrary vector in H has the form

$$\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

v_1 v_2

This calculation shows that $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, where $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are the vectors indicated above. Thus $\boldsymbol{H}$ is a subspace of $\boldsymbol{R^4}$ by Theorem 3. □

Example18 illustrates a useful technique of expressing a subspace $\boldsymbol{H}$ as the set of linear combinations of some small collection of vectors. If $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$, we can think of the vectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}$ in the spanning set as "handles" that allow us to hold on to the subspace $\boldsymbol{H}$. *Calculations with the infinitely many vectors in* $\boldsymbol{H}$ *are often reduced to operations with the finite number of vectors in the spanning set.*



Exercises

In exercises 1-13 a set of objects is given together with operations of addition and scalar multiplication. Determine which sets are vector spaces under the given operations. For those that are not, list all axioms that fail to hold.

1. The set of all triples of real numbers *(x, y, z)* with the operations
$$(x,y,z)+(x',y',z') = (x+x',y+y',z+z') \text{ and } k(x,y,z) = (kx,y,z)$$

2. The set of all triples of real numbers *(x, y, z)* with the operations
$$(x,y,z)+(x',y',z') = (x+x',y+y',z+z') \text{ and } k(x,y,z) = (0,0,0)$$

3. The set of all pairs of real numbers *(x, y)* with the operations
$$(x,y)+(x',y') = (x+x',y+y') \text{ and } k(x,y) = (2kx,2ky)$$

4. The set of all pairs of real numbers of the form *(x, 0)* with the standard operations on $\boldsymbol{R^2}$.

5. The set of all pairs of real numbers of the form *(x, y)*, where $x \geq 0$, with the standard operations on $\boldsymbol{R^2}$.

6. The set of all n-tuples of real numbers of the form *(x, x, …, x)* with the standard operations on $\boldsymbol{R^n}$.

7. The set of all pairs of real numbers *(x, y)* with the operations.
$$(x,y)+(x',y') = (x+x'+1,y+y'+1) \text{ and } k(x,y) = (kx,ky)$$

8. The set of all 2x2 matrices of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with matrix addition and scalar multiplication.

9. The set of all 2x2 matrices of the form $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with matrix addition and scalar multiplication.

10. The set of all pairs of real numbers of the form *(1, x)* with the operations
$$(1,y)+(1,y') = (1,y+y') \text{ and } k(1,y) = (1,ky)$$

11. The set of polynomials of the form $\boldsymbol{a} + b\boldsymbol{x}$ with the operations
$$(\boldsymbol{a_0}+a_1\boldsymbol{x})+(\boldsymbol{b_0}+b_1\boldsymbol{x}) = (\boldsymbol{a_0}+\boldsymbol{b_0})+(a_1+b_1)\boldsymbol{x} \text{ and } k(\boldsymbol{a_0}+a_1\boldsymbol{x}) = (k\boldsymbol{a_0})+(ka_1)\boldsymbol{x}$$

12. The set of all positive real numbers with operations $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{xy}$ and $k\boldsymbol{x} = \boldsymbol{x}^k$

13. The set of all real numbers *(x, y)* with operations

$$(x, y)+(x', y')=(xx', yy') \text{ and } k(x, y)=(kx, ky)$$

14. Determine which of the following are subspaces of $\boldsymbol{M_{nn}}$.
(a) all n x n matrices $\boldsymbol{A}$ such that tr $(\boldsymbol{A}) = 0$
(b) all n x n matrices $\boldsymbol{A}$ such that $\boldsymbol{A}^T = -\boldsymbol{A}$
(c) all n x n matrices $\boldsymbol{A}$ such that the linear system $\boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution
(d) all n x n matrices $\boldsymbol{A}$ such that $\boldsymbol{AB} = \boldsymbol{BA}$ for a fixed n x n matrix $\boldsymbol{B}$

15. Determine whether the solution space of the system $\boldsymbol{Ax} = \boldsymbol{0}$ is a line through the origin, a plane through the origin, or the origin only. If it is a plane, find an equation for it; if it is a line, find parametric equations for it.

(a) $\boldsymbol{A}=\begin{bmatrix} -1 & 1 & 1 \\ 3 & -1 & 0 \\ 2 & -4 & -5 \end{bmatrix}$ (b) $\boldsymbol{A}=\begin{bmatrix} 1 & -2 & 3 \\ -3 & 6 & 9 \\ -2 & 4 & -6 \end{bmatrix}$

(c) $\boldsymbol{A}=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ (d) $\boldsymbol{A}=\begin{bmatrix} 1 & 2 & -6 \\ 1 & 4 & 4 \\ 3 & 10 & 6 \end{bmatrix}$

16. Determine if the set "all polynomial in $\boldsymbol{p_n}$ such that $\boldsymbol{p(0) = 0}$" is a subspace of $\boldsymbol{P_n}$ for an appropriate value of n. Justify your answer.

17. Let $\boldsymbol{H}$ be the set of all vectors of the form $\begin{bmatrix} s \\ 3s \\ 2s \end{bmatrix}$. Find a vector $\boldsymbol{v}$ in $\boldsymbol{R^3}$ such that $\boldsymbol{H}$ = Span $\{\boldsymbol{v}\}$. Why does this show that $\boldsymbol{H}$ is a subspace of $\boldsymbol{R^3}$?

18. Let $\boldsymbol{W}$ be the set of all vectors of the form $\begin{bmatrix} 5b+2c \\ b \\ c \end{bmatrix}$, where b and c are arbitrary. Find vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ such that $\boldsymbol{W}$ = Span $\{\boldsymbol{u}, \boldsymbol{v}\}$. Why does this show that $\boldsymbol{W}$ is a subspace of $\boldsymbol{R^3}$?

19. Let $\boldsymbol{W}$ be the set of all vectors of the form $\begin{bmatrix} s+3t \\ s-t \\ 2s-t \\ 4t \end{bmatrix}$. Show that $\boldsymbol{W}$ is a subspace of $\boldsymbol{R^4}$.

20. Let $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \boldsymbol{v_3} = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$, and $\boldsymbol{w} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$.

(a) Is $\boldsymbol{w}$ in $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$? How many vectors are in $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$?
(b) How many vectors are in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$?
(c) Is $\boldsymbol{w}$ in the subspace spanned by $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$? Why?

In exercises 21 and 22, let $\boldsymbol{W}$ be the set of all vectors of the form shown, where a, b and c represent arbitrary real numbers. In each case, either find a set $\boldsymbol{S}$ of vectors that spans $\boldsymbol{W}$ or give an example to show that $\boldsymbol{W}$ is not a vector space.

21. $\begin{bmatrix} 3a+b \\ 4 \\ a-5b \end{bmatrix}$

22. $\begin{bmatrix} a-b \\ b-c \\ c-a \\ b \end{bmatrix}$

23. Show that $\boldsymbol{w}$ is in the subspace of $\boldsymbol{R^4}$ spanned by $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}$, where

$$\boldsymbol{w} = \begin{bmatrix} -9 \\ 7 \\ 4 \\ 8 \end{bmatrix}, \boldsymbol{v_1} = \begin{bmatrix} 7 \\ -4 \\ -2 \\ 9 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} -4 \\ 5 \\ -1 \\ -7 \end{bmatrix}, \boldsymbol{v_3} = \begin{bmatrix} -9 \\ 4 \\ 4 \\ -7 \end{bmatrix}$$

24. Determine if $\boldsymbol{y}$ is in the subspace of $\boldsymbol{R^4}$ spanned by the columns of $\boldsymbol{A}$, where

$$\boldsymbol{y} = \begin{bmatrix} 6 \\ 7 \\ 1 \\ -4 \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} 5 & -5 & -9 \\ 8 & 8 & -6 \\ -5 & -9 & 3 \\ 3 & -2 & -7 \end{bmatrix}$$

Lecture 21

Null Spaces, Column Spaces, and Linear Transformations

Subspaces arise in as set of all solutions to a system of homogenous linear equations as the set of all linear combinations of certain specified vectors. In this lecture, we compare and contrast these two descriptions of subspaces, allowing us to practice using the concept of a subspace. In applications of linear algebra, subspaces of $\boldsymbol{R^n}$ usually arise in one of two ways:

- as the set of all solutions to a system of homogeneous linear equations or
- as the set of all linear combinations of certain specified vectors.

Our work here will provide us with a deeper understanding of the relationships between the solutions of a linear system of equations and properties of its coefficient matrix.

Null Space of a Matrix

Consider the following system of homogeneous equations:

$$\begin{aligned} x_1 - 3x_2 - 2x_3 &= 0 \\ -5x_1 + 9x_2 + x_3 &= 0 \end{aligned} \qquad (1)$$

In matrix form, this system is written as $\boldsymbol{Ax = 0}$, where

$$\boldsymbol{A} = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \qquad (2)$$

Recall that the set of all $\boldsymbol{x}$ that satisfy (1) is called the solution set of the system (1). Often it is convenient to relate this set directly to the matrix $\boldsymbol{A}$ and the equation $\boldsymbol{Ax = 0}$. We call the set of $\boldsymbol{x}$ that satisfy $\boldsymbol{Ax = 0}$ the **null space** of the matrix $\boldsymbol{A}$. The reason for this name is that if matrix A is viewed as a linear operator that maps points of some vector space V into itself, it can be viewed as mapping all the elements of this solution space of AX = 0 into the null element "0". Thus the null space N of A is that subspace of all vectors in V which are imaged into the null element “0" by the matrix A.

NULL SPACE

Definition The **null space** of an $m \times n$ matrix $\boldsymbol{A}$, written as Nul $\boldsymbol{A}$, is the set of all solutions to the homogeneous equation $\boldsymbol{Ax = 0}$. In set notation,

Nul $\boldsymbol{A}$ = $\{\boldsymbol{x}$: $\boldsymbol{x}$ is in $\boldsymbol{R^n}$ and $\boldsymbol{Ax = 0}\}$

OR

$$Nul(A) = \{x / \forall x \in \square, Ax = 0\}$$

A more dynamic description of Nul $\boldsymbol{A}$ is the set of all $\boldsymbol{x}$ in $\boldsymbol{R^n}$ that are mapped into the zero vector of $\boldsymbol{R^m}$ via the linear transformation $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$, where A is a matrix of transformation. See Figure1

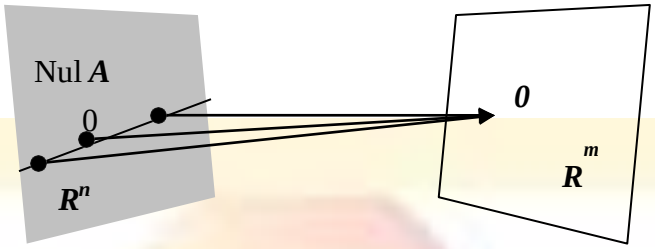


Figure 1

Example 1 Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and let $\boldsymbol{u} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$. Determine if $\boldsymbol{u} \in \text{Nul}\, \boldsymbol{A}$.

Solution To test if $\boldsymbol{u}$ satisfies $\boldsymbol{Au} = \boldsymbol{0}$, simply compute

$$\boldsymbol{Au} = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 5-9+4 \\ -25+27-2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$ Thus $\boldsymbol{u}$ is in Nul $\boldsymbol{A}$.

Example Determine the null space of the following matrix:

$$A = \begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix}$$

Solution To find the null space of A we need to solve the following system of equations:

$$\begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x_1 + 0x_2 \\ -8x_1 + 20x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 4x_1 + 0x_2 = 0 \qquad \Rightarrow x_1 = 0$$

and $$\Rightarrow -8x_1 + 20x_2 = 0 \qquad \Rightarrow x_2 = 0$$

We can find Null space of a matrix with two ways i.e. with matrices or with system of linear equations. We have given this in both matrix form and (here first we convert the matrix into system of equations) equation form. In equation form it is easy to see that by solving these equations together the only solution is $x_1 = x_2 = 0$. In terms of vectors from

$\mathbb{R}^2$ the solution consists of the single vector $\{0\}$ and hence the null space of A is $\{0\}$.

Activity Determine the null space of the following matrices:

1. $0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

2. $M = \begin{pmatrix} 1 & -5 \\ -5 & 25 \end{pmatrix}$

In earlier (previous) lectures, we developed the technique of elementary row operations to solve a linear system. We know that performing elementary row operations on an augmented matrix does not change the solution set of the corresponding linear system ***Ax=0***. Therefore, we can say that it does not change the null space of ***A***. We state this result as a theorem:

Theorem 1 Elementary row operations do not change the null space of a matrix.

Or

Null space N(A) of a matrix A can not be changed (always same) by changing the matrix with elementary row operations.

Example Determine the null space of the following matrix using the elementary row operations: (Taking the matrix from the above Example)

$$A = \begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix}$$

Solution First we transform the matrix to the reduced row echelon form:

$$\begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ -8 & 20 \end{pmatrix} \quad \frac{1}{4}R_1$$

$$\sim \begin{pmatrix} 1 & 0 \\ 0 & 20 \end{pmatrix} \quad R_2 + 8R_1$$

$$\sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \frac{1}{20}R_2$$

which corresponds to the system

$$x_1 = 0$$
$$x_2 = 0$$

Since every column in the coefficient part of the matrix has a leading entry that means our system has the **trivial solution only**:

$$x_1 = 0$$
$$x_2 = 0$$

This means **the null space consists only of the zero vector**.

We can observe and compare both the above examples which show the same result.

Theorem 2 The null space of an $m \times n$ matrix A is a subspace of R^n. Equivalently, the set of all solutions to a system $Ax = 0$ of m homogeneous linear equations in n unknowns is a subspace of R^n.

Or simply, the null space is the space of all the vectors of a Matrix A of any order those are mapped (assign) onto zero vector in the space R^n *(i.e. Ax = 0).*

Proof We know that the subspace of A consists of all the solution to the system $\boldsymbol{Ax = 0}$.

First, we should point out that the zero vector, $\mathbf{0}$, in $\boldsymbol{R^n}$ will be a solution to this system and so we know that the null space is not empty. This is a good thing since a vector space (subspace or not) must contain at least one element.

Now we know that the null space is not empty. Consider $\boldsymbol{u}$, $\boldsymbol{v}$ be two any vectors (elements) (in) from the null space and let c be any scalar. We just need to show that the sum ($\boldsymbol{u+v}$) and scalar multiple ($c.\boldsymbol{u}$) of these are also in the null space.

Certainly Nul $\boldsymbol{A}$ is a subset of $\boldsymbol{R^n}$ because $\boldsymbol{A}$ has n columns. To show that Nul(A) is the subspace, we have to check three conditions whether they are satisfied or not. If Nul(A) satisfies the all three condition, we say Nul(A) is a subspace otherwise not.

First, zero vector "0" must be in the space and subspace. If zero vector does not in the space we can not say that is a vector space (generally, we use space for vector space).

And we know that zero vector maps on zero vector so 0 is in Nul(A). Now choose any vectors u, v from Null space and using definition of Null space (i.e. Ax=0)

$$\boldsymbol{Au = 0} \text{ and } \boldsymbol{Av = 0}$$

Now the other two conditions are vector addition and scalar multiplication. For this we proceed as follow:

Let start with vector addition:

To show that $\boldsymbol{u + v}$ is in Nul $\boldsymbol{A}$, we must show that $\boldsymbol{A\,(u + v) = 0}$. Using the property of matrix multiplication, we find that

$$\boldsymbol{A\,(u + v) = Au + Av = 0 + 0 = 0}$$

Thus $\boldsymbol{u + v}$ is in Nul $\boldsymbol{A}$, and Nul $\boldsymbol{A}$ is closed under vector addition.

For Matrix multiplication, consider any scalar , say c,

$$\mathbf{A}\,(c\boldsymbol{u}) = c\,(\boldsymbol{Au}) = c\,(\boldsymbol{0}) = \boldsymbol{0}$$

which shows that $c\boldsymbol{u}$ is in Nul $\boldsymbol{A}$. Thus Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^n}$.

Example 2 The set $\boldsymbol{H}$, of all vectors in $\boldsymbol{R^4}$ whose coordinates a, b, c, d satisfy the equations

$$a - 2b + 5c = d$$
$$c - a = b$$

is a subspace of $\boldsymbol{R^4}$.

Solution Since $a - 2b + 5c = d$

$c - a = b$

By rearranging the equations, we get

$$a - 2b + 5c - d = 0$$
$$-a - b + c = 0$$

We see that $\boldsymbol{H}$ is the set of all solutions of the above system of homogeneous linear equations.
Therefore from the Theorem 2, $\boldsymbol{H}$ is a subspace of $\boldsymbol{R^4}$.

It is important that the linear equations defining the set $\boldsymbol{H}$ are homogeneous. Otherwise, the set of solutions will definitely not be a subspace (because the zero-vector (origin) is not a solution of a non- homogeneous system), geometrically means that a line that not passes through origin can not be a subspace, because subspace must hold the zero vector (origin). Also, in some cases, the set of solutions could be empty. In this case, we can not find any solution of a system of linear equations, geometrically says that lines are parallel or not intersecting.
If the null space having more than one vector, geometrically means that the lines intersect more than one point and must passes through origin (zero vector) .

An Explicit Description of Nul *A*

There is no obvious relation between vectors in Nul $\boldsymbol{A}$ and the entries in $\boldsymbol{A}$. We say that Nul $\boldsymbol{A}$ is defined implicitly, because it is defined by a condition that must be checked. No explicit list or description of the elements in Nul $\boldsymbol{A}$ is given. However, when we solve the equation $\boldsymbol{Ax} = \boldsymbol{0}$, we obtain an explicit description of Nul $\boldsymbol{A}$.

Example 3 Find a spanning set for the null space of the matrix

$$\boldsymbol{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution The first step is to find the general solution of $\boldsymbol{Ax} = \boldsymbol{0}$ in terms of free variables.
After transforming the augmented matrix $[\boldsymbol{A} \quad \boldsymbol{0}]$ to the reduced row echelon form and we get;

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which corresponds to the system

$$x_1 - 2x_2 - x_4 + 3x_5 = 0$$
$$x_3 + 2x_4 - 2x_5 = 0$$
$$0 = 0$$

The general solution is

$x_1 = 2x_2 + x_4 - 3x_5$
$x_2 =$ *free variable*
$x_3 = -2x_4 + 2x_5$
$x_4 =$ *free variable*
$x_5 =$ *free variable*

Next, decompose the vector giving the general solution into a linear combination of vectors where the weights are the free variables. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\uparrow \qquad \uparrow \qquad \uparrow$$

$$\boldsymbol{u} \qquad \boldsymbol{v} \qquad \boldsymbol{w}$$

$$= x_2\boldsymbol{u} + x_4\boldsymbol{v} + x_5\boldsymbol{w} \qquad (3)$$

Every linear combination of ***u, v*** and ***w*** is an element of Nul ***A***. Thus ***{u, v, w}*** is a spanning set for Nul ***A***.

Two points should be made about the solution in Example 3 that apply to all problems of this type. We will use these facts later.

1. The spanning set produced by the method in Example 3 is automatically linearly independent because the free variables are the weights on the spanning vectors. For instance, look at the 2nd, 4th and 5th entries in the solution vector in (3) and note that $x_2\boldsymbol{u} + x_4\boldsymbol{v} + x_5\boldsymbol{w}$ can be *0* only if the weights x_2, x_4 and x_5 are all zero.
2. When Nul *A* contains nonzero vector, the number of vectors in the spanning set for Nul *A* equals the number of free variables in the equation $Ax = 0$.

Example 4 Find a spanning set for the null space of $\boldsymbol{A} = \begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 3 & 6 & 0 & -3 \\ 2 & -3 & -2 & 4 & 4 \\ 3 & -6 & 0 & 6 & 5 \\ -2 & 9 & 2 & -4 & -5 \end{bmatrix}$.

Solution The null space of ***A*** is the solution space of the homogeneous system

$$x_1 - 3x_2 + 2x_3 + 2x_4 + x_5 = 0$$
$$0x_1 + 3x_2 + 6x_3 + 0x_4 - 3x_5 = 0$$
$$2x_1 - 3x_2 - 2x_3 + 4x_4 + 4x_5 = 0$$
$$3x_1 - 6x_2 + 0x_3 + 6x_4 + 5x_5 = 0$$
$$-2x_1 + 9x_2 + 2x_3 - 4x_4 - 5x_5 = 0$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \\ 2 & -3 & -2 & 4 & 4 & 0 \\ 3 & -6 & 0 & 6 & 5 & 0 \\ -2 & 9 & 2 & -4 & -5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \end{bmatrix} \begin{matrix} \\ -2R_1 + R_3 \\ -3R_1 + R_4 \\ 2R_1 + R_5 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \end{bmatrix} \begin{matrix} \\ \\ (1/3)R_2 \\ \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & -12 & 0 & 5 & 0 \\ 0 & 0 & -12 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ -3R_2 + R_3 \\ -3R_2 + R_4 \\ -3R_2 + R_5 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & -12 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ (-1/12)R_3 \\ \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ 12R_3 + R_4 \\ \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 0 & 2 & 11/6 & 0 \\ 0 & 1 & 0 & 0 & -1/6 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ -2R_3 + R_2 \\ \\ -2R_3 + R_1 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 & 4/3 & 0 \\ 0 & 1 & 0 & 0 & -1/6 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} 3R_2 + R_1$$

The reduced row echelon form of the augmented matrix corresponds to the system

$$\begin{aligned} 1\,x_1 + \quad & 2\,x_4 + (4/3)\,x_5 = 0 \\ 1\,x_2 + \quad & (-1/6)\,x_5 = 0 \\ 1\,x_3 + \quad & (-5/12)\,x_5 = 0\,. \\ & 0 = 0 \\ & 0 = 0 \end{aligned}$$

No equation of this system has a form zero = nonzero; Therefore, the system is **consistent**. The system has **infinitely many solutions**:

$x_1 = -2\,x_4 + (-4/3)\,x_5$ $\quad x_2 = +(1/6)\,x_5$ $\quad x_3 = +(5/12)\,x_5$

x_4 = arbitrary $\quad x_5$ = arbitrary

The solution can be written in the vector form:

$$\boldsymbol{c_4} = (-2,0,0,1,0) \qquad \boldsymbol{c_5} = (-4/3,1/6,5/12,0,1)$$

Therefore $\{(-2,0,0,1,0), (-4/3,1/6,5/12,0,1)\}$ is a spanning set for Null space of $\boldsymbol{A}$.

Activity: Find an explicit description of Nul A where:

1. $A = \begin{pmatrix} 3 & 5 & 5 & 3 & 9 \\ 5 & 1 & 1 & 0 & 3 \end{pmatrix}$

2. $A = \begin{pmatrix} 4 & 1 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$

The Column Space of a Matrix Another important subspace associated with a matrix is its column space. Unlike the null space, the column space is defined explicitly via linear combinations.

Definition (Column Space) The column space of an $m \times n$ matrix $\boldsymbol{A}$, written as Col $\boldsymbol{A}$, is the set of all linear combinations of the columns of $\boldsymbol{A}$. If $\boldsymbol{A} = [\boldsymbol{a_1} \ \ldots \ \boldsymbol{a_n}]$, then

$$\text{Col } \boldsymbol{A} = \textit{Span } \{\boldsymbol{a_1}, \ldots, \boldsymbol{a_n}\}$$

Since Span $\{\boldsymbol{a_1}, \ldots, \boldsymbol{a_n}\}$ is a subspace, by Theorem of lecture 20 i.e. if $v_1,...,v_p$ are in a vector space V, then Span $\{v_1,...,v_p\}$ is a subspace of V .

The column space of a matrix is that subspace spanned by the columns of the matrix (columns viewed as vectors). It is that space defined by all linear combinations of the column of the matrix.

Example, in the given matrix,

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 1 & 4 \\ 3 & 1 & 5 \\ 4 & 1 & 6 \end{pmatrix}$$

The column space Col$\boldsymbol{A}$ is all the linear combination of the first (1, 2, 3, 4), the second (1, 1, 1, 1) and the third column (3, 4, 5, 6). That is, Col$\boldsymbol{A}$ = { a·(1, 2, 3, 4) + b·(1, 1, 1, 1) + c·(3, 4, 5, 6) }. In general, **the column space** Col$\boldsymbol{A}$ **contains all the linear combinations of columns of A**.

The next theorem follows from the definition of Col $\boldsymbol{A}$ and the fact that the columns of $\boldsymbol{A}$ are in $\boldsymbol{R^m}$.

Theorem 3 The column space of an $m \times n$ matrix A is a subspace of R^m.

Note that a typical vector in Col $\boldsymbol{A}$ can be written as $\boldsymbol{Ax}$ for some $\boldsymbol{x}$ because the notation $\boldsymbol{Ax}$ stands for a linear combination of the columns of $\boldsymbol{A}$. That is,

$$\text{Col } \boldsymbol{A} = \{\boldsymbol{b}: \boldsymbol{b} = \boldsymbol{Ax} \text{ for some } \boldsymbol{x} \text{ in } \boldsymbol{R^n}\}$$

The notation $\boldsymbol{Ax}$ for vectors in Col $\boldsymbol{A}$ also shows that Col $\boldsymbol{A}$ is the range of the linear transformation $\boldsymbol{x} \to \boldsymbol{Ax}$.

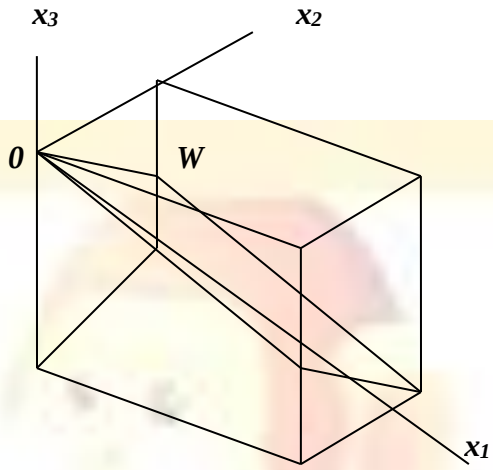
Example 6 Find a matrix $\boldsymbol{A}$ such that $\boldsymbol{W}$ = Col $\boldsymbol{A}$. $\boldsymbol{W} = \left\{ \begin{bmatrix} 6a - b \\ a + b \\ -7a \end{bmatrix} : a,b \text{ in } R \right\}$

Solution First, write $\boldsymbol{W}$ as a set of linear combinations.

$$\boldsymbol{W} = \left\{ a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : a,b \text{ in } R \right\} = \text{Span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Second, use the vectors in the spanning set as the columns of $\boldsymbol{A}$. Let $\boldsymbol{A} = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$.

Then $\boldsymbol{W}$ = Col $\boldsymbol{A}$, as desired.



We know that the columns of $\boldsymbol{A}$ span $\boldsymbol{R^m}$ if and only if the equation $\boldsymbol{Ax = b}$ has a solution for each $\boldsymbol{b}$. We can restate this fact as follows:
The column space of an $m \times n$ matrix A is all of R^m if and only if the equation $Ax = b$ has a solution for each b in R^m.

Theorem 4 A system of linear equations $\boldsymbol{Ax} = \boldsymbol{b}$ is consistent if and only if $\boldsymbol{b}$ in the column space of $\boldsymbol{A}$.

Example 6 A vector $\boldsymbol{b}$ in the column space of $\boldsymbol{A}$. Let $\boldsymbol{Ax} = \boldsymbol{b}$ is the linear system

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

. Show that $\boldsymbol{b}$ is in the column space of $\boldsymbol{A}$, and express $\boldsymbol{b}$ as a linear combination of the column vectors of $\boldsymbol{A}$.

Solution Augmented Matrix is given by

$$\begin{bmatrix} -1 & 3 & 2 & 1 \\ 1 & 2 & -3 & -9 \\ 2 & 1 & -2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & -2 & -1 \\ 0 & 5 & -1 & -8 \\ 0 & 7 & 2 & -1 \end{bmatrix} \begin{matrix} -1R_1 \\ -1R_1 + R_2 \\ -2R_1 + R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & -2 & -1 \\ 0 & 1 & -1/5 & -8/5 \\ 0 & 0 & 17/5 & 51/5 \end{bmatrix} \begin{matrix} 1/5R_2 \\ -7R_2 + R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} (5/17)R_3 \\ (1/5)R_3 + R_2 \\ 2R_3 + R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} 3R_2 + R_1$$

$\Rightarrow x_1 = 2, x_2 = -1, x_3 = 3$. Since the system is consistent, ***b*** is in the column space of ***A***.

Moreover, $$2\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

Example Determine whether **b** is in the column space of **A** and if so, express **b** as a linear combination of the column vectors of **A**:

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{pmatrix}: \quad b = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

Solution

The coefficient matrix $Ax = b$ is:

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

The augmented matrix for the linear system that corresponds to the matrix equation $Ax = b$ is:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 3 & 2 \end{array}\right)$$

We reduce this matrix to the Reduced Row Echelon Form:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 3 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 1 \\ 2 & 1 & 3 & 2 \end{array}\right) \quad R_2 + (-1)R_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 4 \end{array}\right) \quad R_3 + (-2)R_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 4 \end{array}\right) \quad (-1)R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 3 \end{array}\right) \quad R_3 + R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right) \quad \frac{1}{3} R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \quad R_2 + R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \quad R_1 + R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \quad R_1 + (-1) R_2$$

The new system for the equation $Ax = b$ is

$$\begin{aligned} x_1 + x_3 &= 0 \\ x_2 + x_3 &= 0 \\ 0 &= 1 \end{aligned}$$

Equation $0 = 1$ cannot be solved, therefore, the system has **no solution** (i.e. the system is **inconsistent**).

Since the equation $A\mathbf{x} = \mathbf{b}$ has no solution, therefore **b** is not in the column space of ***A***.

Activity Determine whether **b** is in the column space of A and if so, express **b** as a linear combination of the column vectors of **A**:

1.

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 9 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix}; \quad b = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$$

2. $$A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & -1 & -1 \end{pmatrix}; \quad b = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

3. $A=\begin{pmatrix} 1 & 1 & -2 & 1 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & -3 \\ 0 & 2 & 2 & 1 \end{pmatrix}$; $b=\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$

Theorem 5 If x_0 denotes any single solution of a consistent linear system $Ax=b$ and if $\boldsymbol{v_1, v_2, v_3, ..., v_k}$ form the solution space of the homogeneous system $Ax=0$, then every solution of $Ax=b$ can be expressed in the form $\boldsymbol{x = x_0 + c_1 v_1 + c_2 v_2 + ... + c_k v_k}$ and, conversely, for all choices of scalars $c_1, c_2, c_3, ..., c_k$, the vector x is a solution of $Ax=b$.

General and Particular Solutions: The vector x_0 is called a particular solution of $Ax=b$ .The expression $x_0 + c_1 v_1 + c_2 v_2 + \ldots + c_k v_k$ is called the general solution of $Ax=b$, and the expression $c_1 v_1 + c_2 v_2 + \ldots + c_k v_k$ is called the general solution of $Ax=0$.

Example 7 Find the vector form of the general solution of the given linear system $\boldsymbol{Ax = b}$; then use that result to find the vector form of the general solution of $\boldsymbol{Ax=0}$.

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$
$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1$$
$$5x_3 + 10x_4 + 15x_6 = 5$$
$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6$$

Solution We solve the non-homogeneous linear system. The augmented matrix of this system is given by

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix} \begin{matrix} \\ -2R_1 + R_2 \\ \\ -2R_1 + R_4 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix} \begin{matrix} \\ -1R_2 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \end{bmatrix} \begin{matrix} \\ -5R_2 + R_3 \\ \\ -4R_2 + R_4 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R_{34}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (1/6)R_3$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad -3R_3 + R_2$$

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad 2R_2 + R_1$$

The reduced row echelon form of the augmented matrix corresponds to the system

$$\begin{aligned} 1x_1 + 3x_2 + \quad & 4x_4 + 2x_5 = 0 \\ & 1x_3 + 2x_4 = 0 \\ & 1x_6 = (1/3) \\ & 0 = 0 \end{aligned}$$

No equation of this system has a form zero = nonzero; Therefore, the system is **consistent**. The system has **infinitely many solutions**:

$$x_1 = -3x_2 - 4x_4 - 2x_5 \quad x_2 = r \quad x_3 = -2x_4$$
$$x_4 = s \quad x_5 = t \quad x_6 = 1/3$$

$$x_1 = -3r - 4s - 2t \quad x_2 = r \quad x_3 = -2s$$
$$x_4 = s \quad x_5 = t \quad x_6 = \frac{1}{3}$$

This result can be written in vector form as

$$\begin{bmatrix} x \\ x_2^1 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -3r-4s-2t \\ r \\ -2s \\ s \\ t \\ 1 \\ \overline{3} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ \overline{3} \end{bmatrix} + r\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \qquad \text{(A)}$$

which is the general solution of the given system. The vector $\boldsymbol{x_0}$ in (A) is a particular solution of the given system; the linear combination $r\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ in (A) is the general solution of the homogeneous system.

Activity:

1. Suppose that $x_1 = -1,\ x_2 = 2,\ x_3 = 4,\ x_4 = -3$ is a solution of a non-homogenous linear system $Ax = b$ and that the solution set of the homogenous system $Ax = 0$ is given by this formula:

$x_1 = -3r + 4s,$

$x_2 = r - s,$

$x_3 = r,$

$x_4 = s$

(a) Find the vector form of the general solution of $Ax = 0$.

(b) Find the vector form of the general solution of $Ax = 0$.

Find the vector form of the general solution of the following linear system $\boldsymbol{Ax} = \boldsymbol{b}$; then use that result to find the vector form of the general solution of $\boldsymbol{Ax}=\boldsymbol{0}$:

2.
$$\begin{aligned} x_1 - 2x_2 &= 1 \\ 3x_1 - 9x_2 &= 2 \end{aligned}$$

3.
$$\begin{aligned} x_1 + 2x_2 - 3x_3 + x_4 &= 3 \\ -3x_1 - x_2 + 3x_3 + x_4 &= -1 \\ -x_1 + 3x_2 - x_3 + 2x_4 &= 2 \\ 4x_1 - 5x_2 - 3x_4 &= -5 \end{aligned}$$

The Contrast between Nul *A* and Col *A*

It is natural to wonder how the null space and column space of a matrix are related. In fact, the two spaces are quite dissimilar. Nevertheless, a surprising connection between the null space and column space will emerge later.

Example 8 Let $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$

(a) If the column space of $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, what is k?
(b) If the null space of $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, what is k?

Solution

(a) The columns of $\boldsymbol{A}$ each have three entries, so Col $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, where $k = 3$.
(b) A vector $\boldsymbol{x}$ such that $\boldsymbol{Ax}$ is defined must have four entries, so Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, where $k = 4$.

When a matrix is not square, as in Example 8, the vectors in Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$ live in entirely different "universes". For example, we have discussed no algebraic operations that connect vectors in $\boldsymbol{R}^3$ with vectors in $\boldsymbol{R}^4$. Thus we are not likely to find any relation between individual vectors in Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$.

Example 9 If $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$, find a nonzero vector in Col $\boldsymbol{A}$ and a nonzero vector in Nul $\boldsymbol{A}$

Solution It is easy to find a vector in Col $\boldsymbol{A}$. Any column of $\boldsymbol{A}$ will do, say, $\begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$. To find a nonzero vector in Nul $\boldsymbol{A}$, we have to do some work. We row reduce the augmented matrix $[\boldsymbol{A} \quad \boldsymbol{0}]$ to obtain $[\boldsymbol{A} \quad \boldsymbol{0}] \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$. Thus if $\boldsymbol{x}$ satisfies $\boldsymbol{Ax} = \boldsymbol{0}$, then $x_1 = -9x_3$, $x_2 = 5x_3$, $x_4 = 0$, and x_3 is free. Assigning a nonzero value to x_3 (say), $x_3 = 1$, we obtain a vector in Nul $\boldsymbol{A}$, namely, $\boldsymbol{x} = (-9, 5, 1, 0)$.

Example 10 With $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$, let $\boldsymbol{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$.

(a) Determine if $\boldsymbol{u}$ is in Nul $\boldsymbol{A}$. Could $\boldsymbol{u}$ be in Col $\boldsymbol{A}$?

(b) Determine if $\boldsymbol{v}$ is in Col $\boldsymbol{A}$. Could $\boldsymbol{v}$ be in Nul $\boldsymbol{A}$?

Solution (a) An explicit description of Nul $\boldsymbol{A}$ is not needed here. Simply compute the product

$$\boldsymbol{Au} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Obviously, $\boldsymbol{u}$ is not a solution of $\boldsymbol{Ax = 0}$, so $\boldsymbol{u}$ is not in Nul $\boldsymbol{A}$.
Also, with four entries, $\boldsymbol{u}$ could not possibly be in Col $\boldsymbol{A}$, since Col $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^3}$.
(b) Reduce $[\boldsymbol{A} \quad \boldsymbol{v}]$ to an echelon form:

$$[\boldsymbol{A} \quad \boldsymbol{v}] = \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & 2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

At this point, it is clear that the equation $\boldsymbol{Ax = v}$ is consistent, so $\boldsymbol{v}$ is in Col $\boldsymbol{A}$. With only three entries, $\boldsymbol{v}$ could not possibly be in Nul $\boldsymbol{A}$, since Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^4}$.

The following table summarizes what we have learned about Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$.

1. **Nul A is a subspace of R^n.**	1. **Col A is a subspace of R^m.**
2. **Nul A is implicitly defined; i.e. we are given only a condition ($Ax = 0$) that vectors in Nul A must satisfy.**	2. **Col A is explicitly defined; that is, we are told how to build vectors in Col A.**
3. **It takes time to find vectors in Nul A. Row operations on $[A \quad 0]$ are required.**	3. **It is easy to find vectors in Col A The columns of A are displayed; others are formed from them.**
4. **There is no obvious relation between Nul A and the entries in A.**	4. **There is an obvious relation between Col A and the entries in A, since each column of A is in Col A.**
5. **A typical vector v in Nul A has the property that $Av = 0$.**	5. **A typical vector v in Col A has the property that the equation $Ax = v$ is consistent.**
6. **Given a specific vector v, it is easy to tell if v is in Nul A. Just compute Av.**	6. **Given a specific vector v, it may take time to tell if v is in Col A. Row operations on $[A \quad v]$ are required.**
7. **Nul $A = \{0\}$ if and only if the equation $Ax = 0$ has only the trivial solution.**	7. **Col $A = R^m$ if and only if the equation $Ax = b$ has a solution for every b in R^m.**
8. **Nul $A = \{0\}$ if and only if the linear transformation $x \rightarrow Ax$ is one-to-one.**	8. **Col $A = R^m$ if and only if the linear transformation $x \rightarrow Ax$ maps R^n onto R^m.**

Kernel and Range of A Linear Transformation

Subspaces of vector spaces other than $\boldsymbol{R^n}$ are often described in terms of a linear transformation instead of a matrix. To make this precise, we generalize the definition given earlier in Segment I.

Definition A **linear transformation** $\boldsymbol{T}$ from a vector space $\boldsymbol{V}$ into a vector space $\boldsymbol{W}$ is a rule that assigns to each vector $\boldsymbol{x}$ in $\boldsymbol{V}$ a unique vector $\boldsymbol{T}(\boldsymbol{x})$ in $\boldsymbol{W}$, such that

(i) $\boldsymbol{T}(\boldsymbol{u}+\boldsymbol{v}) = \boldsymbol{T}(\boldsymbol{u}) + \boldsymbol{T}(\boldsymbol{v})$ for all $\boldsymbol{u}, \boldsymbol{v}$ in $\boldsymbol{V}$, and

(ii) $\boldsymbol{T}(c\boldsymbol{u}) = c\,\boldsymbol{T}(\boldsymbol{u})$ for all $\boldsymbol{u}$ in $\boldsymbol{V}$ and all scalars c.

The **kernel** (or **null space**) of such a $\boldsymbol{T}$ is the set of all $\boldsymbol{u}$ in $\boldsymbol{V}$ such that $\boldsymbol{T}(\boldsymbol{u}) = \boldsymbol{0}$ (the zero vector in $\boldsymbol{W}$). The **range** of $\boldsymbol{T}$ is the set of all vectors in $\boldsymbol{W}$ of the form $\boldsymbol{T}(\boldsymbol{x})$ for some $\boldsymbol{x}$ in $\boldsymbol{V}$. If $\boldsymbol{T}$ happens to arise as a matrix transformation, say, $\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$ for some matrix $\boldsymbol{A}$ – then the kernel and the range of $\boldsymbol{T}$ are just the null space and the column space of $\boldsymbol{A}$, as defined earlier. So if $\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, col $\boldsymbol{A}$ = range of $\boldsymbol{T}$.

Definition If $T: V \to W$ is a linear transformation, then the set of vectors in V that $\boldsymbol{T}$ maps into $\boldsymbol{0}$ is called the **kernel** of $\boldsymbol{T}$; it is denoted by $\boldsymbol{ker(T)}$. The set of all vectors in $\boldsymbol{W}$ that are images under $\boldsymbol{T}$ of at least one vector in $\boldsymbol{V}$ is called the **range** of $\boldsymbol{T}$; it is denoted by $\boldsymbol{R(T)}$.

Example If $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is multiplication by the $m \times n$ matrix A, then from the above definition; the kernel of T_A is the null space of $\boldsymbol{A}$ and the range of T_A is the column space of $\boldsymbol{A}$.

Remarks The kernel of $\boldsymbol{T}$ is a subspace of $\boldsymbol{V}$ and the range of $\boldsymbol{T}$ is a subspace of $\boldsymbol{W}$.

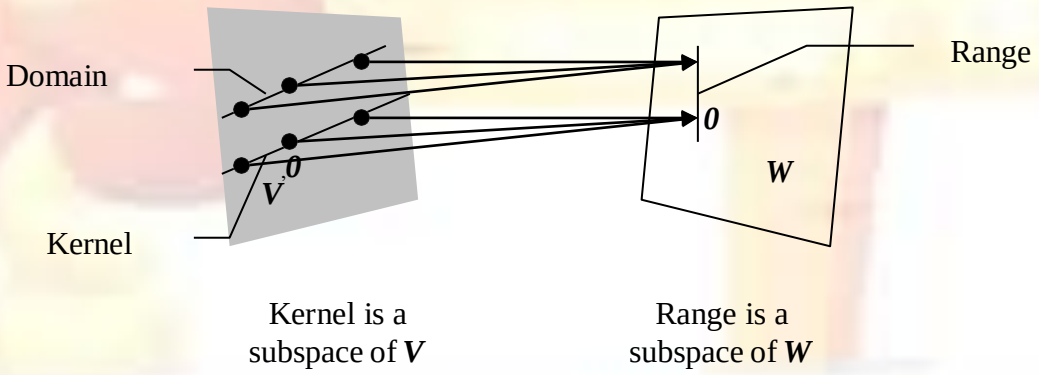


Figure 2 Subspaces associated with a linear transformation.

In applications, a subspace usually arises as either the kernel or the range of an appropriate linear transformation. For instance, the set of all solutions of a homogeneous linear differential equation turns out to be the kernel of a linear transformation. Typically, such a linear transformation is described in terms of one or more derivatives of a

function. To explain this in any detail would take us too far a field at this point. So we present only two examples. The first explains why the operation of differentiation is a linear transformation.

Example 11 Let $\boldsymbol{V}$ be the vector space of all real-valued functions $\boldsymbol{f}$ defined on an interval $[a, b]$ with the property that they are differentiable and their derivatives are continuous functions on $[a, b]$. Let $\boldsymbol{W}$ be the vector space of all continuous functions on $[a, b]$ and let $\boldsymbol{D}:\boldsymbol{V}\to\boldsymbol{W}$ be the transformation that changes $\boldsymbol{f}$ in $\boldsymbol{V}$ into its derivative $\boldsymbol{f}'$. In calculus, two simple differentiation rules are

$$\boldsymbol{D}(\boldsymbol{f}+\boldsymbol{g})=\boldsymbol{D}(\boldsymbol{f})+\boldsymbol{D}(\boldsymbol{g}) \text{ and } \boldsymbol{D}(c\boldsymbol{f})=c\boldsymbol{D}(\boldsymbol{f})$$

That is, $\boldsymbol{D}$ is a linear transformation. It can be shown that the kernel of $\boldsymbol{D}$ is the set of constant functions of $[a, b]$ and the range of $\boldsymbol{D}$ is the set $\boldsymbol{W}$ of all continuous functions on $[a, b]$.

Example 12 The differential equation $y''+wy=0$ (4)

where w is a constant, is used to describe a variety of physical systems, such as the vibration of a weighted spring, the movement of a pendulum and the voltage in an inductance – capacitance electrical circuit. The set of solutions of (4) is precisely the kernel of the linear transformation that maps a function $\boldsymbol{y}=\boldsymbol{f}(\boldsymbol{t})$ into the function $\boldsymbol{f}''(\boldsymbol{t})+w\boldsymbol{f}(\boldsymbol{t})$. Finding an explicit description of this vector space is a problem in differential equations.

Example 13 Let $\boldsymbol{W}=\left\{\begin{bmatrix}a\\b\\c\end{bmatrix} : a-3b-c=0\right\}$. Show that W is a subspace of R^3 in

different ways.

Solution First method: W is a subspace of R^3 by Theorem 2 because W is the set of all solutions to a system of homogeneous linear equations (where the system has only one equation). Equivalently, W is the null space of the $1x3$ matrix $\boldsymbol{A}=[1\ \ -3\ \ -1]$.

Second method: Solve the equation $a-3b-c=0$ for the leading variable a in terms of the free variables b and c.

Any solution has the form $\begin{bmatrix}3b+c\\b\\c\end{bmatrix}$, where b and c are arbitrary, and

$$\begin{bmatrix}3b+c\\b\\c\end{bmatrix}=b\begin{bmatrix}3\\1\\0\end{bmatrix}+c\begin{bmatrix}1\\0\\1\end{bmatrix}$$

$\boldsymbol{v}_1$ $\boldsymbol{v}_2$

This calculation shows that $W = \text{Span}\{v_1, v_2\}$. Thus W is a subspace of R^3 by Theorem i.e. if $v_1,...,v_p$ are in a vector space V, then Span $\{v_1,...,v_p\}$ is a subspace of V. We could also solve the equation $a - 3b - c = 0$ for b or c and get alternative descriptions of W as a set of linear combinations of two vectors.

Example 14 Let $\boldsymbol{A} = \begin{bmatrix} 7 & -3 & 5 \\ -4 & 1 & -5 \\ -5 & 2 & -4 \end{bmatrix}$, $\boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, *and* $\boldsymbol{W} = \begin{bmatrix} 7 \\ 6 \\ -3 \end{bmatrix}$

Suppose you know that the equations $\boldsymbol{Ax} = \boldsymbol{v}$ and $\boldsymbol{Ax} = \boldsymbol{w}$ are both consistent. What can you say about the equation $\boldsymbol{Ax} = \boldsymbol{v} + \boldsymbol{w}$?

Solution Both $\boldsymbol{v}$ and $\boldsymbol{w}$ are in Col $\boldsymbol{A}$. Since Col $\boldsymbol{A}$ is a vector space, $\boldsymbol{v} + \boldsymbol{w}$ must be in Col $\boldsymbol{A}$. That is, the equation $\boldsymbol{Ax} = \boldsymbol{v} + \boldsymbol{w}$ is consistent.

Activity

1. Let $\boldsymbol{V}$ and $\boldsymbol{W}$ be any two vector spaces. The mapping $T: V \to W$ such that $\boldsymbol{T}(\boldsymbol{v}) = \boldsymbol{0}$ for every $\boldsymbol{v}$ in $\boldsymbol{V}$ is a linear transformation called the **zero transformation**. Find the kernel and range of the zero transformation.

2. Let $\boldsymbol{V}$ be any vector space. The mapping $I: V \to V$ defined by $\boldsymbol{I(v)} = \boldsymbol{v}$ is called the **identity operator** on $\boldsymbol{V}$. Find the kernel and range of the identity operator.

Exercises

1. Determine if $\boldsymbol{w} = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$ is in Nul $\boldsymbol{A}$, where $\boldsymbol{A} = \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix}$.

In exercises 2 and 3, find an explicit description of Nul $\boldsymbol{A}$, by listing vectors that span the null space.

2. $\begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & -2 \end{bmatrix}$

3. $\begin{bmatrix} 1 & -2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -9 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

In exercises 4-7, either use an appropriate theorem to show that the given set, $\boldsymbol{W}$ is a vector space, or find a specific example to the contrary.

4. $\left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a+b+c = 2 \right\}$

5. $\left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : \begin{matrix} a - 2b = 4c \\ 2a = c + 3d \end{matrix} \right\}$

6. $\begin{bmatrix} b - 2d \\ 5 + d \\ b + 3d \\ d \end{bmatrix} : b,d \text{ real}$

7. $\begin{bmatrix} -a + 2b \\ a - 2b \\ 3a - 6b \end{bmatrix} : a,b \text{ real}$

In exercises 8 and 9, find $\boldsymbol{A}$ such that the given set is Col $\boldsymbol{A}$.

8. $\left\{ \begin{bmatrix} 2s + 3t \\ r + s - 2t \\ 4r + s \\ 3r - s - t \end{bmatrix} : r,s,t \text{ real} \right\}$

9. $\left\{ \begin{bmatrix} b - c \\ 2b + c + d \\ 5c - 4d \\ d \end{bmatrix} : b,c,d \text{ real} \right\}$

For the matrices in exercises 10-13, (a) find k such that Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, and (b) find k such that Col $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$.

10. $\boldsymbol{A} = \begin{bmatrix} 2 & -6 \\ -1 & 3 \\ -4 & 12 \\ 3 & -9 \end{bmatrix}$

11. $\boldsymbol{A} = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 0 & -5 \\ 0 & -5 & 7 \\ -5 & 7 & -2 \end{bmatrix}$

12. $\boldsymbol{A} = \begin{bmatrix} 4 & 5 & -2 & 6 & 0 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$

13. $\boldsymbol{A} = \begin{bmatrix} 1 & -3 & 9 & 0 & -5 \end{bmatrix}$

14. Let $\boldsymbol{A} = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Determine if $\boldsymbol{w}$ is in Col $\boldsymbol{A}$. Is $\boldsymbol{w}$ in Nul $\boldsymbol{A}$?

15. Let $\boldsymbol{A} = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Determine if $\boldsymbol{w}$ is in Col $\boldsymbol{A}$. Is $\boldsymbol{w}$ in Nul $\boldsymbol{A}$?

16. Define $\boldsymbol{T}$: $\boldsymbol{P_2} \rightarrow \boldsymbol{R^2}$ by $\boldsymbol{T}(\boldsymbol{p}) = \begin{bmatrix} \boldsymbol{p}(0) \\ \boldsymbol{p}(1) \end{bmatrix}$. For instance, if $\boldsymbol{p}(t) = \boldsymbol{3} + 5\boldsymbol{t} + 7\boldsymbol{t^2}$, then $\boldsymbol{T}(\boldsymbol{p}) = \begin{bmatrix} 3 \\ 15 \end{bmatrix}$.

a. Show that $\boldsymbol{T}$ is a linear transformation.
b. Find a polynomial $\boldsymbol{p}$ in $\boldsymbol{P_2}$ that spans the kernel of $\boldsymbol{T}$, and describe the range of $\boldsymbol{T}$.

17. Define a linear transformation $\boldsymbol{T}$: $\boldsymbol{P_2} \rightarrow \boldsymbol{R^2}$ by $\boldsymbol{T}(\boldsymbol{p}) = \begin{bmatrix} \boldsymbol{p}(0) \\ \boldsymbol{p}(1) \end{bmatrix}$. Find polynomials $\boldsymbol{p_1}$ and $\boldsymbol{p_2}$ in $\boldsymbol{P_2}$ that span the kernel of $\boldsymbol{T}$, and describe the range of $\boldsymbol{T}$.

18. Let $\boldsymbol{M_{2x2}}$ be the vector space of all 2x2 matrices, and define $\boldsymbol{T}$: $\boldsymbol{M_{2x2}} \rightarrow \boldsymbol{M_{2x2}}$ by $\boldsymbol{T}(\boldsymbol{A}) = \boldsymbol{A} + \boldsymbol{A^T}$, where $\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

(a) Show that $\boldsymbol{T}$ is a linear transformation.
(b) Let $\boldsymbol{B}$ be any element of $\boldsymbol{M_{2x2}}$ such that $\boldsymbol{B^T}=\boldsymbol{B}$. Find an $\boldsymbol{A}$ in $\boldsymbol{M_{2x2}}$ such that $\boldsymbol{T}(\boldsymbol{A}) = \boldsymbol{B}$.
(c) Show that the range of $\boldsymbol{T}$ is the set of $\boldsymbol{B}$ in $\boldsymbol{M_{2x2}}$ with the property that $\boldsymbol{B^T}=\boldsymbol{B}$.
(d) Describe the kernel of $\boldsymbol{T}$.

19. Determine whether $\boldsymbol{w}$ is in the column space of $\boldsymbol{A}$, the null space of $\boldsymbol{A}$, or both, where

(a) $\boldsymbol{w} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -3 \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} 7 & 6 & -4 & 1 \\ -5 & -1 & 0 & -2 \\ 9 & -11 & 7 & -3 \\ 19 & -9 & 7 & 1 \end{bmatrix}$ (b) $\boldsymbol{w} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{bmatrix}$

20. Let $\boldsymbol{a_1}, \ldots, \boldsymbol{a_5}$ denote the columns of the matrix $\boldsymbol{A}$, where

$$\boldsymbol{A} = \begin{bmatrix} 5 & 1 & 2 & 2 & 0 \\ 3 & 3 & 2 & -1 & -12 \\ 8 & 4 & 4 & -5 & 12 \\ 2 & 1 & 1 & 0 & -2 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{a_1} & \boldsymbol{a_2} & \boldsymbol{a_4} \end{bmatrix}$$

(a) Explain why $\boldsymbol{a_3}$ and $\boldsymbol{a_5}$ are in the column space of $\boldsymbol{B}$

(b) Find a set of vectors that spans Nul $\boldsymbol{A}$

(c) Let $\boldsymbol{T}$: $\boldsymbol{R^5} \rightarrow \boldsymbol{R^4}$ be defined by $\boldsymbol{T(x)} = \boldsymbol{Ax}$. Explain why $\boldsymbol{T}$ is neither one-to-one nor onto.

Lecture 22

Linearly Independent Sets; Bases

First we revise some definitions and theorems from the Vector Space:

Definition Let V be an arbitrary nonempty set of objects on which two operations are defined, addition and multiplication by scalars.

If the following axioms are satisfied by all objects u, v, w in V and all scalars l and m, then we call V a vector space.

Axioms of Vector Space

For any set of vectors u, v, w in V *and scalars* l, m, n*:*

1. $u + v$ is in V
2. $u + v = v + u$
3. $u + (v + w) = (u + v) + w$
4. There exist a zero vector 0 such that
   $0 + u = u + 0 = u$
5. There exist a vector $-u$ in V such that
   $-u + u = 0 = u + (-u)$
6. $(l\ u)$ is in V
7. $l\ (u + v) = l\ u + l\ v$
8. $m\ (n\ u) = (m\ n)\ u = n\ (m\ u)$
9. $(l + m)\ u = I\ u + m\ u$
10. $1u = u$ where 1 is the multiplicative identity

Definition A subset W of a vector space V is called a subspace of V if W itself is a vector space under the addition and scalar multiplication defined on V.

Theorem If W is a set of one or more vectors from a vector space V, then W is subspace of V if and only if the following conditions hold:

(a) If u and v are vectors in W, then $u + v$ is in W
(b) If k is any scalar and u is any vector in W, then $k\ u$ is in W.

Definition The null space of an $m \times n$ matrix A *(Nul A)* is the set of all solutions of the hom equation $Ax = 0$

$$Nul\ A = \{x: x \text{ is in } R^n \text{ and } Ax = 0\}$$

Definition The column space of an m x n matrix A *(Col A)* is the set of all linear combinations of the columns of A.

If $A = [a_1 \ \ldots \ a_n]$,
then
$Col\ A = Span\ \{ a_1, \ldots , a_n \}$

Since we know that a set of vectors $S = \{v_1, v_2, v_3, ...v_p\}$ spans a given vector space $\boldsymbol{V}$ if every vector in $\boldsymbol{V}$ is expressible as a linear combination of the vectors in $\boldsymbol{S}$. In general there may be more than one way to express a vector in $\boldsymbol{V}$ as linear combination of vectors in a spanning set. We shall study conditions under which each vector in $\boldsymbol{V}$ is expressible as a linear combination of the spanning vectors in exactly one way. Spanning sets with this property play a fundamental role in the study of vector spaces.
In this Lecture, we shall identify and study the subspace $\boldsymbol{H}$ as "efficiently" as possible. The key idea is that of linear independence, defined as in $\boldsymbol{R^n}$.

Definition An ***indexed set*** of vectors $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$ in $\boldsymbol{V}$ is said to be **linearly independent** if the vector equation

$$c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + \ldots + c_p\boldsymbol{v_p} = 0 \qquad (1)$$

has only the trivial solution, i.e. $c_1 = 0, \ldots, c_p = 0$.
The set $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$ is said to be **linearly dependent** if (1) has a nontrivial solution, that is, if there are some weights, $c_1, \ldots, c_p$, not all zero, such that (1) holds. In such a case, (1) is called a **linear dependence relation** among $\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}$. Alternatively, to say that the $\boldsymbol{v}$'s are linearly dependent is to say that the zero vector **0** can be expressed as a *nontrivial linear combination* of the $\boldsymbol{v}$'s.

If the trivial solution is the only solution to this equation then the vectors in the set are called **linearly independent** and the set is called a **linearly independent set**. If there is another solution then the vectors in the set are called **linearly dependent** and the set is called a **linearly dependent set**.

Just as in $\boldsymbol{R^n}$, a set containing a single vector $\boldsymbol{v}$ is linearly independent if and only if $\boldsymbol{v} \neq \boldsymbol{0}$. Also, a set of two vectors is linearly dependent if and only if one of the vectors is a multiple of the other. And any set containing the zero-vector is linearly dependent.

Determining whether a set of vectors $a_1, a_2, a_3, ...a_n$ is linearly independent is easy when one of the vectors is **0**: if, say, $a_1 = 0$, then we have a simple solution to $x_1a_1 + x_2a_2 + x_3a_3 + \ldots + x_na_n = 0$ given by choosing x_1 to be any nonzero value and putting all the other x's equal to 0. Consequently, *if a set of vectors contains the zero vector, it must always be linearly dependent*. Equivalently, *any set of linearly independent vectors cannot contain the zero vector*.

Another situation in which it is easy to determine linear independence is when there are more vectors in the set than entries in the vectors. If $n > m$, then the n vectors $a_1, a_2, a_3, ...a_n$ in $\boldsymbol{R^m}$ are columns of an $m \times n$ matrix A. The vector equation $x_1a_1 + x_2a_2 + x_3a_3 + \ldots + x_na_n = 0$ is equivalent to the matrix equation $A\mathbf{x} = \mathbf{0}$ whose corresponding linear system has more variables than equations. Thus there must be at least one free variable in the solution, meaning that there are nontrivial solutions

to $x_1a_1 + x_2a_2 + x_3a_3 + \ldots + x_na_n = 0$: If $n > m$, then the set $\{a_1, a_2, a_3, \ldots a_n\}$ of vectors in $\boldsymbol{R^m}$ *must* be linearly dependent.

When n is small we have a clear geometric picture of the relation amongst linearly independent vectors. For instance, the case $n = 1$ produces the equation $x_1a_1 = 0$, and as long as $a_1 \neq 0$, we only have the trivial solution $x_1 = 0$. *A single nonzero vector always forms a linearly independent set*.

When $n = 2$, the equation takes the form $x_1a_1 + x_2a_2 = 0$. If this were a linear dependence relation, then one of the x's, say x_1, would have to be nonzero. Then we could solve the equation for a_1 and obtain a relation indicating that a_1 is a scalar multiple of a_2. Conversely, if one of the vectors is a scalar multiple of the other, we can express this in the form $x_1a_1 + x_2a_2 = 0$. Thus, *a set of two nonzero vectors is linearly dependent if and only if they are scalar multiples of each other*.

Example **(linearly independent set)**

Show that the following vectors are linearly independent:

$$v_1 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solution Let there exist scalars c_1, c_2, c_3 in $\boldsymbol{R}$ such that

$$c_1v_1 + c_3v_2 + c_3v_3 = 0$$

Therefore,

$$\Rightarrow c_1\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} + c_3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2c_1 \\ c_1 \\ c_1 \end{bmatrix} + \begin{bmatrix} 2c_2 \\ c_2 \\ -2c_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ c_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2c_1 + 2c_2 \\ c_1 + c_2 \\ c_1 - 2c_2 + c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The above can be written as:

$-2c_1 + 2c_2 = 0$ .....(1) $\Rightarrow -c_1 + c_2 = 0$.....(4) (*dividing by* 2 *onboth sides of* (1))

$c_1 + c_2 = 0$ .....(2)

$c_1 - 2c_2 + c_3 = 0$.....(3)

Solving (2) *and* (4) *implies*:

$$\begin{array}{l} c_1 + c_2 = 0 \\ -c_1 + c_2 = 0 \\ \hline 0 + 2c_2 = 0 \\ \Rightarrow c_2 = 0 \end{array}$$

Solving (2) *implies*: $c_1 + 0 = 0 \Rightarrow c_1 = 0$

Solving (3) *implies*: $0 + 0 + c_3 = 0 \Rightarrow c_3 = 0$

$\Rightarrow c_1 = c_2 = c_3 = 0$; *scalars* $c_1, c_2, c_3 \in R$ *are all zero*

$\therefore$ The system has trivial solution.

Hence the given vectors v_1, v_2, v_3 *are linearly independent.*

Example **(linearly dependent set)**

If $v_1 = \{2, -1, 0, 3\}, v_2 = \{1, 2, 5, -1\}$ *and* $v_3 = \{7, -1, 5, 8\}$, then the set of vectors $S = \{v_1, v_2, v_3\}$ is linearly dependent, since $3v_1 + v_2 - v_3 = 0$

Example **(linearly dependent set)**

The polynomials $p_1 = -x + 1$, $p_2 = -2x^2 + 3x + 5$, *and* $p_3 = -x^2 + 3x + 1$ form a linearly dependent set in p_2 since $3p_1 - p_2 + 2p_3 = 0$.

Note The linearly independent or linearly dependent sets can also be determined using the Echelon Form or the Reduced Row Echelon Form methods.

Theorem 1 An indexed set { $\boldsymbol{v_1}$, ... , $\boldsymbol{v_p}$ } of two or more vectors, with $\boldsymbol{v_1 \neq 0}$, is linearly dependent if and only if some $\boldsymbol{v_j}$ (with $j > 1$) is a linear combination of the preceding vectors, $\boldsymbol{v_1, \ldots, v_{j-1}}$.

The main difference between linear dependence in $\boldsymbol{R^n}$ and in a general vector space is that when the vectors are not n – tuples, the homogeneous equation (1) usually cannot be written as a system of *n* linear equations. That is, the vectors cannot be made into the columns of a matrix $\boldsymbol{A}$ in order to study the equation $\boldsymbol{Ax = 0}$. We must rely instead on the definition of linear dependence and on Theorem 1.

Example 1 Let $\boldsymbol{p_1(t) = 1}$, $\boldsymbol{p_2(t) = t}$ and $\boldsymbol{p_3(t) = 4 - t}$. Then { $\boldsymbol{p_1, p_2, p_3}$} is linearly dependent in $\boldsymbol{P}$ because $\boldsymbol{p_3 = 4p_1 - p_2}$.

Example 2 The set *{**Sin** t, **Cos** t}* is linearly independent in $\boldsymbol{C}$ *[0, 1]* because ***Sin*** *t* and ***Cos*** *t* are not multiples of one another as vectors in $\boldsymbol{C}$ *[0, 1]*. That is, there is no scalar *c* such that ***Cos*** *t = c.* ***Sin*** *t* for all *t* in *[0, 1]*. (Look at the graphs of ***Sin*** *t* and ***Cos*** *t*.) However, *{**Sin** t **Cos** t, **Sin** 2t}* is linearly dependent because of he identity:
Sin *2t = 2* ***Sin*** *t* ***Cos*** *t*, for all *t*.

Useful results

- A set containing the zero vector is linearly dependent.
- A set of two vectors is linearly dependent if and only if one is a multiple of the other.
- A set containing one nonzeoro vector is linearly independent. i.e. consider the set containing one nonzeoro vector $\{v_1\}$ so $\{v_1\}$ is linearly independent when $v_1 \neq 0$.
- A set of two vectors is linearly independent if and only if neither of the vectors is a multiple of the other.

Activity Determine whether the following sets of vectors are linearly independent or linearly dependent:

1. $i=(1,0,0,0),\ j=(0,1,0,0),k=(0,0,0,1)$ in $\mathbb{R}^4$.
2. $v_1=(2,0,-1),\ v_2=(-3,-2,-5),\ v_3=(-6,1,-1),\ v_4=(-7,0,2)$ in $\mathbb{R}^3$.
3. $i=(1,0,0,...,0),\ j=(0,1,0,...,0),k=(0,0,0,...,1)$ in $\mathbb{R}^m$.
4. $3x^2+3x+1,\ 4x^2+x,\ 3x^2+6x+5,\ -x^2+2x+7$ in p_2

Definition Let $\boldsymbol{H}$ be a subspace of a vector space $\boldsymbol{V}$. An indexed set of vectors $\boldsymbol{B} = \{\boldsymbol{b_1},\ldots,\boldsymbol{b_p}\}$ in **V** is a **basis** for $\boldsymbol{H}$ if

(i) $\boldsymbol{B}$ is a linearly independent set, and
(ii) the subspace spanned by $\boldsymbol{B}$ coincides with $\boldsymbol{H}$; that is,
$\boldsymbol{H} = Span\ \{\boldsymbol{b_1},\ldots,\boldsymbol{b_p}\}$

The definition of a basis applies to the case when $\boldsymbol{H} = \boldsymbol{V}$, because any vector space is a subspace of itself. Thus a basis of $\boldsymbol{V}$ is a linearly independent set that spans $\boldsymbol{V}$. Observe that when $\boldsymbol{H} \neq \boldsymbol{V}$, condition (ii) includes the requirement that each of the vectors $\boldsymbol{b_1},\ldots,\boldsymbol{b_p}$ must belong to $\boldsymbol{H}$, because $Span\ \{\boldsymbol{b_1},\ldots,\boldsymbol{b_p}\}$ contains $\boldsymbol{b_1},\ldots,\boldsymbol{b_p}$, as we saw in lecture 21.

Example 3 Let A be an invertible $n \times n$ matrix – say, $A = [\boldsymbol{a_1}\ \ldots\ \boldsymbol{a_n}]$. Then the columns of A form a basis for $\boldsymbol{R^n}$ because they are linearly independent and they span $\boldsymbol{R^n}$, by the Invertible Matrix Theorem.

Example 4 Let $\boldsymbol{e_1},\ldots,\boldsymbol{e_n}$ be the columns of the $n \times n$ identity matrix, I_n. That is,

$$\boldsymbol{e}_1=\begin{bmatrix}1\\0\\ \vdots \\0\end{bmatrix},\quad \boldsymbol{e}_2=\begin{bmatrix}0\\1\\ \vdots \\0\end{bmatrix},\quad \ldots\quad \boldsymbol{e}_n=\begin{bmatrix}0\\0\\ \vdots \\1\end{bmatrix}$$

The set $\{\boldsymbol{e_1},\ \ldots,\ \boldsymbol{e_n}\}$ is called the standard basis for $\boldsymbol{R^n}$ (Fig. 1).

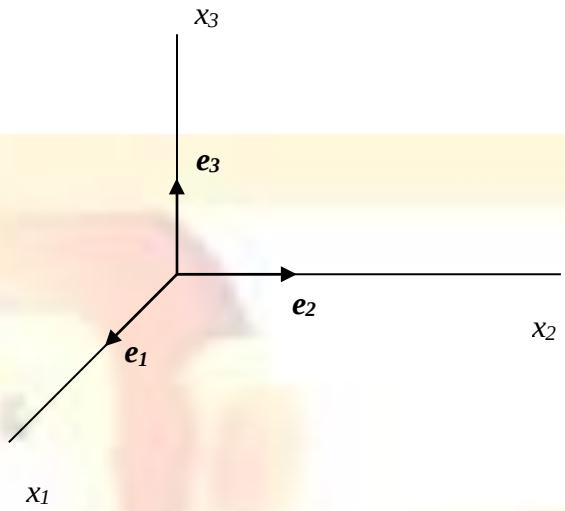


Figure 1 - The standard basis for R^3

Example 5 Let $\boldsymbol{v_1} = \begin{bmatrix} 3 \\ 0 \\ -6 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}$, and $\boldsymbol{v_3} = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$. Determine if $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ is a basis for $\boldsymbol{R^3}$.

Solution Since there are exactly three vectors here in $\boldsymbol{R^3}$, we can use one of any methods to determine whether they are basis for $\mathbb{R}^3$ or not. For this, let solve with help of matrices. First form a matrix of vectors i.e. matrix $A = [\boldsymbol{v_1} \quad \boldsymbol{v_2} \quad \boldsymbol{v_3}]$. If this matrix is invertible (i.e. $|A| \neq 0$ determinant should be non zero).

For instance, a simple computation shows that $\det A = 6 \neq 0$. Thus A is invertible. As in example 3, the columns of A form a basis for $\boldsymbol{R^3}$.

Example 6 Let $\boldsymbol{S} = \{\boldsymbol{1}, \boldsymbol{t}, \boldsymbol{t^2}, \ldots, \boldsymbol{t^n}\}$. Verify that $\boldsymbol{S}$ is a basis for $\boldsymbol{P_n}$. This basis is called the **standard basis** for $\boldsymbol{P_n}$.

Solution Certainly $\boldsymbol{S}$ spans $\boldsymbol{P_n}$. To show that $\boldsymbol{S}$ is linearly independent, suppose that $c_0, \ldots, c_n$ satisfy

$$c_0\boldsymbol{.1} + c_1\boldsymbol{t} + c_2\boldsymbol{t^2} + \ldots\ldots + c_n\boldsymbol{t^n} = \boldsymbol{0}\ \boldsymbol{(t)} \qquad (2)$$

This equality means that the polynomial on the left has the same values as the zero polynomial on the right. A fundamental theorem in algebra says that the only polynomial in $\boldsymbol{P_n}$ with more than n zeros is the zero polynomial. That is, (2) holds for all **t** only if $c_0 = \ldots = c_n = 0$. This proves that S is linearly independent and hence is a basis for $\boldsymbol{P_n}$. See Figure 2.

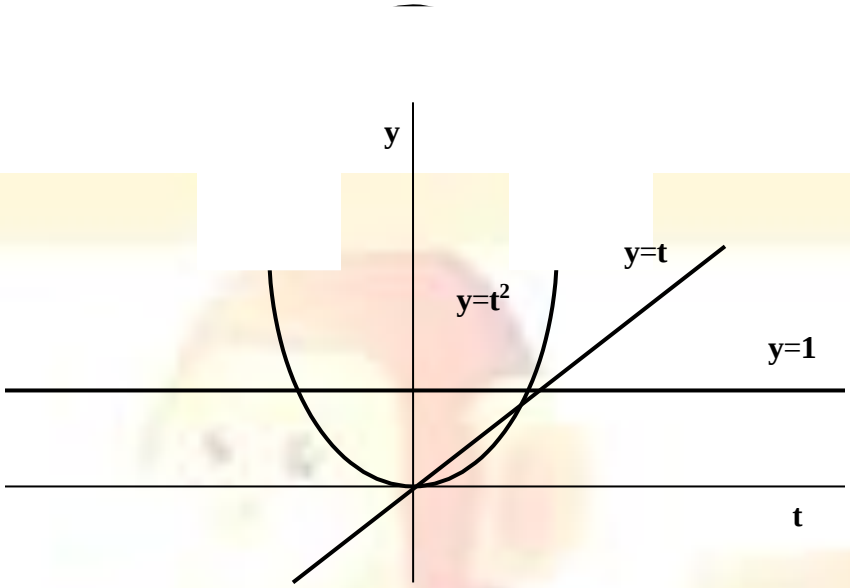


Figure 2 – The standard basis for P_2

Problems involving linear independence and spanning in $\boldsymbol{P_n}$ are handled best by a technique to be discussed later.

Example 7 Check whether the set of vectors *{(2, -3, 1), (4, 1, 1), (0, -7, 1)}* is basis for $\boldsymbol{R^3}$?

Solution The set $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ of vectors in $\boldsymbol{R^3}$ spans $\boldsymbol{V} = \boldsymbol{R^3}$ if

$$c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\boldsymbol{v_3} = d_1\boldsymbol{w_1} + d_2\boldsymbol{w_2} + d_3\boldsymbol{w_3} \qquad (*)$$

with $\boldsymbol{w_1} = (1,0,0)$, $\boldsymbol{w_2} = (0,1,0)$, $\boldsymbol{w_3} = (0,0,1)$ has at least one solution for every set of values of the coefficients d_1, d_2, d_3. Otherwise (i.e., if no solution exists for at least some values of d_1, d_2, d_3), $\boldsymbol{S}$ does not span $\boldsymbol{V}$. With our vectors $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}$, (*) becomes

$$c_1(2,-3,1) + c_2(4,1,1) + c_3(0,-7,1) = d_1(1,0,0) + d_2(0,1,0) + d_3(0,0,1)$$

Rearranging the left hand side yields

$$\begin{aligned} 2c_1 + 4c_2 + 0c_3 &= 1d_1 + 0d_2 + 0d_3 \\ -3c_1 + 1c_2 - 7c_3 &= 0d_1 + 1d_2 + 0d_3 \qquad (A) \\ 1c_1 + 1c_2 + 1c_3 &= 0d_1 + 0d_2 + 1d_3 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

We now find the determinant of coefficient matrix $\begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{bmatrix}$ to determine whether the system is consistent (so that $\boldsymbol{S}$ spans $\boldsymbol{V}$), or inconsistent ($\boldsymbol{S}$ does not span $\boldsymbol{V}$).

Now $\det \begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{bmatrix} = 2(8) - 4(4) + 0 = 0$

Therefore, the system (A) is inconsistent, and, consequently, the set $\boldsymbol{S}$ does not span the space $\boldsymbol{V}$.

Example 8 Check whether the set of vectors
$\{-4 + 1\,\boldsymbol{t} + 3\,\boldsymbol{t}^2,\ 6 + 5\,\boldsymbol{t} + 2\,\boldsymbol{t}^2,\ 8 + 4\,\boldsymbol{t} + 1\,\boldsymbol{t}^2\}$ is a basis for $\boldsymbol{P_2}$?

Solution The set $\boldsymbol{S} = \{\boldsymbol{p_1}(t), \boldsymbol{p_2}(t), \boldsymbol{p_3}(t)\}$ of vectors in $\boldsymbol{P_2}$ spans $\boldsymbol{V} = \boldsymbol{P_2}$ if

$$c_1\,\boldsymbol{p_1}(t) + c_2\,\boldsymbol{p_2}(t) + c_3\,\boldsymbol{p_3}(t) = d_1\,\boldsymbol{q_1}(t) + d_2\,\boldsymbol{q_2}(t) + d_3\,\boldsymbol{q_3}(t) \qquad (*)$$

with $\boldsymbol{q_1}(t) = \boldsymbol{1} + 0\,\boldsymbol{t} + 0\,\boldsymbol{t}^2$, $\boldsymbol{q_2}(t) = \boldsymbol{0} + 1\,\boldsymbol{t} + 0\,\boldsymbol{t}^2$, $\boldsymbol{q_3}(t) = \boldsymbol{0} + 0\,\boldsymbol{t} + 1\,\boldsymbol{t}^2$ has at least one solution for every set of values of the coefficients d_1, d_2, d_3. Otherwise (i.e., if no solution exists for at least some values of d_1, d_2, d_3), $\boldsymbol{S}$ does not span $\boldsymbol{V}$. With our vectors $\boldsymbol{p_1}(t), \boldsymbol{p_2}(t), \boldsymbol{p_3}(t)$, (*) becomes:

$$c_1(-4 + 1\,\boldsymbol{t} + 3\,\boldsymbol{t}^2) + c_2(6 + 5\,\boldsymbol{t} + 2\,\boldsymbol{t}^2) + c_3(8 + 4\,\boldsymbol{t} + 1\,\boldsymbol{t}^2) =$$
$$d_1(1 + 0\,\boldsymbol{t} + 0\,\boldsymbol{t}^2) + d_2(0 + 1\,\boldsymbol{t} + 0\,\boldsymbol{t}^2) + d_3(0 + 0\,\boldsymbol{t} + 1\,\boldsymbol{t}^2)$$

Rearranging the left hand side yields

$$(-4c_1 + 6c_2 + 8c_3)\boldsymbol{1} + (1c_1 + 5c_2 + 4c_3)\,\boldsymbol{t} + (3c_1 + 2c_2 + 1c_3)\,\boldsymbol{t}^2 =$$
$$(1d_1 + 0d_2 + 0d_3)\boldsymbol{1} + (0d_1 + 1d_2 + 0d_3)\,\boldsymbol{t} + (0d_1 + 0d_2 + 1d_3)\,\boldsymbol{t}^2$$

In order for the equality above to hold for all values of t, the coefficients corresponding to the same power of **t** on both sides of the equation must be equal. This yields the following system of equations:

$$\begin{aligned} -4c_1 + 6c_2 + 8c_3 &= 1d_1 + 0d_2 + 0d_3 \\ 1c_1 + 5c_2 + 4c_3 &= 0d_1 + 1d_2 + 0d_3 \qquad (A) \\ 3c_1 + 2c_2 + 1c_3 &= 0d_1 + 0d_2 + 1d_3 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

We now find the determinant of coefficient matrix $\begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$ to determine whether the system is consistent (so that $\boldsymbol{S}$ spans $\boldsymbol{V}$), or inconsistent ($\boldsymbol{S}$ does not span $\boldsymbol{V}$).

Now $\det \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = -26 \neq 0$. Therefore, the system (A) is consistent, and, consequently, the set $\boldsymbol{S}$ spans the space $\boldsymbol{V}$.

The set $\boldsymbol{S} = \{\boldsymbol{p_1}(t), \boldsymbol{p_2}(t), \boldsymbol{p_3}(t)\}$ of vectors in $\boldsymbol{P_2}$ is linearly independent if the only solution of

$$c_1\,\boldsymbol{p_1}(t) + c_2\,\boldsymbol{p_2}(t) + c_3\,\boldsymbol{p_3}(t) = \boldsymbol{0} \qquad (**)$$

is $c_1, c_2, c_3 = 0$. In this case, the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$. Otherwise (i.e., if a solution with at least some nonzero values exists), $\boldsymbol{S}$ is linearly dependent. With our vectors $\boldsymbol{p_1(t)}, \boldsymbol{p_2(t)}, \boldsymbol{p_3(t)}$, (2) becomes: $c_1(-4 + 1\,t + 3\,t^2) + c_2(6 + 5\,t + 2\,t^2) + c_3(8 + 4\,t + 1\,t^2) = 0$ Rearranging the left hand side yields

$$(-4c_1 + 6c_2 + 8c_3)1 + (1c_1 + 5c_2 + 4c_3)\,t + (3c_1 + 2c_2 + 1c_3)\,t^2 = 0$$

This yields the following homogeneous system of equations:

$$\begin{array}{l} -4c_1 + 6c_2 + 8c_3 = 0 \\ 1c_1 + 5c_2 + 4c_3 = 0 \\ 3c_1 + 2c_2 + 1c_3 = 0 \end{array} \Rightarrow \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

As $det \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = -26 \neq 0$. Therefore the set $\boldsymbol{S} = \{\boldsymbol{p_1(t)}, \boldsymbol{p_2(t)}, \boldsymbol{p_3(t)}\}$ is linearly independent. Consequently, the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$.

Example 9 The set $\boldsymbol{S} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for the vector space $\boldsymbol{V}$ of all 2 x 2 matrices.

Solution To verify that $\boldsymbol{S}$ is linearly independent, we form a linear combination of the vectors in $\boldsymbol{S}$ and set it equal to zero:

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This gives $\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, which implies that $c_1 = c_2 = c_3 = c_4 = 0$. Hence $\boldsymbol{S}$ is linearly independent.

To verify that $\boldsymbol{S}$ spans $\boldsymbol{V}$ we take any vector $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ in $\boldsymbol{V}$ and we must find scalars $c_1, c_2,$ c_3, and c_4 such that

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We find that $c_1 = a$, $c_2 = b$, $c_3 = c$, and $c_4 = d$ so that $\boldsymbol{S}$ spans $\boldsymbol{V}$.

The basis $\boldsymbol{S}$ in this example is called the standard basis for $\boldsymbol{M_{22}}$. More generally, the standard basis for $\boldsymbol{M_{mn}}$ consists of mn different matrices with a single 1 and zeros for the remaining entries

Example 10 Show that the set of vectors

$$\left\{ \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \right\}$$

is a basis for the vector space V of all 2 x 2 matrices (i.e. $\boldsymbol{M_{22}}$).

Solution The set $S = \{v_1, v_2, v_3, v_4\}$ of vectors in M_{22} spans $V = M_{22}$ if

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = d_1 w_1 + d_2 w_2 + d_3 w_3 + d_4 w_4 \quad (*)$$

with $w_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, w_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, w_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, w_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

has at least one solution for every set of values of the coefficients d_1, d_2, d_3, d_4. Otherwise (i.e., if no solution exists for at least some values of d_1, d_2, d_3, d_4), S does not span V. With our vectors v_1, v_2, v_3, v_4, (*) becomes:

$$c_1 \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix} + c_2 \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

$$= d_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + d_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + d_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Rearranging the left hand side yields

$$\begin{bmatrix} 3c_1 + 0c_2 + 0c_3 + 1c_4 & 6c_1 - 1c_2 - 8c_3 + 0c_4 \\ 3c_1 - 1c_2 - 12c_3 - 1c_4 & -6c_1 + 0c_2 - 4c_3 + 2c_4 \end{bmatrix} =$$

$$\begin{bmatrix} 1d_1 + 0d_2 + 0d_3 + 0d_4 & 0d_1 + 1d_2 + 0d_3 + 0d_4 \\ 0d_1 + 0d_2 + 1d_3 + 0d_4 & 0d_1 + 0d_2 + 0d_3 + 1d_4 \end{bmatrix}$$

The matrix equation above is equivalent to the following system of equations

$$\begin{aligned} 3c_1 + 0c_2 + 0c_3 + 1c_4 &= 1d_1 + 0d_2 + 0d_3 + 0d_4 \\ 6c_1 - 1c_2 - 8c_3 + 0c_4 &= 0d_1 + 1d_2 + 0d_3 + 0d_4 \\ 3c_1 - 1c_2 - 12c_3 - 1c_4 &= 0d_1 + 0d_2 + 1d_3 + 0d_4 \\ -6c_1 + 0c_2 - 4c_3 + 2c_4 &= 0d_1 + 0d_2 + 0d_3 + 1d_4 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix}$$

We now find the determinant of coefficient matrix $A = \begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix}$ to determine whether the system is consistent (so that S spans V), or inconsistent (S does not span V).

Now $det\ (A) = 48 \neq 0$. Therefore, the system (A) is consistent, and, consequently, the set $\boldsymbol{S}$ spans the space $\boldsymbol{V}$.

Now, the set $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}\}$ of vectors in $\boldsymbol{M_{22}}$ is linearly independent if the only solution of $c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\boldsymbol{v_3} + c_4\boldsymbol{v_4} = 0$ is $c_1, c_2, c_3, c_4 = 0$. In this case the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$. Otherwise (i.e., if a solution with at least some nonzero values exists), $\boldsymbol{S}$ is linearly dependent. With our vectors $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}$, we have

$$c_1\begin{bmatrix}3 & 6\\ 3 & -6\end{bmatrix} + c_2\begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix} + c_3\begin{bmatrix}0 & -8\\ -12 & -4\end{bmatrix} + c_4\begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

Rearranging the left hand side yields

$$\begin{bmatrix}3c_1 + 0c_2 + 0c_3 + 1c_4 & 6c_1 - 1c_2 - 8c_3 + 0c_4\\ 3c_1 - 1c_2 - 12c_3 - 1c_4 & -6c_1 + 0c_2 - 4c_3 + 2c_4\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

The matrix equation above is equivalent to the following homogeneous equation.

$$\begin{bmatrix}3 & 0 & 0 & 1\\ 6 & -1 & -8 & 0\\ 3 & -1 & -12 & -1\\ -6 & 0 & -4 & 2\end{bmatrix}\begin{bmatrix}c_1\\ c_2\\ c_3\\ c_4\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\\ 0\end{bmatrix}$$

As $\det(A) = 48 \neq 0$

Therefore the set $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}\}$ is linearly independent. Consequently, the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$.

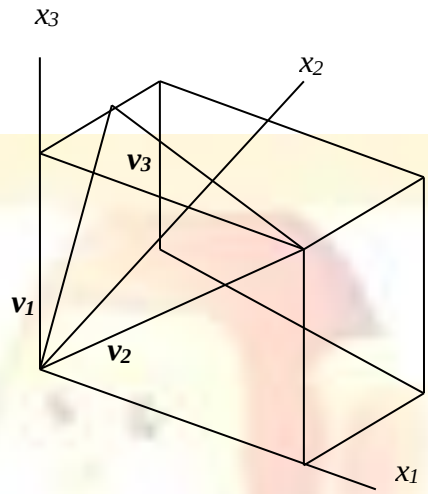
Example 11 Let $\boldsymbol{v}_1 = \begin{bmatrix}1\\ -2\\ -3\end{bmatrix}$, $\boldsymbol{v}_2 = \begin{bmatrix}-3\\ 5\\ 7\end{bmatrix}$, $\boldsymbol{v}_3 = \begin{bmatrix}-4\\ 5\\ 6\end{bmatrix}$, and $\boldsymbol{H} = Span\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$.

Note that $\boldsymbol{v_3} = 5\boldsymbol{v_1} + 3\boldsymbol{v_2}$ and show that Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ = Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$. Then find a basis for the subspace $\boldsymbol{H}$.

Solution

Every vector in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ belongs to $\boldsymbol{H}$ because

$$c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + 0\boldsymbol{v_3}$$



Now let $\boldsymbol{x}$ be any vector in $\boldsymbol{H}$ – say, $\boldsymbol{x} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\boldsymbol{v_3}$. Since $\boldsymbol{v_3} = 5\boldsymbol{v_1} + 3\boldsymbol{v_2}$, we may substitute

$$\boldsymbol{x} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\,(5\boldsymbol{v_1} + 3\boldsymbol{v_2})$$
$$= (c_1 + 5c_3)\,\boldsymbol{v_1} + (c_2 + 3c_3)\,\boldsymbol{v_2}$$

Thus $\boldsymbol{x}$ is in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, so every vector in $\boldsymbol{H}$ already belongs to Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$. We conclude that $\boldsymbol{H}$ and *Span* $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ are actually the same set of vectors. It follows that $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis of $\boldsymbol{H}$ since $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is obviously linearly independent.

Activity Show that the following set of vectors is basis for $\mathbb{R}^3$:

1.

$$v_1 = (1, 0, 0),\ v_2 = (0, 2, 1),\ v_1 = (3, 0, 1)$$

2.

$$v_1 = (1, 2, 3),\ v_2 = (0, 1, 1),\ v_1 = (0, 1, 3)$$

The Spanning Set Theorem

As we will see, a basis is an "efficient" spanning set that contains no unnecessary vectors. In fact, a basis can be constructed from a spanning set by discarding unneeded vectors.

Theorem 2 **(The Spanning Set Theorem)** Let $\boldsymbol{S} = \{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$ be a set in $\boldsymbol{V}$ and let $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$.

- **a.** If one of the vectors in $\boldsymbol{S}$ – say, $\boldsymbol{v_k}$ – is a linear combination of the remaining vectors in $\boldsymbol{S}$, then the set formed from $\boldsymbol{S}$ by removing $\boldsymbol{v_k}$ still spans $\boldsymbol{H}$.
- **b.** If $\boldsymbol{H} \neq \{\boldsymbol{0}\}$, some subset of $\boldsymbol{S}$ is a basis for $\boldsymbol{H}$.

Since we know that span is the set of all linear combinations of some set of vectors and basis is a set of linearly independent vectors whose span is the entire vector space. The spanning set is a set of vectors whose span is the entire vector space. "The Spanning set theorem" is that a spanning set of vectors always contains a subset that is a basis.

Remark Let $V = R^m$ and let $S = \{v_1, v_2, \ldots, v_n\}$ be a set of nonzero vectors in V.

Procedure

The procedure for finding a subset of S that is a basis for W = span S is as follows:

Step 1 Write the Equation,

$$c_1v_1 + c_2v_2 + \ldots + c_n v_n = 0 \qquad (3)$$

Step 2 Construct the augmented matrix associated with the homogeneous system of Equation (1) and transforms it to reduced row echelon form.

Step 3 The vectors corresponding to the columns containing the leading 1's form a basis for W = span S.

Thus if $S = \{v_1, v_2, \ldots, v_6\}$ and the leading 1's occur in columns *1*, *3*, and *4*, then $\{v_1, v_3, v_4\}$ is a basis for span S.

Note In step 2 of the procedure above, it is sufficient to transform the augmented matrix to row echelon form.

Example 12 Let $S = \{v_1, v_2, v_3, v_4, v_5\}$ be a set of vectors in R^4, where $v_1 = (1,2,-2,1)$, $v_2 = (-3,0,-4,3)$, $v_3 = (2,1,1,-1)$, $v_4 = (-3,3,-9,6)$, and $v_5 = (9,3,7,-6)$. Find a subset of S that is a basis for W = span S.

Solution Step 1 Form Equation (3),

$c_1(1,2,-2,1) + c_2(-3,0,-4,3) + c_3(2,1,1,-1) + c_4(-3,3,-9,6) + c_5(9,3.7,-6) = (0,0,0,0)$.

Step 2 Equating corresponding components, we obtain the homogeneous system

$$\begin{aligned} c_1 - 3c_2 + 2c_3 - 3c_4 + 9c_5 &= 0 \\ 2c_1 + c_3 + 3c_4 + 3c_5 &= 0 \\ -2c_1 - 4c_2 + c_3 - 9c_4 + 7c_5 &= 0 \\ c_1 + 3c_2 - c_3 + 6c_4 - 6c_5 &= 0 \end{aligned}$$

The reduced row echelon form of the associated augmented matrix is

$$\begin{bmatrix} 1 & 0 & ½ & 3/2 & 3/2 & : 0 \\ 0 & 1 & -1/2 & 3/2 & -5/2 & : 0 \\ 0 & 0 & 0 & 0 & 0 & : 0 \\ 0 & 0 & 0 & 0 & 0 & : 0 \end{bmatrix}$$

Step 3 The leading 1's appear in columns 1 and 2, so $\{v_1, v_2\}$ is a basis for W = span S.

Two Views of a Basis When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent. If an additional vector is deleted, it will not be a linear combination of the remaining vectors and hence the smaller set will no longer span ***V***. Thus a basis is a spanning set that is as small as possible.

A basis is also a linearly independent set that is as large as possible. If ***S*** is a basis for ***V***, and if ***S*** is enlarged by one vector – say, ***w*** – from ***V***, then the new set cannot be linearly independent, because *S* spans *V*, and *w* is therefore a linear combination of the elements in ***S***.

Example 13 The following three sets in $\boldsymbol{R^3}$ show how a linearly independent set can be enlarged to a basis and how further enlargement destroys the linear independence of the set. Also, a spanning set can be shrunk to a basis, but further shrinking destroys the spanning property.

$$\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix}\right\} \qquad \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix}\right\} \qquad \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix},\begin{bmatrix}7\\8\\9\end{bmatrix}\right\}$$

Linearly independent but does not span $\boldsymbol{R^3}$ — A basis for $\boldsymbol{R^3}$ — Spans $\boldsymbol{R^3}$ but is linearly dependent

Example 14 Let $\boldsymbol{v}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\boldsymbol{H} = \left\{\begin{bmatrix}s\\s\\0\end{bmatrix} : \sin R\right\}$. then every vector in ***H*** is a linear combination of $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ because $\begin{bmatrix}s\\s\\0\end{bmatrix} = s\begin{bmatrix}1\\0\\0\end{bmatrix} + s\begin{bmatrix}0\\1\\0\end{bmatrix}$. Is $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ a basis for ***H***?

Solution Neither $\boldsymbol{v_1}$ nor $\boldsymbol{v_2}$ is in ***H***, so $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ cannot a basis for ***H***. In fact, $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis for the plane of all vectors of the form $(c_1, c_2, 0)$, but ***H*** is only a line.

Activity Find a Basis for the subspace W in $\mathbb{R}^3$ spanned by the following sets of vectors:

1. $v_1 = (1,0,2), v_2 = (3,2,1), v_3 = (1,0,6), v_4 = (3,2,1)$
2. $v_1 = (1,2,2), v_2 = (3,2,1), v_3 = (1,1,7), v_4 = (7,6,4)$

Exercises

Determine which set in exercises 1-4 are bases for R^2 or R^3. Of the sets that are not bases, determine which one are linearly independent and which ones span R^2 or R^3. Justify your answers.

1. $\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$

2. $\begin{bmatrix}1\\-3\\0\end{bmatrix}, \begin{bmatrix}-2\\9\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\-3\\5\end{bmatrix}$

3. $\begin{bmatrix}1\\2\\-3\end{bmatrix}, \begin{bmatrix}-4\\-5\\6\end{bmatrix}$

4. $\begin{bmatrix}1\\-4\\3\end{bmatrix}, \begin{bmatrix}0\\3\\-1\end{bmatrix}, \begin{bmatrix}3\\-5\\4\end{bmatrix}, \begin{bmatrix}0\\2\\-2\end{bmatrix}$

5. Find a basis for the set of vectors in R^3 in the plane $x + 2y + z = 0$.

6. Find a basis for the set of vectors in R^2 on the line $y = 5x$.

7. Suppose $R^4 = \text{Span}\{v_1, v_2, v_3, v_4\}$. Explain why $\{v_1, v_2, v_3, v_4\}$ is a basis for R^4.

8. Explain why the following sets of vectors are not bases for the indicated vector spaces. (Solve this problem by inspection).
(a) $u_1 = (1, 2)$, $u_2 = (0, 3)$, $u_3 = (2, 7)$ for R^2
(b) $u_1 = (-1, 3, 2)$, $u_2 = (6, 1, 1)$ for R^3
(c) $p_1 = 1 + x + x^2$, $p_2 = x - 1$ for P_2
(d) $A = \begin{bmatrix}1 & 1\\2 & 3\end{bmatrix}, B = \begin{bmatrix}6 & 0\\-1 & 4\end{bmatrix}, C = \begin{bmatrix}3 & 0\\1 & 7\end{bmatrix}, D = \begin{bmatrix}5 & 1\\4 & 2\end{bmatrix}, E = \begin{bmatrix}7 & 1\\2 & 9\end{bmatrix}$ for M_{22}

9. Which of the following sets of vectors are bases for R^2?
(a) (2, 1), (3, 0) (b) (4, 1), (-7, -8) (c) (0, 0), (1, 3) (d) (3, 9), (-4, -12)

10. Let V be the space spanned by $v_1 = \text{Cos}^2 x$, $v_2 = \text{Sin}^2 x$, $v_3 = \cos 2x$.
(a) Show that $S = \{v_1, v_2, v_3\}$ is not a basis for V (b) Find a basis for V

In exercises 11-13, determine a basis for the solution space of the system.

11.
$$\begin{aligned} x_1 + x_2 - x_3 &= 0\\ -2x_1 - x_2 + 2x_3 &= 0\\ -x_1 + x_3 &= 0 \end{aligned}$$

12.
$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0\\ x_1 + 5x_3 &= 0\\ x_2 + x_3 &= 0 \end{aligned}$$

13.
$$x + y + z = 0$$
$$3x + 2y - 2z = 0$$
$$4x + 3y - z = 0$$
$$6x + 5y + z = 0$$

14. Determine bases for the following subspace of $\boldsymbol{R^3}$

(a) the plane $3x - 2y + 5z = 0$ (b) the plane $x - y = 0$

(c) the line $x = 2t$, $y = -t$, $z = 4t$ (d) all vectors of the form (a, b, c), where $b = a + c$

15. Find a standard basis vector that can be added to the set $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ to produce a basis for $\boldsymbol{R^3}$.

(a) $\boldsymbol{v_1} = (-1, 2, 3)$, $\boldsymbol{v_2} = (1, -2, -2)$ (b) $\boldsymbol{v_1} = (1, -1, 0)$, $\boldsymbol{v_2} = (3, 1, -2)$

16. Find a standard basis vector that can be added to the set $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ to produce a basis for $\boldsymbol{R^4}$.

$\boldsymbol{v_1} = (1, -4, 2, -3)$, $\boldsymbol{v_2} = (-3, 8, -4, 6)$

Lecture No.23